

SECTION 1

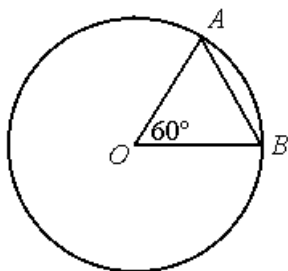
Time-30minutes

38 Questions

$$5y = 15$$

$$x = 2y$$

$$1. \quad x \quad 5$$



O is the center of the circle and the perimeter of $\triangle AOB$ is 6.

$$2. \quad \text{The circumference of the circle} \quad 12$$

Ken's monthly take-home pay is w dollars. After he pays for food and rent, he has x dollars left

$$3. \quad x \quad w - x$$

$$4. \quad \frac{\frac{13}{15} + \frac{7}{8} + \frac{3}{4}}{3} \quad 1$$

$$(x - 2y)(x + 2y) = 4$$

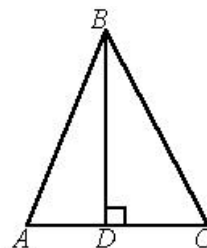
$$5. \quad x^2 - 4y^2 \quad 8$$

$$6. \quad \frac{0.3}{1.5} \quad \frac{2}{10}$$

The operation $\blacklozenge$ is defined for all positive numbers r

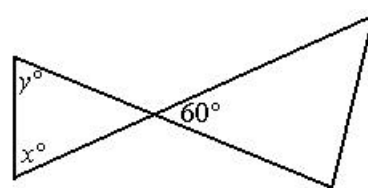
$$\text{and } t \text{ by } r \blacklozenge t = \frac{(r-t)^2 + rt}{t}$$

$$7. \quad 71 \blacklozenge 37 \quad 37 \blacklozenge 71$$



$$8. \quad \frac{BD}{AB} \quad \frac{DC}{BC}$$

$$9. \quad (250)(492) \quad \frac{492,000}{4}$$



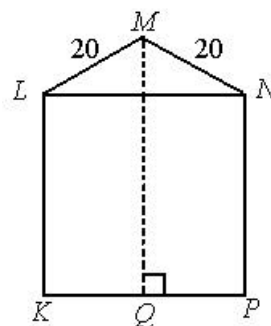
$$10. \quad x \quad y$$

11. The number of prime numbers between 70 and 76 The number of prime numbers between 30 and 36

$$6 < x < 7$$

$$y = 8$$

$$12. \quad \frac{x}{y} \quad 0.85$$



$KLNP$ is a square with perimeter 128.

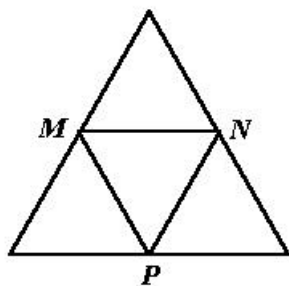
$$13. \quad MQ \quad 42$$

$$14. \quad \frac{2+3x}{2} \quad 1+3x$$

The median salary for professional group *A* is \$40,610. The median salary for professional group *B* is \$40,810.

15. The median salary for groups *A* and *B* combined is \$40,710
16. The water level in a tank is lowered by 6 inches, then raised by $8\frac{1}{2}$ inches, and then lowered by 4 inches. If the water level was x inches before the changes in level, which of the following represents the water level, in inches, after the changes?

- (A) $x - 1\frac{1}{2}$
- (B) $x + 1\frac{1}{2}$
- (C) $x - 6\frac{1}{2}$
- (D) $x + 6\frac{1}{2}$
- (E) $x - 18\frac{1}{2}$



17. In the figure above, *M*, *N*, and *P* are midpoints of the sides of an equilateral triangle whose perimeter is 18. What is the perimeter of the shaded region?

- (A) 2
- (B) 3
- (C) $4\frac{1}{2}$

- (D) 6
- (E) 9

18. Which of the following sets of number is has the greatest standard deviation?

- (A) 2, 3, 4
- (B) 2.5, 3, 3.5
- (C) 1, 1.25, 1.5
- (D) -2, 0, 2
- (E) 20, 21, 21.5

19. If x , y , and z represent consecutive integers, and $x < y < z$, which of the following equals y ?

- I. $x + 1$
- II. $\frac{x+z}{2}$
- III. $\frac{x+y+z}{3}$

- (A) I only
- (B) I and II only
- (C) I and III only
- (D) II and III only
- (E) I, II and III

20. When 9 students took a zoology quiz with a possible score of 0 to 10, inclusive, their average (arithmetic mean) score was 7.5. If a tenth student takes the same quiz, what will be the least possible average score on the quiz for all 10 students?

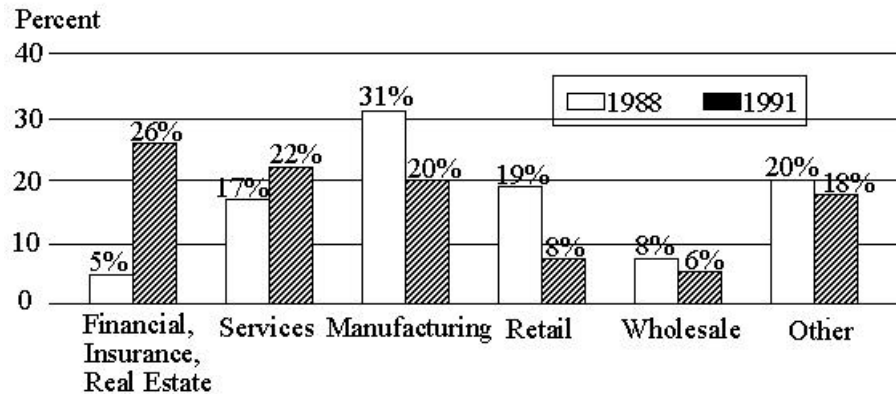
- (A) 6.5
- (B) 6.75
- (C) 7.0
- (D) 7.25
- (E) 7.5

Questions 21-25 refer to the following graph.

CORPORATE SUPPORT FOR THE ARTS BY SECTOR IN 1988 AND 1991

Total for 1988: \$630 million

Total for 1991: \$520 million



21. The two corporate sectors that increased their support for the arts from 1988 to 1991 made a total contribution in 1991 of approximately how many million dollars?
- (A) 112
(B) 125
(C) 200
(D) 250
(E) 315
22. How many of the six corporate sectors listed each contributed more than \$60 million to the arts in both 1988 and 1991?
- (A) One
(B) Two
(C) three
(D) Four
(E) Five
23. Approximately how many million dollars more did the wholesale sector contribute to the arts in 1988 than in 1991?
- (A) 10.4
(B) 12.6
(C) 14.0
(D) 16.5
(E) 19.2
24. From 1988 to 1991, which corporate sector decreased its support for the arts by the greatest dollar amount?
- (A) Services
(B) Manufacturing
(C) Retail
(D) Wholesale
(E) Other
25. Of the retail sector's 1991 contribution to the arts, $\frac{1}{4}$ went to symphony orchestras and $\frac{1}{2}$ of the remainder went to public television. Approximately how many million dollars more did the retail sector contribute to public television that year than to symphony orchestras?
- (A) 5.2
(B) 6.3
(C) 10.4
(D) 13.0
(E) 19.5
26. If $x = a^5$ and $y = a^6$, $a \neq 0$, which of the following is equivalent to a^{13} ?
- (A) xy
(B) x^2y

(C) $\frac{x^3}{y}$

(D) $\frac{x^4}{y}$

(E) $\frac{y^3}{x}$

27. The probabilities that each of two independent

experiments will have a successful outcome are $\frac{8}{15}$

and $\frac{2}{3}$, respectively. What is the probability that both experiments will have successful outcomes?

(A) $\frac{4}{5}$

(B) $\frac{6}{5}$

(C) $\frac{2}{15}$

(D) $\frac{16}{45}$

(E) $\frac{64}{225}$

28. If x is 1, 2, or 3 and y is either 2 or 4, then the product xy can have how many different possible values?

(A) Three

(B) Four

(C) Five

(E) Six

(E) Seven

29. If the radius of a circular region were decreased by 20 percent, the area of the circular region would decrease by what percent?

(A) 16%

(B) 20%

(C) 36%

(D) 40%

(E) 44%

30. Workers at Companies X and Y are paid the same

base hourly rate. Workers at company X are paid 1.5 times the base hourly rate for each hour worked per week in excess of the first 37, while workers at Company Y are paid 1.5 times the base hourly rate for each hour worked per week in excess of the first 40. In a given week, how many hours must a Company X worker work in order to receive the same pay as a company Y worker who works 46 hours?

(A) 46

(B) 45

(C) 44

(D) 43

(E) 42

SECTION 2

Time- 30 Minutes

38 Questions

1. As businesses become aware that their advertising must ----- the everyday concerns of consumers, their commercials will be characterized by a greater degree of -----.
 (A) allay...pessimism
 (B) address...realism
 (C) evade....verisimilitude
 (D) engage...fancy
 (E) change...sincerity
2. Because the lawyer's methods were found to be -----, the disciplinary committee ----- his privileges.
 (A) unimpeachable...suspended
 (B) ingenious...withdrew
 (C) questionable...expanded
 (D) unscrupulous...revoked
 (E) reprehensible...augmented
3. People of intelligence and achievement can nonetheless be so ----- and lacking in ----- that they gamble their reputations by breaking the law to further their own ends.
 (A) devious...propensity
 (B) culpable...prosperity
 (C) obsequious...deference
 (D) truculent... independence
 (E) greedy... integrity
4. A number of scientists have published articles ----- global warming, stating ----- that there is no solid scientific evidence to support the theory that the Earth is warming because of increases in greenhouse gases.
 (A) debunking...categorically
 (B) rejecting...paradoxically
 (C) deploring...optimistically
 (D) dismissing...hesitantly
 (E) proving...candidly
5. The senator's attempt to convince the public that she is not interested in running for a second term is as ----- as her opponent's attempt to disguise his intention to run against her.
 (A) biased
 (B) unsuccessful
 (C) inadvertent
 (D) indecisive
 (E) remote
6. MacCrory's conversation was -----: she could never tell a story, chiefly because she always forgot it, and she was never guilty of a witticism, unless by accident.
 (A) scintillating
 (B) unambiguous
 (C) perspicuous
 (D) stultifying
 (E) facetious
7. Despite its many -----, the whole-language philosophy of teaching reading continues to gain ----- among educators.
 (A) detractors...notoriety
 (B) adherents...prevalence
 (C) critics...currency
 (D) enthusiasts...popularity
 (E) practitioners... credibility
8. CENSUS: POPULATION::
 (A) interrogation : guilt
 (B) survey : price
 (C) interview : personality
 (D) questionnaire : explanation
 (E) inventory : stock
9. AUTHENTICITY : FRAUDULENT::
 (A) morality : utopian
 (B) intensity : vigorous
 (C) sincerity : hypocritical
 (D) particularity : unique
 (E) plausibility : narrated
10. VARNISH : GLOSSY::
 (A) sharpen : blunt
 (B) measure : deep
 (C) sand : smooth
 (D) approximate : precise
 (E) anchor : unstable

11. AMENITY : COMFORTABLE

- (A) tact : circumspect
- (B) nuisance : aggravated
- (C) honorarium : grateful
- (D) favorite : envious
- (E) lounge : patient

12. PAIN : ANALGESIC::

- (A) energy : revitalization
- (B) interest : stimulation
- (C) symptom : palliative
- (D) despair : anxiety
- (E) reward : incentive

13. VOICE:SHOUT::

- (A) ear : overhear
- (B) eye : see
- (C) hand : clutch
- (D) nerve : feel
- (E) nose : inhale

14. PONTIFICATE: SPEAK::

- (A) strut : walk
- (B) stare : look
- (C) patronize : frequent
- (D) eulogize : mourn
- (E) reciprocate : give

15. BIBLIOPHILE : BOOKS::

- (A) environmentalist : pollution
- (B) zoologist : animals
- (C) gourmet : food
- (D) calligrapher : handwriting
- (E) aviator : aircraft

16. INDIGENT : WEALTH::

- (A) presumptuous : independence
- (B) imperturbable : determination
- (C) inevitable : inescapability
- (D) indigestible : sustenance
- (E) redundant : indispensability

This passage is based on an article published in 1990.

Eight times within the past million years, something in the Earth's climatic equation has changed.

allowing snow in the mountains and the northern
Line latitudes to accumulate from one season to the next

(5) instead of melting away. Each time, the enormous ice sheets resulting from this continual buildup lasted tens of thousands of years until the end of each particular glacial cycle brought a warmer climate. Scientists speculated that these glacial cycles were ultimately
(10) driven by astronomical factors: slow, cyclic changes in the eccentricity of the Earth's orbit and in the tilt and orientation of its spin axis. But up until around 30 years ago, the lack of an independent record of ice-age timing made the hypothesis untestable.

(15) Then in the early 1950's Emiliani produced the first complete record of the waxings and wanings of past glaciations. It came from a seemingly odd place, the seafloor. Single-cell marine organisms called "foraminifera" house themselves in shells made
(20) from calcium carbonate. When the foraminifera die, sink to the bottom, and become part of seafloor sediments, the carbonate of their shells preserves certain characteristics of the seawater they inhabited. In particular, the ratio of a heavy, isotope of oxygen
(25) (oxygen-18) to ordinary oxygen (oxygen-16) in the carbonate preserves the ratio of the two oxygens in water molecules.

It is now understood that the ratio of oxygen isotopes in seawater closely reflects the proportion of
(30) the world's water locked up in glaciers and ice sheets. A kind of meteorological distillation accounts for the link. Water molecules containing the heavier isotope tend to condense and fall as precipitation slightly sooner than molecules containing the lighter isotope.

(35) Hence, as water vapor evaporated from warm oceans moves away from its source, its oxygen-18 returns more quickly to the oceans than does its oxygen-16. What falls as snow on distant ice sheets and mountain glaciers is relatively depleted of oxygen-18. As the
(40) oxygen-18-poor ice builds up the oceans become relatively enriched in the isotope. The larger the ice sheets grow, the higher the proportion of oxygen-18 becomes in seawater- and hence in the sediments.

Analyzing cores drilled from seafloor sediments,
(45) Emiliani found that the isotopic ratio rose and fell in rough accord with the Earth's astronomical cycles. Since that pioneering observation, oxygen-isotope measurements have been made on hundreds of cores

A chronology for the combined record enables scientists to show that the record contains the very same periodicities as the orbital processes. Over the past 800,000 years, the global ice volume has peaked every 100,000 years, matching the period of the orbital eccentricity variation. In addition, “wrinkles” superposed on each cycle – small decreases or surges in ice volume – have come at intervals of roughly 23,000 and 41,000 years, in keeping with the precession and tilt frequencies of the Earth’s spin axis.

17. Which of the following best expresses the main idea of the passage?

- (A) Marine sediments have allowed scientists to amass evidence tending to confirm that astronomical cycles drive the Earth’s glacial cycles.
- (B) the ratio between two different isotopes of oxygen in seawater correlates closely with the size of the Earth’s ice sheets.
- (C) Surprisingly, single-cell marine organisms provide a record of the Earth’s ice ages.
- (D) The Earth’s astronomical cycles have recently been revealed to have an unexpectedly large impact on the Earth’s climate.
- (E) The earth has experienced eight periods of intense glaciation in the past million years, primarily as a result of substantial changes in its orbit.

18. The passage asserts that one reason that oceans become enriched in oxygen – 18 as ice sheets grow is because

- (A) water molecules containing oxygen –18 condense and fall as precipitation slightly sooner than those containing oxygen –16
- (B) the ratio of oxygen- 18 to oxygen- 16 in water vapor evaporated from oceans is different from that of these isotopes in seawater
- (C) growing ice sheets tend to lose their oxygen- 18 as the temperature of the oceans near them gradually decreases
- (D) less water vapor evaporates from oceans during glacial periods and therefore less oxygen-18 is removed from the seawater
- (E) the freezing point of seawater rich in oxygen-18

is slightly lower than that of seawater poor in oxygen- 18

19. According to the passage, the large ice sheets typical of glacial cycles are most directly caused by

- (A) changes in the average temperatures in the tropics and over open oceans
- (B) prolonged increases in the rate at which water evaporates from the oceans
- (C) extreme seasonal variations in temperature in northern latitudes and in mountainous areas
- (D) steadily increasing precipitation rates in northern latitudes and in mountainous areas
- (E) the continual failure of snow to melt completely during the warmer seasons in northern latitudes and in mountainous areas

20. It can be inferred from the passage that which of the following is true of the water locked in glaciers and ice sheets today?

- (A) It is richer in oxygen- 18 than frozen water was during past glacial periods.
- (B) It is primarily located in the northern latitudes of the Earth.
- (C) Its ratio of oxygen isotopes is the same as that prevalent in seawater during the last ice age.
- (D) It is steadily decreasing in amount due to increased thawing during summer months.
- (E) In comparison with seawater, it is relatively poor in oxygen-18.

21. The discussion of the oxygen-isotope ratios in paragraph three of the passage suggests that which of the following must be assumed if the conclusions described in lines 49-58 are to be validly drawn?

- (A) The Earth's overall annual precipitation rates do not dramatically increase or decrease over time.
- (B) The various chemicals dissolved in seawater have had the same concentrations over the past million years.
- (C) Natural processes unrelated to ice formation do not result in the formation of large quantities of oxygen- 18.
- (D) Water molecules falling as precipitation usually fall on the open ocean rather than on continents

- or polar ice packs.
- (E) Increases in global temperature do not increase the amount of water that evaporates from the oceans.
22. The passage suggests that the scientists who first constructed a coherent, continuous picture of past variations in marine-sediment isotope ratios did which of the following?
- (A) Relied primarily on the data obtained from the analysis of Emiliani's core samples.
- (B) Combined data derived from the analysis of many different core samples.
- (C) Matched the data obtained by geologists with that provided by astronomers.
- (D) Evaluated the isotope-ratio data obtained in several areas in order to eliminate all but the most reliable data.
- (E) Compared data obtained from core samples in many different marine environments with data samples derived from polar ice caps.
23. The passage suggests that the scientists mentioned in line 8 considered their reconstruction of past astronomical cycles to be
- (A) unreliable because astronomical observations have been made and recorded for only a few thousand years
- (B) adequate enough to allow that reconstruction's use in explaining glacial cycles if a record of the latter could be found
- (C) in need of confirmation through comparison with an independent source of information about astronomical phenomena
- (D) incomplete and therefore unusable for the purposes of explaining the causes of ice ages
- (E) adequate enough for scientists to support conclusively the idea that ice ages were caused by astronomical changes
- Although Victor Turner's writings have proved fruitful for fields beyond anthropology, his definition of ritual is overly restrictive. Ritual, he says, is "prescribed formal behavior for occasions not given over to technological routine, having reference to beliefs in mystical beings or powers," "Technological routine"
- refers to the means by which a social group provides for its material needs. Turner's differentiating ritual from technology helps us recognize that festivals and (10) celebrations may have little purpose other than play, but it obscures the practical aims, such as making crops grow or healing patients, of other rituals. Further, Turner's definition implies a necessary relationship between ritual and mystical beliefs. However, not all (15) rituals are religious; some religions have no reference to mystical beings; and individuals may be required only to participate in, not necessarily believe in, a ritual. Turner's assumption that ritual behavior follows belief thus limits the usefulness of his definition in (20) studying ritual across cultures.
24. According to the passage, which of the following does Turner exclude from his conception of ritual?
- (A) Behavior based on beliefs
- (B) Behavior based on formal rules
- (C) Celebrations whose purpose is play
- (D) Routines directed toward practical ends
- (E) Festivals honoring supernatural beings
25. The passage suggests that an assumption underlying Turner's definition of ritual is that
- (A) anthropological concepts apply to other fields
- (B) festivals and ceremonies are related cultural phenomena
- (C) there is a relationship between play and practical ends
- (D) rituals refer only to belief in mystical beings or powers
- (E) mystical beings and powers have certain common attributes across cultures
26. It can be inferred that the author of the passage believes each of the following concerning rituals EXCEPT:
- (A) Some are unrelated to religious belief.
- (B) Some are intended to have practical consequences.
- (C) Some have no purpose other than play.
- (D) They sometimes involve reference to mystical beings.
- (E) They are predominantly focused on agricultural ends.

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| <p>27. Which of the following best describes the organization of the passage?</p> <p>(A) Factual data are presented and a hypothesis is proposed.</p> <p>(B) A distinction is introduced then shown not to be a true distinction.</p> <p>(C) A statement is quoted, and two assumptions on which it is based are clarified.</p> <p>(D) A definition is challenged, and two reasons for the challenge are given.</p> <p>(E) An opinion is offered and then placed within a historical framework.</p> <p>28. SLOUCH:</p> <p>(A) stand erect</p> <p>(B) move unhesitatingly</p> <p>(C) stretch languidly</p> <p>(D) scurry</p> <p>(E) totter</p> <p>29. CLAIM:</p> <p>(A) renounce</p> <p>(B) repeal</p> <p>(C) deter</p> <p>(D) hinder</p> <p>(E) postpone</p> <p>30. EXPEDITE:</p> <p>(A) impeach</p> <p>(B) deflect</p> <p>(C) resist</p> <p>(D) retard</p> <p>(E) remove</p> <p>31. VALEDICTION:</p> <p>(A) greeting</p> <p>(B) promise</p> <p>(C) accusation</p> <p>(D) denigration</p> <p>(E) aphorism</p> <p>32. FACTORABLE</p> <p>(A) absorbent</p> <p>(B) magnifiabl</p> <p>(C) simulated</p> | <p>(D) irreducible</p> <p>(E) ambiguous</p> <p>33. CONVOKE:</p> <p>(A) disturb</p> <p>(B) impress</p> <p>(C) adjourn</p> <p>(D) extol</p> <p>(E) applaud</p> <p>34. REND:</p> <p>(A) sink</p> <p>(B) unite</p> <p>(C) find</p> <p>(D) spend</p> <p>(E) unleash</p> <p>35. CONTRAVENE:</p> <p>(A) condescend</p> <p>(B) embark</p> <p>(C) support</p> <p>(D) offend</p> <p>(E) amass</p> <p>36. NADIR:</p> <p>(A) summit</p> <p>(B) impasse</p> <p>(C) sanctuary</p> <p>(D) weak point</p> <p>(E) direct route</p> <p>37. ABSTRACT:</p> <p>(A) deny</p> <p>(B) organize</p> <p>(C) elaborate</p> <p>(D) deliberate</p> <p>(E) produce</p> <p>38. MENDACIOUS:</p> <p>(A) assured</p> <p>(B) honest</p> <p>(C) intelligent</p> <p>(D) fortunate</p> <p>(E) gracious</p> |
|---|--|

SECTION 3

Time - 30 minutes

25 Questions

1. The painter Peter Brandon never dated his works, and their chronology is only now beginning to take shape in the critical literature. A recent dating of a Brandon self-portrait to 1930 is surely wrong. Brandon was 63 years old in 1930, yet the painting shows a young, dark-haired man-obviously Brandon, but clearly not a man of 63.

Which of the following, if justifiably assumed, allows the conclusion to be properly drawn?

- (A) There is no securely dated self-portrait of Brandon that he painted when he was significantly younger than 63.
 - (B) In refraining from dating his works, Brandon intended to steer critical discussion of them away from considerations of chronology.
 - (C) Until recently, there was very little critical literature on the works of Brandon.
 - (D) Brandon at age 63 would not have portrayed himself in a painting as he had looked when he was a young man.
 - (E) Brandon painted several self-portraits that showed him as a man past the age of 60.
2. Dance critic from Europe: The improved quality of ballet in the United States is the result of more Europeans' teaching ballet in the United States than ever before. I know the proportion of teachers who were born and trained in Europe has gone up among ballet teachers in the United States, because last year, on my trip to New York, more of the ballet teachers I met were from Europe-born and trained there -than ever before.
- Which of the following identifies a questionable assumption made by the dance critic's reasoning?
- (A) The argument overlooks the possibility that some ballet teachers in the United States could have been born in Europe but trained in the United States.
 - (B) The argument assumes that the ballet teachers whom the critic met last year on the critic's trip to New York were a generally typical group of such teachers.

- (C) The argument assumes that the teaching of ballet in the United States is superior to the teaching of ballet in Europe
- (D) Other possible reasons for the improved mental attitudes of United States dancers are not examined.
- (E) The argument assumes that dancers born and trained in Europe are typically more talented than dancers born and trained in the United States.

Questions 3-8

A volunteer who sends packages to hospital patients is preparing three packages containing exactly five items each from a supply of eighteen available items-four games, six jigsaw puzzles, and eight novels. The packages must conform to the following.

conditions:

The three packages together contain all of the novels. Each package contains at least one jigsaw puzzle. No package contains more games than novels.

3. Which of the following can be a complete and accurate list of the contents of one of the packages?
- (A) Five jigsaw puzzles
 - (B) One game, four novels
 - (C) One jigsaw puzzle, four novels
 - (D) Two games, two jigsaw puzzles, two novels
 - (E) Three games, one jigsaw puzzle, one novel
4. If the first two packages contain exactly two games each, then the third package must contain exactly
- (A) one jigsaw puzzle and four novels
 - (B) two jigsaw puzzles and three novels
 - (C) four jigsaw puzzles and one novel
 - (D) one game, one jigsaw puzzle, and three novels
 - (E) two games, one jigsaw puzzle and two novels
5. If one of the packages contains exactly three jigsaw puzzles and none of the packages contains more than three novels, which of the following must be true?
- (A) The package that contains three jigsaw puzzles also contains exactly one game.
 - (B) One of the two packages that do not contain three jigsaw puzzles contains exactly two games.
 - (C) One of the two packages that do not contain three

- jigsaw puzzles contains exactly two jigsaw puzzles.
- (D) Each of the two packages that do not contain three jigsaw puzzles contains exactly one game.
- (E) Each of the two packages that do not contain three jigsaw puzzles contains exactly three novels.
6. If the first two packages contain exactly two jigsaw puzzles each, which of the following can be a complete and accurate list of the contents of the third package?
- (A) One game, four novels
- (B) Two games, three novels
- (C) Two jigsaw puzzles, three novels
- (D) One game, three jigsaw puzzles, one novel
- (E) Two games, two jigsaw puzzles, one novel
7. If each of the packages contains at least one game, then it must be true that one of the package contains exactly
- (A) two games
- (B) two jigsaw puzzles
- (C) one novel
- (D) two novels
- (E) four novels
8. If each of the packages contains a different number of novels from the others, which of the following can be true?
- (A) There are exactly three games among the items in one of the packages.
- (B) There are exactly two jigsaw puzzles among the items in one of the packages.
- (D) There are exactly four games among the items in the three packages together.
- (E) There are exactly four jigsaw puzzles among the items in the three packages together.
9. Mayor Four years ago when we reorganized the city police department in order to save money, critics claimed that the reorganization would make the police less responsive to citizens and would thus lead to more crime. The police have compiled theft statistics from the years following the reorganization that show that the critics were wrong. There was an overall decrease in reports of thefts of all kinds, including small thefts.
- Which of the following, if true, most seriously challenges the mayor's argument?
- (A) When city police are perceived as unresponsive, victims of theft are less likely to report thefts to the police.
- (B) The mayor's critics generally agree that police statistics concerning crime reports provide the most reliable available data on crime rates.
- (C) In other cities where police departments have been similarly reorganized, the numbers of reported thefts have generally risen following reorganization.
- (D) The mayor's reorganization of the police department failed to save as much money as it was intended to save.
- (E) During the four years immediately preceding the reorganization, reports of all types of theft had been rising steadily in comparison to reports of other crimes.
10. It takes a particular talent to be a successful business manager. Business courses can help people to solve management problems, but such courses can do so only for those people with managerial talent. Such people should take business courses to acquire ideas that they can subsequently use to good advantage if management problems happen to arise.
- If the statements above are true, which of the following must also be true on the basis of them?
- (A) People who are helped by business courses in solving management problems also have managerial talent.
- (B) People who are already skilled at solving management problems are unlikely to benefit from business courses.
- (C) Most ideas that are used successfully in solving management problems are those acquired in business courses.
- (D) People who lack managerial talent are more likely to take business courses than are people who have managerial talent.
- (E) Those people who have never taken business courses are unable to solve management problems when such problems arise.

11. When a driver is suspected of having had too much to drink, testing the driver's ability to walk a straight line gives a more reliable indication of fitness to drive than does testing the driver's blood-alcohol level.

Which of the following, if true, best supports the claim made in the statement above?

- (A) Not all observers will agree whether or not an individual has succeeded in walking a straight line.
- (B) Because of genetic differences and variations in acquired tolerance to alcohol, some individuals suffer more serious motor impairment from a given high blood-alcohol level than do others.
- (C) Tests designed to measure blood-alcohol levels are accurate, inexpensive, and easy to administer.
- (D) More than half the drivers involved in fatal accidents have blood-alcohol levels that exceed the legal limit, whereas in less-serious accidents the proportion of legally intoxicated drivers is lower.
- (E) Some individuals with high blood-alcohol levels are capable of walking a straight line but are not capable of driving safely.

12. That sales can be increased by the presence of sunlight within a store has been shown by the experience of the only Savefast department store with a large skylight. The skylight allows sunlight into half of the store, reducing the need for artificial light. The rest of the store uses only artificial light. Since the store opened two years ago, the departments on the sunlit side have had substantially higher sales than the other departments. Which of the following, if true, most strengthens the argument?

- (A) On particularly cloudy days, more artificial light is used to illuminate the part of the store under the skylight.
- (B) When the store is open at night, the departments in the part of the store under the skylight have sales that are no higher than those of other departments.
- (C) Many customers purchase items from

departments in both parts of the store on a single shopping trip.

- (D) Besides the skylight, there are several significant architectural differences between the two parts of the store.
- (E) The departments in the part of the store under the skylight are the departments that generally have the highest sales in other stores in the Savefast chain.

Questions 13-17

A humanities course must discuss six out of eight topics—faith, knowledge, love, madness, revolution, skepticism, technology, and utopia—one at a time, each for one of six periods numbered consecutively from 1 through 6. The ordering of topics must meet these conditions:

- If faith is not discussed, utopia must be discussed last.
- If technology is discussed, it must be discussed immediately before or else immediately after love.
- If faith is discussed, it must be discussed immediately before skepticism and immediately after madness.
- Knowledge or else revolution must be discussed first.

13. Which of the following is an acceptable sequence of topics discussed, in order from first through sixth?

- (A) Knowledge, love, madness, faith, skepticism, technology, love
- (B) Knowledge, madness, utopia, skepticism, technology
- (C) Love, technology, revolution, madness, faith, skepticism
- (D) Revolution, madness, faith, skepticism, love, technology
- (E) Revolution, madness, skepticism, faith, technology, love

14. If exactly one topic is discussed between faith and love, that topic could be

- (A) knowledge
- (B) revolution
- (C) skepticism
- (D) technology
- (E) utopia

15. If neither faith nor madness is discussed and if revolution is discussed fourth, then skepticism must be discussed

- (A) first
- (B) second
- (C) third
- (D) fourth
- (E) fifth

16. If revolution and utopia are the first two topics discussed, the two topics not discussed could be

- (A) faith and love
- (B) faith and technology
- (C) knowledge and skepticism
- (D) love and madness
- (E) love and technology

17. If knowledge is not discussed, the other topic not discussed could be

- (A) faith
- (B) love
- (C) madness
- (D) revolution
- (E) skepticism

Questions 18-22

A jeweler is setting eight gemstones—garnet, jade, malachite, opal, ruby, sapphire, turquoise, and zircon—around a circular bracelet. There are eight adjacent positions, numbered consecutively 1 through 8 around the bracelet, in which to set the stones, with position 8 adjacent to position 1. The setting of the stones must conform to the following conditions:

The ruby is adjacent to the zircon.

The garnet is adjacent to the zircon.

The jade is adjacent to the opal.

The jade is not adjacent to the malachite.

If the turquoise is set in position 2, the opal is set in position 3; otherwise, the opal is set in position 2.

18. Which of the following can be the order, from position 1 through position 8, of the stones set around the bracelet?

- (A) Jade, opal, malachite, ruby, zircon, garnet, sapphire, turquoise

(B) Jade, opal, sapphire, turquoise, garnet, ruby, zircon, malachite

(C) Malachite, turquoise, opal, jade, ruby, zircon, garnet, sapphire

(D) Turquoise, opal, jade, sapphire, garnet, zircon, ruby, malachite

(E) Turquoise, sapphire, opal, jade, garnet, zircon, ruby, malachite

19. If the turquoise is set in position 8, which of the following must be true?

- (A) The garnet is set in position 5.
- (B) The jade is set in position 1.
- (C) The jade is set in position 3.
- (D) The malachite is set in position 1.
- (E) The sapphire is set in position 1.

20. Which of the following is a position in which the zircon can be set?

- (A) 1
- (B) 2
- (C) 3
- (D) 4
- (E) 5

21. If the malachite is set in position 5, which of the following can be true?

- (A) The garnet is set in position 3.
- (B) The jade is set in position 4.
- (C) The opal is set in position 3.
- (D) The sapphire is set in position 6.
- (E) The zircon is set in position 1.

22. If the turquoise is set in position 2, which of the following can be true?

- (A) The garnet is set in position 1.
- (B) The jade is set in position 1.
- (C) The malachite is set in position 5.
- (D) The ruby is set in position 5.
- (E) The sapphire is set in position 4.

23. To protect beachfront buildings from ocean storms, ocean resorts have built massive seawalls between beaches and the buildings. Not only do the seawalls block off some buildings' ocean view, but the beaches themselves become ever narrower, because

sand can no longer creep inland as storms erode it at the water's edge.

If the information is correct, which of the following conclusions is most strongly supported on the basis of it?

- (A) Since the ferocity of ocean storms is increasing, increasingly high seawalls must be built between beaches and beachfront property.
- (B) Even when beaches are heavily used by people, they are necessary to the survival of the many wild species that use them.
- (C) Seawalls constructed to protect beachfront buildings will not themselves eventually be damaged by storms and will not require, if they are to protect the buildings, expensive repair or replacement.
- (D) The conservation of beaches for future generations should be the overriding goal of shore management at ocean coasts.
- (E) Trying to protect beachfront buildings by constructing seawalls is counterproductive in the long run for an oceanfront community wishing to maintain itself as a beach resort.

24. A study found that 70 percent of children surveyed in 1970 had at one time had cavities, whereas only 50 percent of those surveyed in 1985 had ever had cavities. The researchers concluded that the level of dental disease in children had declined between 1970 and 1985.

Which of the following, if true, would most seriously undermine the researchers' conclusion presented above?

- (A) Cavities are the most common kind of dental disease to which children are subject.
- (B) The children surveyed came from a broad variety of income backgrounds.
- (C) The children surveyed were selected from among students of teachers cooperating with the researchers.
- (D) The accuracy of cavity detection techniques has improved dramatically since 1970.
- (E) The children surveyed in 1985 were younger on average than those surveyed in 1970.

25. David: Since attempting to preserve every species

that is currently endangered is prohibitively expensive, the endangered species whose value to humanity is the greatest should be accorded the highest priority for preservation.

Karen: Such a policy would be unsound because it is impossible to predict the future value of a species, nor is it always possible to assess the present value of species whose contributions to humanity, though significant, are indirect.

Which of the following is the main point of Karen's reply to David?

- (A) Although it would be desirable to preserve all endangered species, doing so is not economically feasible.
- (B) Even if the value to humanity of a given species is known, that value should not be a factor in any decision on whether to expend effort to preserve that species.
- (C) Species whose contributions to humanity are direct should have a higher priority for preservation efforts than species whose contributions to humanity are only indirect.
- (D) Since the methods for deciding which species have the most value to humanity are imperfect, informed decisions cannot be made on the basis of the assessment of such value.
- (E) The preservation of endangered species whose value to humanity can be reliably predicted is more important than the preservation of species whose value for humanity is unpredictable.

SECTION 4

Time – 30 minutes

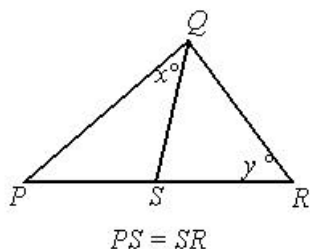
30 Questions

$$x = y = z$$

$$1. \quad x^3 \qquad xyz$$

$$x < 0$$

$$2. \quad 3x^2 \qquad 3x^3$$



$$3. \quad x \qquad y$$

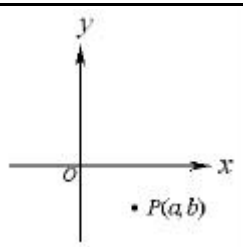
$$4. \quad \frac{24}{23} + \frac{101}{100} \qquad 2$$

The points $P(2,0)$, $Q(0,2)$, $R(4,2)$ and $S(2,4)$ are in the rectangular coordinate system.

$$5. \quad \begin{array}{cc} \text{The distance from} & \text{The distance from} \\ P \text{ to } Q & R \text{ to } S \end{array}$$

The probability that events E and F will both occur is 0.42

$$6. \quad \begin{array}{cc} \text{The probability that} & 0.58 \\ \text{event } E \text{ will occur} & \end{array}$$



$$7. \quad a \qquad b$$

$$8. \quad (1 + \sqrt{2})^2 \qquad 3$$

$$9. \quad \begin{array}{cc} (109)(87-14) & (109)(87)-(109)(14) \\ \text{Carol's age, in years, can be expressed by reversing} & \\ \text{the digits in her father's age, in years. The sum of} & \\ \text{the digits in each age is 10.} & \end{array}$$

$$10. \quad \begin{array}{cc} \text{The positive difference} & 36 \\ \text{between Carol's age, in} & \\ \text{years, and her father's} & \\ \text{age, in years} & \end{array}$$

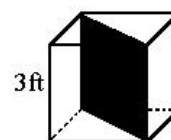
$$11. \quad \begin{array}{cc} 0 < p < 1 & \\ p^4 - p^6 & p^3 - p^5 \end{array}$$

$$3 - 2x^2 - [-x(1 + 2x)] = -5$$

$$12. \quad x \qquad -8$$

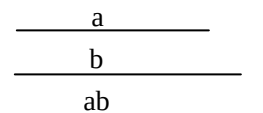
a and b are positive integers.

$$13. \quad \frac{a}{b} \qquad \frac{a+3}{b+3}$$



A solid cubical block of wood has dimensions as shown in the figure, and the block is to be cut in half as indicated by the shaded region.

$$14. \quad \begin{array}{cc} \text{The total surface area} & 36 \text{ square feet} \\ \text{of one of the resulting} & \\ \text{halves of the block} & \end{array}$$



The lengths of the line segments are a , b , and ab , respectively. The line segments are drawn to scale.

$$15. \quad a \qquad 1$$

$$16. \quad \begin{array}{c} \text{The average(arithmetic mean) number of students in} \\ \text{3 economics classes at a certain college is 24. If the} \\ \text{total number of students in 2 of the classes} \\ \text{combined is 38, how many students are in the} \\ \text{remaining class?} \end{array}$$

- (A) 14
(B) 19
(C) 24
(D) 31
(E) 34

17. If the cube of n is 180 greater than the square of n , then $n =$

(A) 10
(B) 9
(C) 8
(D) 7
(E) 6

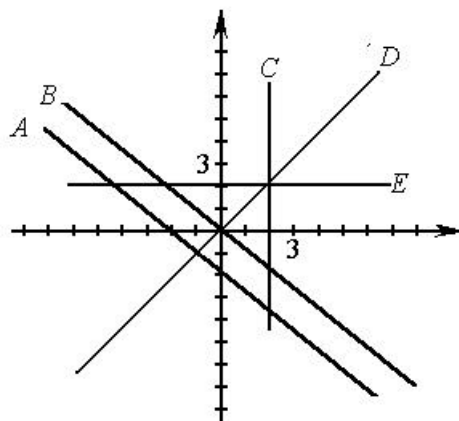


18. The circular clock above shows a time of exactly 3:30. What is the value of x ?

(A) 60
(B) 75
(C) 85
(D) 90
(E) 105

19. What percent of the integers between 200 and 999, inclusive, end with the digits "03"?

(A) 1%
(B) 25
(C) 3%
(D) 4%
(E) 5%



20. Which of the lines in the figure above contains only points (x,y) with $x = y$?

(A) A
(B) B
(C) C
(D) D
(E) E

Questions 21-35 refer to the following information about student enrollment in a certain small college.

DISTRIBUTION OF ENROLLMENT

BY CLASS AND SEX

(Total enrollment: 1,400)

	Males	Females
Freshmen	303	259
Sophomores	215	109
Juniors	182	88
Seniors	160	84
Total	860	540

PERCENT OF TOTAL ENROLLMENT

MAJORING IN EACH OF THE FOLLOWING ACADEMIC AREAS

(No student is majoring in more than one area.)

Area	Percent
Humanities	33%
Social Sciences	30%
Physical Sciences	24%

21. the ratio of the number of male freshmen to the number of female sophomores is approximately
- (A) 2 to 1
(B) 3 to 1
(C) 3 to 2
(D) 4 to 1
(E) 5 to 3
22. How many of the enrolled students are not majoring in humanities, social sciences, or physical sciences?
- (A) 87
(B) 122
(C) 182
(D) 230
(E) 322
23. Which of the following can be inferred from the tables?
- I. The number of males majoring in physical sciences is greater than the number of females majoring in that area.
II. Students majoring in either social sciences or physical sciences constitute more than 50 percent of the total enrollment.
III. The ratio of the number of males to the number of females in the senior class is less than 2 to 1.
- (A) I only
(B) II only
(C) I and II
(D) I and III
(E) II and III
24. How many students are either juniors or males or both?
- (A) 678
(B) 766
(C) 948
(D) 1,130
(E) 1,312
25. If the total enrollment is 12 percent greater than it was five years ago, what was the total enrollment five years ago?
- (A) 1,180
- (B) 1,192
(C) 1,220
(D) 1,232
(E) 1,250
26. If the ratio of the number of English books to the number of all other books on a bookshelf is 4 to 1, what percent of the books on the bookshelf are English books?
- (A) 20%
(B) 25%
(C) 50%
(D) 75%
(E) 80%
- 3, 7, 9, 14, x
27. The numbers in the list above are ordered from least to greatest. If the average (arithmetic mean) is 2 greater than the median, what is the value of x ?
- (A) 22
(B) 20
(C) 17
(D) 16
(E) 15
28. A developer has land that has x feet of lake frontage. The land is to be subdivided into lots, each of which is to have either 80 feet or 100 feet of lake frontage. If $\frac{1}{9}$ of the lots are to have 80 feet of frontage each and the remaining 40 lots are to have 100 feet of frontage each, what is the value of x ?
- (A) 400
(B) 3,200
(C) 3,700
(D) 4,400
(E) 4,760
29. If $\frac{a}{b} = \frac{3}{2}$, which of the following must be true?
- I $\frac{b}{a} = \frac{2}{3}$
II $\frac{a-b}{a} = \frac{1}{3}$

$$\text{III } a + b = 5$$

- (A) I only
 (B) II only
 (C) III only
 (D) I and II
 (E) II and III

30. What is the least integer value of n such that

$$\frac{1}{2^n} < 0.001 ?$$

- (A) 10
 (B) 11
 (C) 500
 (D) 501
 (E) there is no such least value.

SECTION 5

Time-30 minutes

38 Questions

- That she was _____ rock climbing did not diminish her _____ to join her friends on a rock-climbing expedition.
 (A) attracted to ...eagerness
 (B) timid about ... reluctance
 (C) fearful of ... determination
 (D) curious about ... aspiration
 (E) knowledgeable about ... hope
- Data concerning the effects on a small population of high concentrations of a potentially hazardous chemical are frequently used to _____ the effects on a large population of lower amounts of the same chemical.
 (A) verify
 (B) redress
 (C) predict
 (D) realize
 (E) augment
- Conceptually, it is hard to reconcile a defense attorney's _____ to ensure that false testimony is not knowingly put forward with the attorney's mandate to mount the most _____ defense conceivable for the client.
 (A) efforts ... cautious
 (B) duty ... powerful
 (C) inability ... eloquent
 (D) failure ... diversified
 (E) promises ... informed
- The term "modern" has always been used broadly by historians, and recent reports indicate that its meaning has become more _____ than ever.
 (A) precise
 (B) pejorative
 (C) revisionist
 (D) acceptable
 (E) amorphous
- He would _____ no argument, and to this end he enjoined us to _____.

(A) brook ... silence (B) acknowledge ... neglect (C) broach ... abstinence (D) fathom ... secrecy (E) tolerate ... defiance 6. Originally, most intellectual criticism of mass culture was ____ in character, being based on the assumption that the wider the appeal, the more ____ the product. (A) unpredictable ... undesirable (B) ironic ... popular (C) extreme ... outlandish (D) frivolous ... superfluous (E) negative ... shoddy 7. Surprisingly, given the dearth of rain that fell on the corn crop, the yield of the harvest was ____; consequently, the corn reserves of the country have not been ____. (A) inadequate ... replenished (B) encouraging ... depleted (C) compromised ... salvaged (D) abundant ... extended (E) disappointing ... harmed 8. REPELLENT: ATTRACT:: (A) elastic: stretch (B) sensitive: cooperate (C) progressive: change (D) flammable: ignite (E) ephemeral: endure 9. ANARCHIST: GOVERNMENT:: (A) legislator: taxation (B) reformer: bureaucracy (C) jurist: law (D) SUFFRAGIST : VOTING (E) abolitionist: slavery 10. ADMONISH: DENOUNCE:: (A) challenge: overcome (B) reward: praise (C) control: contain (D) persuade: convince (E) punish: pillory	11. JOKE: PUNCH LINE:: (A) sermon: congregation (B) conceit: allegory (C) rhetoric: persuasion (D) conspiracy: arrest (E) plot: denouement 12. VEER: DIRECTION:: (A) align: connection (B) filter: contamination (C) convert: belief (D) deflect: motivation (E) substantiate: authenticity 13. REPROBATE: MISBEHAVE:: (A) sycophant: fawn (B) critic: rebuke (C) ruffian: tease (D) cynic: brood (E) narcissist: covet 14. IMPERVIOUS: PENETRATE:: (A) ineluctable: avoid (B) ineradicable: damage (C) boorish: flatter (D) irrepressible: censure (E) disruptive: restrain 15. CONSENSUS: FACTIONALISM:: (A) ritual: orthodoxy (B) reality: plausibility (C) reason: thought (D) clarity: confusion (E) leadership: subordination 16. MARTINET: DISCIPLINE:: (A) illusionist: misdirection (B) dilettante: commitment (C) renegade: allegiance (D) pedant: learning (E) hack: writing Benjamin Franklin established that lightning is the transfer of positive or negative electrical charge between regions of a cloud or from cloud to earth.
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line Such transfers require that electrically neutral clouds, with uniform charge distributions, become electrified by separation of charges into distinct regions. The greater this separation is, the greater the voltage, or electrical potential of the cloud. Scientists still do not now the precise distribution of charges in thunderclouds nor how separation adequate to support the huge voltages typical of lightning bolts arises. According to one theory, the precipitation hypothesis, charge separation occurs as a result of precipitation. Larger droplets in a thundercloud precipitate downward past smaller suspended droplets. Collisions among droplets transfer negative charge to precipitating droplets, leaving the suspended droplets with a positive charge, thus producing a positive dipole in which the lower region of the thundercloud is filled with negatively charged raindrops and the upper with positively charged suspended droplets.

17. The passage is primarily concerned with discussing which of the following?

- (A) A central issue in the explanation of how lightning occurs
- (B) Benjamin Franklin's activities as a scientist
- (C) Research into the strength and distribution of thunderstorms
- (D) The direction of movement of electrical charges in thunderclouds
- (E) The relation between a cloud's charge distribution and its voltage

18. The passage suggests that lightning bolts typically

- (A) produce a distribution of charges called a positive dipole in the clouds where they originate
- (B) result in the movement of negative charges to the centers of the clouds where they originate
- (C) result in the suspension of large, positively charged raindrops at the tops of the clouds where they originate
- (D) originate in clouds that have large numbers of negatively charged droplets in their upper regions
- (E) originate in clouds in which the positive and negative charges are not uniformly distributed

19. According to the passage, Benjamin Franklin contributed to the scientific study of lightning by

- (A) testing a theory proposed earlier, showing it to be false, and developing an alternative, far more successful theory of his own
- (B) making an important discovery that is still important for scientific investigations of lightning
- (C) introducing a hypothesis that, though recently shown to be false, proved to be a useful source of insights for scientists studying lightning
- (D) developing a technique that has enabled scientists to measure more precisely the phenomena that affect the strength and location of lightning bolts
- (E) predicting correctly that two factors previously thought unrelated to lightning would eventually be shown to contribute jointly to the strength and location of lightning bolts

20. Which of the following, if true, would most seriously undermine the precipitation hypothesis, as it is set forth in the passage?

- (A) Larger clouds are more likely than smaller clouds to be characterized by complete separation of positive and negative charges.
- (B) In smaller clouds lightning more often occurs within the cloud than between the cloud and the earth.
- (C) Large raindrops move more rapidly in small clouds than they do in large clouds.
- (D) Clouds that are smaller than average in size rarely, if ever, produce lightning bolts.
- (E) In clouds of all sizes negative charges concentrate in the center of the clouds when the clouds become electrically charged

line Before Laura Gilpin (1891-1979), few women in the history of photography had so devoted themselves to chronicling the landscape. Other women had photographed the land, but none can be regarded as a landscape photographer with a sustained body of work documenting the physical terrain. Anne Brigman often photographed woodlands and coastal areas, but

They were generally settings for her artfully placed subjects. Dorothea Lange's landscapes were always
(10) conceived of as counterparts to her portraits of rural women.

At the same time that Gilpin's interest in landscape work distinguished her from most other women photographers, her approach to landscape photography set
(15) her apart from men photographers who, like Gilpin, documented the western United States. Western American landscape photography grew out of a male tradition, pioneered by photographers attached to government and commercial survey teams that went
(20) west in the 1860's and 1870's. These explorer-photographers documented the West that their employers wanted to see: an exotic and majestic land shaped by awesome natural forces, unpopulated and ready for American settlement. The next generation
(25) of male photographers, represented by Ansel Adams and Eliot Porter, often worked with conservationist groups rather than government agencies or commercial companies, but they nonetheless preserved the "heroic" style and maintained the role of respectful
(30) outsider peering in with reverence at a fragile natural world.

For Gilpin, by contrast, the landscape was neither an empty vista awaiting human settlement nor a jewel-like scene resisting human intrusion, but a
(35) peopled landscape with a rich history and tradition of its own, an environment that shaped and molded the lives of its inhabitants. Her photographs of the Rio Grande, for example, consistently depict the river in terms of its significance to human culture: as a source
(40) of irrigation water, a source of food for livestock, and a provider of town sites. Also instructive is Gilpin's general avoidance of extreme close-ups of her natural subjects: for her, emblematic details could never suggest the intricacies of the interrelationship between
(45) people and nature that made the landscape a compelling subject. While it is dangerous to draw conclusions about a "feminine" way of seeing from the work of one woman, it can nonetheless be argued that Gilpin's unique approach to landscape photography was anal-
(50) ogous to the work of many women writers who, far more than their male counterparts, described the landscape in terms of its potential to sustain human life. Gilpin never spoke of herself as a photographer

with a feminine perspective: she eschewed any
(55) discussion of gender as it related to her work and maintained little interest in interpretations that relied on the concept of a "woman's eye." Thus it is ironic that her photographic evocation of a historical landscape should so clearly present a distinctively feminine approach to landscape photography.

21. Which of the following best expresses the main idea of the passage?

- (A) Gilpin's landscape photographs more accurately documented the Southwest than did the photographs of explorers and conservationists.
- (B) Gilpin's style of landscape photography substantially influenced the heroic style practiced by her male counterparts.
- (C) The labeling of Gilpin's style of landscape photography as feminine ignores important ties between it and the heroic style.
- (D) Gilpin's work exemplifies an arguably feminine style of landscape photography that contrasts with the style used by her male predecessors.
- (E) Gilpin's style was strongly influenced by the work of women writers who described the landscape in terms of its relationship to people.

22. It can be inferred from the passage that the teams mentioned in line 19 were most interested in which of the following aspects of the land in the western United States?

- (A) Its fragility in the face of increased human intrusion
- (B) Its role in shaping the lives of indigenous peoples
- (C) Its potential for sustaining future settlements
- (D) Its importance as an environment for RARE PLANTS AND ANIMALS
- (E) Its unusual vulnerability to extreme natural forces

23. The author of the passage claims that which of the following is the primary reason why Gilpin generally avoided extreme close-ups of natural subjects?

- (A) Gilpin believed that pictures of natural details could not depict the interrelationship between the

- land and humans.
- (B) Gilpin considered close-up photography to be too closely associated with her predecessors.
- (C) Gilpin believed that all of her photographs should include people in them.
- (D) Gilpin associated close-up techniques with photography used for commercial purposes.
- (E) Gilpin feared that pictures of small details would suggest an indifference to the fragility of the land as a whole.
24. The passage suggests that a photographer who practiced the heroic style would be most likely to emphasize which of the following in a photographic series focusing on the Rio Grande ?
- (A) Indigenous people and their ancient customs relating to the river
- (B) The exploits of navigators and explorers
- (C) Unpopulated, pristine parts of the river and its surroundings
- (D) Existing commercial ventures that relied heavily on the river
- (E) The dams and other monumental engineering structures built on the river
25. It can be inferred from the passage that the first two generations of landscape photographers in the western United States had which of the following in common?
- (A) They photographed the land as an entity that had little interaction with human culture.
- (B) They advanced the philosophy that photographers should resist alliances with political or commercial groups.
- (C) They were convinced that the pristine condition of the land needed to be preserved by government action.
- (D) They photographed the land as a place ready for increased settlement.
- (E) They photographed only those locations where humans had settled.
26. Based on the description of her works in the passage, which of the following would most likely be a subject for a photograph taken by Gilpin?
- (A) A vista of a canyon still untouched by human culture
- (B) A portrait of a visitor to the West against a desert backdrop
- (C) A view of historic Native American dwellings carved into the side of a natural cliff
- (D) A picture of artifacts from the West being transported to the eastern United States for retail sale
- (E) An abstract pattern created by the shadows of clouds on the desert
27. The author of the passage mentions women writers in line 50 most likely in order to
- (A) counter a widely held criticism of her argument
- (B) bolster her argument that Gilpin's style can be characterized as a feminine style
- (C) suggest that Gilpin took some of her ideas for photographs from landscape descriptions by women writers
- (D) clarify the interrelationship between human culture and the land that Gilpin was attempting to capture
- (E) offer an analogy between photographic close-ups and literary descriptions of small details
28. FICTTTIOUS:
- (A) classical
- (B) natural
- (C) factual
- (D) rational
- (E) commonplace
29. BRIDLED:
- (A) without recourse
- (B) without restraint
- (C) without meaning
- (D) without curiosity
- (E) without subtlety
30. CAPTIVATE:
- (A) repulse
- (B) malign
- (C) proscribe
- (D) send out

(E) deliver from	(A) discontinuous
31. DISSIPATE:	(B) natural
(A) accumulate	(C) nascent
(B) emerge	(D) rational
(C) overwhelm	(E) dominant
(D) adhere	38. CATHOLIC:
(E) invigorate	(A) narrow
32. OSTRACIZE:	(B) soft
(A) clarify	(C) trivial
(B) subdue	(D) calm
(C) welcome	(E) quick
(D) renew	
(E) crave	
33. LOATH:	
(A) clever	
(B) reasonable	
(C) fortunate	
(D) eager	
(E) confident	
34. VITIATE:	
(A) ingratiate	
(B) convince	
(C) regulate	
(D) fortify	
(E) constrict	
35. LAVISH:	
(A) insist	
(B) criticize	
(C) undermine	
(D) stint	
(E) waste	
36. VITUPERATIVE:	
(A) complimentary	
(B) demagogic	
(C) hopeful	
(D) admirable	
(E) veracious	
37. MORIBUND:	

SECTION 6

Time-30 Minutes

25 Questions

Questions 1-7

A scientist will perform six experiments - P, R, T, X, Y, and Z - during a three-month period, August through October. In each of the three months, exactly two of the experiments will be performed. Each experiment will start on the first day of a month and be completed during that month. The order in which the experiments are performed will also be governed by the following restrictions:

R must be performed in August or in September.

T must be performed in September or in October.

T cannot be performed in the same month in which X is performed.

X must be performed in an earlier month than the month in which Z is performed.

1. Which of the following can be the schedule for the six experiments?

<u>August</u>	<u>September</u>	<u>October</u>
(A) P, R	T, X	Y, Z
(B) R, T	X, Y	P, Z
(C) R, X	T, Y	P, Z
(D) X, Y	P, Z	R, T
(E) Y, Z	R, T	P, X

2. Any of the following experiments can be performed in August EXCEPT

- (A) P
(B) R
(C) X
(D) Y
(E) Z

3. If T is performed in September, which of the following must be true?

- (A) P is performed in August.
(B) R is performed in September.
(C) X is performed in August.
(D) Y is performed in September.
(E) Z is performed in October.

4. If R is performed in the same month as Z, which of the following can be the pair of experiments performed in October?

- (A) P and X
(B) P and Y
(C) R and Z
(D) T and Y
(E) X and Y

5. If T is performed in the month before Z is performed, which of the following is a pair of experiments that can be performed in the same month as each other?

- (A) P and R
(B) P and Y
(C) R and Y
(D) R and Z
(E) X and Y

6. If P is performed in the same month as Y, which of the following must be true?

- (A) R is performed in the same month as T.
(B) R is performed in the same month as X.
(C) T is performed in August.
(D) X is performed in August.
(E) Y is performed in October.

7. If X is performed in the month before Y is performed, which of the following must be true?

- (A) P is performed in August.
(B) R is performed in September.
(C) T is performed in September.
(D) X is performed in August.
(E) Z is performed in October.

8. Roger: Reading a lot as a child causes
nearsightedness-difficulty seeing things at a distance.

Louise: I disagree. Any correlation between nearsightedness and reading results from the fact that children who have trouble seeing things at a distance are likeliest to prefer those activities, such as reading, that involve looking at things close up.

Louise disputes Roger's claim by

- (A) demonstrating that an absurd conclusion would

follow if Roger's claim were accepted (B) arguing that what Roger claims to be a cause of a given phenomenon is actually its effect (C) using an analogy to expose a flaw in Roger's reasoning (D) pointing out that Roger's claim is self-contradictory (E) attempting to demonstrate that Roger uses the term "nearsightedness" in an ambiguous way 9. Years ago, consumers in Frieland began paying an energy tax in the form of two Frieland pennies for each unit of energy consumed that came from nonrenewable sources. Following the introduction of this energy tax, there was a steady reduction in the total yearly consumption of energy from nonrenewable sources. If the statements in the passage are true, then which of the following must on the basis of them be true? (A) There was a steady decline in the yearly revenues generated by the energy tax in Frieland. (B) There was a steady decline in the total amount of energy consumed each year in Frieland. (C) There was a steady increase in the use of renewable energy sources in Frieland (D) The revenues generated by the energy tax were used to promote the use of energy from renewable sources. (E) The use of renewable energy sources in Frieland greatly increased relative to the use of nonrenewable energy sources.	following must sit at the same table as each other? (A) J and T (B) L and R (C) N and K (D) N and T (E) O and N 11. Which of the following can sit at a table with L and R? (A) J (B) K (C) N (D) O (E) T 12. If N sits at the same table as S, which of the following can be true ? (A) J sits at a table with only one other person. (B) L sits at a table with only one other person. (C) K sits at the same table as O. (D) J sits at the same table as N. (E) L sits at the same table as S. 13. Each of the following is a pair of people who can sit at the same table as each other EXCEPT (A) J and O (B) K and S (C) L and R (D) N and S (E) O and T
<u>Questions 10-14</u> A seating arrangement is being planned for a group of eight people - three women: J, K, and L; two men: N and O ;and three children: R, S, and T, Each of the eight will sit at exactly one of three tables according to the following conditions: No table can have more than three people sitting at it . Each table must have one of the children sitting at it . O and S must sit at the same table as each other. K and L cannot sit at the same table as each other. N and R cannot sit at the same table as each other. 10. If O sits at the same table as K, which of the	14. If O and S are the only people sitting at one of the tables, which of the following can be the group of people sitting at one of the other two tables? (A) J, K, and N (B) K, L, and T (C) K, N, and T (D) K, R, and T (E) L, N, and R 15. Despite a dramatic increase in the number of people riding bicycles for recreation in Parkville. a recent report by the Parkville Department of Transportation shows that the number of accidents involving bicycles has decreased for the third consecutive year.

Which of the following, if true during the last three years, best reconciles the apparent discrepancy in the facts above?

- (A) The Parkville Department of Recreation confiscated abandoned bicycles and sold them at auction to any interested Parkville residents.
- (B) Increased automobile and bus traffic in Parkville has been the leading cause of the most recent increase in automobile accidents.
- (C) Because of the local increase in the number of people bicycling for recreation, many out-of-town bicyclists ride in the Parkville area.
- (D) The Parkville Police Department enforced traffic rules for bicycle riders much more vigorously and began requiring recreational riders to pass a bicycle safety course.
- (E) The Parkville Department of Transportation canceled a program that required all bicycles to be inspected and registered each year.

16. Do strong electric currents, by means of the electromagnetic fields that accompany them, cause cancer in people who live and work nearby? Telephone line workers, who work near such currents every day, can provide a test case. They show elevated levels of brain cancer, therefore, the hypothesis of electromagnetic causation is supported. Which of the following if true, most seriously weakens the argument?

- (A) Burying power lines and other measures to protect the public from such electromagnetic fields would be prohibitively expensive.
- (B) Telephone line workers are exposed to levels of chemical solvents high enough to cause brain cancer.
- (C) High exposure to strong electromagnetic fields is correlated with a slightly higher-than-normal incidence of childhood leukemia, which is a form of cancer.
- (D) Public health officials who found that a group of different illnesses in people living near a power substation could not reliably be attributed to its electromagnetic field were accused of covering up the facts.
- (E) Telephone line workers, like most people, have

electrical appliances at home, and most electrical appliances, when turned on, are surrounded by and electromagnetic field of some measurable level.

Questions 17-20

A library is equipped with a system of pneumatic tubes for sending documents from one to another of exactly six departments-G, H, L, M, S, and T. A tube line is a pair of tubes that connects one department with exactly one other department, with documents moving in one direction in one tube and in the opposite direction in the other tube. The library's system consists of the following seven tube lines and no others.

- Line 1 connects H and L.
- Line 2 connects H and S.
- Line 3 connects L and T.
- Line 4 connects S and T.
- Line 5 connects M and T.
- Line 6 connects L and M.
- Line 7 connects G and H.

Use of the system is subject to the following restrictions:

Documents to be sent between departments that are not connected by a tube line can be transferred from one line to another at departments served by two or more lines, until the document reaches its destination.

A document cannot use any tube line more than once on its way to its destination, nor can the document return to its department of origin on its way to its destination.

17. Any of the following is an acceptable pathway for a document to be sent from S to M, listing all lines used in order from the line first used to the line last used, EXCEPT

- (A) line 4. line 5
- (B) line 2. line 3. line 5
- (C) line 2. line 1, line 6
- (D) line 4. line 1, line 6
- (E) line 2, line 1, line 3, line 5.

18. Which of the following is a complete and accurate list of the lines any one of which could be the second line used by a document sent from T to G?

- (A) Lines 1, 2, and 3

- | | |
|---|---|
| <p>(B) Lines 1, 2, and 4</p> <p>(C) Lines 1, 2, and 6</p> <p>(D) Lines 2, 3, and 4</p> <p>(E) Lines 2, 3, and 6</p> <p>19. If line 3 cannot be used, a document to be sent from T to H that uses as few tube lines as possible must use line</p> <p>(A) 1</p> <p>(B) 2</p> <p>(C) 5</p> | <p>(D) 6</p> <p>(E) 7</p> <p>20. A pathway from M to H that includes as many tube lines as possible must include lines</p> <p>(A) 1 and 2</p> <p>(B) 1 and 3</p> <p>(C) 3 and 4</p> <p>(D) 4 and 5</p> <p>(E) 5 and 6</p> |
|---|---|

Questions 21-23

Eight figure skaters -four women: Fiona, Gloria, Heidi, and Jill; and four men: Ravi, Shigeru, Toby, and Vernon-will participate in a one-day skating exhibition consisting of four consecutively performed sets - set 1 through set 4. Each set will be performed in exactly one pair of skaters, one man and one woman. Each skater will performed by exactly one of the sets, subject to the following constraints:

Ravi skates in an earlier set than Vernon does.

Fiona skates in either set 1 or set 4.

Jill does not skate with Toby.

Shigeru skates with either Fiona or Gloria.

21. Which of the following could be the pairs of skaters who skate in each set, from set 1 through set 4?

Set 1	Set 2	Set 3	Set 4
(A) Fiona. Ravi	Jill, Toby	Gloria, Shigeru	Heidi. Vernon
(B) Gloria. Shigeru	Heidi, Ravi	Fiona, Toby	Jill, Vernon
(C) Heidi. Shigeru	Gloria, Ravi	Jill. Vernon	Fiona. Toby
(D) Heidi, Toby	Gloria, Shigeru	Jill. Ravi	Fiona, Vernon
(E) Jill. Vernon	Heidi, Ravi	Gloria. Shigeru	Fiona. Toby

22. If Gloria skates with Toby in set 1, which of the following must be true?

- (A) Vernon skates in set 2.
- (B) Shigeru skates in set 4.
- (C) Ravi skates in set 3.
- (D) Jill skates in set 4.
- (E) Heidi skates in set 3.

23. If Heidi skates in set 1 and Toby skates in set 2, which of the following must be true?

- (A) Fiona skates with Ravi.
- (B) Gloria skates with Ravi.

- (C) Gloria skates with Shigeru.
- (D) Gloria skates with Vernon.
- (E) Jill skates with Vernon.

24. Neither the Sami nor the Kephrian delegations attended the international conference. Beforehand, the delegations of Daqua and Kephria, allies whose governments had grievances against Tessia, officially announced that one or both of the two would stay away if the Tessian delegation attended the conference. In response, the Sami delegation officially announced that it would definitely attend if both the Daquan and Kephrian delegations stayed away.

If the statements given are all true and all the delegations adhered to their official announcements, it must also be true that the

- (A) Daquan delegation attended the conference
- (B) Daquan delegation did not attend the conference
- (C) Sami government had no grievance against Tessia
- (D) Tessian delegation did not attend the conference
- (E) Tessian delegation made no official announcement regarding its attendance at the conference

25. On turning 65 years old, everyone living in the town of Malton becomes eligible to receive a card that guarantees discounts on most goods and services sold in the town. Census records for 1990 show that 2,450 inhabitants of Malton turned 64 in that year. Yet . in 1991 over 3,000 people applied for and properly received discount cards. So clearly some of Malton's population growth between 1990 and 1992 must be attributable to migration into the city by people in their mid -60's
Which of the following is an assumption on which the argument depends?

- (A) The town of Malton has no complete census records for 1991.
- (B) The overall size of the population of Malton grew by over 500 during 1990.
- (C) Fewer people applied for and received discount cards in 1991 than did so in 1992.
- (D) Among the people 65 years old or older who moved into Malton in 1991, there was no one who did not apply for a discount card .
- (E) In general, people who applied for and received discount cards in 1991 first became eligible to do so in that year

SECTION 1

Time –30 minutes

38 Questions

1. A computer program can provide information in ways that force students to --- learning instead of being merely ---- of knowledge.
 (A) shore up .. reservoirs
 (B) accede to .. consumers
 (C) participate in .. recipients
 (D) compensate for.. custodians
 (E) profit from .. beneficiaries
2. The form and physiology of leaves vary according to the ---- in which they develop: for example, leaves display a wide range of adaptations to different degrees of light and moisture.
 (A) relationship
 (B) species
 (C) sequence
 (D) patterns
 (E) environment
3. One theory about intelligence sees ---- as the logical structure underlying thinking and insists that since animals are mute, they must be ---- as well.
 (A) behavior.. inactive
 (B) instinct.. cooperative
 (C) heredity.. thoughtful
 (D) adaptation.. brutal
 (E) language.. mindless
4. Though ---- in her personal life, Edna St. Vincent Millay was nonetheless ---- about her work, usually producing several pages of complicated rhyme in a day.
 (A) jaded.. feckless
 (B) verbose.. ascetic
 (C) vain.. humble
 (D) impulsive.. disciplined
 (E) self-assured.. sanguine
5. The children's ---- natures were in sharp contrast to the even-tempered dispositions of their parents.
 (A) mercurial
 (B) blithe
 (C) phlegmatic
 (D) introverted
 (E) artless
6. By ---- scientific rigor with a quantitative approach, researchers in the social sciences may often have --- their scope to those narrowly circumscribed topics that are well suited to quantitative methods.
 (A) undermining.. diminished
 (B) equating.. enlarged
 (C) vitiating.. expanded
 (D) identifying.. limited
 (E) imbuing.. broadened
7. As early as the seventeenth century, philosophers called attention to the ---- character of the issue, and their twentieth-century counterparts still approach it with ----.
 (A) absorbing.. indifference
 (B) unusual.. composure
 (C) complex.. antipathy
 (D) auspicious.. caution
 (E) problematic.. uneasiness
8. TRIPOD: CAMERA::
 (A) scaffolding: ceiling
 (B) prop: set
 (C) easel: canvas
 (D) projector: film
 (E) frame: photograph
9. AQUATIC: WATER::
 (A) cumulus: clouds
 (B) inorganic: elements
 (C) variegated: leaves
 (D) rural: soil
 (E) arboreal: trees
10. EMOLLIENT: SUPPLENESS::
 (A) unguent: elasticity
 (B) precipitant: absorption
 (C) additive: fusion
 (D) desiccant: dryness
 (E) retardant: permeability

11. DRAW: DOODLE::

- (A) talk: whisper
- (B) travel: ramble
- (C) run: walk
- (D) calculate: add
- (E) eat: gobble

12. CONSPICUOUS: SEE:

- (A) repulsive: forget
- (B) prohibited: discount
- (C) deceptive: delude
- (D) impetuous: disregard
- (E) transparent: understand

13. IMMATURE: DEVELOPED::

- (A) accessible: exposed
- (B) theoretical: conceived
- (C) tangible: identified
- (D) irregular: classified
- (E) incipient: realized

14. PERSPICACITY: ACUTE::

- (A) adaptability: prescient
- (B) decorum: complacent
- (C) caprice: whimsical
- (D) discretion: literal
- (E) ignorance: pedantic

15. PLAYFUL: BANTER::

- (A) animated: originality
- (B) exaggerated: hyperbole
- (C) insidious: effrontery
- (D) pompous: irrationality
- (E) taciturn: solemnity

16. QUARANTINE: CONTAGION::

- (A) blockage: obstacle
- (B) strike: concession
- (C) embargo: commerce
- (D) vaccination: inoculation
- (E) prison: reform

Influenced by the view of some twentieth-century feminists that women's position within the family is one of the central factors determining women's social position, some historians have underestimated the signi-

(5)ficance of the woman suffrage movement. These historians contend that nineteenth-century suffragism was less radical and, hence, less important than, for example, the moral reform movement or domestic feminism—two nineteenth-century movements in which women struggled for more power and autonomy within the family. True, by emphasizing these struggles, such historians have broadened the conventional view of nineteenth-century feminism, but they do a historical disservice to suffragism. Nineteenth-century feminists and anti-feminist alike perceived the suffragists' demand for enfranchisement as the most radical element in women's protest, in part because suffragists were demanding power that was not based on the institution of the family, women's traditional sphere. When evaluating nineteenth-century feminism as a social force, contemporary historians should consider the perceptions of actual participants in the historical events.

17. The author asserts that the historians discussed in the passage have

- (A) influenced feminist theorists who concentrate on the family
- (B) honored the perceptions of the women who participated in the women suffrage movement
- (C) treated feminism as a social force rather than as an intellectual tradition
- (D) paid little attention to feminist movements
- (E) expanded the conventional view of nineteenth-century feminism

18. The author of the passage asserts that some twentieth-century feminists have influenced some historians view of the

- (A) significance of the woman suffrage movement
- (B) importance to society of the family as an institution
- (C) degree to which feminism changed nineteenth-century society
- (D) philosophical traditions on which contemporary feminism is based
- (E) public response to domestic feminism in the nineteenth century

19. The author of the passage suggests that which of the following was true of nineteenth-century feminists?

- (A) Those who participated in the moral reform movement were motivated primarily by a desire to reconcile their private lives with their public positions.
- (B) Those who advocated domestic feminism, although less visible than the suffragists, were in some ways the more radical of the two groups.
- (C) Those who participated in the woman suffrage movement sought social roles for women that were not defined by women's familial roles.
- (D) Those who advocated domestic feminism regarded the gaining of more autonomy within the family as a step toward more participation in public life.
- (E) Those who participated in the nineteenth-century moral reform movement stood midway between the positions of domestic feminism and suffragism.

20. The author implies that which of the following is true of the historians discussed in the passage?

- (A) They argue that nineteenth-century feminism was not as significant a social force as twentieth-century feminism has been.
- (B) They rely too greatly on the perceptions of the actual participants in the events they study.
- (C) Their assessment of the relative success of nineteenth-century domestic feminism does not adequately take into account the effects of antifeminist rhetoric.
- (D) Their assessment of the significance of nineteenth-century suffragism differs considerably from that of nineteenth-century feminists.
- (E) They devote too much attention to nineteenth-century suffragism at the expense of more radical movements that emerged shortly after the turn of the century.

Many objects in daily use have clearly been influenced by science, but their form and function, their dimensions and appearance, were determined by technologists artisans, designers, inventors, and engineers—using non-

(5) scientific modes of thought. Many features and qualities of the objects that a technologist thinks about cannot be reduced to unambiguous verbal descriptions; they are dealt with in the mind by a visual, nonverbal process. In the development of Western technology, it has been non-

(10) verbal thinking, by and large, that has fixed the outlines and filled in the details of our material surroundings. Pyramids, cathedrals, and rockets exist not because of geometry or thermodynamics, but because they were first a picture in the minds of those who built them.

(15) The creative shaping process of a technologist's mind can be seen in nearly every artifact that exists. For example, in designing a diesel engine, a technologist might impress individual ways of nonverbal thinking on the machine by continually using an intuitive sense of right-

(20) ness and fitness. What would be the shape of the combustion chamber? Where should the valves be placed? Should it have a long or short piston? Such questions have a range of answers that are supplied by experience, by physical requirements, by limitations of available

(25) space, and not least by a sense of form. Some decisions, such as wall thickness and pin diameter, may depend on scientific calculations, but the nonscientific component of design remains primary.

Design courses, then, should be an essential element

(30) in engineering curricula. Nonverbal thinking, a central mechanism in engineering design, involves perceptions, the stock-in-trade of the artist, not the scientist. Because perceptive processes are not assumed to entail "hard thinking," nonverbal thought is sometimes seen as a prim-

(35) itive stage in the development of cognitive processes and inferior to verbal or mathematical thought. But it is paradoxical that when the staff of the *Historic American Engineering Record* wished to have drawings made of machines and isometric views of industrial processes for

(40) its historical record of American engineering, the only college students with the requisite abilities were not engineering students, but rather students attending architectural schools.

It courses in design, which in a strongly analytical

(45) engineering curriculum provide the background required for practical problem- solving, are not provided, we can expect to encounter silly but costly errors occurring in advanced engineering systems. For example, early models of high-speed railroad cars loaded with sophisticated

(50) controls were unable to operate in a snowstorm because

a fan sucked snow into the electrical system. Absurd random failures that plague automatic control systems are not merely trivial aberrations; they are a reflection of the chaos that results when design is assumed to be primarily a problem in mathematics.

21. In the passage, the author is primarily concerned with

- (A) identifying the kinds of thinking that are used by technologists
- (B) stressing the importance of nonverbal thinking in engineering design
- (C) proposing a new role for nonscientific thinking in the development of technology
- (D) contrasting the goals of engineers with those of technologists
- (E) criticizing engineering schools for emphasizing science in engineering curricula

22. It can be inferred that the author thinks engineering curricula are

- (A) strengthened when they include courses in design
- (B) weakened by the substitution of physical science courses for courses designed to develop mathematical skills
- (C) strong because nonverbal thinking is still emphasized by most of the courses
- (D) strong despite the errors that graduates of such curricula have made in the development of automatic control systems
- (E) strong despite the absence of nonscientific modes of thinking

23. Which of the following statements best illustrates the main point of lines 1-28 of the passage?

- (A) When a machine like a rotary engine malfunctions, it is the technologist who is best equipped to repair it.
- (B) Each component of an automobile—for example, the engine or the fuel tank—has a shape that has been scientifically determined to be best suited to that component's function
- (C) A telephone is a complex instrument designed by technologists using only nonverbal thought
- (D) The designer of a new refrigerator should

consider the designs of other refrigerators before deciding on its final form.

- (E) The distinctive features of a suspension bridge reflect its designer's conceptualization as well as the physical requirements of its site.

24. Which of the following statements would best serve as an introduction to the passage?

- (A) The assumption that the knowledge incorporated in technological developments must be derived from science ignores the many non-scientific decisions made by technologists.
- (B) Analytical thought is no longer a vital component in the success of technological development.
- (C) As knowledge of technology has increased, the tendency has been to lose sight of the important role played by scientific thought in making decisions about form, arrangement, and texture.
- (D) A movement in engineering colleges toward a technician's degree reflects a demand for graduates who have the nonverbal reasoning ability that was once common among engineers.
- (E) A technologist thinking about a machine, reasoning through the successive steps in a dynamic process, can actually turn the machine over mentally.

25. The author calls the predicament faced by the *Historic American Engineering Record* "paradoxical" (lines 36-37) most probably because

- (A) the publication needed drawings that its own staff could not make
- (B) architectural schools offered but did not require engineering design courses for their students
- (C) college students were qualified to make the drawings while practicing engineers were not
- (D) the drawings needed were so complicated that even students in architectural schools had difficulty making them.
- (E) engineering students were not trained to make the type of drawings needed to record the development of their own discipline

26. According to the passage, random failures in automatic control systems are "not merely trivial aberrations" (lines 53) because

(A) automatic control systems are designed by engineers who have little practical experience in the field

(B) the failures are characteristic of systems designed by engineers relying too heavily on concepts in mathematics

(C) the failures occur too often to be taken lightly

(D) designers of automatic control systems have too little training in the analysis of mechanical difficulties

(E) designers of automatic control systems need more help from scientists who have a better understanding of the analytical problems to be solved before such systems can work efficiently

27. The author uses the example of the early models of high-speed railroad cars primarily to

(A) weaken the argument that modern engineering systems have major defects because of an absence of design courses in engineering curricula

(B) support the thesis that the number of errors in modern engineering systems is likely to increase

(C) illustrate the idea that courses in design are the most effective means for reducing the cost of designing engineering systems

(D) support the contention that a lack of attention to the nonscientific aspects of design results in poor conceptualization by engineers

(E) weaken the proposition that mathematics is a necessary part of the study of design

28. IGNITE:

(A) amplify

(B) douse

(C) obscure

(D) blemish

(E) replicate

29. MUTATE:

(A) recede

(B) grow larger

(C) link together

(D) remain the same

(E) decrease in speed

30. FRAGMENT:

(A) ensue

(B) revive

(C) coalesce

(D) balance

(E) accommodate

31. OSTENSIBLE:

(A) gargantuan

(B) inauspicious

(C) intermittent

(D) perpetual

(E) inapparent

32. PROLIXITY:

(A) ceremoniousness

(B) flamboyance

(C) succinctness

(D) inventiveness

(E) lamentation

33. CONCERTED:

(A) meant to obstruct

(B) not intended to last

(C) enthusiastically supported

(D) run by volunteers

(E) individually devised

34. FORBEARANCE:

(A) fragility

(B) impatience

(C) freedom

(D) nervousness

(E) tactlessness

35. COSSETED:

(A) unspoiled

(B) irrepressible

(C) serviceable

(D) prone to change

(E) free from prejudice

36.PROBITY:

- (A) timidity
- (B) sagacity
- (C) impertinence
- (D) uncertainty
- (E) unscrupulousness

37.ESCHEW:

- (A) habitually indulge in
- (B) take without authorization
- (C) leave unsaid
- (D) boast about
- (E) handle carefully

38.REDOUBTABLE:

- (A) trustworthy
- (B) unschooled
- (C) credulous
- (D) not formidable
- (E) not certain

SECTION 2

Time –30 minutes

25 Questions

Questions 1-6

A newsstand will display exactly one copy each of six different magazines—M, O, P, S, T, and V—in a single row on a rack. Each magazine will occupy exactly one of the six positions, numbered consecutively 1 through 6. The magazines must be displayed in accordance with the following rules:

Either P or else T occupies position 1.

Either S or else T occupies position 6.

M and O, not necessarily in that order, occupy consecutively numbered positions.

V and T, not necessarily in that order, occupy consecutively numbered positions.

1.Which of the following is an order in which the six magazines can be arranged, from position 1 through position 6?

- (A) M, O, P, S, V, T
- (B) P, O, S, M, V, T
- (C) P, V, T, O, M, S
- (D) P, V, T, S, O, M
- (E) T, P, V, M, O, S

2.If P occupies position 3, which of the following must be true?

- (A) M occupies position 4.
- (B) O occupies position 2.
- (C) S occupies position 5.
- (D) T occupies position 6.
- (E) V occupies position 2.

3.If O and T, not necessarily in that order, occupy consecutively numbered positions, then T can be in position

- (A) 1
- (B) 2
- (C) 4
- (D) 5
- (E) 6

4. Which of the following can be true?

- (A) M occupies position 4 and P occupies position 5.
- (B) P occupies position 4 and V occupies position 5.
- (C) S occupies position 2 and P occupies position 3.
- (D) P occupies position 2.
- (E) S occupies position 5.

5. If V occupies position 4, then T must occupy the position that is numbered exactly one lower than the position occupied by

- (A) M (B) O (C) P
- (D) S (E) V

6. If S and V, not necessarily in that order, occupy consecutively numbered positions, which of the following must be true?

- (A) M occupies position 4.
- (B) O occupies position 2.
- (C) P occupies position 1.
- (D) S occupies position 6.
- (E) T occupies position 6.

7. Patel: Although enrollment in the region's high school has been decreasing for several years, enrollment at the elementary school has grown considerably. Therefore, the regional school board proposes building a new elementary school.

Quintero: Another solution would be to convert some high school classrooms temporarily into classrooms for elementary school students.

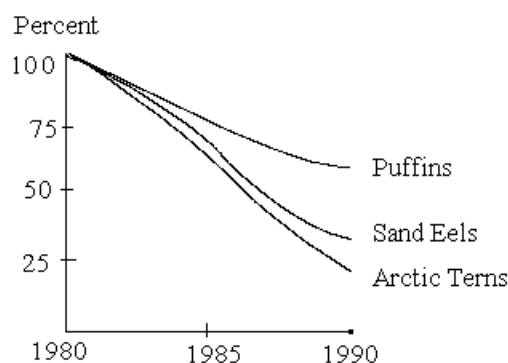
Which of the following, if true, most helps to support Quintero's alternative proposal?

- (A) Some rooms at the high school cannot be converted into rooms suitable for the use of elementary school students.
- (B) The cost of building a high school is higher than the cost of building an elementary school.
- (C) Although the birth rate has not increased, the number of families sending their children to the region's high school has increased markedly.
- (D) A high school atmosphere could jeopardize the safety and self-confidence of elementary school students.

(E) Even before the region's high school population began to decrease, several high school classrooms rarely needed to be used.

Question 8 is based on the following graph

POPULATIONS OF ARCTIC TERNS, PUFFINS, AND SAND EELS OF ALAIR ISLAND AS A PERCENT OF 1980 POPULATION



8. Which of the following, if true, most helps explain the difference in the rates of decline between 1980 and 1990 in population of puffins and arctic terns, two kinds of seabirds for which sand eels serve as a primary source of food?

- (A) Puffins switched in part from their preferred food of sand eels to rockfish and other fish, but arctic terns did not.
- (B) The marked decline in the populations of puffins and arctic terns that occurred on Alair Island did not occur on other similar islands nearby, where there are substantial populations of both species.
- (C) The decline in sand eels was due to changes in environmental conditions that affected the reproduction of eels rather than to overfishing by people.
- (D) The main diet of puffin and arctic tern chicks on Alair Island in 1980 consisted of young sand eels.
- (E) Unusual severe weather that disrupted the breeding cycle of the sand eels of Alair Island in 1989 also damaged the nests of puffins but not those of arctic terns.

9. Peter: More than ever before in Risland, college graduates with science degrees are accepting permanent jobs in other fields. That just goes to show that scientists in Risland are not being paid enough.

Lila: No, it does not. These graduates are not working in science for the simple reason that there are not enough jobs in science in Risland to employ all of these graduates.

Which of the following, if true in Risland, would most undermine the reasoning in Peter's argument?

- (A) The college graduates with science degrees who are not working in science are currently earning lower salaries than they would earn as scientists.
- (B) Fewer college students than ever before are receiving degrees in science.
- (C) The number of jobs in science has steadily risen in the last decade.
- (D) A significant number of college graduates with science degrees worked at low-paying jobs while they were in college.
- (E) Every year some recent college graduates with science degrees accept permanent jobs in nonscientific fields.

Questions 10-15

Exactly six lectures will be given one at a time at a one-day conference. Two of the lectures—S and T—will be given by resident speakers, the other four—W, X, Y, and Z—will be given by visiting speakers. At least two but no more than four of the lectures will be given before lunch; the remaining lectures will be given after lunch. The following conditions must be observed:

S will be the fourth lecture.

Exactly one of the lectures by a resident will be given before lunch.

Y will be given at some time before T is given.

If W is given before lunch, Y will be given after lunch.

10. Which of the following can be the order of lectures and lunch at the conference?

- (A) W, X, Lunch, Y, S, T, Z
- (B) X, Y, T, Lunch, S, Z, W
- (C) Y, T, Lunch, S, W, X, Z
- (D) Z, T, W, S, Lunch, Y, X
- (E) Z, W, Y, S, Lunch, X, T

11. If exactly two lectures are given before lunch, they must be

- (A) X and T
- (B) Y and T
- (C) Z and T
- (D) Z and W
- (E) Z and Y

12. If exactly three lectures, including Y and Z, are given before lunch, which of the following can be true?

- (A) T is the second lecture.
- (B) T is the fifth lecture.
- (C) W is the third lecture.
- (D) X is the first lecture.
- (E) X is the third lecture.

13. If T is the sixth lecture, which of the following must be true?

- (A) X is the first lecture.
- (B) X is the second lecture.
- (C) Exactly two lectures are given before lunch.
- (D) Exactly three lectures are given before lunch.
- (E) Exactly four lectures are given before lunch.

14.If S and Z are both given after lunch, which of the following must be true?

- (A) X is given before lunch.
- (B) X is given after lunch.
- (C) Y is given before lunch.
- (D) T is the third lecture.
- (E) Z is the fifth lecture.

15.Which of the following lectures CANNOT be given immediately before lunch?

- (A) S
- (B) T
- (C) X
- (D) Y
- (E) Z

Questions 16-22

A circus has seven fenced enclosures, numbered 1 through 7, for two animals: a lion and a tiger. Each enclosure is connected to adjacent enclosures by interior gates. There are exactly eight such gates, each connecting one enclosure to exactly one other enclosure: enclosure 1 is connected to enclosures 2, 3 and 4; enclosure 3 to enclosures 1, 2, 4, and 5; and enclosure 5 to enclosures 3, 6, and 7. These gates provide the only connections between enclosures. Occasionally a trainer moves the animals. Taking either animals from one enclosure to an adjacent enclosure through a gate is called a "transfer." The following conditions are strictly observed:

The two animals cannot be together in any enclosure or gate.

Transfers cannot occur simultaneously

In moving either one animal or both to a specified enclosure or enclosures, the minimum number of transfers needed to achieve the specified result are used.

16.If the lion is in enclosure 1 and the tiger is in enclosure 3, and the lion is to be moved to enclosure 7, the tiger could be in which of the following enclosures when all of the transfers have been completed?

- (A) 1
- (B) 3
- (C) 4
- (D) 5
- (E) 6

17.If the tiger is in enclosure 5 and the lion is in enclosure 3, moving the tiger to which of the following enclosures requires exactly two transfers?

- (A) 2
- (B) 3
- (C) 4
- (D) 6
- (E) 7

18.If the lion is in enclosure 6 and the tiger is in enclosure 7, and the lion is to be moved to enclosure 7 and the tiger to enclosure 6, then which of the following must be true?

- (A) The lion is transferred to enclosure 3 at some time during the move.
- (B) The tiger is transferred to enclosure 5 twice.
- (C) One of the two animals is transferred to enclosure 3 twice.
- (D) Three transfers to enclosure 5 are made.
- (E) At least one transfer is made to either enclosure 2 or enclosure 4.

19.If the lion is in enclosure 3 and the tiger is in enclosure 4, and the lion is to be moved to enclosure 5 and the tiger to enclosure 7, then exactly how many transfers must be made?

- (A) Four
- (B) Five
- (C) Six
- (D) Seven
- (E) Eight

20.If the lion is in enclosure 1 and the tiger is in enclosure 7, and the lion is to be transferred to enclosure 3 and the tiger to enclosure 1, then which of the following CANNOT be true?

- (A) The lion is transferred to enclosure 2 in the first transfer.
- (B) The lion is transferred to enclosure 3 in the second transfer.
- (C) The lion is transferred to enclosure 4 in the second transfer.
- (D) The tiger is transferred to enclosure 5 in the first transfer.
- (E) The tiger is transferred to enclosure 3 in the second transfer.

21. If the lion is in enclosure 1 and the tiger is in enclosure 3, and the lion is to be moved to enclosure 6 and the tiger to enclosure 5, then the second transfer could be a transfer of the
- (A) lion to enclosure 2
 - (B) lion to enclosure 5
 - (C) tiger to enclosure 4
 - (D) tiger to enclosure 5
 - (E) tiger to enclosure 7

22. If the lion is in enclosure 3 and the tiger is in enclosure 6, and the lion is to be moved to enclosure 6 and the tiger to enclosure 3, then which of the following must be true?

- (A) Exactly five enclosures are used in the move
- (B) One animal is transferred exactly twice as many times as the other animal.
- (C) All of the transfers of the lion are completed before any transfer of the tiger occurs.
- (D) At one point one of the animals is transferred to either enclosure 2 or enclosure 4.
- (E) At one point neither the lion nor the tiger is in enclosure 3, enclosure 5, or enclosure 6

23. Counselor: Every year a popular newsmagazine publishes a list of United States colleges, ranking them according to an overall numerical score that is a composite of ratings according to several criteria. However, the overall scores generally should not be used by students as the basis for deciding to which colleges to apply.

Which of the following, if true, most helps to justify the counselor's recommendation?

- (A) The vast majority of people who purchase the magazine in which the list appears are not

college-bound students.

- (B) Colleges that are ranked highest in the magazine's list use this fact in advertisements aimed at attracting students.
- (C) The rankings seldom change from one year to the next.
- (D) The significance that particular criteria have for any two students is likely to differ according to the students' differing needs.
- (E) Some college students who are pleased with their schools considered the magazine's rankings before deciding which college to attend.

24. A thorough search of Edgar Allan Poe's correspondence has turned up not a single letter in which he mentions his reputed morphine addiction. On the basis of this evidence it is safe to say that Poe's reputation for having been a morphine addict is undeserved and that reports of his supposed addiction are untrue.

Which of the following is assumed by the argument above?

- (A) Reports claiming that Poe was addicted to morphine did not begin to circulate until after his death.
- (B) None of the reports of Poe's supposed morphine addiction can be traced to individuals who actually knew Poe.
- (C) Poe's income from writing would not have been sufficient to support a morphine addiction.
- (D) Poe would have been unable to carry on an extensive correspondence while under the influence of morphine.
- (E) Fear of the consequences would not have prevented Poe from indicating in his correspondence that he was addicted to morphine.

25. Adelle: The government's program to reduce the unemployment rate in the province of Carthena by encouraging job creation has failed, since the rate there has not changed appreciably since the program began a year ago.

Fran: But the unemployment rate in Carthena had been rising for three years before the program began, so the program is helping.

Which of the following, if true, most strongly counters Fran's

objection to Adelle's argument?

- (A) The government is advised by expert economists, some of whom specialize in employment issues.
- (B) The unemployment rate in the province of Carthena has historically been higher than that of the country as a whole.
- (C) The current government was elected by a wide margin, because of its promises to reduce the unemployment rate in Carthena.
- (D) Around the time the government program began, large numbers of unemployed Carthena residents began leaving the province to look for work elsewhere.
- (E) The unemployment rate in Carthena had been relatively stable until shortly before the current government took office.

SECTION 3

Time –30 minutes
30 Questions

$$x^2 - 1 = y$$

$$x = 3$$

1. y^2 80

The gross receipts from the sale of t tickets, at \$17 per ticket, total \$16,660.

2. t 1,000

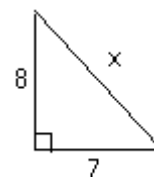
Points T and U are on a circle with center O

3. OT TU

A box contains 20 marbles all of which are solid colored; 5 of the marbles are green and 10 of the marbles are red.

4. The probability that a marble selected at random from the box will be green The probability that a marble selected at random from the box will be neither red nor green

5. Eleven thousand plus eleven hundred plus eleven 11,111



6. x 15

The cost c of an order of n special envelopes is given by $c = (\$0.50)n + \15.00 .

7. The cost of an order of 500 special envelopes \$260

The average (arithmetic mean) of 7, 9, and x is greater than 9.

8. x 11

$$a > 0$$

9. $(4\sqrt{5a})^2$ $40a$

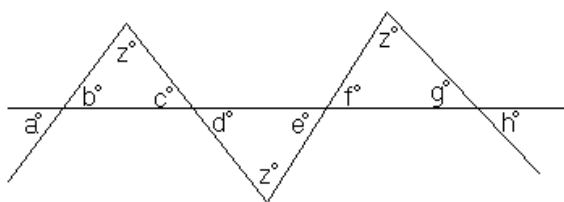
10. $\frac{0.27}{0.53}$ $\frac{0.027}{0.053}$

Each of the numbers x , y , w , and z (not necessarily distinct) can have any of the values 2, 3, 9, or 14.

11. $\frac{x}{y}$ wz

12. $a^7 + a^{15}$ $a = -219$ $a^8 + a^{14}$

13. $x^2 + 2x + 1$ x^2



14. d $a > h$ e

w , x , y , and z are consecutive positive integers and $w < x < y < z$.

15. The remainder when $(w + x)(x + y)(y + z)$ is divided by 2 1

16. A certain machine drills 30 holes in 8 minutes. At that constant rate, how many holes will 4 such machines drill in $1\frac{1}{3}$ hours?

- (A) 300
- (B) 900
- (C) 960
- (D) 1,200
- (E) 2,560

17. Tina, Ed, and Lauren agree to share the cost of a gift and to make their contributions in proportion to their ages. Ed's age is $\frac{1}{2}$ of Tina's age, and Lauren's age is $\frac{1}{3}$ of Ed's age. If Lauren's share of the cost is \$ 2.50, what is the cost of the gift?

(A) \$25

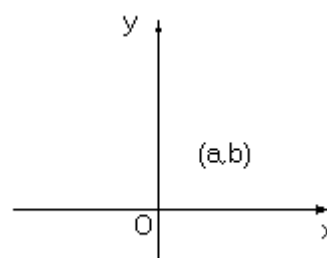
- (B) \$20
- (C) \$15
- (D) \$12
- (E) \$10

18. Three solid cubes of lead, each with edges 10 centimeters long, are melted together in a level, rectangular-shaped pan. The base of the pan has inside dimensions of 20 centimeters by 30 centimeters, and the pan is 15 centimeters deep. If the volume of the solid lead is approximately the same as the volume of the molted lead, approximately how many centimeters deep is the melted lead in the pan?

- (A) 2.5
- (B) 3
- (C) 5
- (D) 7.5
- (E) 9

19. Which of the following CANNOT be the sum of two integers that have a product of 30?

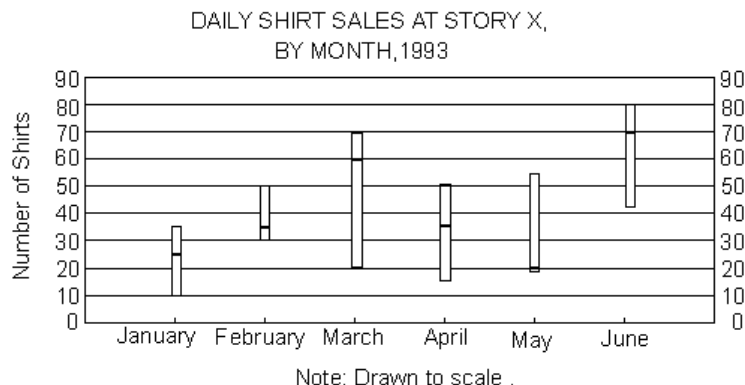
- (A) 31
- (B) 17
- (C) -11
- (D) -13
- (E) -21



20. In the rectangular coordinate system above, if point (a, b) , shown, and the two points $(4a, b)$ and $(2a, 2b)$, not shown, were connected by straight lines, then the area of the resulting triangular region, in terms of a and b , would be

- (A) $\frac{ab}{2}$
- (B) ab
- (C) $\frac{3ab}{2}$
- (D) $2ab$
- (E) $4ab$

Questions 21-22 refer to the following graph.



The top and bottom of each bar indicate, respectively, the highest and lowest daily number of shirts sold during the month. The heavy line across each bar indicates the average (arithmetic mean) number of shirts sold per day during the month.

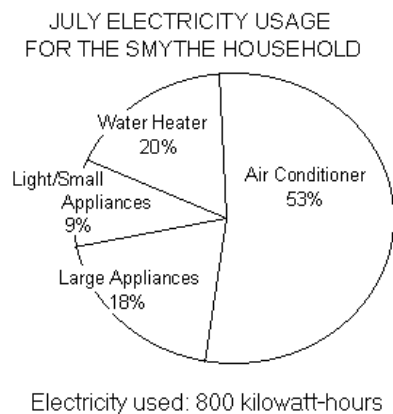
21. What was the range in the daily number of shirts sold during March?

- (A) 20
- (B) 45
- (C) 50
- (D) 60
- (E) 70

22. The average (arithmetic mean) number of shirts sold per day during February was approximately what percent greater than the average number sold during January?

- (A) 10%
- (B) 20%
- (C) 30%
- (D) 40%
- (E) 70%

Questions 23-25 refer to the following graph.



23. For which two uses of electricity was the ratio of the amounts of electricity used most nearly 3 to 1?

- (A) Water heater and lights/small appliances
- (B) Large appliances and lights/small appliances
- (C) Air conditioner and water heater
- (D) Air conditioner and lights/small appliances
- (E) Air conditioner and large appliances

24. The electricity used by the water heater was measured separately and its cost per kilowatt-hour was one-half the cost per kilowatt-hour of the rest of the electricity used. The cost of the electricity used by the water heater was most nearly what fraction of the total cost of all the electricity used?

- (A) $\frac{1}{11}$
- (B) $\frac{1}{9}$
- (C) $\frac{1}{8}$
- (D) $\frac{1}{5}$

(E) It cannot be determined from the information given.

25. In November the Smythe household used the same total amount of electricity as in July, but the water heater used 33 percent of this total amount. By approximately what percent did the amount of electricity used by the water heater increase from July to November?

- (A) 13%
(B) 33%
(C) 50%
(D) 65%
(E) 130%
26. One integer will be randomly selected from the integers 11 to 60, inclusive. What is the probability that the selected integer will be a perfect square or a perfect cube?
(A) 0.1
(B) 0.125
(C) 0.16
(D) 0.5
(E) 0.9
27. The measures of two angles of a parallelogram differ by 52 degrees. The number of degrees in the smaller angle is
(A) 38
(B) 52
(C) 64
(D) 76
(E) 128
28. The odds in favor of winning a game can be found by computing the ratio of the probability of winning to the probability of not winning. If the probability that Pat will win a game is $\frac{4}{9}$, what are the odds that Pat will win the game?
(A) 4 to 5
(B) 4 to 9
(C) 5 to 4
(D) 5 to 9
(E) 9 to 5
29. If a , b , c , and d are consecutive integers such that $a < b < c < d$, then in terms of a , the sum $a + b + d =$
(A) $a + 4$
(B) $2a + 3$
(C) $3a + 2$
(D) $3a + 3$
(E) $3a + 4$
30. $2^x + 2^x =$
(A) 2^{x+1}
(B) 2^{x+2}
(C) 2^{2x}
(D) 4^x
(E) 4^{2x}

SECTION 4

Time –30 minutes

38 Questions

1. Since most if not all learning occurs through---, relating one observation to another, it would be strange indeed if the study of other cultures did not also illuminate the study of our own.
(A) assumptions
(B) experiments
(C) comparisons
(D) repetitions
(E) impressions
2. The new ---- of knowledge has created ---- people: everyone believes that his or her subject cannot and possibly should not be understood by others.
(A) specialization.. barriers between
(B) decline.. associations among
(C) redundancy.. complacency in
(D) disrepute.. concern for
(E) promulgation.. ignorance among
3. If a species of parasite is to survive, the host organisms must live long enough for the parasite to ---- ; if the host species becomes----, so do its parasites.
(A) atrophy.. healthy
(B) reproduce.. extinct
(C) disappear.. widespread
(D) succumb.. nonviable
(E) mate.. infertile
4. The author argues for serious treatment of such arts as crochet and needlework, finding in too many art historians a cultural blindness --- to their ---- textiles as a medium in which women artists predominate.
(A) traceable.. prejudice against
(B) opposed.. distrust of
(C) referring.. need for
(D) reduced.. respect for
(E) corresponding.. expertise in
5. Those who fear the influence of television deliberately ----- its persuasive power, hoping that they might keep knowledge of its potential to effect social change from being widely disseminated.
(A) promote
(B) underplay
(C) excuse
(D) laud
(E) suspect
6. Because the high seriousness of their narratives resulted in part from their metaphysics, Southern writers were praised for their ---- bent.
(A) technical
(B) discursive
(C) hedonistic
(D) philosophical
(E) scientific
7. Far from being ----, Pat was always ---- to appear acquiescent.
(A) unctuous.. loath
(B) brazen.. reluctant
(C) ignoble.. concerned
(D) obsequious.. eager
(E) gregarious.. willing
8. CHUCKLE: LAUGHING::
(A) uproar: shouting
(B) whisper: speaking
(C) hum: whistling
(D) lecture: conversing
(E) murmur: mimicking
9. PARAGRAPH: ESSAY::
(A) object: verb
(B) phrase: preposition
(C) interjection: parenthesis
(D) clause: sentence
(E) colloquialism: expression
10. STUPOR: ALERT::
(A) rebellion: defiant
(B) despair: hopeful
(C) expectation: unfulfilled
(D) circumspection: careful
(E) ennui: listless

11. PAEAN: JOY::

- (A) dirge: grief
- (B) oratory: persuasion
- (C) aria: opera
- (D) chant: choir
- (E) lecture: instruction

12. RENEGADE: ALLEGIANCE::

- (A) revolutionary: reform
- (B) aesthete: discernment
- (C) apostate: faith
- (D) politician: challenge
- (E) criminal: imprisonment

13. DEVOTED: ZEALOUS::

- (A) affectionate: demonstrative
- (B) animated: lively
- (C) rabid: extreme
- (D) objective: indifferent
- (E) careful: fastidious

14. VESTIGE: REMAINDER::

- (A) figurine: statue
- (B) knife: cutlery
- (C) hub: wheel
- (D) angle: slope
- (E) inventory: goods

15. EPHEMERAL: ENDURE::

- (A) insensitive: cooperate
- (B) infirm: react
- (C) ineffectual: proceed
- (D) inelastic: stretch
- (E) inflammable: ignite

16. MISDEMEANOR: CRIME::

- (A) interview: conversation
- (B) lapse: error
- (C) oath: promise
- (D) rebuke: criticism
- (E) vendetta: feud

One explanation for the tendency of animals to be more vigilant in smaller groups than in larger ones assumes that the vigilant behavior—looking up, for

example—is aimed at predators. If individuals on the (5) edge of a group are more vigilant because they are at greater risk of being captured, then individuals on average would have to be more vigilant in smaller groups, because the animals on the periphery of a group form a greater proportion of the whole group as the size of the (10) group diminishes.

However, a different explanation is necessary in cases where the vigilant behavior is not directed at predators. J. Krebs has discovered that great blue herons look up more often when in smaller flocks than when in larger (15) ones, solely as a consequence of poor feeding conditions. Krebs hypothesizes that the herons in smaller flocks are watching for herons that they might follow to better feeding pools, which usually attract larger numbers of the birds.

17. It can be inferred from the passage that in species in which vigilant behavior is directed at predators, the tendency of the animals to be more vigilant in smaller groups than in larger ones would most likely be minimized if which of the following were true?

- (A) The vigilance of animals on the periphery of a group always exceeded that of animals located in its interior, even when predators were not in the area.
- (B) The risk of capture for individuals in a group was the same, whether they were located in the interior of the group or on its periphery.
- (C) Animals on the periphery of a group tended to be less capable of defending themselves from attack by predators than animals located in the interior of the group.
- (D) Animals on the periphery of a group tended to bear marks that were more distinctive to predators than animals located in the interior of the group.
- (E) Animals on the periphery of a group tended to have shorter life spans than animals located in the interior of the group.

18. Which of the following best describes the relationship of the second paragraph to the first?

- (A) The second paragraph relies on different evidence in drawing a conclusion similar to that expressed in the first paragraph.

- (B) The second paragraph provides further elaboration on why an assertion made at the end of the first paragraph proves to be true in most cases.
- (C) The second paragraph provides additional information in support of a hypothesis stated in the first paragraph.
- (D) The second paragraph provides an example of a case in which the assumption described in the first paragraph is unwarranted.
- (E) The second paragraph describes a phenomenon that has the same cause as the phenomenon described in the first paragraph.
19. It can be inferred from the passage that the author of the passage would be most likely to agree with which of the following assertions about vigilant behavior?
- (A) The larger the group of animals, the higher the probability that individuals in the interior of the group will exhibit vigilant behavior.
- (B) Vigilant behavior exhibited by individuals in small groups is more effective at warding off predators than the same behavior exhibited by individuals in larger groups.
- (C) Vigilant behavior is easier to analyze in species that are preyed upon by many different predators than in species that are preyed upon by relatively few of them.
- (D) The term "vigilant," when used in reference to the behavior of animals, does not refer exclusively to behavior aimed at avoiding predators.
- (E) The term "vigilant," when used in reference to the behavior of animals, usually refers to behavior exhibited by large groups of animals.
20. The passage provides information in support of which of the following assertions?
- (A) The avoidance of predators is more important to an animal's survival than is the quest for food.
- (B) Vigilant behavior aimed at predators is seldom more beneficial to groups of animals than to individual animals.
- (C) Different species of animals often develop different strategies for dealing with predators.
- (D) The size of a group of animals does not necessarily reflect its success in finding food.
- (E) Similar behavior in different species of animals does not necessarily serve the same purpose.
- The earliest controversies about the relationship between photography and art centered on whether photography's fidelity to appearances and dependence on a machine allowed it to be a fine art as distinct from
- (5) merely a practical art. Throughout the nineteenth century, the defense of photography was identical with the struggle to establish it as a fine art. Against the charge that photography was a soulless, mechanical copying of reality, photographers asserted that it was instead a privileged
- (10) way of seeing, a revolt against commonplace vision, and no less worthy an art than painting.
- Ironically, now that photography is securely established as a fine art, many photographers find it pretentious or irrelevant to label it as such. Serious photographers variously claim to be finding, recording, impartially observing, witnessing events, exploring themselves—anything but making works of art. In the nineteenth century, photography's association with the real world placed it in an ambivalent relation to art; late in the twentieth
- (20) century, an ambivalent relation exists because of the Modernist heritage in art. That important photographers are no longer willing to debate whether photography is or is not a fine art, except to proclaim that their own work is not involved with art, shows the extent to which
- (25) they simply take for granted the concept of art imposed by the triumph of Modernism: the better the art, the more subversive it is of the traditional aims of art.
- Photographers' disclaimers of any interest in making art tell us more about the harried status of the contemporary
- (30) notion of art than about whether photography is or is not art. For example, those photographers who suppose that, by taking pictures, they are getting away from the pretensions of art as exemplified by painting remind us of those Abstract Expressionist painters who imagined
- (35) they were getting away from the intellectual austerity of classical Modernist painting by concentrating on the physical act of painting. Much of photography's prestige today derives from the convergence of its aims with those of recent art, particularly with the dismissal of abstract
- (40) art implicit in the phenomenon of Pop painting during the 1960's. Appreciating photographs is a relief to sensi-

bilities tired of the mental exertions demanded by abstract art. Classical Modernist painting—that is, abstract art as developed in different ways by Picasso, (45)Kandinsky, and Matisse—presupposes highly developed skills of looking and a familiarity with other paintings and the history of art. Photography, like Pop painting, reassures viewers that art is not hard; photography seems to be more about its subjects than about art.

(50) Photography, however, has developed all the anxieties and self-consciousness of a classic Modernist art. Many professionals privately have begun to worry that the promotion of photography as an activity subversive of the traditional pretensions of art has gone so far that the (55)public will forget that photography is a distinctive and exalted activity—in short, an art.

21. In the passage, the author is primarily concerned with

- (A) defining the Modernist attitude toward art
- (B) explaining how photography emerged as a fine art after the controversies of the nineteenth century
- (C) explaining the attitudes of serious contemporary photographers toward photography as art and placing those attitudes in their historical context
- (D) defining the various approaches that serious contemporary photographers take toward their art and assessing the value of each of those approaches
- (E) identifying the ways that recent movements in painting and sculpture have influenced the techniques employed by serious photographers

22. Which of the following adjectives best describes “the concept of art imposed by the triumph of Modernism” as the author represents it in lines 25-27?

- (A) Objective
- (B) Mechanical
- (C) Superficial
- (D) Dramatic
- (E) Paradoxical

23. The author introduces Abstract Expressionist painters (lines 34) in order to

- (A) provide an example of artists who, like serious contemporary photographers, disavowed

- traditionally accepted aims of modern art
- (B) call attention to artists whose works often bear a physical resemblance to the works of serious contemporary photographers
- (C) set forth an analogy between the Abstract Expressionist painters and classical Modernist painters
- (D) provide a contrast to Pop artists and others who created works that exemplify the Modernist heritage in art
- (E) provide an explanation of why serious photography, like other contemporary visual forms, is not and should not pretend to be an art

24. According to the author, the nineteenth-century defenders of photography mentioned in the passage stressed that photography was

- (A) a means of making people familiar with remote locales and unfamiliar things
- (B) a technologically advanced activity
- (C) a device for observing the world impartially
- (D) an art comparable to painting
- (E) an art that would eventually replace the traditional arts

25. According to the passage, which of the following best explains the reaction of serious contemporary photographers to the question of whether photography is an art?

- (A) The photographers' belief that their reliance on an impersonal machine to produce their art requires the surrender of the authority of their personal vision
- (B) The photographers' fear that serious photography may not be accepted as an art by the contemporary art public
- (C) The influence of Abstract Expressionist painting and Pop Art on the subject matter of the modern photograph
- (D) The photographers' belief that the best art is subversive of art as it has previously been defined
- (E) The notorious difficulty of defining art in its relation to realistic representation

26. According to the passage, certain serious contemporary photographers expressly make which of the following claims about their photographs?

- (A) Their photographs could be created by almost anyone who had a camera and the time to devote to the activity.
- (B) Their photographs are not examples of art but are examples of the photographers' impartial observation of the world.
- (C) Their photographs are important because of their subjects but not because of the responses they evoke in viewers.
- (D) Their photographs exhibit the same ageless principles of form and shading that have been used in painting.
- (E) Their photographs represent a conscious glorification of the mechanical aspects of twentieth-century life.

27. It can be inferred from the passage that the author most probably considers serious contemporary photography to be a

- (A) contemporary art that is struggling to be accepted as fine art
- (B) craft requiring sensitivity but by no means an art
- (C) mechanical copying of reality
- (D) modern art that displays the Modernist tendency to try to subvert the prevailing aims of art
- (E) modern art that displays the tendency of all Modernist art to become increasingly formal and abstract

28. PREOCCUPATION:

- (A) finality
- (B) innocence
- (C) liberality
- (D) unconcern
- (E) tolerance

29. CHROMATIC:

- (A) opaque
- (B) colorless
- (C) lengthy
- (D) profound
- (E) diffuse

30. PEDESTRIAN:

- (A) widely known
- (B) strongly motivated
- (C) discernible
- (D) uncommon
- (E) productive

31. EQUIVOCATE:

- (A) communicate straightforwardly
- (B) articulate persuasively
- (C) instruct exhaustively
- (D) study painstakingly
- (E) reproach sternly

32. DENUDE:

- (A) crowd out
- (B) skim over
- (C) change color
- (D) cover
- (E) sustain

33. RANCOR:

- (A) deference
- (B) optimism
- (C) courage
- (D) superiority
- (E) goodwill

34. OSSIFIED:

- (A) vulnerable to destruction
- (B) subject to illusion
- (C) worthy of consideration
- (D) capable of repetition
- (E) amenable to change

35. CONTROVERT:

- (A) substantiate
- (B) transform
- (C) ameliorate
- (D) simplify
- (E) differentiate

36. PROTRACT:

- (A) thrust
- (B) reverse

- (C) curtail
(D) disperse
(E) forestall

37. ABRADE:

- (A) unfasten
(B) prolong
(C) augment
(D) extinguish
(E) transmit

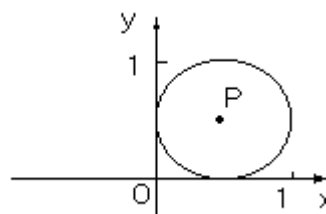
38. APOLOGIST:

- (A) egotist
(B) wrongdoer
(C) freethinker
(D) detractor
(E) spendthrift

SECTION 5

Time –30 minutes

30 Questions

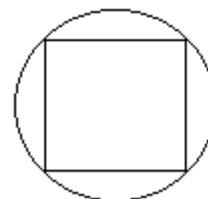


In the rectangular coordinate system, the circle with center P is tangent to both the x - and y -axes.

1. The x -coordinate of P The y -coordinate of P

2. $\frac{3}{5} + \frac{2}{3}$ 1

3. $|x^2|$ $|x|^2$



The square is inscribed in the circle.

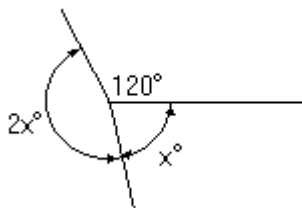
4. The length of a diagonal of the square The length of a diameter of the circle

5. $x < y < 20$
 $x + y$ 35

In college M the average (arithmetic mean) number of students per course is 30 and the ratio of the number of students to the number of faculty is 20 to 1.

6. The total number of students in College M 600

7. $x > 0$
 $\frac{590 + x}{800}$ $\frac{600 + x}{790}$



8. x 80

Integer n will be randomly selected from the integers 1 to 13, inclusive.

9. The probability that n will be even The probability that n will be odd

$$p + q = 1$$

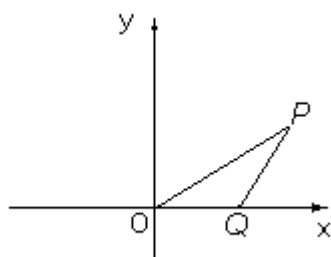
$$0 < p < q$$

10. $\frac{1}{pq}$ 1

$$2x + 3y = 29$$

$$3x + 4y = 41$$

11. $x + y$ 12



$$PQ = OQ = 5$$

12. The area of region OPQ 10

13. $(\sqrt{5} + \sqrt{5})^2$ 20

$$\frac{n}{4} + \frac{r}{8} = \frac{s}{8} + \frac{t}{6}$$

$n, r, s,$ and t are positive integers.

14. $2n + r$ $2s + t$

In the xy -coordinate system, the point (x, y) lies on the circle with equation $x^2 + y^2 = 1$

15. $x + y$ 1.01

16. A health food store prepares a breakfast food that consists of oats, raisins, and nuts mixed in the ratio 9:2:1, respectively, by weight. If the nuts in the mixture weigh 9.2 pounds, how many pounds does the total mixture weigh?

- (A) 82.2
(B) 92.2
(C) 101.2
(D) 110.4
(E) 165.6

17. $3 - 2[5 - 7(3 + 2)] =$

- (A) -30
(B) -10
(C) 23
(D) 63
(E) 77

18. For every positive integer n greater than 1, $n!$ is defined as the product of the first n positive integers. For example, $4! = (1)(2)(3)(4) = 24$. What is the value of $\frac{12!}{10!}$?

- (A) 2
(B) 66
(C) 121
(D) 132
(E) 144

19. A market survey showed that 76 percent of the visitors at a certain resort came from Pacific or southwestern states. Of these, $\frac{2}{3}$ were from

California, and 87 percent of the Californians were from southern California. Approximately what percent of the visitors at the resort were from southern California?

- (A) 40%
(B) 45%
(C) 50%
(D) 55%
(E) 65%

20 If $\frac{5^4 - 1}{n}$ is an integer and n is an integer, then n

could be each of the following EXCEPT

- (A) 4
- (B) 6
- (C) 13
- (D) 25
- (E) 26

Questions 21-22 refer to the following table.

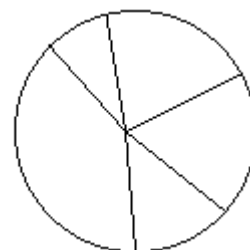
DISTRIBUTION OF A FAMILY'S
ANNUAL BUDGET OF \$42,000

Category	Percent
Food	20%
Housing	18%
Entertainment	5%
Savings	15%
Goods and Services	42%

21. What is the ratio of the amount budgeted annually for food to the amount budgeted annually

for savings?

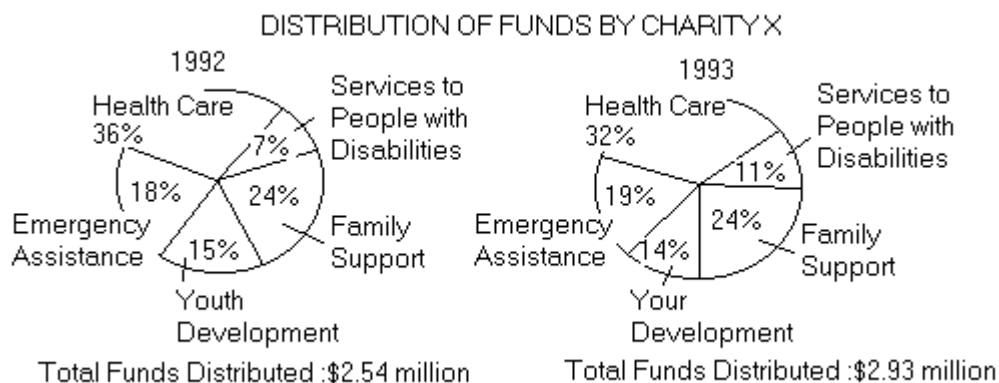
- (A) 4 to 3
- (B) 4 to 7
- (C) 5 to 3
- (D) 7 to 3
- (E) 7 to 4



22. If a pie graph (such as the one above) were drawn to scale to represent the budget distribution into the five categories, what would be the measure of the central angle of the sector representing savings?

- (A) 15°
- (B) 30°
- (C) 36°
- (D) 42°
- (E) 54°

Questions 23-25 refer to the following graphs.



23. The funds distributed in 1992 for youth development were approximately

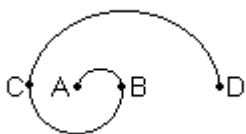
- (A) \$38,000
- (B) \$170,000
- (C) \$380,000
- (D) \$450,000
- (E) \$1,700,000

24. The increase in the amount of money distributed for family support from 1992 to 1993 was closest to which of the following?

- (A) \$0
- (B) \$24,000
- (C) \$40,000
- (D) \$60,000
- (E) \$94,000

25. If all of the emergency assistance funds in 1993 were distributed among 40 groups, which of the following is closest to the average (arithmetic mean) amount distributed per group?

(A) \$10,000
(B) \$11,000
(C) \$12,000
(D) \$13,000
(E) \$14,000



26. The curve above consists of three semicircles: AB , BC , and CD . The diameter of AB is 2, the diameter of BC is twice the diameter of AB , and the diameter of CD is twice the diameter of BC . What is the total length of the curve?

(A) $2p$
(B) $4p$
(C) $6p$
(D) $7p$
(E) $8p$

27. What is the cost, in cents, of using a certain fax machine to send n pages of a report if the total cost for sending the first k pages is r cents and the cost for sending each additional page is s cents?

(Assume that $n > k$.)

(A) $r + s(n - k)$
(B) $r + s(n + k)$
(C) $rs(n + k)$
(D) $kr + s(n - k)$
(E) $kr + ns$

28. A rectangular solid has a square base and altitude of 7. If the volume of the solid is 252, then the perimeter of the square base is

(A) 9
(B) 24
(C) 28
(D) 36
(E) 49

29. In a series of races, 10 toy cars are raced, 2 cars at a time. If each car must race each of the other cars exactly twice, how many races must be held?

(A) 40
(B) 90
(C) 100
(D) 180
(E) 200

30. $(2^{10} - 2^9)(2^8 - 2^7) =$

(A) 2
(B) 2^2
(C) 2^4
(D) 2^8
(E) 2^{16}

SECTION 6

Time –30 minutes

25 Questions

1. Soft Drink Manufacturer : Our new children's soft drink, RipeCal, is fortified with calcium. Since calcium is essential for developing healthy bones, drinking RipeCal regularly will help make children healthy.

Consumer Advocate : But RipeCal also contains large amounts of sugar, and regularly consuming large amounts of sugar is unhealthful, especially for children.

In responding to the soft drink manufacturer, the consumer advocate does which of the following ?

- (A) Challenges the manufacturer's claim about the nutritional value of calcium in children's diets
- (B) Argues that the evidence cited by the manufacturer, when properly considered, leads to a conclusion opposite to that reached by the manufacturer.
- (C) Implies that the manufacturer of a product is typically unconcerned with the nutritional value of that product.
- (D) Questions whether a substance that is healthful when eaten in moderation can be unhealthful when eaten in excessive amounts.
- (E) Presents additional facts that call into question the conclusion drawn by the manufacturer.

2. Over a period of several months, researchers attached small lights to the backs of wetas—flightless insects native to New Zealand—enabling researchers for the first time to make comprehensive observations of the insects' nighttime activities. Thus, since wetas forage only at night, the researchers' observations will significantly improve knowledge of the normal foraging habits of wetas.

Which of the following is an assumption on which the argument depends ?

- (A) Researchers were interested only in observing the wetas' foraging habits and so did not keep

track of other types of behavior.

- (B) No pattern of behavior that is exhibited by wetas during the nighttime is also exhibited by wetas during the daytime.
- (C) Attaching the small lights to the wetas' backs did not greatly alter the wetas' normal nighttime foraging habits.
- (D) Wet as typically forage more frequently during the months in which the researchers studied them than they do at other times.
- (E) The researchers did not use other observational techniques to supplement their method of using small lights to track the nighttime behavior of wetas.

Questions 3-8

On each day of the Monday through Friday workweek, exactly two of three employees—George, Hilda, and Irene—must remain on duty in the office to answer telephones during the noon to 1 p.m. lunch break. In the scheduling of telephone duty the following constraints must be met :

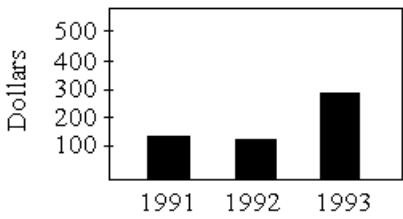
Hilda and Irene have telephone duty together on Tuesday.

George and Hilda have telephone duty together on Thursday.

No one can have telephone duty on more than four of the five days of the week.

3. Which of the following can be the week's telephone duty schedule ?

	<u>Monday</u>	<u>Tuesday</u>	<u>Wednesday</u>	<u>Thursday</u>	<u>Friday</u>
(A)	George Hilda	George Irene	Hilda Irene	George Hilda	Hilda Irene
(B)	George Hilda	Hilda Irene	George Hilda	Hilda Irene	George Irene
(C)	George Irene	Hilda Irene	George Hilda	George Hilda	George Hilda
(D)	Hilda Irene	Hilda Irene	George Hilda	George Hilda	Hilda Irene
(E)	Hilda Irene	Hilda Irene	George Hilda	George Irene	George Irene

- 4.If Hilda has telephone duty for exactly two days of the week, which of the following must be true ?
 (A)George and Irene have telephone duty together on Wednesday.
 (B)George and Hilda have telephone duty together on Friday.
 (C)Hilda and Irene have telephone duty together on Wednesday.
 (D)Hilda and Irene have telephone duty together on Friday.
 (E)Irene has telephone duty for exactly three days of the week.
- 5.If Hilda and Irene have telephone duty together on Monday and on Wednesday, which of the following must be true ?
 (A)George and Hilda have telephone duty together on Friday.
 (B)George and Irene have telephone duty together on Friday.
 (C)George has telephone duty on exactly three of the days of the week.
 (D)Hilda has telephone duty on exactly three of the days of the week.
 (E)Irene has telephone duty on exactly three of the days of the week.
- 6.If George and Hilda have telephone duty together on Monday and George and Irene have telephone duty together on Friday, any of the following can be true EXCEPT :
 (A)George and Hilda have telephone duty together on Wednesday.
 (B)George and Irene have telephone duty together on Wednesday.
 (C)George has telephone duty for four days of the week.
 (D)Irene has telephone duty for four days of the week.
 (E)Hilda and Irene have telephone duty together for two days of the week.
- 7.If there is one pair of employees who have telephone duty together for three of the five days, which of the following must be true ?
 (A)George has telephone duty on Wednesday.
 (B)George and Hilda have telephone duty together for three days of the week.
 (C)Hilda and Irene have telephone duty together for three days of the week.
 (D)One of the three employees has telephone duty for exactly two days of the week.
 (E)Exactly one of the workers has telephone duty for exactly three days of the week.
- 8.Any of the following can be true EXCEPT :
 (A)One pair of employees has telephone duty together for exactly one day of the week
 (B)One pair of employees has telephone duty together for exactly four days of the week
 (C)The pair of employees that has telephone duty together on Monday also has telephone duty together on Wednesday
 (D)The pair of employees that has telephone duty together on Tuesday also has telephone duty together on Wednesday
 (E)The pair of employees that has telephone duty together on Thursday also has telephone duty together on Friday
- Question 9 is based on the following graph
- AVERAGE DAILY ICE CREAM SALES
FOR THE MONTH OF JANUARY
AT AN AIRPORT IN THE
NORTHEASTERN UNITED STATES
- 
- | Year | Average Daily Sales (Dollars) |
|------|-------------------------------|
| 1991 | 150 |
| 1992 | 120 |
| 1993 | 280 |
- 9.Which of the following, if true, contributes most to an explanation of the change in ice cream sales?
 (A)Because of low demand for its transcontinental flights departing from the airport, one airline ceased offering such flights as of December 31,1993.
 (B)There were fewer airline passengers who were traveling to destinations outside the United States in January1993 than there were in January1992

- (C)The average daily number of passengers at the airport in the month of January was the same for each of the three years
- (D)In January 1993 a blizzard forced all flights out of the airport to be canceled for three days, stranding many passengers at the airport.
- (E)There were five percent fewer commuter flights scheduled to depart from the airport in January 1993 than there were in January of either 1991 or 1992

10. People whose bodies cannot produce the substance cytochrome P450 are three times as likely to develop Parkinson's disease, a disease that affects the brain, as are people whose bodies do produce this substance. Since cytochrome P450 protects the brain from toxic chemicals, toxic chemicals probably play a role in the development of Parkinson's disease.

Which of the following, if true, most strongly supports the argument ?

- (A)It will soon be possible for cytochrome P450 to be synthesized for the treatment of people whose bodies cannot produce this substance.
- (B)Many people whose bodies are unable to produce cytochrome P450 lack the ability to produce certain other substances as well.
- (C)Cytochrome P450 has no effect on the brain other than to protect it from toxic chemicals.
- (D)People with Parkinson's disease often exhibit a marked lessening in the severity of their symptoms when they are treated with dopamine, a chemical produced naturally in the brain.
- (E)Many people with Parkinson's disease have the ability to produce cytochrome P450 naturally.

11. The early universe contained only the lightest elements, hydrogen and helium. Heavier elements, such as carbon, form only in nuclear reactions in stars and are dispersed when the stars explode. A recently discovered gas cloud contained carbon several billion years ago, when the universe was no more than two billion years old.

If the statements above are true, which of the following must, on the basis of them, also be true ?

- (A)The earliest stars contained only hydrogen.
- (B)Some stars were formed before the universe was two billion years old.
- (C)The carbon in the gas cloud later formed part of some stars.
- (D)No stars identified to date are as old as the gas cloud.
- (E)The gas cloud also contained hydrogen and helium.

12. Sleep deprivation is a known cause of workplace error, and many physicians frequently go without sleep for periods of 24 hours or more. However, few of these physicians have, in the course of a routine examination by a peer, been diagnosed with sleep deprivation. So there is little cause for concern that habitual sleep deprivation will cause widespread physician error.

The answer to which of the following questions would be most helpful in evaluating the argument ?

- (A)Do physicians who have been diagnosed with sleep disorders also show signs of other ills not related to sleep deprivation ?
- (B)Is the ability to recognize the symptoms of sleep deprivation in others significantly impaired by habitual sleep deprivation ?
- (C)Do factors other than habitual sleep deprivation ever lead to errors in the workplace on the part of physicians ?
- (D)Of people who have recently been treated by physicians, what percentage believe that many physicians have occasionally suffered from sleep deprivation ?
- (E)Is the incidence of sleep deprivation higher among physicians than it is among other health care workers ?

Questions 13-18

In a benefit concert each of exactly seven solo performers—Harris, Jones, McIntyre, Nelson, Strapp, Trevino, and Williams—will sing. The concert director is deciding the order in which the performers will sing and is assigning each to exactly one of seven time slots, numbered consecutively 1 through 7, according to the following conditions :

Harris must sing at some time before McIntyre sings.
Strapp must sing at some time before Jones sings.
Trevino must sing either immediately before or
immediately after Nelson sings.
Williams must be assigned to time slot 3.

13. Which of the following could be the order, from first through seventh, in which the performers sing ?
- (A) Harris, McIntyre, Williams, Trevino, Nelson, Strapp, Jones
- (B) Jones, Harris, Williams, Strapp, McIntyre, Nelson, Trevino
- (C) Strapp, McIntyre, Williams, Nelson, Trevino, Harris, Jones
- (D) Trevino, Harris, Williams, Strapp, Nelson, Jones, McIntyre
- (E) Trevino, Nelson, Harris, Strapp, Williams, McIntyre, Jones
14. If there are exactly four performers who are to sing after Nelson sings but before Strapp sings, Harris must be assigned to time slot
- (A) 1
- (B) 2
- (C) 4
- (D) 5
- (E) 6
15. If Williams is to sing immediately after Harris sings and immediately before Trevino sings, which of the following performers could be assigned to time slot 6 ?
- (A) Harris
- (B) Jones
- (C) Nelson
- (D) Strapp
- (E) Trevino
16. If Jones is to sing immediately before Harris sings, which of the following must be assigned to time slot 7 ?
- (A) Harris
- (B) Jones
- (C) McIntyre
- (D) Strapp
- (E) Trevino

- 17.If McIntyre is to sing immediately before Strapp sings, Trevino could be assigned to which of the following time slots ?
- (A)2
(B)4
(C)5
(D)6
(E)7
- 18.If McIntyre is assigned to time slot 4, which of the following must be true ?
- (A)Harris is assigned to a lower-numbered time slot than Strapp.
(B)Jones is assigned to a lower-numbered time slot than Trevino.
(C)Nelson is assigned to a lower-numbered time slot than McIntyre.
(D)Strapp is assigned to a lower-numbered time slot than Williams.
(E)Trevino is assigned to a lower-numbered time slot than Jones.

Questions 19-22

Along a street that is currently without trees, seven trees are to be planted in the pattern

$$\begin{array}{cccc} \underline{1} & \underline{3} & \underline{5} & \underline{7} \\ & \underline{2} & \underline{4} & \underline{6} \end{array}$$

Where each number designates the position of a tree. No more than two kinds of trees can be planted. If maples are used, no maple can be planted adjacent to or immediately diagonally opposite another maple. Two trees are adjacent to each other if the numbers of their positions differ by two, and immediately diagonally opposite if their numbers differ by one. The following trees, of three

kinds, are available for planting:

- Three red oaks
Four maples
Four sycamores

19. Which of the following can be the trees planted along the side of the street that has four trees, in order of their positions beginning with position 1?

- (A)Maple, sycamore, maple, sycamore
- (B)Maple, sycamore, red oak, maple
- (C)Red oak, maple, maple, red oak
- (D)Sycamore, sycamore, maple, maple
- (E)Sycamore, sycamore, red oak, red oak

20.If red oaks are used, then which of the following must be true ?

- (A)The other trees used are all maples.
- (B)The other trees used are all sycamores.
- (C)The red oaks are in positions 1, 2, and 3.
- (D)The red oaks are in positions 3, 4, and 5.
- (E)The red oaks are in positions 4, 5, and 6

21.Among the trees left over after the planting is done there must be

- (A)at least one maple
- (B)at least one red oak
- (C)at least one sycamore
- (D)at most one maple
- (E)at most one red oak

22.If maples are planted, the side of the street that has four trees must have

- (A)red oaks in positions 1 and 7
- (B)red oaks in positions 3 and 5
- (C)sycamores in positions 1 and 3
- (D)sycamores in positions 1 and 7
- (E)sycamores in positions 3 and 5

23.A list of the fifteen operas most frequently performed in recent times includes no works by the nineteenth-century German composer Richard Wagner. Although music producers tend to produce what audiences want, relative infrequency of performance probably does not indicate lack of popularity in Wagner's case, since Wagner's operas are notoriously expensive to perform on stage.

Which of the following, if true, most strongly supports the conclusion of the argument above ?

- (A)The list of most frequently performed operas does not include operas produced by small amateur groups.
- (B)Some opera companies are backed by patrons who are willing to commit large sums of

money in order to enjoy lavish productions.

- (C)All of the fifteen most frequently performed operas of recent times are works that have been popular for at least 75 years.
- (D)More recordings have been produced recently of the works of Wagner than of the works of any other composer of opera.
- (E)Operatic works of all kinds have been increasing in popularity in recent years.

24.The bodies of dwarf individuals of mammalian species are generally smaller in relation to those of nondwarf individuals than are the teeth of the dwarf individuals in relation to those of the nondwarf individuals. Fragmentary skeletal remains of an adult dwarf woolly mammoth were recently found. The teeth are three-fourths the size of the teeth of an average adult nondwarf woolly mammoth.

The statements above, if true, most strongly support which of the following ?

- (A)The body of the dwarf woolly mammoth was less than three-fourths the size of the body of an average adult nondwarf woolly mammoth.
- (B)None of the teeth of the dwarf woolly mammoth that were recently discovered was as large as any of the teeth of nondwarf woolly mammoths that have been discovered.
- (C)The teeth of most adult dwarf individuals of mammalian species are three-fourths the size of the teeth of the adult nondwarf individuals of the same species.
- (D)Dwarf woolly mammoths had the same number of teeth as did nondwarf woolly mammoths.
- (E)Dwarf individuals of most mammalian species are generally no more than three-fourths the size of the adult nondwarf individuals of those species.

25.Excluding purchases by businesses, the average amount spent on a factory-new car has risen 30 percent in the last five years. In the average household budget, the proportion spent on car purchases has remained unchanged in that period. Therefore the average household budget must have increased by 30 percent over the last five years.

Which of the following is an assumption on which the argument relies ?

- (A) The average number of factory-new cars purchased per household has remained unchanged over the last five years.
- (B) The average amount spent per car by businesses buying factory-new cars has risen 30 percent in the last five years.
- (C) The proportion of the average household budget spent on all car-related expenses has remained unchanged over the last five years.
- (D) The proportion of the average household budget spent on food and housing has remained unchanged over the last five years.
- (E) The total amount spent nationwide on factory-new cars has increased by 30 percent over the last five years.

SECTION 1

Time – 30 minutes

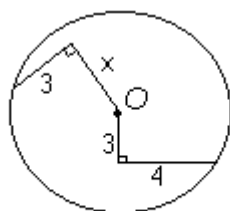
30 Questions

1. $2(8-7)$ $2(7-8)$

$$x + y = 2$$

2. x y

3. $\frac{3}{7} + \frac{2}{7}$ 1

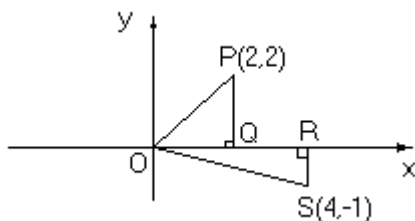


O is the center of the circle

4. x 5

On Elm Street there are 6 houses on one side of the street and 4 houses on the other. Each pair of houses on Elm Street is connected by exactly one telephone line.

5. The total number of such lines that connect houses on opposite sides of Elm Street 12

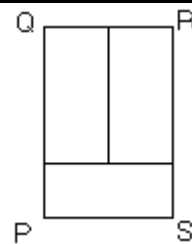


6. The area of triangular region OPQ The area of triangular region ORS

7. $(0.01)(0.07)(70)$ 0.49

$$x < y < z$$

8. $\frac{x+y+z}{3}$ z



The three small rectangles have the same dimensions

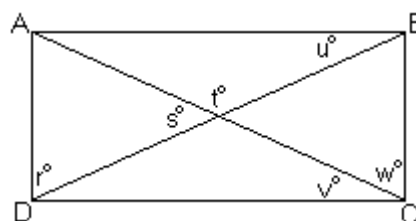
9. $\frac{PS}{RS}$ $\frac{1}{2}$

In a certain city, 20°F was the average (arithmetic mean) of the low temperatures of $x^\circ\text{F}$, 25°F , and 37°F on three consecutive days.

10. x 0

$$m = 4x + 4y, x \neq -y$$

11. $\frac{2m}{x+y}$ 8



$ABCD$ is a rectangle with diagonals AC and DB .

12. $r + u + v$ $r + u + v + w$

n is a positive integer.

13. n^{100} 100^n

$f(t) = kt$ for all t , where k is a constant, and

$$f(3) = \frac{1}{2}$$

14. k $f(1)$

$$100x < y$$

$$1,000x < 2y$$

15. $1,100x$ y

16. Mr. Gifford wishes to put 372 eggs into cartons that can hold 12 eggs each. If he has 50 empty cartons and completely fills as many of them as possible with the 327 eggs, how many of the cartons will remain empty?

(A) 12
(B) 15
(C) 19
(D) 28
(E) 31

17. Which of the following numbers is greatest?

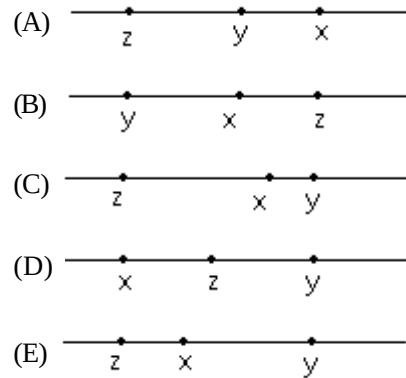
(A) -0.225
(B) -0.0225
(C) -0.323
(D) -0.0325
(E) -0.3205

18. If a certain automobile gets between 20 and 24 miles per gallon of gasoline, inclusive, what would be the maximum amount of gasoline, in gallons, this automobile would consume on a trip of 360 miles?

(A) 20.0
(B) 18.0
(C) 16.4
(D) 16.0
(E) 15.0

19. If $y - x = 2$ and $y - z = 3$, which of the following best represents the relative positions of x , y , and z on the

number line? (Note: The figures are drawn to scale.)

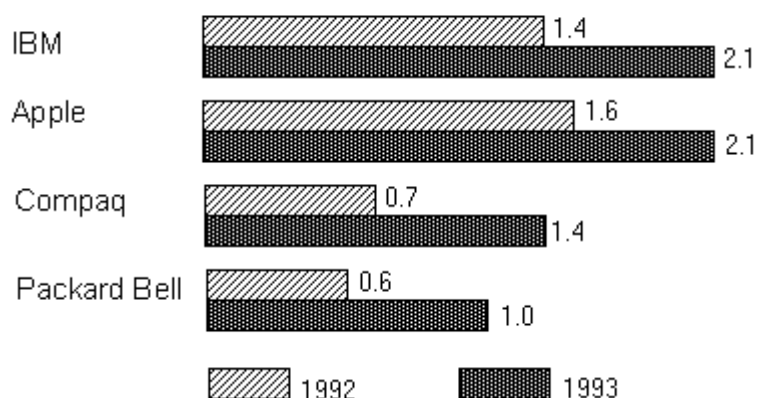


20. Two beads are to be independently and randomly selected, one from each of two bags. If $\frac{2}{7}$ of the beads in one bag and $\frac{3}{7}$ of the beads in the other bag are yellow, what is the probability that both beads selected will be yellow?

(A) $\frac{2}{3}$
(B) $\frac{5}{7}$
(C) $\frac{6}{7}$
(D) $\frac{4}{49}$
(E) $\frac{6}{49}$

Questions 21-23 refer to the graph below.

NUMBER OF SELECTED PERSONAL COMPUTERS SOLD.
1992-1993(in millions)



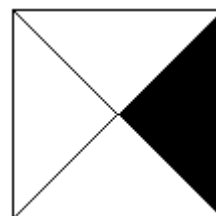
21. By what percent did the number of personal computers sold by Compaq increase from 1992 to 1993?
- (A) 50%
(B) 65%
(C) 75%
(D) 100%
(E) 110%
22. In 1992, Packard Bell accounted for what percent of the computers sold by the four companies listed?
- (A) 6%
(B) 9%
(C) 10%
- (D) 12%
(E) 14%
23. If the ratio of the number of personal computers sold by IBM Compaq, and Tandy (not shown) in 1993 was 6 to 4 to 1, respectively, approximately how many personal computers were sold by Tandy in 1993?
- (A) 350,000
(B) 400,000
(C) 450,000
(D) 500,000
(E) 550,000

Questions 24-25 refer to the following table.

EXPECTED NUMBER OF GRADUATES IN THE UNITED STATES
BY DEGREE GRANTED BETWEEN 1995 AND 2001(in thousands)

	High School Diploma	Associate Degree	Bachelor's Degree	Master's Degree	Doctoral Degree
1995	2,329	458	1,047	324	36
1997	2,452	463	1,015	324	36
1999	2,587	477	1,010	325	36
2001	2,865	489	1,037	327	36

24. For the categories given, which category accounts for approximately $\frac{1}{4}$ of the total number of graduates expected for each of the years shown?
- (A) High school diploma
(B) Associate degree
(C) Bachelor's degree
(D) Master's degree
(E) Doctoral degree
25. The number of associate degrees expected to be granted in 2001 is most nearly what percent greater than the number of associate degrees expected to be granted in 1995?
- (A) 2%
(B) 3%
(C) 5%
(D) 7%
(E) 9%
26. If the area of the shaded region of the square above is 20, what is the perimeter of the square?
- (A) $4\sqrt{5}$
(B) $8\sqrt{5}$
(C) $16\sqrt{5}$
(D) 80
(E) 400



27. If $x = \frac{1}{y}$ and $y = \frac{1}{1-x}$, then $y =$

- (A) 2
- (B) $\frac{1}{2}$
- (C) $-\frac{1}{2}$
- (D) -1
- (E) -2

28. If 720 is the product of the consecutive integers beginning with 2 and ending with n , what is the value of $n-1$?

- (A) 5
- (B) 6
- (C) 8
- (D) 11
- (E) 23

29. When it was found that 150 more tickets for the school play were sold than the seating capacity of the auditorium. It was decided to have two performances. if the total number of tickets sold was equal to the total number who attended and if the auditorium was $\frac{2}{3}$ full for each of the two performances, what is the seating capacity of the auditorium?

- (A) 100
- (B) 200
- (C) 225
- (D) 300
- (E) 450

30. If $n = pqr$, where p , q , and r are three different positive prime numbers, how many different positive divisors does n have, including 1 and n ?

- (A) 3
- (B) 5
- (C) 6
- (D) 7
- (E) 8

SECTION 2

Time -30 minutes

38 Questions

1. Though ---- to some degree, telling a small lie sometimes enables one to avoid ---- another's feelings.
(A) necessary.. mollifying
(B) regrettable.. harming
(C) unfortunate.. exaggerating
(D) attractive.. considering
(E) difficult.. resisting
2. Perhaps because scientists have been so intrigued by dogs' superior senses of smell and hearing, researchers have long ---- their eyesight, assuming that they inhabit a drab, black-and-white world, devoid of color.
(A) studied
(B) coveted
(C) appreciated
(D) resented
(E) underestimated
3. Despite a string of dismal earnings reports, the two-year-old strategy to return the company to profitability is beginning to ----.
(A) falter
(B) disappoint
(C) compete
(D) work
(E) circulate
4. The President reached a decision only after lengthy -----, painstakingly weighing the ----opinions expressed by cabinet members.
(A) deliberation.. divergent
(B) confrontation.. unanimous
(C) relegation.. consistent
(D) speculation.. conciliatory
(E) canvassing.. arbitrary
5. Although just barely ---- as a writer of lucid prose, Jones was an extremely ---- editor who worked superbly with other writers in helping them improve the clarity of their writing.

- (A) deficient.. muddling
 (B) proficient.. contentious
 (C) adequate.. capable
 (D) appalling.. competent
 (E) engaging.. inept
6. The accusations we bring against others should be ---- ourselves; they should not ---- complacency and easy judgments on our part concerning our own moral conduct.
 (A) definitions of.. produce
 (B) instructions to.. equate
 (C) denigrations of.. exclude
 (D) warnings to.. justify
 (E) parodies of.. satirize
7. Although the meanings of words may necessarily be liable to change, it does not follow that the lexicographer is therefore unable to render spelling, in a great measure, ----.
 (A) arbitrary
 (B) superfluous
 (C) interesting
 (D) flexible
 (E) constant
8. ELEGIAC: SORROW::
 (A) polemical: resolution
 (B) fictional: humor
 (C) devotional: reverence
 (D) didactic: inspiration
 (E) literary: emotion
9. ROSTRUM: ORATOR::
 (A) stage: audience
 (B) bench: judge
 (C) shelf: clerk
 (D) municipality: citizen
 (E) crosswalk: pedestrian
10. MISUNDERSTOOD: CLARIFY
 (A) fanatical: espouse
 (B) popular: renounce
 (C) fantastic: shock
 (D) erroneous: retract
 (E) conspicuous: flaunt

11. REFINERY: PETROLEUM::

- (A) mill: grain
 (B) mine: ore
 (C) warehouse: merchandise
 (D) generator: electricity
 (E) forest: lumber

12. TEDIOUS: ENERGY::

- (A) avaricious: satisfaction
 (B) fractious: irritation
 (C) disturbing: composure
 (D) improbable: ambition
 (E) informed: intelligence

13. GRACEFUL: MOVEMENT::

- (A) euphonious: sound
 (B) forbidding: countenance
 (C) ephemeral: duration
 (D) melodramatic: emotion
 (E) vibrant: color

14. BRAVURA: PERFORMANCE::

- (A) extravagant: expenditure
 (B) elaborate: oration
 (C) foreseeable: outcome
 (D) thorough: analysis
 (E) resplendent: appearance

15. BADGER: BOTHER::

- (A) persecute: injure
 (B) haunt: remember
 (C) belabor: mention
 (D) quibble: argue
 (E) censure: evaluate

16. CONGRUENT: DIMENSIONS::

- (A) convenient: time
 (B) coordinate: axis
 (C) conglomerate: parts
 (D) coincident: chance
 (E) coeval: age

It is possible for students to obtain advanced degrees in English while knowing little or nothing about traditional scholarly methods. The consequences of this neglect of

traditional scholarship are particularly unfortunate for the
(5) study of women writers. If the canon—the list of authors
whose works are most widely taught—is ever to include
more women, scholars must be well trained in historical
scholarship and textual editing. Scholars who do not know
how to read early manuscripts, locate rare books, establish
(10) a sequence of editions, and so on are bereft of crucial tools
for revising the canon.

To address such concerns, an experimental version of
the traditional scholarly methods course was designed to
raise students' consciousness about the usefulness of
(15) traditional learning for any modern critic or theorist. To
minimize the artificial aspects of the conventional course,
the usual procedure of assigning a large number of small
problems drawn from the entire range of historical periods
was abandoned, though this procedure has the obvious
(20) advantage of at least superficially familiarizing students
with a wide range of reference sources. Instead students
were engaged in a collective effort to do original work on
a neglected eighteenth-century writer, Elizabeth Griffith, to
give them an authentic experience of literary scholarship
(25) and to inspire them to take responsibility for the quality of
their own work.

Griffith's work presented a number of advantages for
this particular pedagogical purpose. First, the body of
extant scholarship on Griffith was so tiny that it could all
(30) be read in a day; thus students spent little time and effort
mastering the literature and had a clear field for their own
discoveries. Griffith's play *The Platonic Wife* exists in three
versions, enough to provide illustrations of editorial issues
but not too many for beginning students to manage. In addi-
(35) tion, because Griffith was successful in the eighteenth cen-
tury, as her continued productivity and favorable reviews
demonstrate, her exclusion from the canon and virtual dis-
appearance from literary history also helped raise issues
concerning the current canon.

(40) The range of Griffith's work meant that each student
could become the world's leading authority on a particular
Griffith text. For example, a student studying Griffith's
Wife in the Right obtained a first edition of the play and
studied it for some weeks. This student was suitably
(45) shocked and outraged to find its title transformed into *A*
Wife in the Night in Watt's *Bibliotheca Britannica*. Such
experiences, inevitable and common in working on a writer
to whom so little attention has been paid, serve to vaccinate
the student ---I hope for a lifetime---against credulous use

of reference sources.

17. The author of the passage is primarily concerned with
(A) revealing a commonly ignored deficiency
(B) proposing a return to traditional terminology
(C) describing an attempt to correct a shortcoming
(D) assessing the success of a new pedagogical approach
(E) predicting a change in a traditional teaching strategy
18. It can be inferred that the author of the passage expects
that the experience of the student mentioned as having
studied *Wife in the Right* would have which of the fol-
lowing effects?
(A) It would lead the student to disregard information
found in the *Bibliotheca Britannica*.
(B) It would teach the student to question the accuracy
of certain kinds of information sources when
studying neglected authors.
(C) It would teach the student to avoid the use of refer-
ence sources in studying neglected authors.
(D) It would help the student to understand the impor-
tance of first editions in establishing the author-
ship of plays.
(E) It would enhance the student's appreciation of the
works of authors not included in the canon.
19. The author of the passage suggests that which of the
following is a disadvantage of the strategy employed in
the experimental scholarly methods course?
(A) Students were not given an opportunity to study
women writers outside the canon.
(B) Students' original work would not be appreciated
by recognized scholars.
(C) Little scholarly work has been done on the work
of Elizabeth Griffith.
(D) Most of the students in the course had had little
opportunity to study eighteenth-century literature.
(E) Students were not given an opportunity to encoun-
ter certain sources of information that could
prove useful in their future studies.
20. Which of the following best states the "particular
pedagogical purpose" mentioned in line 28?
(A) To assist scholars in revising the canon of authors

- (B) To minimize the trivial aspects of the traditional scholarly methods course
- (C) To provide students with information about Griffith's work
- (D) To encourage scholarly rigor in students' own research
- (E) To reestablish Griffith's reputation as an author
21. Which of the following best describes the function of the last paragraph in relation to the passage as a whole?
- (A) It summarizes the benefits that students can derive from the experimental scholarly methods course.
- (B) It provides additional reasons why Griffith's work raises issues having to do with the canon of authors.
- (C) It provides an illustration of the immediate nature of the experiences students can derive from the experimental scholarly methods course.
- (D) It contrasts the experience of a student in the experimental scholarly methods course with the experience of a student in the traditional course
- (E) It provides information that emphasizes the suitability of Griffith's work for inclusion in the canon of authors.
22. It can be inferred that which of the following is most likely to be among the "issues" mentioned in line 38?
- (A) Why has the work of Griffith, a woman writer who was popular in her own century, been excluded from the canon?
- (B) In what ways did Griffith's work reflect the political climate of the eighteenth century?
- (C) How was Griffith's work received by literary critics during the eighteenth century?
- (D) How did the error in the title of Griffith's play come to be made?
- (E) How did critical reception of Griffith's work affect the quantity and quality of that work?
23. It can be inferred that the author of the passage considers traditional scholarly methods courses to be
- (A) irrelevant to the work of most students
- (B) inconsequential because of their narrow focus
- (C) unconcerned about the accuracy of reference sources
- (D) too superficial to establish important facts about authors
- (E) too wide-ranging to approximate genuine scholarly activity
- Experiments show that insects can function as pollinators of cycads, rare, palmlike tropical plants. Furthermore, cycads removed from their native habitats—and therefore from insects native to those habitats—are usually infertile. Nevertheless, anecdotal reports of wind pollination in cycads cannot be ignored. The structure of cycads male cones is quite consistent with the wind dispersal of pollen, clouds of which are released from some of the larger cones. The male cone of *Cycas circinalis* for example, sheds almost 100 cubic centimeters of pollen, most of which is probably dispersed by wind. Still, many male cycad cones are comparatively small and thus produce far less pollen. Furthermore, the structure of most female cycad cones seems inconsistent with direct pollination by wind. Only in the *Cycas* genus are the females' ovules accessible to airborne pollen, since only in this genus are the ovules surrounded by a loose aggregation of megasporophylls rather than by a tight cone.
24. According to the passage, the size of a male cycad cone directly influences which of the following?
- (A) The arrangement of the male cone's structural elements
- (B) The mechanism by which pollen is released from the male cone.
- (C) The degree to which the ovules of female cycads are accessible to airborne pollen
- (D) The male cone's attractiveness to potential insect pollinators
- (E) The amount of pollen produced by the male cone
25. The passage suggests that which of the following is true of the structure of cycad cones?
- (A) The structure of cycad cones provides conclusive evidence in favor of one particular explanation of cycad pollination.
- (B) The structure of cycad cones provides evidence concerning what triggers the first step in the pollination process.
- (C) An irresolvable discrepancy exists between what the structure of most male cycad cones suggests

- about cycad pollination and what the structure of most female cones suggests about that process.
- (D) The structure of male cycad cones rules out a possible mechanism for cycad pollination that is suggested by the structure of most female cycad cones.
- (E) The structure of male cycad cones is consistent with a certain means of cycad pollination, but that means is inconsistent with the structure of most female cycad cones.
26. The evidence in favor of insect pollination of cycads presented in lines 2-4 would be more convincing if which of the following were also true?
- (A) Only a small variety of cycad species can be successfully transplanted.
- (B) Cycads can sometimes be pollinated by means other than wind or insects.
- (C) Insects indigenous to regions to which cycads are transplanted sometimes feed on cycads.
- (D) Winds in the areas to which cycads are usually transplanted are similar to winds in cycads' native habitats.
- (E) The transplantation of cycads from one region to another usually involves the accidental removal and introduction of insects as well.
27. The passage suggests that which of the following is true of scientific investigations of cycad pollination?
- (A) They have not yet produced any systematic evidence of wind pollination in cycads.
- (B) They have so far confirmed anecdotal reports concerning the wind pollination of cycads.
- (C) They have, until recently, produced little evidence in favor of insect pollination in cycads.
- (D) They have primarily been carried out using cycads transplanted from their native habitats.
- (E) They have usually concentrated on describing the physical characteristics of the cycad reproductive system.
28. PROCRASTINATION:
- (A) diligence
- (B) complacency
- (C) reasonableness
- (D) allegiance
- (E) rehabilitation
29. CIRCUITY
- (A) straightforwardness
- (B) inventiveness
- (C) authenticity
- (D) insightfulness
- (E) practicality
30. CONCLUDE:
- (A) foster
- (B) frequent
- (C) emanate from
- (D) empower to
- (E) embark on
31. RITE:
- (A) coherent interpretation
- (B) improvised act
- (C) deductive approach
- (D) casual observation
- (E) unnecessary addition
32. BLATANT:
- (A) indecisive
- (B) perceptive
- (C) unobtrusive
- (D) involuntary
- (E) spontaneous
33. PONTIFICATE:
- (A) request rudely
- (B) glance furtively
- (C) behave predictably
- (D) work efficiently
- (E) speak modestly
34. POSIT:
- (A) deceive
- (B) begrudge
- (C) deny
- (D) consent
- (E) reinforce

35. FETTER:

- (A) justify
- (B) comfort
- (C) intrude
- (D) liberate
- (E) optimize

36. SYNERGIC:

- (A) natural in origin
- (B) fragile in structure
- (C) untainted
- (D) inessential
- (E) antagonistic

37. DEPRIVATION:

- (A) sanity
- (B) awareness
- (C) surfeit
- (D) fecundity
- (E) health

38. CORPOREAL:

- (A) unreliable
- (B) unscientific
- (C) indistinguishable
- (D) inanimate
- (E) immaterial

SECTION 3

Time –30 minutes

25 Questions

1. Armtech, a temporary-employment agency, previously gave its employees 2.5 paid vacation days after each 700 hours worked. Armtech's new policy is to give its employees 5.0 paid vacation days after each 1,200 hours worked. Therefore, this new policy is more generous to Armtech employees in giving them more vacation days per hour worked than the old policy did.

Which of the following is an assumption on which the argument depends ?

- (A) Most current Armtech employees approve of the company's new vacation policy.
- (B) A few Armtech employees leave the company before having worked 700 hours.
- (C) Most Armtech employees were not aware that the company planned to change its vacation policy until after it had already done so.
- (D) A significant portion of Armtech employees stay with the company long enough to work for 1,200 hours.
- (E) Armtech's new vacation policy closely matches the vacation policies of competing temporary employment agencies.

- 2 The global population of frogs has declined in recent years while the amount of ultraviolet radiation reaching the Earth has increased. Since the genetic material in frog eggs is harmed when exposed to ultraviolet radiation, and since the eggs themselves are not protected by shells or leathery coverings but are gelatinous, the frog population decline is probably due, at least in part, to the ultraviolet radiation increase.

Which of the following, if true, provides the strongest support for the argument ?

- (A) Even in those regions where there has been no significant increase in ultraviolet radiation, only a small proportion of the frog eggs that are laid ever hatch.
- (B) In areas where there has been the least decline in frog populations, populations of species of insects that frogs eat have decreased.

- (C) The eggs of frog species whose populations are declining tend to have higher concentrations of damaging pesticides than do the eggs of frog species whose populations have not declined.
- (D) In many places where turtles, which lay eggs with tough, leathery coverings, share habitats with frogs, turtle populations are also in decline.
- (E) Populations of frog species that hide their eggs beneath rocks or under sand have declined considerably less than have populations of frog species that do not cover their eggs.

Questions 3-8

A doctor is scheduling one appointment each with five patients—J, K, L, M, and N. The five appointments will be consecutive and are numbered 1 through 5, from earliest to latest. The doctor must schedule at least four of the patients for appointments preferred by those patients and cannot schedule any patient for an appointment unacceptable to that patient. The following is a complete list of what the patients prefer and, if they do not receive their preferences, will accept :

- J prefers an appointment earlier than appointment 3, but will accept any appointment.
- K prefers appointment 2, but will accept any appointment except appointment 1.
- L prefers appointment 1, but will accept appointment 5.
- M prefers and will accept only an appointment later than appointment 3.
- N prefers and will accept only appointment 3.

3. Which of the following lists the patients in an order in which their scheduled appointments can occur, from appointment 1 through appointment 5 ?

- (A) J, K, N, L, M
- (B) J, M, N, K, L
- (C) K, J, N, M, L
- (D) L, J, K, N, M
- (E) L, J, N, M, K

4. If J is scheduled for appointment 2, which of the following can be true ?

- (A) K is scheduled for appointment 3.
- (B) K is scheduled for appointment 4.

- (C) L is scheduled for appointment 4.
- (D) L is scheduled for appointment 5.
- (E) M is scheduled for appointment 1.

5. If L is scheduled for appointment 5, which of the following must be true ?

- (A) J is scheduled for appointment 1.
- (B) J is scheduled for appointment 2.
- (C) J is scheduled for appointment 4.
- (D) K is scheduled for appointment 4.
- (E) N is scheduled for appointment 5.

6. Which of the following is a complete and accurate list of patients any one of whom can be the patient scheduled for appointment 2 ?

- (A) K
- (B) J, K
- (C) J, M
- (D) J, K, L
- (E) K, L, M

7. If M is scheduled for appointment 5, which of the following can be true of the scheduling ?

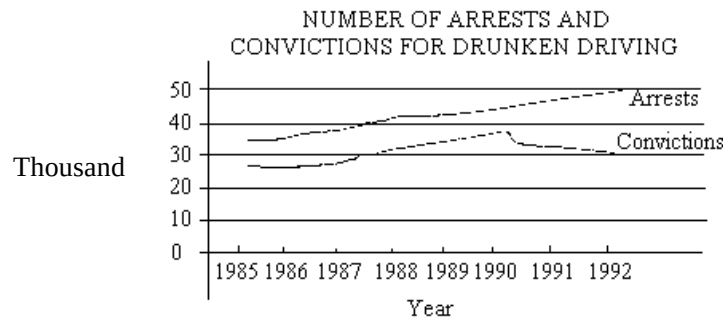
- (A) J's appointment is appointment 1.
- (B) N's appointment is appointment 1.
- (C) J's appointment is earlier than K's appointment.
- (D) K's appointment is earlier than L's appointment.
- (E) N's appointment is earlier than L's appointment.

8. If K's appointment is scheduled for a time later than N's appointment, which of the following must be true ?

- (A) J is scheduled for appointment 4.
- (B) K is scheduled for appointment 5.
- (C) L is scheduled for appointment 1.
- (D) M is scheduled for appointment 4.
- (E) N is scheduled for appointment 2.

Questions 9-10 are based on the following graph.

In January of 1990 a certain country enacted a strict new law to deter people from drunken driving. The law imposes mandatory jail sentences for anyone convicted of drunken driving.



9. Which of the following, if true about the years 1990 through 1992, most helps to explain the data illustrated in the graph?
- (A) Most of the people arrested for and convicted of drunken driving were repeat offenders.
 - (B) Many of the people arrested for and convicted of drunken driving participated in alcohol-education programs in order to reduce their jail sentences.
 - (C) Juries in drunken driving cases became increasingly reluctant to convict people on whom mandatory jail sentences would be imposed.
 - (D) Since the law was enacted, the number of deaths attributed to drunken driving has declined significantly.
 - (E) The majority of the residents of the country supported the strict law to deter people from drunken driving.

10. Which of the following, if true, strengthens the claim that the changes in the ratio of arrests to convictions since the beginning of 1990 are due to an increase in the number of people arrested for drunken driving who were not drunk?

- (A) Before 1990 only people driving erratically were stopped by the police on suspicion of drunken driving, but since the beginning of 1990 police have been allowed to stop drivers randomly and to arrest any driver whom they suspect of having drunk any alcohol.
- (B) Since the beginning of 1990 new technology has enabled police who stop a driver to establish immediately whether the driver is drunk,

whereas before 1990 police had to rely on observations of a driver's behavior to make a judgment about that driver's drunkenness.

- (C) After 1990 the number of police officers assigned to patrol for drunken drivers increased only very slightly compared to the number of police officers assigned to patrol for drunken drivers in the years 1985 through 1989.
- (D) In 1990 a greater number of drivers were ignorant of the laws concerning drunken driving than were ignorant of the drunken driving laws in 1989.
- (E) After 1990 teenagers and young adults constituted a greater proportion of those arrested for drunken driving than in the years 1985 through 1989.

- 11 To improve productivity, manufacturing companies have recently begun restructuring work to produce more goods with fewer assembly-line workers, and the companies have laid off many workers as a consequence. The workers laid off have been those with the least seniority (time on the job), generally the younger workers.

The statements above, if true, most strongly support which of the following as a conclusion?

- (A) The products manufactured by the companies are not undergoing design changes while the manufacturing jobs are being restructured.
- (B) When assembly-line workers have made suggestions for improvements in manufacturing processes, some suggestions have been

implemented, but many have not.

- (C) Assembly-line workers now need increased reading and mathematical skills to do their jobs.
- (D) Some of the innovations in assembly-line processes and procedures that were made to increase productivity have instead proved to be counterproductive.
- (E) The manufacturing companies are increasing the average age of their assembly-line workforce while still seeking to increase production.

12. During the nineteenth century, Britain's urban population increased as its rural population diminished. A historian theorizes that, rather than industrialization's being the cause, this change resulted from a series of migrations to urban areas, each occasioned by a depression in the agrarian economy. To test this hypothesis, the historian will compare economic data with population census data.

The historian's hypothesis would be most strongly supported if which of the following were found to be true?

- (A) The periods of greatest growth in the industrial economy were associated with a relatively rapid decline in the rural population.
- (B) The periods of greatest weakness in the agrarian economy were associated with relatively slow growth in the population as a whole.
- (C) Periods when the agrarian economy was comparatively strong and the industrial economy comparatively weak were associated with a particularly rapid decline in the rural population.
- (D) Periods when the agrarian and industrial economies were both strong were associated with particularly rapid growth in the urban population.
- (E) The periods of greatest strength in the agrarian economy were associated with relatively slow growth in the urban population.

Questions 13-16

On each of the three consecutive days Monday through Wednesday, exactly two employees are to staff a com-

pany's information booth. The three available employees—Feng, Gómez, and Hull—will staff the booth in accordance with the following conditions:

Gómez and Hull must each staff the booth on at least one of the days, but Feng must staff it on at least two of the days.

The booth cannot be staffed by the same two employees on any two consecutive days.

If Hull staffs the booth on Monday, Gómez must be the other employee staffing the booth on Monday.

13. Which of the following can be the schedule of employees staffing the booth on the three days?

<u>Monday</u>	<u>Tuesday</u>	<u>Wednesday</u>
(A) Feng, Gómez	Feng, Gómez	Feng, Hull
(B) Feng, Gómez	Feng, Hull	Gómez, Hull
(C) Feng, Hull	Feng, Gómez	Gómez, Hull
(D) Gómez, Hull	Feng, Gómez	Gómez, Hull
(E) Gómez, Hull	Feng, Hull	Feng, Hull

14. If Gómez staffs the booth on Monday and Tuesday, which of the following must be true?

- (A) Feng staffs the booth on Monday.
- (B) Feng staffs the booth on Tuesday.
- (C) Feng staffs the booth on Wednesday.
- (D) Hull staffs the booth on Monday.
- (E) Hull staffs the booth on Tuesday.

15. If Hull staffs the booth on Monday and Wednesday, which of the following must be true?

- (A) Feng and Gómez staff the booth on Tuesday.
- (B) Feng and Hull staff the booth on Monday.
- (C) Feng and Hull staff the booth on Tuesday.
- (D) Gómez and Hull staff the booth on Tuesday.
- (E) Gómez and Hull staff the booth on Wednesday.

16. If Hull staffs the booth on only one of the days, which of the following can be true?

- (A) Feng and Hull staff the booth on Monday.
- (B) Feng and Hull staff the booth on Wednesday.
- (C) Gómez and Hull staff the booth on Monday.
- (D) Gómez and Hull staff the booth on Tuesday.
- (E) Gómez and Hull staff the booth on Wednesday.

Questions 17-22

A science teacher is selecting projects for each of two classes from a group of exactly seven projects—R, S, T, V, X, Y, and Z. The teacher will assign projects to Class 1 and Class 2 according to the following conditions :

Each project must be assigned to exactly one class.

Four of the projects must be assigned to Class 1 and three to Class 2.

R must be assigned to Class 2.

The class to which V is assigned cannot be the same class as the one to which Y is assigned.

If V is assigned to Class 1, X must be assigned to Class 1.

If Z is assigned to Class 2, Y must be assigned to Class 1.

17. Which of the following could be the projects assigned to the two classes ?

Class 1 Class 2

- (A) R, V, X, Y S, T, Z
 (B) S, T, V, Z R, X, Y
 (C) S, T, X, Y R, V, Z
 (D) S, T, X, Z R, V, Y
 (E) S, V, X, Y R, T, Z

18. If X is assigned to Class 2, which of the following must be true ?

- (A) R is assigned to Class 1.
 (B) S is assigned to Class 2.
 (C) T is assigned to Class 2.
 (D) Y is assigned to Class 1.
 (E) Z is assigned to Class 2.

19. If Z is assigned to Class 2, which of the following must be true ?

- (A) S is assigned to Class 2.
 (B) T is assigned to Class 2.
 (C) V is assigned to Class 1.
 (D) X is assigned to Class 1.
 (E) Y is assigned to Class 2.

20. If Y is assigned to Class 2, any of the following could be assigned together to one of the classes EXCEPT

- (A) R and T
 (B) S and T
 (C) S and Y
 (D) T and Z
 (E) X and Z

21. If T is assigned to the same class as V, which of the following must be assigned to the same class as each other ?

- (A) R and T
 (B) S and X
 (C) S and Y
 (D) X and Y
 (E) Y and Z

22. If V is assigned to a different class from Z, which of the following must be true ?

- (A) S is assigned to Class 1.
 (B) S is assigned to Class 2.
 (C) T is assigned to Class 2.
 (D) V is assigned to Class 2.
 (E) X is assigned to Class 1.

23. Politician : Each year, small businesses create more jobs than do large established businesses. Therefore, in order to reduce unemployment in the long term, we should provide incentives for starting small businesses rather than for expanding established large businesses.

Which of the following, if true, casts the most doubt on the politician's argument ?

- (A) In general, people employed by small businesses report higher job satisfaction than do people employed by large businesses.
 (B) Among the currently unemployed are many people with sufficient job skills to perform the jobs that small businesses would create.
 (C) Providing an effective incentive for starting a business generally costs significantly less than providing an effective incentive for expanding a large business.
 (D) A high proportion of small businesses fail within three years of starting because of their owners' inexperience.

- (E) The average large business contributes more money to politicians' campaign funds than the average small business does.

24 In the workplace, influenza is typically spread by infected individuals to others with whom they work in close quarters. A new medication that suppresses the symptoms of influenza therefore will actually increase the number of influenza cases, because this medication will allow people who would otherwise be home in bed to return to work while infected.

Which of the following, if true, most seriously challenges the prediction ?

- (A) Coughing, a symptom of influenza that the new medication suppresses, is a primary mechanism in the spread of this illness.
- (B) Some medications that are used to suppress symptoms of influenza are also used by many people to treat symptoms that are caused not by influenza but by other illnesses.
- (C) Many workers who now remain at home when infected with influenza do so because the symptoms of influenza prevent them from performing their jobs effectively.
- (D) Most adults who are immunized against influenza in order to avoid being infected are over 65 years old and retired and thus do not work outside the home.
- (E) Symptoms of an illness are often the body's means of curing itself of the illness, and therefore suppression of symptoms can prolong the illness that causes them.

25 Editorial :

Critics of nuclear power complain about the allegedly serious harm that might result from continued operation of existing nuclear power plants. But such concerns do not justify closing these plants ; after all, their operation has caused no more harm than that caused by pollution generated by coal-and oil-burning power plants, the most important other sources of energy.

Which of the following is an assumption on which the argument depends ?

- (A) Existing nuclear power plants should be closed only

if it can be conclusively demonstrated that their continued operation is likely to cause harm more serious than the harm their operation has already caused.

- (B) Closing existing nuclear power plants would require greatly increased reliance on coal-and oil-burning power plants.
- (C) The harm that has resulted from operation of existing coal-and oil-burning power plants has been significant.
- (D) The harm that a nuclear power plant is likely to cause as it continues to operate can be reliably predicted from the past history of nuclear power plants.
- (E) The only harm that has resulted from operation of existing coal-and oil-burning power plants has resulted from the pollution generated by these plants.

SECTION 4

Time –30 minutes

25 Questions

Questions 1-7

A museum will display seven statues—P, Q, R, S, T, U, and W—in two of its galleries, gallery 1 and gallery 2. Exactly four of the statues will be displayed in gallery 1 and exactly three of the statues will be displayed in gallery 2. The statues will be displayed according to the following conditions:

U cannot be displayed in a gallery with W

Neither S nor T can be displayed in a gallery with R.

- 1.If U is displayed in gallery 2, which of the following must be true?
 - (A) P is displayed in gallery 1.
 - (B) R is displayed in gallery 2.
 - (C) S is displayed in gallery 1.
 - (D) T is displayed in gallery 2.
 - (E) W is displayed in gallery 1.
- 2.If S is displayed in gallery 2, the other two statues displayed in gallery 2 can be
 - (A) P and Q
 - (B) P and T
 - (C) Q and T
 - (D) T and W
 - (E) U and W
- 3.If P is displayed in gallery 1 and W is displayed in gallery 2, then the display in gallery 1 can include any of the following pairs of statues EXCEPT
 - (A) Q and R
 - (B) Q and T
 - (C) Q and U
 - (D) R and U
 - (E) S and T
- 4.If P and Q are displayed in gallery 1, which of the following is a statue that must also be displayed in gallery 1?
 - (A) R
 - (B) S
 - (C) T
 - (D) U
 - (E) W
- 5.If S is displayed in gallery 1, which of the following must be true?
 - (A) P is displayed in gallery 1.
 - (B) Q is displayed in gallery 1.
 - (C) R and U are displayed in the same gallery as each other.
 - (D) P and Q are not displayed in the same gallery as each other.
 - (E) Q and R are not displayed in the same gallery as each other.
- 6.If T is displayed in gallery 2, which of the following is a pair of statues that CANNOT be displayed in the same gallery as each other?
 - (A) P and S
 - (B) Q and R
 - (C) Q and W
 - (D) R and U
 - (E) T and W
- 7.If Q is displayed in the same gallery as S. Which of the following must be true?
 - (A) P is displayed in gallery 1.
 - (B) R is displayed in gallery 2.
 - (C) Q and S are displayed in gallery 2.
 - (D) P is displayed in the same gallery as W.
 - (E) R is displayed in the same gallery as U.
- 8 Drug manufacturer: Although our company requires that patients who use our new drug also purchase from us nonreusable kits for weekly blood testing, the expense of those kits is an entirely necessary one: weekly blood testing must be done to monitor the drug's potential side effects, which can be very dangerous.

Which of the following, if true, most seriously weakens the manufacturer's argument?
 - (A) The expense of purchasing the blood-test kits has not prevented any patients from obtaining them or the drug.
 - (B) Medical laboratories can perform the blood test-

ing at a lower cost to patients or their insurers than the price the manufacturer charges for the kits.

- (C) A one-year supply of the drug and the weekly blood-test kits can cost patients or their insurers over \$10,000.
- (D) Most government and other health insurance programs will not reimburse patients for the full cost of both the drug and the blood-test kits.
- (E) Patients who suffer one or more of the dangerous side effects of the drug can incur heavy expenses for the treatment of those side effects.

9 Virginia and her brother William disagree over when their father was born: Virginia claims it was in 1935 and William claims it was in 1933. The hospital where their father was born has no records for 1933 but has complete records for 1935—records that do not include a birth record for their father. Therefore, he must have been born in 1933.

The argument depends on which of the following assumptions?

- (A) Either Virginia's claim or William's claim is correct.
- (B) The records of the hospital where their father was born date back to 1933.
- (C) Virginia and William know the day and the month of their father's birth.
- (D) There are urgent practical reasons why Virginia and William must know the date of their father's birth.
- (E) None of their other relatives knows the year in which Virginia and William's father was born.

10. RESULTS OF TWO SURVEYS OF OPINIONS REGARDING THE EFFECTS OF SCIENCE ON HUMAN SOCIETY

<u>Responses</u>	<u>August 1991</u>	<u>August 1992</u>
Mostly beneficial	25%	81%
Equally harmful and beneficial	37%	9%

Mostly harmful 20% 7%

No opinion 18% 3%

Which of the following, if true, contributes most to explaining the shift in opinions about the effects of science on human society?

- (A) The surveys questioned people who regularly watch prime-time television, and an innovative weekly prime-time television series called "Wonders of Science" had been steadily winning viewers since its widely seen premiere in January 1992.
- (B) The surveys questioned college-educated adults, and a report called "The State of the Nation's Schools," published in June 1992, noted an increase in students' interest in science courses since 1982.
- (C) The surveys were conducted in a suburban shopping area near a company that ceased operation in April 1992 as a result of lawsuits arising from unexpected toxic effects of the company's products.
- (D) Both survey forms were mailed to equally large samples of the population; after returning the 1991 survey forms, respondents were sent discount coupons for food products, and after returning the 1992 survey forms, respondents were sent a pamphlet on recycling.
- (E) The surveys questioned first-year college students across the country, and the people who did the questioning were all research scientists.

Questions 11-17

A science reporter will make a trip to visit exactly six archaeological sites—Quin, Ram, Sud, Tunin, Vara, and Xilat. The reporter must visit the sites one at a time in accordance with the following conditions:

- The reporter visits each site exactly once.
- The reporter's trip begins at Quin or else at Xilat.
- The reporter's trip ends at Vara or else at Xilat.
- The reporter visits Vara immediately after visiting Sud.
- The reporter visits Sud at some time after visiting Ram.

11. Which of the following is a list of the sites in an order in which the reporter can visit them, from the first site visited to the last site visited?

- (A) Quin, Ram, Sud, Vara, Xilat, Tunin
 (B) Quin, Sud, Vara, Tunin, Ram, Xilat
 (C) Ram, Sud, Vara, Tunin, Quin, Xilat
 (D) Xilat, Ram, Sud, Tunin, Quin, Vara
 (E) Xilat, Tunin, Ram, Quin, Sud, Vara

12.If Sud is visited immediately after Quin is visited, which of the following can be the second site visited?

- (A) Quin
 (B) Sud
 (C) Tunin
 (D) Vara
 (E) Xilat

13.If Tunin is visited as late in the trip as possible, which of the following must be the third site visited?

- (A) Quin
 (B) Ram
 (C) Sud
 (D) Vara
 (E) Xilat

14.If Tunin is visited before Xilat is visited and if exactly one site is visited between the visit to Tunin and the visit to Xilat, which of the following must be true?

- (A) Quin is visited second.
 (B) Ram is visited third.
 (C) Sud is visited fourth.
 (D) Vara is visited fifth.
 (E) Xilat is visited sixth.

15.If Xilat is visited immediately after Ram is visited.

Which of the following must be true?

- (A) Quin is visited at some time after Tunin is visited.
 (B) Ram is visited at some time after Quin is visited.
 (C) Tunin is visited at some time after Ram is visited.
 (D) Tunin is visited at some time after Sud is visited.
 (E) Xilat is visited at some time after Sud is visited.

16.If Ram is the fourth site visited, which of the following must be true?

- (A) Quin is the first site visited.

- (B) Tunin is the second site visited.
 (C) Tunin is the third site visited.
 (D) Vara is the sixth site visited.
 (E) Xilat is the sixth site visited.

17.Which of the following can be true?

- (A) Quin is the fifth site visited
 (B) Ram is the fifth site visited.
 (C) Sud is the second site visited.
 (D) Xilat is the second site visited.
 (E) Xilat is the fifth site visited.

Questions 18-22

Eight representatives—Gold, Herrera, Jones, Karami, Lowell, Nakamura, Orson, and Porter—will be scheduled to present information at four project meetings: W, X, Y and Z. Each representative will be scheduled for exactly one meeting, and at least one representative will be scheduled for each meeting. The meetings will be held one at a time, one after another. The order of the meetings and the schedule of representatives for the meetings must meet the following conditions:

Meeting W is held first, and exactly three representatives are scheduled for it.

Meeting X is held at some time before meeting Y.

Gold and Herrera are both scheduled for meeting X.

Karami is scheduled for meeting Z.

Orson is scheduled for the same meeting as Porter.

18.If the meetings are scheduled in the order W, X, Y, Z, which of the following can be the schedule of representatives for the meetings?

- | | <u>W</u> | <u>X</u> | <u>Y</u> | <u>Z</u> |
|-----|--------------------------|--------------------|--------------------|--------------------|
| (A) | Gold
Herrera
Jones | lowell
Nakamura | Orson
Porter | Karami |
| (B) | Jones
Lowell | Gold
Herrera | Orson
Porter | Karami
Nakamura |
| (C) | Jones
Loweil | Gold
Herrera | Nakamura
Porter | Orson
Karami |
| (D) | Jones
Lowell | Gold
Herrera | Orson
Porter | Karami
Nakamura |
| (E) | Jones
Lowell | Gold
Herrera | Orson
Porter | Karami
Nakamura |

19.If Orson is scheduled for meeting Y, which of the following can be true?

- (A) Gold is scheduled for the same meeting as Jones.
- (B) Herrera is scheduled for the same meeting as Lowell.
- (C) Jones is scheduled for the second meeting.
- (D) Karami is scheduled for the third meeting.
- (E) Lowell is scheduled for the fourth meeting.

20.If Gold and Jones are both scheduled for the third meeting, which of the following must be true?

- (A) Herrera is scheduled for the first meeting.
- (B) Lowell is scheduled for the first meeting.
- (C) Porter is scheduled for the first meeting.
- (D) Karami is scheduled for the same meeting as Nakamura.
- (E) Lowell is scheduled for the same meeting as Nakamura.

21.If Nakamura is scheduled for the third meeting and Karami is scheduled for the fourth meeting, which of the following must be true?

- (A) Herrera is scheduled for the second meeting.
- (B) Jones is scheduled for the second meeting.
- (C) Lowell is scheduled for meeting Y.
- (D) Nakamura is scheduled for meeting Z.
- (E) Porter is scheduled for meeting Y.

22.If no other representative is scheduled for the meeting for which Jones is scheduled, any of the following can be true EXCEPT:

- (A) Jones is scheduled for the third meeting.
- (B) Lowell is scheduled for the second meeting.
- (C) Nakamura is scheduled for the fourth meeting.
- (D) Lowell is scheduled for meeting Z.
- (E) Nakamura is scheduled for meeting Y.

23 The town of San Leonardo has recently enacted a law banning smoking in all restaurants within town limits. Since many smokers who normally dine in San Leonardo's restaurants will not want to refrain from smoking during their meals, San Leonardo's restaurants will undoubtedly lose many patrons and considerable income.

Which of the following, if true, most helps to strengthen the argument above?

- (A) Most residents of San Leonardo who eat in restaurants are not smokers.
- (B) Most smokers who dine in the company of non-smokers are willing to refrain from smoking during their meals.
- (C) If the law banning smoking in restaurants had not been enacted, it is likely that a more stringent law banning smoking in all public places in San Leonardo would have been enacted instead.
- (D) Prior to the enactment of the law banning smoking in San Leonardo's restaurant, the town had a law that required most restaurants to have nonsmoking sections.
- (E) None of the other communities adjacent to San Leonardo, which have restaurants comparable to those of San Leonardo, has enacted and enforces any antismoking legislation.

24. Children whose biological parents both have Tic Syndrome Z (TSZ), which is characterized by the involuntary contraction of certain muscles, are about four times more likely to develop such contractions than are children whose biological parents do not have TSZ. It is likely, therefore, that predisposition to TSZ is an inherited trait.

Which of the following, if true, would most strengthen the conclusion above?

- (A) Children whose parents have TSZ are more likely to develop TSZ if they are under unusual stress at school or at home than if they are not under such stress.
- (B) Children whose biological parents do not have TSZ are more likely to develop TSZ if they are raised by adoptive parents with TSZ than if they are raised by their biological parents.
- (C) Children whose biological parents have TSZ are as likely to develop TSZ if they are raised by adoptive parents who do not have TSZ as if they are raised by their biological parents.
- (D) Children whose biological parents have TSZ and who develop TSZ usually avoid developing a severe form of the syndrome if they seek

treatment for TSZ shortly after developing the first signs of it.

- (E) Children with TSZ whose biological parents do not have TSZ are less likely to have the syndrome diagnosed when symptoms first appear than are children with TSZ whose biological parents have TSZ.

25. Playing eighteenth-century music on the instruments of that period provides valuable information about how the music originally sounded. Eighteenth-century instruments cannot be played without being restored, however, and restoring such an instrument destroys all of the information that researchers could obtain from it about eighteenth-century instrument-making techniques.

If the statements above are true, which of the following must be true on the basis of them?

- (A) Eighteenth-century instruments cannot be used to provide information about the original techniques used in playing such instruments if they have been restored.
- (B) Eighteenth-century instruments that have been restored can provide information only about how eighteenth-century music originally sounded
- (C) Eighteenth-century instruments are the only source of information about the instrument-making techniques of that period.
- (D) An eighteenth-century instrument that has not been restored can provide more information than can one that has been restored.
- (E) An eighteenth-century instrument cannot serve as a source of new information about eighteenth-century instrument-making techniques once it can be played

SECTION 5

Time –30 minutes

30 Questions

To the nearest hundredth, $p = 3.14$ and $\sqrt{10} = 3.16$

1. p^2 10

A marble is to be drawn at random from a bag that contains 2 yellow marbles, 4 blue marbles, 6 green marbles, and no other marbles.

2. The probability that the marble drawn will be green The probability that the marble drawn will be yellow or blue

3. $\frac{250}{\frac{1}{3}}$ $\frac{250}{\frac{1}{2}}$

$$\begin{aligned} x + y + n &= 15 \\ x + y + k &= 9 \end{aligned}$$

4. $n - k$ 6

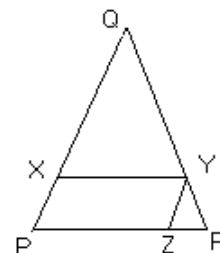
In the rectangular coordinate system, line k passes through the points (0,0) and (4,8); line m passes through the points (0,1) and (4,9).

5. The slope of line k The slope of line m

The vertices of an equilateral triangle are on a circle.

6. The length of a side of the triangle The diameter of the circle

7. $\frac{1}{11}$ 0.09



XQY and ZYR are equilateral triangles, and the ratio of ZR to PR is 1 to 4.

8. The perimeter of $\triangle XQY$ The perimeter of parallelogram $PXYZ$

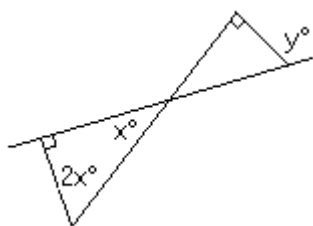
GRE10-2

9.	$x + y$	$2(x + y)$
----	---------	------------

In a certain store, computer X costs 30 percent more than computer Y , and computer Y costs 30 percent more than computer Z .

- | | |
|---|---|
| 10. The cost of computer X minus the cost of computer Y . | The cost of computer Y minus the cost of computer Z . |
|---|---|
-

	$x^2 y < 0$	
11.	xy	0



12.	y	$4x$
-----	-----	------

m is a positive integer less than 4.

13.	$(m + 2)^m$	m^{m+2}
-----	-------------	-----------

$$\frac{1-x}{x-1} = \frac{1}{x}$$

$$x \neq 1$$

14.	x	$-\frac{1}{2}$
-----	-----	----------------

The median of 10, 15, x , and y is 18.5, and $x < y$.

15.	x	22
-----	-----	------

16. The cost, in dollars, for an appliance repair at a certain company is $1.2p + 20h$, where p is the wholesale price of the parts, in dollars, and h is the number of hours it takes to repair the appliance. What is the cost of repairing an appliance if the wholesale price of the parts is \$15 and it takes 2 hours to repair it?

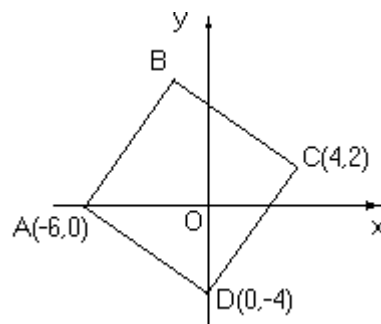
- (A) \$12
(B) \$18
(C) \$20
(D) \$40
(E) \$58

17. For what value of x will $8 + (x-3)^2$ have the least value?

- (A) -3
(B) 0
(C) 3
(D) 5
(E) 8

18. How many integers from 3 to 30, inclusive, are odd?

- (A) 13
(B) 14
(C) 15
(D) 16
(E) 17



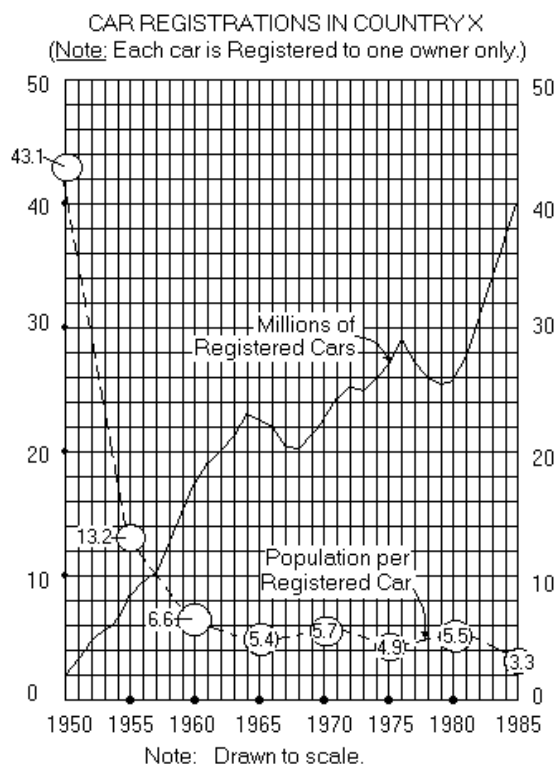
19. In the figure above, $ABCE$ is a square. What are the coordinates of point B ?

- (A) (-4,2)
(B) (-2,4)
(C) (-2,6)
(D) (4,-6)
(E) (6,-2)

20. $3.7(10^7) =$

- (A) 370,000
(B) 3,700,000
(C) 37,000,000
(D) 370,000,000
(E) 3,700,000,000

Questions 21-25 refer to the following graph.



21. For how many of the years shown after 1950 was there a decrease from the previous year in the number of registered cars?

- (A) Nine
- (B) Eight
- (C) Seven
- (D) Six
- (E) Five

22. The ratio of the population per registered car in 1985 to that in 1975 was most nearly

- (A) 0.55
- (B) 0.65
- (C) 0.75
- (D) 0.85
- (E) 0.95

23. From 1972 to 1985, the percent increase in the number of registered cars was most nearly

- (A) 60%
- (B) 50%
- (C) 45%
- (D) 35%
- (E) 15%

24. From 1950 to 1985, the population of Country X increased by approximately how many million people?

- (A) 45
- (B) 80
- (C) 165
- (D) 200
- (E) It cannot be determined from the information given.

25. If the number of registered cars were to increase yearly through the year 2000 at the same average annual rate shown for the period 1981-1985, for which of the following years would the number of registered cars be closest to 76 million?

- (A) 1995
- (B) 1996
- (C) 1997
- (D) 1998
- (E) 1999

26. A rectangular field is 400 feet long and 300 feet wide. If a square field has the same perimeter as the rectangular field, what is the length, in feet, of each side of the square field?

- (A) 175
- (B) 350
- (C) $200\sqrt{2}$
- (D) $350\sqrt{2}$
- (E) $100\sqrt{3}$

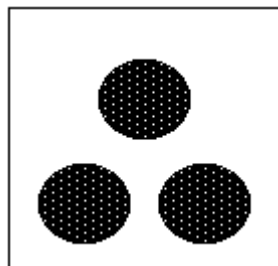
	Distance(in miles) From Centerville
Freight Train	$-10t + 115$
Passenger Train	$-20t + 150$

27. The expressions in the table above give the distance of each of two trains from Centerville at t hours after 12:00 noon. At what time will the trains be equidistant from Centerville?

- (A) 1:30 p.m.
 (B) 3:30 p.m.
 (C) 5:10 p.m.
 (D) 8:50 p.m.
 (E) 11:30 p.m.

28. In 1982, if the 1.8 billion dollars collected as child support payments was only 10 percent of the total court-ordered payments due, approximately how many billion dollars of court-ordered payments for child support were not collected?

- (A) 1.6
 (B) 14.4
 (C) 16.2
 (D) 17.2
 (E) 18.0



30

29. If each shaded circular region in the figure above has radius 5, then the total area of the shaded regions is what fraction of the area of the square region?

- (A) $\frac{p}{12}$
 (B) $\frac{p}{36}$
 (C) $\frac{p}{60}$
 (D) $\frac{1}{6}$
 (E) $\frac{1}{3}$

30. If $\frac{1}{4x} + \frac{1}{y} = \frac{1}{3} \left(\frac{1}{x} + \frac{1}{y} \right)$, what is the ratio of x to y ?

- (A) 3 to 4
 (B) 2 to 3
 (C) 1 to 2
 (D) 1 to 8
 (E) 1 to 9s

SECTION 6

Time –30 minutes

38 Questions

- Some activists believe that because the health-care system has become increasingly— to those it serves, individuals must — bureaucratic impediments in order to develop and promote new therapies.
 (A) attuned.. avoid
 (B) inimical.. utilize
 (C) unresponsive ..circumvent
 (D) indifferent.. supplement
 (E) sensitized.. forsake
- The acts of vandalism that these pranksters had actually — were insignificant compared with those they had — but had not attempted.
 (A) hidden .. renounced
 (B) advocated .. meditated
 (C) inflicted .. dismissed
 (D) committed .. effected
 (E) perpetrated .. contemplated
- Though one cannot say that Michelangelo was an impractical designer, he was, of all nonprofessional architects known, the most — in that he was the least constrained by tradition or precedent.
 (A) pragmatic
 (B) adventurous
 (C) empirical
 (D) skilled
 (E) learned
- Before adapting to changes in values, many prefer to —, to — the universally agreed-on principles that have been upheld for centuries.
 (A) innovate .. protect
 (B) resist .. defend
 (C) ponder .. subvert
 (D) vacillate .. publicize
 (E) revert .. ignore
- Although the records of colonial New England are — in comparison with those available in France or England, the records of other English colonies in America are even more —.

- (A) sporadic.. irrefutable
 (B) sparse.. incontrovertible
 (C) ambiguous.. authoritative
 (D) sketchy.. fragmentary
 (E) puzzling .. unquestionable
6. High software prices are frequently said to ----- widespread illegal copying, although the opposite --- that high prices are the cause of the copying -- is equally plausible.
 (A) contribute to
 (B) result from
 (C) correlate with
 (D) explain
 (E) precede
7. Because early United States writers thought that the mark of great literature was grandiosity and elegance not to be found in common speech, they ---- the vernacular.
 (A) dissected
 (B) avoided
 (C) misunderstood
 (D) investigated
 (E) exploited
8. OBSTRUCT: PROGRESS::
 (A) reveal: information
 (B) polish: illumination
 (C) implicate: guilt
 (D) inspire: artistry
 (E) stunt: growth
9. INTERVIEW: APPLICANT::
 (A) recital: pianist
 (B) exercise: athlete
 (C) audition: actor
 (D) manuscript: writer
 (E) flight plan: pilot
10. COMBUSTIBLE: IGNITE::
 (A) impermeable: saturate
 (B) impenetrable: pierce
 (C) malleable: shape
 (D) rigid: stretch
 (E) sterile: extract
11. SLACKEN: TENSION::
 (A) rarefy: expansion
 (B) blunt: sharpness
 (C) obscure: cloudiness
 (D) quicken: animation
 (E) oscillate: rotation
12. BIGOT: TOLERANCE::
 (A) scoundrel: misdeed
 (B) liar: honesty
 (C) brat: annoyance
 (D) outcast: respect
 (E) snitch: information
13. IMPROVEMENTS: MASTERY::
 (A) efforts: exertion
 (B) savings: wealth
 (C) performance: talent
 (D) practice: intention
 (E) diversification: proficiency
14. DILETTANTE: SUPERFICIALITY::
 (A) partisan: bias
 (B) crusader: passivity
 (C) libertarian: authority
 (D) champion: restlessness
 (E) sage: argumentativeness
15. WINNOW: CHAFF::
 (A) ferment: alcohol
 (B) skim: cream
 (C) pare: fruit
 (D) refine: oil
 (E) filter: impurities
16. STANZA: LINE::
 (A) essay: theme
 (B) scene: monologue
 (C) play: vignette
 (D) volume: issue
 (E) concert: program

(This passage is adapted from an article published in 1981.)

The term "remote sensing" refers to the techniques of measurement and interpretation of phenomena from a distance. Prior to the mid-1960's the interpretation of film images was the primary means for remote sensing of the

(5) Earth's geologic features. With the development of the optomechanical scanner, scientists began to construct digital multispectral images using data beyond the sensitivity range of visible light photography. These images are constructed by mechanically aligning pictorial representations of such (10) phenomena as the reflection of light waves outside the visible spectrum, the refraction of radio waves, and the daily changes in temperature in areas on the Earth's surface.

Digital multispectral imaging has now become the basic tool in geologic remote sensing from satellites.

(15) The advantage of digital over photographic imaging is evident: the resulting numerical data are precisely known, and digital data are not subject to the vagaries of difficult-to-control chemical processing. With digital processing, it is possible to combine a large number of spectral images. The

(20) acquisition of the first multispectral digital data set from the multispectral scanner (MSS) aboard the satellite Landsat in 1972 consequently attracted the attention of the entire geologic community. Landsat MSS data are now being applied to a variety of geologic problems that are (25) difficult to solve by conventional methods alone. These include specific problems in mineral and energy resource exploration and the charting of glaciers and shallow seas.

A more fundamental application of remote sensing is to augment conventional methods for geologic mapping of

(30) large areas. Regional maps present compositional, structural, and chronological information for reconstructing geologic evolution. Such reconstructions have important practical applications because the conditions under which rock units and other structural features are formed influence (35) the occurrence of ore and petroleum deposits and affect the thickness and integrity of the geologic media in which the deposits are found.

Geologic maps incorporate a large, varied body of specific field and laboratory measurements, but the maps must

(40) be interpretative because field measurements are always limited by rock exposure, accessibility and labor resources. With remote-sensing techniques it is possible to obtain much geologic information more efficiently than it can be obtained on the ground. These techniques also facilitate

(45) overall interpretation. Since detailed geologic mapping is generally conducted in small areas, the continuity of regional features that have intermittent and variable expressions is often not recognized, but in the comprehensive views of Landsat images these continuities are apparent.

(50) However, some critical information cannot be obtained through remote sensing, and several characteristics of the Landsat MSS impose limitations on the acquisition of diagnostic data. Some of these limitations can be overcome by designing satellite systems specifically for geologic purposes; but, to be most effective, remote-sensing data must still be combined with data from field surveys and laboratory tests, the techniques of the earlier twentieth century.

17. By using the word "interpretative" in line 40, the author is indicating which of the following?

- (A) Some maps are based more on data from aerial photography than on data from field operations.
- (B) Some maps are based almost exclusively on laboratory measurements.
- (C) Some maps are based on incomplete data from field observations.
- (D) Some maps show only large geologic features.
- (E) Some maps can be three-dimensional.

18. With which of the following statements about geologic mapping would the author be most likely to agree?

- (A) Geologic mapping is basically an art and not a science.
- (B) Geologic mapping has not changed significantly since the early 1960's.
- (C) Geologic mapping will have limited practical applications until remote-sensing systems are perfected.
- (D) A developmental milestone in geologic mapping was reached in 1972.
- (E) Without the present variety of remote-sensing techniques, geologic mapping could not be done.

19. According to the passage, measurements of which of the following can be provided by the optomechanical scanner but not by visible-light photography?

- (A) The amount of visible light reflected from oceans
- (B) The density of foliage in remote areas on the Earth's surface
- (C) Daily temperature changes of areas on the Earth's

surface. (D) The degree of radioactivity emitted by exposed rocks on the Earth's surface. (E) Atmospheric conditions over large landmasses 20. It can be inferred from the passage that a major disadvantage of photographic imaging in geologic mapping is that such photography (A) cannot be used at night (B) cannot focus on the details of a geologic area (C) must be chemically processed (D) is always enhanced by digital reconstruction (E) cannot reflect changes over extended periods of time 21. It can be inferred from the passage that Landsat images differ from conventional geologic maps in that Landsat images (A) reveal the exact size of petroleum deposits and ore deposits (B) indicate the continuity of features that might not otherwise be interpreted as continuous (C) predict the movements of glaciers (D) provide highly accurate data about the occurrence of mineral deposits (E) reveal the integrity of the media in which petroleum deposits and ore deposits are found 22. The passage provides information about each of the following topics EXCEPT (A) the principal method of geologic remote sensing prior to the mid-1960's (B) some of the phenomena measured by digital multi-spectral images in remote sensing (C) some of the practical uses of regional geologic maps (D) the kinds of problems that are difficult to solve solely through conventional methods of geologic mapping (E) the specific limitations of the Landsat multi-spectral scanner 23. The passage suggests which of the following about the "conventional methods" mentioned in line 29? (A) They consist primarily of field surveys and laboratory measurements. (B) They are not useful in providing information	necessary for reconstructing geologic evolution (C) They have rarely been used by geologists since 1972. (D) They are used primarily to gather compositional information about geologic features. (E) They are limited primarily because of difficulties involved in interpreting film images. Although the development of new infrastructure (such public facilities as power plants, schools, and bridges) is usually determined by governmental planning, sometimes this development can be planned more flexibly and realistically by private investors who anticipate profit from the collection of user fees. Such profits can contribute to the financing of more infrastructure if demand proves great enough, whereas the reluctance of developers to invest in such projects can signal that additional infrastructure is not needed. During the economic boom of the 1980's, for example, the state of Virginia authorized private developers to build a \$300 million toll road. These developers obtained the needed right-of-way from property owners, but by 1993 they still had not raised the necessary financing. The unwillingness of investors to finance this project does not negate the viability of privately financed roads; rather, it illustrates a virtue of private financing. If a road appears unlikely to attract enough future traffic to pay for the road, then it should not be built. 24. The primary purpose of the passage is to (A) build a case for increasing the development of new infrastructure (B) advocate an alternative to government financing of infrastructure (C) explain the failure of a privately financed venture (D) suggest the types of infrastructure most appropriate for private financing (E) argue against government restrictions on developing new infrastructure 25. The passage implies that the "governmental planning" mentioned in line 3 may lead to which of the following problems? (A) Improper use of profits derived from user fees (B) Unduly slow development of necessary new infrastructure (C) Unrealistic decisions about developing new
---	--

- infrastructure
- (D) Incorrect predictions about profits to be gained from user fees
- (E) Obstruction of private financing for the development of new infrastructure
26. According to the passage, which of the following is true of the toll road mentioned in line 12?
- (A) After it was built, it attracted too little traffic to pay for its construction.
- (B) It was partially financed by the state of Virginia.
- (C) Its development was authorized during an economic boom.
- (D) Its construction was controversial among local residents.
- (E) Its developers were discouraged by governmental restrictions on acquiring the necessary land.
27. The passage suggests that which of the following would occur if a privately financed bridge that proved to be profitable failed after a number of years to meet the demands of traffic?
- (A) Private developers who financed the bridge would rely on governmental authorities to develop new infrastructure.
- (B) User fees would be increased so that usage would become more costly.
- (C) Governmental authorities would be reluctant to rely on private contractors to develop a new bridge.
- (D) The success of the project would be jeopardized by public dissatisfaction with the project's adequacy.
- (E) Profits generated by user fees would be used to help finance the construction of new infrastructure to alleviate the traffic problem.
28. EVOKE:
- (A) try to hinder
- (B) fail to elicit
- (C) refuse to implore
- (D) pretend to agree
- (E) attempt to calm
29. OSTENTATION:
- (A) austerity
- (B) wisdom
- (C) illumination
- (D) superficiality
- (E) agitation
30. BRISTLE:
- (A) cower
- (B) feint
- (C) equivocate
- (D) coerce
- (E) apprise
31. AGGRANDIZE:
- (A) conciliate
- (B) undermine
- (C) relegate
- (D) remain unapologetic
- (E) remain inexplicit
32. ENDEMIC:
- (A) undeniable
- (B) intermittent
- (C) anomalous
- (D) foreign
- (E) unexpected
33. BELLICOSE:
- (A) enervated
- (B) disloyal
- (C) honest
- (D) likely to be generous
- (E) inclined to make peace
34. ABJURE:
- (A) affirm
- (B) cajole
- (C) insist
- (D) pronounce
- (E) shout
35. SALUTARY
- (A) unexpected
- (B) transitory
- (C) unhealthy
- (D) disoriented
- (E) dilapidated

36. LUGUBRIOUSNESS:

- (A) orderliness
- (B) shallowness
- (C) believability
- (D) cheerfulness
- (E) dedication

37. PRESCIENCE:

- (A) acuity
- (B) myopia
- (C) vacillation
- (D) tardiness
- (E) inhibition

38. INVETERATE:

- (A) arbitrary
- (B) occasional
- (C) obvious
- (D) progressive
- (E) compelling

SECTION 1

Time –30 minutes

25 Questions

Questions 1-5

Five participants at an international conference are planning to take a car trip together. Two persons? the driver and one passenger? will sit in the front seat of the car, and three persons will sit in the back seat. The names of the five participants and all of the languages that each of them speaks are as follows:

Mohsen: Farsi and Hebrew

Orlando: Italian and Russian

Shelly: Hebrew and Russian

Theo: German and Italian

Ursula: Farsi, German, and Hebrew

The participants must be seated in the car according to the following restrictions:

The driver must be Orlando or else Shelly.

Two persons can be seated side by side only if at least one of the languages they speak is the same.

1. Which of the following is an acceptable seating arrangement, with the driver listed first under "Front Seat" and the passengers in the back seat listed from one side to the other side?

Front SeatBack Seat

- | | |
|---------------------|-----------------------|
| (A) Mohsen, Ursula | Theo, Orlando, Shelly |
| (B) Orlando, Mohsen | Shelly, Theo, Ursula |
| (C) Orlando, Shelly | Mohsen, Ursula, Theo |
| (D) Shelly, Mohsen | Ursula, Orlando, Theo |
| (E) Shelly, Orlando | Theo, Mohsen, Ursula |

2. If Mohsen sits in the front seat, which of the following can be true?

- (A) Orlando will be the driver.
- (B) Orlando will sit next to Ursula.
- (C) Shelly will sit in the middle position in the back
- (D) Shelly will be the driver.
- (E) Ursula will sit in the middle position in the back seat.

3. If Theo sits in the front seat, which of the following must be true?

- (A) Mohsen and Shelly will sit side by side.
- (B) Mohsen and Ursula will sit side by side.
- (C) Orlando and Theo will sit side by side.
- (D) Orlando and Ursula will sit side by side.
- (E) Shelly and Ursula will sit side by side.

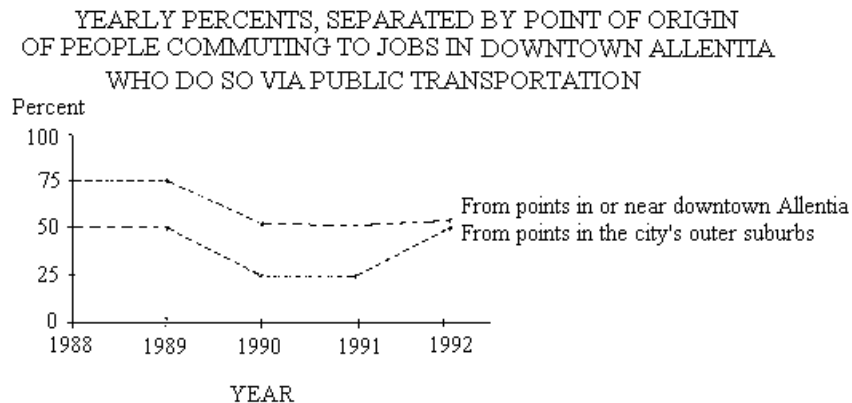
4. If both persons sitting in the front seat speak Hebrew, then it must be true that

- (A) exactly one person sitting in the back seat speaks Russian
- (B) neither speaker of Farsi is sitting in the front seat
- (C) no one sitting in the front seat speaks Russian
- (D) no one sitting in the back seat speaks Hebrew
- (E) a speaker of Russian is sitting in the middle position in the back seat

5. Which of the following must be true if Orlando is the driver?

- (A) If Shelly sits in the front seat, Ursula will sit in the middle position in the back seat.
- (B) If Shelly sits in the back seat, she will sit next to Ursula.
- (C) If Theo sits in the front seat, Ursula will sit in the middle position in the back seat.
- (D) If Theo sits in the back seat, he will sit between Mohsen and Ursula.
- (E) If Ursula sits in the back seat, she will sit in the middle position in the back seat.

Questions 6-7 are based on the following graph



6. Which of the following, if true about early 1990, would most help to explain the decrease, in 1990, of the percent of people commuting to jobs in downtown Allentia who do so via public transportation?

- (A) The termination of a governmental subsidy to the public transportation system that serves both the city and its suburbs caused a substantial increase in fares.
- (B) Many new trains and buses were put into service in the public transportation system both within the city and between the city and its suburbs.
- (C) Security was improved in the passenger waiting areas and on the public trains and buses used within the city as well as on those used between the city and its suburbs.
- (D) Legislation was passed that increased the frequency of public transportation service within the city as well as between the city and its suburbs.
- (E) The number of points served by the public transportation system both within the city and

between the city and its suburbs was increased substantially by adding new routes.

7. Which of the following, if true about early 1992, could most contribute to an explanation of the change, between 1991 and 1992, in the percent of those who commute via public transportation from the outer suburbs of Allentia, as compared to the change for the other group of commuters?

- (A) The price per gallon for gasoline declined by five percent.
- (B) The cost of using public transportation, per mile traveled, increased.
- (C) The number of people who commuted to work via public transportation from points in or near downtown Allentia increased.
- (D) The frequency of public transportation service between the city and its suburbs decreased.
- (E) The cost per mile of getting to and from work by car tripled.

8. A new and more aggressive form of the fungus that caused the Irish potato famine of the nineteenth century has recently arisen. However, since this new form of the fungus can be killed by increased application of currently used fungicides, it is unlikely that the fungus will cause widespread food shortages in countries that currently rely on potatoes for sustenance.

Which of the following, if true, most calls into question the conclusion in the argument above?

- (A) Though potatoes are an important staple crop in many parts of the world, people in most countries rely primarily on wheat or rice for sustenance.
- (B) Potato farmers in many countries to which the new form of the fungus has spread cannot afford to increase their spending on fungicides.
- (C) The new form of the fungus first began to spread when contaminated potato seeds were inadvertently exported from a major potato-exporting country.
- (D) Potato farmers in most countries use several insecticides on their crops in addition to fungicides of the sort that kill the new form of the fungus.
- (E) Most governments have funds set aside that can be used to alleviate the effects of large-scale disasters such as severe food shortages and floods.

Questions 9-16

The organizers of a music festival are scheduling exactly six master classes, one class per day for six consecutive days. Three of the classes will be given by violinists and three by pianists. The only musicians who can teach the classes are the violinists F, G, H, and J, and the pianists R, S, T, W, and Z. The festival's organizers must observe the following constraints:

No musician will teach more than one class.

F will not teach unless the first three classes are taught by violinists.

If J teaches a class, it will be the sixth.

R will teach only if T teaches the first class.

No pianist will teach on a day that immediately precedes or immediately follows a day on which W teaches.

9. Which of the following can be the musicians scheduled to teach the master classes, in the order in which they will teach, from first to sixth?

- (A) F, J, G, T, Z, S
- (B) F, W, H, T, G, Z
- (C) G, F, H, T, S, Z
- (D) S, G, W, H, R, J
- (E) T, G, W, H, R, S

10. If R is scheduled to teach the second class, which of the following could be scheduled to teach the third class?

- (A) F
- (B) G
- (C) J
- (D) T
- (E) W

11. Which of the following must be true about the schedule of master classes?

- (A) J is not scheduled to teach if R is scheduled to teach.
- (B) J is not scheduled to teach if T is scheduled to teach.
- (C) J is not scheduled to teach if W is scheduled to teach.
- (D) W is not scheduled to teach if F is scheduled to teach.
- (E) Z is not scheduled to teach if W is scheduled to teach.

12. If pianists are scheduled to teach the fourth, fifth, and sixth classes, which of the following must be true?

- (A) F is scheduled to teach the first class.
- (B) G is scheduled to teach the first class.
- (C) H is scheduled to teach an earlier class than the class Z is scheduled to teach.
- (D) R is scheduled to teach an earlier class than the class T is scheduled to teach.
- (E) S is scheduled to teach an earlier class than the class T is scheduled to teach.

13. Which of the following must be true about the schedule of the master classes?

- (A) If F is scheduled to teach a class, then H is also scheduled to teach a class.
- (B) If J is scheduled to teach a class, then R is also scheduled to teach a class.
- (C) If J is scheduled to teach a class, then S is also scheduled to teach a class.
- (D) If T is scheduled to teach a class, then R is also scheduled to teach a class.
- (E) If W is scheduled to teach a class, then Z is also scheduled to teach a class.

14. If classes are scheduled so that the classes taught by pianists and the classes taught by violinists alternate with one another, which of the following can be true?

- (A) F is scheduled to teach the fourth class.
- (B) G is scheduled to teach the first class.
- (C) H is scheduled to teach the third class.
- (D) R is scheduled to teach the fifth class.
- (E) W is scheduled to teach the second class.

15. If a violinist is scheduled to teach the first class and another violinist is scheduled to teach the sixth class, which of the following can be true?

- (A) F is scheduled to teach the second class.
- (B) H is scheduled to teach the sixth class.
- (C) R is scheduled to teach the fourth class.
- (D) T is scheduled to teach the second class.
- (E) W is scheduled to teach the third class.

16. Which of the following CANNOT be true about the schedule of the master classes?

- (A) F is scheduled to teach the third class.
- (B) G is scheduled to teach the first class.
- (C) T is scheduled to teach the sixth class.
- (D) W is scheduled to teach the sixth class.
- (E) Z is scheduled to teach the fifth class.

Questions 17-22.

In a small office suite, six offices are arranged in a straight line, one after another, and are consecutively numbered 1 through 6. Exactly six people—P, Q, R, S, T and U—are to be assigned to these six offices, exactly one person to an office, according to the following

conditions:

P must be assigned to an office immediately adjacent to the office to which T is assigned.

Q cannot be assigned to an office immediately adjacent to the office to which S is assigned.

R must be assigned either to office 1 or to office 6.

S must be assigned to a lower-numbered office than the office to which U is assigned.

17. Which of the following can be the list of the six people in the order of their offices, from office 1 through office 6?

- (A) Q, U, S, T, P, R
- (B) R, P, T, S, U, Q
- (C) R, S, Q, U, P, T
- (D) S, T, Q, P, U, R
- (E) T, P, S, R, Q, U

18. If T is assigned to office 6, then U must be assigned to office

- (A) 1
- (B) 2
- (C) 3
- (D) 4
- (E) 5

19. If Q is assigned to office 2, then the person assigned to office 6 must be

- (A) P
- (B) R
- (C) S
- (D) T
- (E) U

20. If Q is assigned to office 1, which of the following CANNOT be true?

- (A) P is assigned to office 3.
- (B) P is assigned to office 4.
- (C) S is assigned to office 4.
- (D) T is assigned to office 2.
- (E) T is assigned to office 3.

21. If U is assigned to office 3, then Q must be assigned to office

- (A) 1 or 2
- (B) 1 or 6

- (C) 2 or 5
- (D) 4 or 5
- (E) 4 or 6

22.If S is assigned to office 2, which of the following can be true?

- (A) P is assigned to office 1.
- (B) Q is assigned to office 3.
- (C) R is assigned to office 6.
- (D) T is assigned to office 5.
- (E) U is assigned to office 4.

23.As government agencies, faced with budget difficulties, reduce their funding for scientific research, a greater amount of such research is being funded by private foundations. This shift means that research projects likely to produce controversial results will almost certainly comprise a smaller proportion of all funded research projects, since private foundations, concerned about their public image, tend to avoid controversy.

Which of the following is an assumption on which the argument depends?

- (A) Only research that is conducted without concern for the possibility of generating controversy is likely to produce scientifically valid results.
- (B) Private foundations that fund scientific research projects usually recognize that controversial results from those projects cannot always be avoided.
- (C) Scientists who conduct research projects funded by private foundations are unlikely to allow the concerns of the funding organizations to influence the manner in which they conduct the research.
- (D) Many government agencies are more concerned about their public image than are most private foundations.
- (E) Government agencies are more willing than are private foundations to fund research projects that are likely to produce controversial results.

24.Juries in criminal trials do not base verdicts on uncorroborated testimony given by any one witness. Rightly so, because it is usually prudent to be highly skeptical

of unsubstantiated claims made by any one person. But then, to be consistent, juries should end an all-too-common practice: convicting defendants on the basis of an uncorroborated full confession.

Which of the following, if true, most strengthens the argument above?

- (A) Juries often acquit in cases in which a defendant retracts a full confession made before trial.
- (B) The process of jury selection is designed to screen out people who have a firm opinion about the defendant's guilt in advance of the trial.
- (C) Defendants sometimes make full confessions when they did in fact do what they are accused of doing and have come to believe that the prosecutor has compelling proof of this.
- (D) Highly suggestible people who are accused of wrongdoing sometimes become so unsure of their own recollection of the past that they can come to accept the accusations made against them.
- (E) Many people believe that juries should not convict defendants who have not made a full confession.

25.Although spinach is rich in calcium, it also contains large amounts of oxalic acid, a substance that greatly impedes calcium absorption by the body. Therefore, other calcium-containing foods must be eaten either instead of or in addition to spinach if a person is to be sure of getting enough calcium.

Which of the following, if true, most seriously weakens the argument above?

- (A) Rice, which does not contain calcium, counteracts the effects of oxalic acid on calcium absorption.
- (B) Dairy products, which contain even more calcium than spinach does, are often eaten by people who eat spinach on a regular basis.
- (C) Neither the calcium nor the oxalic acid in spinach is destroyed when spinach is cooked.
- (D) Many leafy green vegetables other than spinach that are rich in calcium also contain high concentrations of oxalic acid.
- (E) Oxalic acid has little effect on the body's ability to absorb nutrients other than calcium.

SECTION 2

Time – 30 minutes

30 Questions

1. $\frac{1}{4} - \frac{1}{5}$ $\frac{1}{20}$

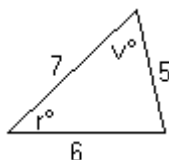
2. $x - y - 3 = 0$
 x y

The average (arithmetic mean) of 3 numbers is 37.5.
 3. The sum of the 3 numbers 100

4. $\frac{1}{x} + 1$ $x > 0$ $\frac{1}{x+1}$

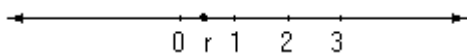
5. The perimeter of a square with sides of length 5 The perimeter of a rectangle with length 10 and width 2

x is positive number and y is 30 percent of x
 6. 25. percent of y 55 percent of x



7. r v

8. $|x|$ $x < 0$ x



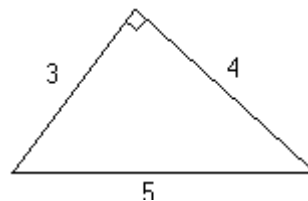
9. r $\frac{1}{r}$

m and n are positive integers.
 10. $m + n$ mn

A bicycle registration costs \$2.250 in City X and \$3.00 in City Y. At these rates, the cost of 4 registrations in City X is k percent of the cost of 3 registrations in City Y.

11. k 90

12. $\left(\frac{1}{x}\right)^2$ x^2



13. The area of the triangular region 6

An identification code read from left to right consists of 2 digits, a dash, 3 digits, a dash, and then 4 digits. Each digit can be any number from 0 through 9.

14. The number of different identification codes possible 10^9

In a rectangular coordinate system, line k has x -intercept 4 and slope -2 .

15. The y -intercept of k 2

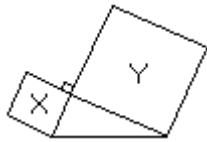
16. Of the following, which is the closest approximation to $\frac{(1.5)(19.9)(4.012)}{3.02}$?

- (A) 400
- (B) 120
- (C) 100
- (D) 40
- (E) 10

17. If $(x - 1)^2 = (x - 2)^2$, then $x =$

- (A) $-\frac{5}{8}$
- (B) $\frac{2}{3}$
- (C) $\frac{4}{3}$

- (D) $\frac{2}{3}$
 (E) $\frac{5}{2}$



18. In the figure above, the areas of square regions X and Y are 1 and 4, respectively. What is the area of the triangular region?
- (A) 2
 (B) 1
 (C) $\frac{3}{4}$
 (D) $\frac{1}{2}$
 (E) $\frac{1}{4}$
19. If erasers cost \$0.25 each, at most how many erasers

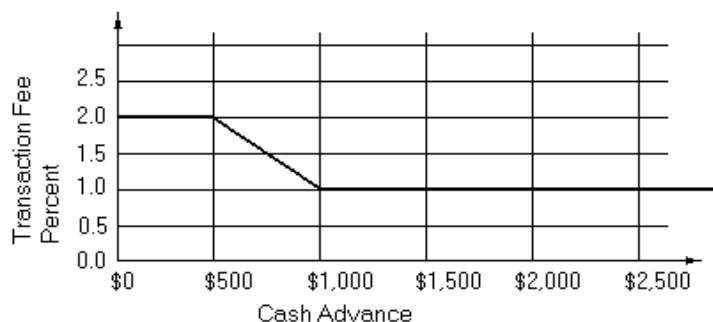
can be purchased for n dollars, where n is an integer?

- (A) $\frac{n}{25}$
 (B) $\frac{n}{4}$
 (C) $4n$
 (D) $25n$
 (E) $\frac{25n}{4}$

20. Three salespeople are paid commissions in proportion to the amount of their sales, which total \$25,000, \$40,000, and \$60,000, respectively. If a total of \$20,000 is allocated for these three commissions, what is the amount of the largest commission paid?
- (A) \$8,000
 (B) \$8,400
 (C) \$9,600
 (D) \$10,000
 (E) \$12,000

Questions 21-23 refer to the following information.

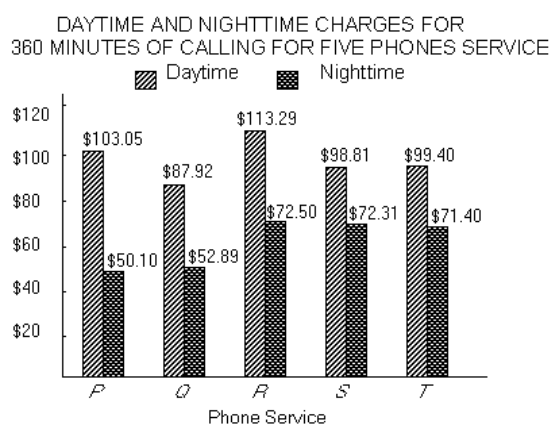
For a cash advance, a certain credit card company charges a transaction fee equal to a percent of the total amount of the cash advance, according to the graph below.



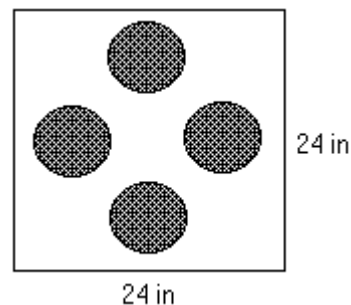
21. When compared with the transaction fee for a \$1,000 cash advance, the transaction fee for a \$500 cash advance is
- (A) \$5 more
 (B) \$10 more
 (C) the same
 (D) \$5 less
 (E) \$10 less
22. For which of the following cash advance amounts is the transaction fee approximately \$4?
- (A) \$190
 (B) \$420
 (C) \$750
 (D) \$1,200
 (E) \$1,580

23. For a total of \$1,500 that is advanced in separate cash amounts, for which of the following is the total of the transaction fees the LEAST?
- (A) Two cash advances of \$750
 (B) Three cash advances of \$500
 (C) Six cash advances of \$250
 (D) Two cash advances, one of \$1,100 and one of \$400
 (E) Two cash advances, one of \$1,250 and one of \$250

Questions 24-25 refer to the following information



24. What is the median nighttime charge for 360 minutes of calling?
- (A) \$63.84
 (B) \$71.40
 (C) \$72.50
 (D) \$87.92
 (E) \$113.29
25. The daytime charge for 360 minutes of calling for phone service *T* is approximately what percent more than the nighttime charge?
- (A) 7%
 (B) 14%
 (C) 28%
 (D) 33%
 (E) 40%



26. A square dart board has four dark circular regions of radius 3 inches as shown in the design above. Each point on the dart board is equally likely to be hit by a dart that hits the board. What is the probability that a dart that hits the board will hit one of the circular regions?

- (A) $\frac{p}{16}$
 (B) $\frac{p}{48}$
 (C) $\frac{p}{64}$
 (D) $\frac{1}{3}$
 (E) $\frac{1}{4}$

27. If x increased by 50 percent is equal to 20, then $x =$

- (A) $\frac{40}{3}$
 (B) 10
 (C) $\frac{20}{3}$
 (D) 5
 (E) $\frac{3}{4}$

28. In the rectangular coordinate plane, point *A* has coordinates $(-4, 0)$, point *B* has coordinates $(0, 4)$, point *C* has coordinates $(4, 0)$, and point *D* has coordinates $(0, -4)$. What is the area of quadrilateral *ABCD*?

- (A) 8
 (B) 16
 (C) 24
 (D) 32
 (E) 64

29. An experiment has three possible outcomes, I , J , and K . The probabilities of the outcomes are 0.25, 0.35, and 0.40, respectively. If the experiment is to be performed twice and the successive outcomes are independent, what is the probability that K will not be an outcome either time?

- (A) 0.36
- (B) 0.40
- (C) 0.60
- (D) 0.64
- (E) 0.80

30. If the inside diameter of a cylindrical garden hose is 1 inch, what is the length, in inches, of a straight hose that can hold a maximum of 1 gallon of water? (1 gallon = 231 cubic inches)

- (A) 231p
- (B) $\frac{231}{p}$
- (C) 924
- (D) 924p
- (E) $\frac{924}{p}$

SECTION 3

Time –30 minutes

38 Questions

1. It is assumed that scientists will avoid making ---- claims about the results of their experiments because of the likelihood that they will be exposed when other researchers cannot ---- their findings.
 - (A) hypothetical.. evaluate
 - (B) fraudulent.. duplicate
 - (C) verifiable.. contradict
 - (D) radical.. contest
 - (E) extravagant.. dispute
2. As long as the nuclear family is ---- a larger kinship group through contiguous residence on undivided land, the pressure to ---- and thus to get along with relatives is strong.
 - (A) nurtured among.. abstain
 - (B) excluded from.. compromise
 - (C) embedded in .. share
 - (D) scattered throughout.. reject
 - (E) accepted by .. lead
3. In contrast to the substantial muscular activity required for inhalation, exhalation is usually a ---- process.
 - (A) slow
 - (B) passive
 - (C) precise
 - (D) complex
 - (E) conscious
4. The documentary film about high school life was so realistic and ---- that feelings of nostalgia flooded over the college-age audience.
 - (A) logical
 - (B) pitiful
 - (C) evocative
 - (D) critical
 - (E) clinical
5. Although Georgia O'Keeffe is best known for her affinity with the desert landscape, her paintings of urban subjects ---- her longtime residency in New York City.

(A) condemn (B) obfuscate (C) attest to (D) conflict with (E) contend with 6. Even though the survey was designated as an inter-disciplinary course, it involved no real ---- of subject matter. (A) encapsulation (B) organization (C) synthesis (D) discussion (E) verification 7. The failure of many psychotherapists to ---- the results of pioneering research could be due in part to the specialized nature of such findings: even ---- findings may not be useful. (A) understand.. baffling (B) envision.. accessible (C) utilize.. momentous (D) reproduce.. duplicated (E) affirm.. controversial 8. EARPLUG: NOISE:: (A) saw: wood (B) detonation: explosion (C) clothes: covering (D) liquid: flask (E) shield: impact 9. REVISE: MANUSCRIPT:: (A) retouch: picture (B) replicate: experiment (C) repair: hammer (D) replace: book (E) restore: masterpiece 10. DAREDEVIL: AUDACITY:: (A) malcontent: dissatisfaction (B) perfectionist: patience (C) cynic: indiscretion (D) melancholic: bitterness (E) hedonist: ambition	11. CALCIUM: MINERAL:: (A) sugar: carbohydrate (B) salt: solution (C) enzyme: food (D) milk: cheese (E) calorie: diet 12. DIRGE: GRIEF:: (A) diatribe: uneasiness (B) parody: cruelty (C) paean: praise (D) testimonial: veracity (E) anthem: seriousness 13. ABANDON: INHIBITION:: (A) ascendancy: effort (B) prickliness: sensation (C) surrender: resignation (D) reversal: instigation (E) tranquillity: agitation 14. INAUGURATION: OFFICIAL:: (A) instruction: lecturer (B) election: politician (C) pilgrimage: devotee (D) dispute: arbitrator (E) matriculation: student 15. SCORN: REJECT:: (A) adulate: flatter (B) conjecture: forecast (C) pledge: renege (D) allege: declare (E) disparage: ignore 16. PROFLIGATE: SOLVENT:: (A) mercurial: committed (B) caustic: rational (C) indecisive: confused (D) cautious: uncertain (E) practical: seemly As people age, their cells become less efficient and less able to replace damaged components. At the same time their tissues stiffen. For example, the lungs and the heart muscle expand less successfully, the blood vessels
--	---

(5) become increasingly rigid, and the ligaments and tendons tighten.

Few investigators would attribute such diverse effects to a single cause. Nevertheless, researchers have discovered that a process long known to discolor and toughen (10) foods may also contribute to age-related impairment of both cells and tissues. That process is nonenzymatic glycosylation, whereby glucose becomes attached to proteins without the aid of enzymes. When enzymes attach glucose to proteins (enzymatic glycosylation), they do so (15) at a specific site on a specific protein molecule for a specific purpose. In contrast, the nonenzymatic process adds glucose haphazardly to any of several sites along any available peptide chain within a protein molecule.

This nonenzymatic glycosylation of certain proteins (20) has been understood by food chemists for decades, although few biologists recognized until recently that the same steps could take place in the body. Nonenzymatic glycosylation begins when an aldehyde group (CHO) of glucose and an amino group (NH₂) of a protein are (25) attracted to each other. The molecules combine, forming what is called a Schiff base within the protein. This combination is unstable and quickly rearranges itself into a stabler, but still reversible, substance known as an Amadori product.

(30) If a given protein persists in the body for months or years, some of its Amadori products slowly dehydrate and rearrange themselves yet again, into new glucose-derived structures. These can combine with various kinds of molecules to form irreversible structures named advanced (35) glycosylation end products (AGE's). Most AGE's are yellowish brown and fluorescent and have specific spectrographic properties. More important for the body, many are also able to cross-link adjacent proteins, particularly ones that give structure to tissues and organs. Although (40) no one has yet satisfactorily described the origin of all such bridges between proteins, many investigators agree that extensive cross-linking of proteins probably contributes to the stiffening and loss of elasticity characteristic of aging tissues.

(45) In an attempt to link this process with the development of cataracts (the browning and clouding of the lens of the eye as people age), researchers studied the effect of glucose on solutions of purified crystallin, the major protein in the lens of the eye. Glucose-free solutions (50) remained clear, but solutions with glucose caused the

proteins to form clusters, suggesting that the molecules had become cross-linked. The clusters diffracted light, making the solution opaque. The researchers also discovered that the pigmented cross-links in human (55) cataracts have the brownish color and fluorescence characteristic of AGE's. These data suggest that nonenzymatic glycosylation of lens crystallins may contribute to cataract formation.

17. With which of the following statements concerning the stiffening of aging tissues would the author most likely agree?

- (A) It is caused to a large degree by an increased rate of cell multiplication.
- (B) It paradoxically both helps and hinders the longevity of proteins in the human body.
- (C) It can be counteracted in part by increased ingestion of glucose-free foods.
- (D) It is exacerbated by increased enzymatic glycosylation.
- (E) It probably involves the nonenzymatic glycosylation of proteins.

18. According to the passage, which of the following statements is true of the process that discolors and toughens foods?

- (A) It takes place more slowly than glycosylation in the human body.
- (B) It requires a higher ratio of glucose to protein than glycosylation requires in the human body.
- (C) It does not require the aid of enzymes to attach glucose to protein.
- (D) It proceeds more quickly when the food proteins have a molecular structure similar to that of crystallin proteins.
- (E) Its effectiveness depends heavily on the amount of environmental moisture.

19. According to the passage, which of the following is characteristic of enzymatic glycosylation of proteins?

- (A) AGE's are formed after a period of months or years.
- (B) Proteins affected by the process are made unstable.

- (C) Glucose attachment impairs and stiffens tissues.
- (D) Glucose is attached to proteins for specific purposes.
- (E) Amino groups combine with aldehyde groups to form Schiff bases.
20. According to the passage, which of the following statements is true of Amadori products in proteins?
- (A) They are more plentiful in a dehydrated environment.
- (B) They are created through enzymatic glycosylation.
- (C) They are composed entirely of glucose molecules.
- (D) They are derived from Schiff bases.
- (E) They are derived from AGE's
21. Which of the following best describes the function of the third paragraph of the passage (lines 19-29)?
- (A) It offers evidence that contradicts the findings described in the first two paragraphs.
- (B) It presents a specific example of the process discussed in the first two paragraphs.
- (C) It explains a problem that the researchers mentioned in the second paragraph have yet to solve.
- (D) It evaluates the research discoveries described in the previous paragraph.
- (E) It begins a detailed description of the process introduced in the previous two paragraphs.
22. The passage suggests that which of the following would be LEAST important in determining whether nonenzymatic glycosylation is likely to have taken place in the proteins of a particular tissue?
- (A) The likelihood that the tissue has been exposed to free glucose
- (B) The color and spectrographic properties of structures within the tissue.
- (C) The amount of time that the proteins in the tissue have persisted in the body
- (D) The number of amino groups within the proteins in the tissue
- (E) The degree of elasticity that the tissue exhibits
23. If the hypothesis stated in lines 56-58 is true, it can be inferred that the crystallin proteins in the lenses of people with cataracts
- (A) have increased elasticity
- (B) do not respond to enzymatic glycosylation
- (C) are more susceptible to stiffening than are other proteins
- (D) are at least several months old
- (E) respond more acutely than other proteins to changes in moisture levels
- Writing of the Iroquois nation, Smith has argued that through the chiefs' council, tribal chiefs traditionally maintained complete control over the political affairs of both the Iroquois tribal league and the individual tribes
- (5) belonging to the league, whereas the sole jurisdiction over religious affairs resided with the shamans.
- According to Smith, this division was maintained until the late nineteenth century, when the dissolution of the chiefs' council and the consequent diminishment of the
- (10) chiefs' political power fostered their increasing involvement in religious affairs.
- However, Smith fails to recognize that this division of power between the tribal chiefs and shamans was not actually rooted in Iroquois tradition; rather, it resulted
- (15) from the Iroquois' resettlement on reservations early in the nineteenth century. Prior to resettlement, the chiefs' council controlled only the broad policy of the tribal league; individual tribes had institutions? most important, the longhouse? to govern their own affairs. In the
- (20) longhouse, the tribe's chief influenced both political and religious affairs.
24. The primary purpose of the passage is to
- (A) question the published conclusions of a scholar concerning the history of the Iroquois nation
- (B) establish the relationship between an earlier scholar's work and new anthropological research
- (C) summarize scholarly controversy concerning an incident from Iroquois history.
- (D) trace two generations of scholarly opinion concerning Iroquois social institutions
- (E) differentiate between Iroquois political practices and Iroquois religious practices

25. It can be inferred that the author of the passage regards Smith's argument as
- (A) provocative and potentially useful, but flawed by poor organization
 - (B) eloquently presented, but needlessly inflammatory
 - (C) accurate in some of its particulars, but inaccurate with regard to an important point.
 - (D) historically sound, but overly detailed and redundant
 - (E) persuasive in its time, but now largely outdated
26. The author of the passage implies that which of the following occurred after the Iroquois were resettled on reservations early in the nineteenth century?
- (A) Chiefs became more involved in their tribes' religious affairs.
 - (B) The authority of the chiefs' council over the affairs of individual tribes increased.
 - (C) The political influence of the Iroquois shamans was diminished.
 - (D) individual tribes coalesced into the Iroquois tribal league.
 - (E) The longhouse became a political rather than a religious institution.
27. Which of the following best expresses an opinion presented by the author of the passage?
- (A) Smith has overstated the importance of the political role played by Iroquois tribal chiefs in the nineteenth century.
 - (B) Smith has overlooked the fact that the Iroquois rarely allowed their shamans to exercise political authority.
 - (C) Smith has failed to explain why the chiefs' council was dissolved late in the nineteenth century.
 - (D) Smith has failed to acknowledge the role prior to the nineteenth century of the Iroquois tribal chiefs in religious affairs.
 - (E) Smith has failed to recognize that the very structure of Iroquois social institutions reflects religious beliefs.
28. DRONE:
- (A) behave bestially
 - (B) decide deliberately
 - (C) err intentionally
 - (D) speak animatedly
 - (E) plan inefficiently
29. CERTAINTY:
- (A) obstinacy
 - (B) impetuosity
 - (C) recklessness
 - (D) indecision
 - (E) indifference
30. MORIBUND:
- (A) fully extended
 - (B) automatically controlled
 - (C) loosely connected
 - (D) completely dispersed
 - (E) increasingly vital
31. PROFANE:
- (A) approach expectantly
 - (B) punish mildly
 - (C) appease fully
 - (D) treat reverently
 - (E) admonish sternly
32. PERSONABLE:
- (A) unrefined
 - (B) unselfish
 - (C) unattractive
 - (D) uncommitted
 - (E) undistinguished
33. MIRE:
- (A) straighten
 - (B) fracture
 - (C) extricate
 - (D) elevate
 - (E) augment
34. CONCEPTUAL:
- (A) proven
 - (B) effective
 - (C) manageable
 - (D) concrete
 - (E) punctilious

35. SURFEIT:

- (A) precise length
- (B) delayed increment
- (C) obtainable quantity
- (D) unascertained limit
- (E) insufficient supply

36. TENACITY:

- (A) vacillation
- (B) servility
- (C) temerity
- (D) perversity
- (E) diversity

37. APPOSITE:

- (A) irrelevant
- (B) nameless
- (C) tentative
- (D) disfavored
- (E) lavish

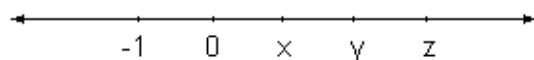
38. STYMIE:

- (A) ponder
- (B) predict
- (C) divulge
- (D) abet
- (E) explain

SECTION 5

Time – 30 minutes

30 Questions



x , y , and z are coordinates of three points on the number line above.

1. xy xz

10 inches of snowfall is equivalent to 1 inch of rainfall.
(1 foot = 12 inches)

2. The number of inches 1
of rainfall equivalent
to 1 foot of snowfall

$$b > 0$$

3. $a - b$ $b - a$

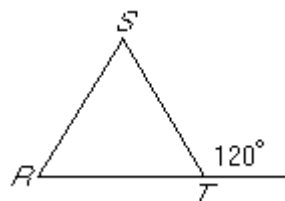
The geometric mean of any two positive numbers x and y is $\sqrt{xy}$.

4. The geometric mean The average (arithmetic
of 4 and 8 mean) of 4 and 8

5. $\frac{16}{35}$ $\frac{4}{9}$

$$1 < x < y$$

6. $x + 4$ y



7. The length of ST The length of RS

$$x + 5 = 3$$

$$y = 2x$$

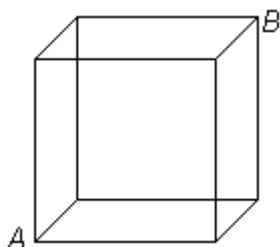
8. x y

1 gallon = 8 pints

1 quart = 2 pints

9. The least number of half-pint bottles needed to hold x quarts of milk

The least number of one-quart bottles needed to hold x gallons of milk



The figure shows a cube with edge of length e .

10. The length of diagonal AB

$$\sqrt{2}e$$

$$y = 3x - 1$$

11. x

$$\frac{y}{3} + 3$$

12. $37 \times \frac{37}{36}$

$$37 + \frac{37}{36}$$

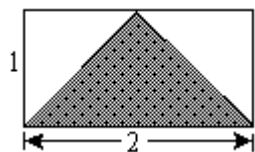
13. 0.01% of 1,000

$$1$$

x and y are positive integers and $x > y$.

14. $\frac{x^2}{y^3}$

$$\frac{y^3}{x^2}$$



15. The perimeter of the shaded region in the rectangle

$$2\sqrt{2} + 2$$

16. $-\left(\frac{-6}{2}\right) =$

- (A) -4
(B) -3
(C) 3
(D) 4

- (E) 8

17. A rectangular parking lot $2x$ feet long and x feet wide is to be enlarged so that the lot will be 2 times as long and 3 times as wide as it is now. The area of the enlarged rectangular lot will be how many times the area of the present lot?

- (A) 6
(B) 5
(C) 4
(D) 3
(E) 2

18. If $2x = 5$ and $3y = 8$, then $\frac{4x}{9y} =$

- (A) $\frac{5}{18}$
(B) $\frac{5}{16}$
(C) $\frac{5}{12}$
(D) $\frac{5}{8}$
(E) $\frac{5}{4}$

19. A certain jar contains 100 jelly beans: 50 white, 30 green, 10 yellow, 5 red, 4 purple, and 1 black. If a jelly bean is to be chosen at random, what is the probability that the jelly bean will be neither purple nor red?

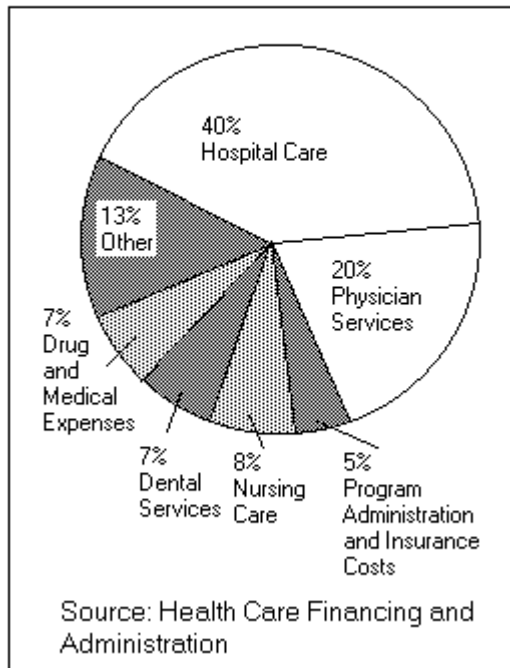
- (A) 0.09
(B) 0.11
(C) 0.55
(D) 0.91
(E) 0.96

20. (:PROBLEMSOLVE) The average (arithmetic mean) of $|x|$ and x equals

- (A) x if $x > 0$, and equals 0 if $x = 0$
(B) $-x$ if $x < 0$, and equals 0 if $x = 0$
(C) 0, regardless of the value of x
(D) x , regardless of the value of x
(E) $|x|$, regardless of the value of x

Questions 21- 22 refer to the following graph.

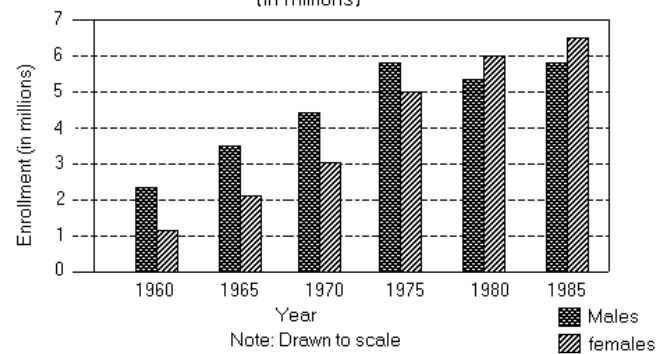
1989 HEALTH CARE EXPENDITURES
(to the nearest whole percent)
Total \$550 billion



21. Expenditures for physician services in 1989 were approximately how many billion dollars more than expenditures for nursing care?
- (A) 38.5
(B) 66.0
(C) 82.5
(D) 110.0
(E) 154.0
22. If health care expenditures accounted for 11 percent of the gross domestic product in 1989, then the gross domestic product was approximately how many billion dollars?
- (A) 605
(B) 5,000
(C) 5,500
(D) 6,050
(E) 6,500

Questions 23-25 refer to the following graph.

COLLEGE AND UNIVERSITY ENROLLMENT
IN THE UNITED STATES FROM 1960 TO 1985
(in millions)



23. The total enrollment in 1985 was approximately how much greater than the total enrollment in 1960?
- (A) 4 million
(B) 5 million
(C) 7 million
(D) 9 million
(E) 11 million
24. For which of the years shown was the ratio of male to female enrollment greatest?
- (A) 1980
(B) 1975
(C) 1970
(D) 1965
(E) 1960
25. For which of the following periods was the percent increase in female enrollment the greatest?
- (A) 1960 to 1965
(B) 1965 to 1970
(C) 1970 to 1975
(D) 1975 to 1980
(E) 1980 to 1985
26. If a certain town has 90 doctors and the ratio of male doctors to female doctors is 3 to 2, then the number of female doctors in this town is
- (A) 18
(B) 30
(C) 36
(D) 45
(E) 54
27. Which of the following points (x, y) is NOT on the

graph of $y < 2x$?

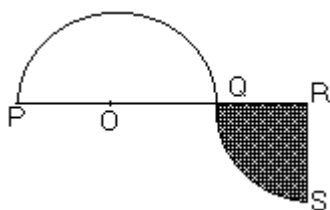
- (A) $(-3, -7)$
- (B) $(3, 3)$
- (C) $(2, -9)$
- (D) $(2, 2)$
- (E) $(2, 5)$

28. If apples sell for \$0.68 per pound and Juanita bought 36 apples for \$8.16, then the average (arithmetic mean) number of apples per pound was

- (A) $\frac{1}{3}$
- (B) $\frac{1}{2}$
- (C) 1
- (D) 2
- (E) 3

29. The sum of the first 50 positive integers is 1,275. What is the sum of the integers from 51 to 100.

- (A) 2,525
- (B) 2,550
- (C) 3,250
- (D) 3,775
- (E) 5,050



30. The figure above shows a semicircle with center O and a quarter circle with center R . If $OQ = 4$ and $QR = 6$, what is the ratio of the area of the shaded region to the area of the semicircular region?

- (A) 2:3
- (B) 4:3
- (C) 8:9
- (D) 9:4
- (E) 9:8

SECTION 6

Time –30 minutes

25 Questions

1. Joyce: Three years ago the traffic commission modified our town's busiest intersection for better visibility, a commendable effort to cut down on traffic accidents there.

Gary: Over the past three years there have been more, not fewer, traffic accidents per week at that intersection, so the modification has increased the likelihood of accidents there

The answer to which of the following questions would be most useful in evaluating Gary's argument?

- (A) What proportion of the town's drivers involved in accidents that occurred prior to the modification suffered personal injury in their accidents?
- (B) How long, on average, had the members of the traffic commission held their offices when the modification was implemented?
- (C) Do a majority of the town's residents approve of the traffic commission's overall performance?
- (D) What measures have nearby towns taken within the last three years in order to improve visibility at dangerous intersections?
- (E) How has the volume of traffic at the town's busiest intersection changed over the last three years?

2. Women make up the majority of the population in the country, and many of the prescriptions written by doctors for tranquilizers are for women patients. The testing of these drugs for efficacy and the calibration of recommended doses, however, was done only on men. Not even the animals used to test toxicity were female.

The statements above, if true, best support which of the following as a conclusion?

- (A) Some tranquilizers are more appropriately prescribed for women than for men.
- (B) There have been no reports of negative side effects from prescribed tranquilizers in women

- (C) Tranquilizers are prescribed for patients in some instances when doctors do not feel confident of their diagnoses.
- (D) The toxicity of drugs to women is less than the

toxicity of the same drugs to men.

(E) Whether the recommended dosages of tranquilizers are optimal for women is not known.

Questions 3-8

A landscape designer has available two trees each of eight different species—linden, maple, pine, quince, redbud, spruce, tupelo, and walnut—from which a selection must be made for planting at two different locations. For planting at each location, the designer will select exactly four trees, representing four different tree species, according to the following conditions:

If quince is planted at location 1, spruce cannot be planted at location 2.

If linden and quince are planted at location 1, pine must be planted at location 2.

If pine is planted at location 2, quince and tupelo must be planted at location 1.

Spruce cannot be planted at location 1 unless linden and pine are planted at location 2.

3. Which of the following is an acceptable selection of species to be planted at the two locations?

Location 1

- (A) Linden, maple, pine, quince
 (B) Linden, pine, quince, walnut
 (C) Maple, quince, spruce, tupelo
 (D) Quince, spruce, tupelo, walnut
 (E) Quince, spruce, tupelo, walnut

Location 2

- Maple, pine, tupelo, walnut
 Linden, maple, tupelo, walnut
 Maple, pine, tupelo, walnut
 Linden, maple, pine, tupelo
 Linden, pine, quince, spruce

4. If maple, spruce, and tupelo are planted at location 1, then which of the following must be the other species planted at location 1?

- (A) Quince
 (B) Redbud
 (C) Spruce
 (D) Tupelo
 (E) Walnut

5. If maple, redbud, and walnut are planted at location 1, then any of the following species can be planted at location 2 EXCEPT

- (A) linden
 (B) pine
 (C) quince

- (D) tupelo
 (E) walnut

6. If linden and quince are planted at location 1, then which of the following must be true?

- (A) Quince is planted at location 2.
 (B) Spruce is planted at location 2.
 (C) Pine is not planted at location 1.
 (D) Spruce is not planted at location 1.
 (E) Tupelo is planted at location 1.

7. If each of the eight species is selected for planting and spruce is planted at location 2, then which of the following must also be planted at location 2?

- (A) Linden

- (B) Pine
- (C) Quince
- (D) Tupelo
- (E) Walnut

8. If each of the eight species must be selected for planting and quince and tupelo are planted at location 1, then which of the following can be the other two species planted at location 1?

- (A) Maple and spruce
- (B) Maple and walnut
- (C) Pine and redbud
- (D) Pine and spruce
- (E) Redbud and walnut

9. Gray wolves have been absent from a large national park for decades. Park officials wish to reestablish the wolves without jeopardizing any existing species of wildlife there. Since the park contains adequate prey for the wolves and since the wolves avoid close contact with people, reintroducing them would serve the officials' purpose without seriously jeopardizing visitors' safety.

Each of the following, if true, strengthens the argument above EXCEPT:

- (A) The park is so large that wolves will not need to venture into areas frequented by people.
- (B) Rabies is very rare in wolves, and there have been no verified cases of serious human injuries from nonrabid wild wolves since records have been kept.
- (C) Ranchers in the region near the park have expressed concern that gray wolves, if reintroduced, would sometimes prey on their livestock.
- (D) Predation by gray wolves on elk in the park is likely to improve the health and viability of the park's elk population as a whole by reducing malnutrition among the elk.
- (E) Wolves do not prey on animals of any endangered species that currently inhabit the park.

10. Osteoporosis is a disease that reduces bone mass, leading to fragile bones that break easily. Current treatments for osteoporosis such as estrogen or calcitonin help prevent further loss of bone but do not increase

bone mass. Since fluoride is known to increase bone mass, administering fluoride to osteoporosis patients would therefore help make their bones less susceptible to breaking.

Which of the following, if true, most seriously weakens the argument above?

- (A) Most people who suffer from osteoporosis are not aware that administration of fluoride can increase bone mass.
- (B) Fluoride is added to drinking water in many locations in order to strengthen the teeth of people who drink the water.
- (C) The risk of contracting osteoporosis and other degenerative bone diseases is lessened by exercise and an adequate intake of calcium.
- (D) Unlike administration of fluoride, administration of estrogen or calcitonin is known to cause undesirable side effects for many people.
- (E) The new bone mass that is added by the administration of fluoride is more brittle and less elastic than normal bone tissue.

11. The closest distance from which an asteroid has been photographed using ground-based radar is 2.2 million miles, the distance from which the asteroid Toutatis was recently photographed. The closest photograph of an asteroid is of Gaspra, which was photographed from a distance of only 10,000 miles.

Which of the following can be properly concluded from the statements above?

- (A) Toutatis is more likely to collide with the Earth than Gaspra is.
- (B) Toutatis, unlike Gaspra, has only recently been discovered.
- (C) Asteroids can be photographed only by using ground-based radar.
- (D) Ground-based radar photography cannot take photographs of objects much beyond 2.2 million miles from Earth.
- (E) The photograph of Gaspra was not taken using ground-based radar.

12. Which of the following most logically completes the argument below?

Alone among living species, human beings experience adolescence, a period of accelerated physical growth prior to full maturity. Whether other hominid species, which are now all extinct and are known only through the fossil record, went through adolescence cannot be known, since

- (A) the minimum acceleration in physical growth that would indicate adolescence might differ according to species
- (B) the fossil record, though steadily expanding, will always remain incomplete
- (C) detecting the adolescent growth spurt requires measurements on the same individual at different ages
- (D) complete skeletons of extinct hominids are extremely rare
- (E) human beings might be the first species to benefit from the survival advantages, if any, conferred by adolescence

Questions 13-17

A circus magician has a "magic box " that has exactly six chambers designed to hold at least two animals each. There are five trained animals—a frog, a hen, a mouse, a parakeet, and a rabbit. On the front of the box, chambers 1, 2, and 3 are arranged in a straight line so that chamber 1 is directly adjacent to chamber 2, and chamber 2 is directly adjacent to chamber 3. On the back of the box, the chambers are also arranged in a straight line with chamber 4 directly adjacent to chamber 5, and chamber 5 directly adjacent to chamber 6. No chamber in the front of the box is directly adjacent to a chamber in the back.

When working with the animals, the magician must obey the following restrictions:

None of the chambers can contain more than two animals at the same time.

The mouse cannot be in the same chamber as any other animal, and any chamber directly adjacent to the chamber that the mouse occupies must remain empty.

Neither the hen nor the frog can be in the same chamber as the parakeet.

13.If the mouse is in chamber 2, and the other four animals are all in chambers, which of the following is a pair of chambers that must be empty?

- (A) 1 and 3
- (B) 1 and 4
- (C) 2 and 4
- (D) 3 and 5
- (E) 3 and 6

14. If the mouse in chamber 2, the parakeet is in chamber 4, and all the other animals are in chambers, then the hen can be in chamber.

- (A) 1
- (B) 2
- (C) 3
- (D) 4
- (E) 5

15.If no chamber contains more than one animal and each of the five animals is in a chamber, then there is a total of how many different chambers any one of which could be the chamber that contains the mouse?

- (A) One
- (B) Two
- (C) Three
- (D) Four
- (E) Five

16.If all five animals are in the chambers, the mouse is in chamber 2, and the frog's chamber is different from and not directly adjacent to the hen's chamber , then the parakeet must be in chamber

- (A) 1
- (B) 3
- (C) 4
- (D) 5
- (E) 6

17.If the hen is in chamber 1 with another animals, one animals is in chamber 5, and two animals are in chamber 3, which of the following pairs of animals must be in chamber 3?

- (A) The frog and the mouse
- (B) The frog and the parakeet
- (C) The frog and the rabbit
- (D) The mouse and the rabbit
- (E) The parakeet and the rabbit

Questions 18-22

Seven science students—John, Kate, Luz, Mark, Nelson, Olga, and Pat—are to be divided into three groups. One group will consist of three students, and the other groups will consist of two students each. The following conditions apply to the assignment of students to groups:

John cannot be assigned to the group to which Luz is assigned

Nelson must be assigned to the group to which Pat is assigned.

Olga must be assigned to the group consisting of three students.

18. Which of the following is an acceptable assignment of students to groups?

(A) John, Kate, and Mark; Luz, Nelson, and Olga; Pat

(B) John, Mark, and Olga; Kate and Luz; Nelson and Pat

(C) Kate, Luz, and Olga; John and Nelson; Mark and Pat

(D) Mark, Nelson, and Pat; John and Kate; Luz and Olga

(E) Nelson, Olga, and Pat; John and Luz; Kate and Mark

19. If one of the groups of two consists of Luz and Mark, which of the following must be true?

(A) John is assigned to the group to which Kate is assigned.

(B) John is assigned to the group to which Nelson is assigned.

(C) Kate is assigned to the group to which Pat is assigned.

(D) Kate is assigned to a group consisting of two students.

(E) Nelson is assigned to the group consisting of three students.

20. If Nelson is assigned to the group to which Olga is assigned, which of the following can be true?

(A) Kate is assigned to the group to which John is assigned.

(B) Kate is assigned to the group to which Mark is assigned.

(C) Luz is assigned to the group to which Olga is assigned.

(D) John is assigned to the group consisting of three students.

(E) Pat is assigned to a group consisting of two students.

21. If Mark is assigned to a group to which neither John nor Pat is assigned, which of the following must be true?

(A) John is assigned to the group to which Kate is assigned.

(B) John is assigned to the group to which Nelson is assigned.

(C) Kate is assigned to the group to which Luz is assigned.

(D) Kate is assigned to the group to which Nelson is assigned.

(E) Kate is assigned to the group to which Olga is assigned.

22. If John and Olga are assigned to different groups from each other, which of the following CANNOT be true?

(A) John is assigned to the group to which Mark is assigned.

(B) Luz is assigned to the group to which Kate is assigned.

(C) Nelson is assigned to the group to which Olga is assigned.

(D) John is assigned to the group consisting of three members.

(E) John and Kate are assigned to different groups from each other.

23. GRAND CITY CENSUS REPORTS

(1950-1980)

<u>Year</u>	<u>Population</u>	<u>Housing Units</u>
1950	500, 000	214, 000
1960	476, 000	218, 000
1970	453, 000	226, 000
1980	425, 000	237, 000

Which of the following, if true, most helps to reconcile the increase in housing units with the decline in population shown in the table above?

- | | |
|---|--|
| <p>(A) The percentage of families that included adult children living at home increased during the 1970's.</p> <p>(B) The number of people moving to Grand City from other cities gradually decreased during the three decades.</p> <p>(C) The number of housing units that were vacant in Grand City fell steadily between 1950 and 1980.</p> <p>(D) The number of adults who lived alone in Grand City housing units increased dramatically between 1950 and 1980.</p> <p>(E) Many housing units that were occupied by only one family in 1950 were occupied by two or more families in 1980.</p> | <p>to have been able to live in deep water.</p> <p>The argument depends on the assumption that</p> <p>(A) no fossils of the coral <i>A. palmata</i> have been found that date from periods when glaciers were not spreading from the polar regions</p> <p>(B) geological disturbances since the last period during which glaciers spread have caused no major downward shift in the location of <i>A. palmata</i> fossils</p> <p>(C) <i>A. palmata</i> now live in shallow waters in most of the same geographical regions as those in which deep-lying <i>A. palmata</i> fossils have been found</p> <p>(D) <i>A. palmata</i> fossils have been found that date from each of the periods during which glaciers are known to have spread from the polar region</p> <p>(E) <i>A. palmata</i> can live at greater depths where the ocean temperature is colder than they can where the ocean temperature is warmer</p> |
|---|--|
24. Fossils of the coral *Acrocora palmata* that date from the last period when glaciers grew and consequently spread from the polar regions are found at ocean depths far greater than those at which *A. palmata* can now survive. Therefore, although the fossilized *A. palmata* appears indistinguishable from *A. palmata* now living, it must have differed in important respects

25. Conservationists have believed that by concentrating their preservation efforts on habitats rich in an easily surveyed group of species, such as birds, they would thereby be preserving areas rich in overall species diversity. This belief rests on a view that a geographical area rich in one group of species will also be rich in the other groups characteristic of the entire regional climate zone.

Which of the following findings about widely scattered tracts 10 kilometers by 10 kilometers in a temperate climate zone would most seriously challenge the conservationists' assumptions?

- (A) The tracts show little damage from human intrusion and from pollution by human activities.
- (B) Where a certain group of species, such as birds, is abundant, there is also an abundance of the species, such as insects, on which that group of species feeds, or in the case of plants, of the land and water resources it requires.
- (C) The area of one of the tracts is generally large enough to contain a representative sample of the organisms in the region.
- (D) There is little overlap between the list of tracts that are rich in species of butterflies and the list of those that are rich in species of birds.
- (E) The highest concentration of individuals of rare species is found where the general diversity of species is greatest.

SECTION 7

Time –30 minutes

38 Questions

1. In the nineteenth century, novelists and unsympathetic travelers portrayed the American West as a land of ---- adversity, whereas promoters and idealists created ---- image of a land of infinite promise.
(A) lurid.. a mundane
(B) incredible.. an underplayed
(C) dispiriting.. an identical
(D) intriguing.. a luxuriant
(E) unremitting.. a compelling
2. Honeybees tend to be more ---- than earth bees: the former, unlike the latter, search for food together and signal their individual findings to one another.
(A) insular
(B) aggressive
(C) differentiated
(D) mobile
(E) social
3. Joe spoke of superfluous and ---- matters with exactly the same degree of intensity, as though for him serious issues mattered neither more nor less than did ----.
(A) vital.. trivialities
(B) redundant.. superficialities
(C) important.. necessities
(D) impractical.. outcomes
(E) humdrum.. essentials
4. The value of Davis' sociological research is compromised by his unscrupulous tendency to use materials---- in order to substantiate his own claims, while ---- information that points to other possible conclusions.
(A) haphazardly.. deploying
(B) selectively.. disregarding
(C) cleverly.. weighing
(D) modestly.. refuting
(E) arbitrarily.. emphasizing
5. Once Renaissance painters discovered how to ---- volume and depth, they were able to replace the medieval convention of symbolic, two-dimensional space with the more ---- illusion of actual space.
(A) reverse.. conventional
(B) portray.. abstract
(C) deny.. concrete
(D) adumbrate.. fragmented
(E) render.. realistic
6. He had expected gratitude for his disclosure, but instead he encountered ---- bordering on hostility.
(A) patience
(B) discretion
(C) openness
(D) ineptitude
(E) indifference
7. The diplomat, selected for her demonstrated patience and skill in conducting such delicate negotiations, ---- to make a decision during the talks because any sudden commitment at that time would have been ----.
(A) resolved.. detrimental
(B) refused.. apropos
(C) declined.. inopportune
(D) struggled.. unconscionable
(E) hesitated.. warranted
8. CONDUCTOR: INSTRUMENTALIST::
(A) director: actor
(B) sculptor: painter
(C) choreographer: composer
(D) virtuoso: amateur
(E) poet: listener
9. QUARRY: ROCK
(A) silt: gravel
(B) sky: rain
(C) cold: ice
(D) mine: ore
(E) jewel: diamond
10. STICKLER: EXACTING::
(A) charlatan: forthright
(B) malcontent: solicitous
(C) misanthrope: expressive
(D) defeatist: resigned
(E) braggart: unassuming

11. WALK: AMBLE::

- (A) dream: imagine
- (B) talk: chat
- (C) swim: float
- (D) look: stare
- (E) speak: whisper

12. JAZZ: MUSIC::

- (A) act: play
- (B) variety: vaudeville
- (C) portraiture: painting
- (D) menu: restaurant
- (E) species: biology

13. REPATRIATE: EMIGRATION::

- (A) reinstate: election
- (B) recall: impeachment
- (C) appropriate: taxation
- (D) repeal: ratification
- (E) appeal: adjudication

14. PLACEBO: INNOCUOUS::

- (A) antibiotic: viral
- (B) vapor: opaque
- (C) salve: unctuous
- (D) anesthetic: astringent
- (E) vitamin: synthetic

15. DISSEMINATE: INFORMATION::

- (A) amend: testimony
- (B) analyze: evidence
- (C) investigate: crime
- (D) prevaricate: confirmation
- (E) foment: discontentment

16. VOICE: QUAVER::

- (A) pace: quicken
- (B) cheeks: dimple
- (C) concentration: focus
- (D) hand: tremble
- (E) eye: blink

Mary Barton, particularly in its early chapters, is a moving response to the suffering of the industrial worker in the England of the 1840's. What is most impressive about the book is the intense and painstaking effort made

(5) by the author, Elizabeth Gaskell, to convey the experience of everyday life in working-class homes. Her method is partly documentary in nature: the novel includes such features as a carefully annotated reproduction of dialect, the exact details of food prices in an account of a tea party, an itemized description of the furniture of the Bartons' living room, and a transcription (again annotated) of the ballad "The Oldham Weaver." The interest of this record is considerable, even though the method has a slightly distancing effect.

(15) As a member of the middle class, Gaskell could hardly help approaching working-class life as an outside observer and a reporter, and the reader of the novel is always conscious of this fact. But there is genuine imaginative re-creation in her accounts of the walk in Green Heys Fields, of tea at the Bartons' house, and of John Barton and his friend's discovery of the starving family in the cellar in the chapter "Poverty and Death." Indeed, for a similarly convincing re-creation of such families' emotions and responses (which are more crucial than the material details on which the mere reporter is apt to concentrate), the English novel had to wait 60 years for the early writing of D. H. Lawrence. If Gaskell never quite conveys the sense of full participation that would completely authenticate this aspect of *Mary Barton*, she (30) still brings to these scenes an intuitive recognition of feelings that has its own sufficient conviction.

The chapter "Old Alice's History " brilliantly dramatizes the situation of that early generation of workers brought from the villages and the countryside to the (35) urban industrial centers. The account of Job Legh, the weaver and naturalist who is devoted to the study of biology, vividly embodies one kind of response to an urban industrial environment: an affinity for living things that hardens, by its very contrast with its environment, into a kind of crankiness. The early chapters? (40) about factory workers walking out in spring into Green Heys Fields; about Alice Wilson, remembering in her cellar the twig-gathering for brooms in the native village that she will never again see; about Job Legh, intent on (45) his impaled insects? capture the characteristic responses of a generation to the new and crushing experience of industrialism. The other early chapters eloquently portray the development of the instinctive cooperation with each other that was already becoming an important tradition among workers.

17. Which of the following best describes the author's attitude toward Gaskell's use of the method of documentary record in *Mary Barton*?
- (A) Uncritical enthusiasm
(B) Unresolved ambivalence
(C) Qualified approval
(D) Resigned acceptance
(E) Mild irritation
18. According to the passage, *Mary Barton* and the early novels of D. H. Lawrence share which of the following?
- (A) Depiction of the feelings of working-class families
(B) Documentary objectivity about working-class circumstances
(C) Richly detailed description of working-class adjustment to urban life
(D) Imaginatively structured plots about working-class characters
(E) Experimental prose style based on working-class dialect
19. Which of the following is most closely analogous to Job Legh in *Mary Barton*, as that character is described in the passage?
- (A) An entomologist who collected butterflies as a child
(B) A small-town attorney whose hobby is nature photography
(C) A young man who leaves his family's dairy farm to start his own business
(D) A city dweller who raises exotic plants on the roof of his apartment building
(E) A union organizer who works in a textile mill under dangerous conditions
20. It can be inferred from examples given in the last paragraph of the passage that which of the following was part of "the new and crushing experience of industrialism" (lines 46-47) for many members of the English working class in the nineteenth century?
- (A) Extortionate food prices
(B) Geographical displacement
(C) Hazardous working conditions
(D) Alienation from fellow workers
(E) Dissolution of family ties
21. It can be inferred that the author of the passage believes that *Mary Barton* might have been an even better novel if Gaskell had
- (A) concentrated on the emotions of a single character
(B) made no attempt to re-create experiences of which she had no firsthand knowledge
(C) made no attempt to reproduce working-class dialects
(D) grown up in an industrial city
(E) managed to transcend her position as an outsider
22. Which of the following phrases could best be substituted for the phrase "this aspect of *Mary Barton*" in line 29 without changing the meaning of the passage as a whole?
- (A) the material details in an urban working-class environment
(B) the influence of *Mary Barton* on Lawrence's early work
(C) the place of *Mary Barton* in the development of the English novel
(D) the extent of the poverty and physical suffering among England's industrial workers in the 1840's.
(E) the portrayal of the particular feelings and responses of working-class characters
23. The author of the passage describes *Mary Barton* as each of the following EXCEPT
- (A) insightful
(B) meticulous
(C) vivid
(D) poignant
(E) lyrical
- As of the late 1980's, neither theorists nor large-scale computer climate models could accurately predict whether cloud systems would help or hurt a warming globe. Some studies suggested that a four percent increase in stratocumulus clouds over the ocean could compensate for a doubling in atmospheric carbon dioxide, preventing a potentially disastrous planetwide temperature increase. On the other hand, an increase in cirrus clouds could increase global warming.
- (10) That clouds represented the weakest element in cli-

mate models was illustrated by a study of fourteen such models. Comparing climate forecasts for a world with double the current amount of carbon dioxide, researchers found that the models agreed quite well if clouds were (15)not included. But when clouds were incorporated, a wide range of forecasts was produced. With such discrepancies plaguing the models, scientists could not easily predict how quickly the world's climate would change, nor could they tell which regions would face dustier droughts or deadlier monsoons. 24. The author of the passage is primarily concerned with (A) confirming a theory (B) supporting a statement (C) presenting new information (D) predicting future discoveries (E) reconciling discrepant findings 25. It can be inferred that one reason the fourteen models described in the passage failed to agree was that (A) they failed to incorporate the most up-to-date information about the effect of clouds on climate (B) they were based on faulty information about factors other than clouds that affect climate. (C) they were based on different assumptions about the overall effects of clouds on climate (D) their originators disagreed about the kinds of forecasts the models should provide (E) their originators disagreed about the factors other than clouds that should be included in the models 26. It can be inferred that the primary purpose of the models included in the study discussed in the second paragraph of the passage was to (A) predict future changes in the world's climate (B) predict the effects of cloud systems on the world's climate (C) find a way to prevent a disastrous planetwide temperature increase (D) assess the percentage of the Earth's surface covered by cloud systems (E) estimate by how much the amount of carbon	dioxide in the Earth's atmosphere will increase 27. The information in the passage suggests that scientists would have to answer which of the following questions in order to predict the effect of clouds on the warming of the globe? (A) What kinds of cloud systems will form over the Earth? (B) How can cloud systems be encouraged to form over the ocean? (C) What are the causes of the projected planetwide temperature increase? (D) What proportion of cloud systems are currently composed of cirrus of clouds? (E) What proportion of the clouds in the atmosphere form over land masses? 28. SUSPEND: (A) force (B) split (C) tilt (D) slide down (E) let fall 29. CREDULITY: (A) originality (B) skepticism (C) diligence (D) animation (E) stoicism 30. MILD: (A) toxic (B) uniform (C) maximal (D) asymptomatic (E) acute 31. IMPLEMENT: (A) distort (B) foil (C) overlook (D) aggravate (E) misinterpret
--	--

32. DIFFIDENCE::

- (A) trustworthiness
- (B) assertiveness
- (C) lack of preparation
- (D) resistance to change
- (E) willingness to blame

(D) ambiguous

(E) extremist

33. BYZANTINE:

- (A) symmetrical
- (B) variegated
- (C) discordant
- (D) straightforward
- (E) unblemished

34. PROCLIVITY:

- (A) confusion
- (B) deprivation
- (C) obstruction
- (D) aversion
- (E) hardship

35. PROTRACT:

- (A) treat fairly
- (B) request hesitantly
- (C) take back
- (D) cut short
- (E) make accurate

36. VAUNTING:

- (A) plucky
- (B) meek
- (C) chaste
- (D) cowardly
- (E) ardent

37. HALE:

- (A) unenthusiastic
- (B) staid
- (C) odious
- (D) infirm
- (E) uncharacteristic

38. SEMINAL:

- (A) derivative
- (B) substantiated
- (C) reductive

SECTION 1

Time-30minutes

38 Questions

1. There is hardly a generalization that can be made about people's social behavior and the values informing it that cannot be -----from one or another point of view, or even -----as simplistic or vapid.
 - (A) accepted...praised
 - (B) intuited...exposed
 - (C) harangued...retracted
 - (D) defended...glorified
 - (E) challenged...dismissed
2. Although any destruction of vitamins caused by food irradiation could be ----- the use of diet supplements, there may be no protection from carcinogens that some fear might be introduced into foods by the process.
 - (A) counterbalanced by
 - (B) attributed to
 - (C) inferred from
 - (D) augmented with
 - (E) stimulated by
3. Though he refused any responsibility for the failure of the negotiations, Stevenson had no right to -----himself: it was his -----that had caused the debacle.
 - (A) blame... skill
 - (B) congratulate...modesty
 - (C) berate...largesse
 - (D) accuse...obstinacy
 - (E) absolve...acrimony
4. The prevailing union of passionate interest in detailed facts with equal devotion to abstract -----is a hallmark of our present society; in the past this union appeared, at best, -----and as if by chance.
 - (A) data...extensively
 - (B) philosophy...cyclically
 - (C) generalization...sporadically
 - (D) evaluation...opportunistically
 - (E) intuition....selectively
5. A century ago the physician's word was ----- to doubt it was considered almost sacrilegious
 - (A) inevitable
 - (B) intractable
 - (C) incontrovertible
 - (D) objective
 - (E) respectable
6. So much of modern fiction in the United States is autobiographical, and so much of the autobiography fictionalized, that the -----sometimes seem largely-----.
 - (A) authors...ignored
 - (B) needs...unrecognized
 - (C) genres...interchangeable
 - (D) intentions...misunderstood
 - (E) misapprehensions...uncorrected
7. Robin's words were not without emotion: they retained their level tone only by a careful ----- imminent extremes.
 - (A) equipoise between
 - (B) embrace of
 - (C) oscillation between
 - (D) limitation to
 - (E) Subjection to
8. OIL : LUBRICATE::
 - (A) preservative : desiccate
 - (B) wine : ferment
 - (C) honey : pollinate
 - (D) antiseptic : disinfect
 - (E) soil : fertilize
9. CONSTRUCT : REMODEL::
 - (A) exhibit : perform
 - (B) compose : edit
 - (C) demolish : repair
 - (D) quantify : estimate
 - (E) predict : assess
10. SPOKE : HUB::
 - (A) radius : center
 - (B) parabola : equation
 - (C) line : point

(D) vector : direction

(E) slope : change

11. ILLUSTRATE : PICTURES::

(A) particularize : details

(B) abridge : texts

(C) parse : sentences

(D) regularize : inconsistencies

(E) economize : words

12. PANTRY : FOOD::

(A) museum : replicas

(B) ship : cargo

(C) office : business

(D) armory : weapons

(E) warehouse : storage

13. MIRTH : LAUGHTER::

(A) uncertainty : nod

(B) approval : applause

(C) danger : alarm

(D) labor : sweat

(E) love : respect

14. ABRADED : FRICTION::

(A) refined : combustion

(B) attenuated : coagulation

(C) diluted : immersion

(D) strengthened : compression

(E) desiccated : dehydration

15. PARSIMONY : MISER::

(A) temerity : despot

(B) belligerence : traitor

(C) remorse : delinquent

(D) equanimity : guardian

(E) rebelliousness : insurgent

16. NTTPICK : CRITICIZE::

(A) mock : imitate

(B) complain : argue

(C) interrogate : probe

(D) fret : vex

(E) cavil : object

Directions: Each passage in this group is followed by questions based on its content. After reading a passage, choose the best answer to each question. Answer all questions following a passage on the basis of what is stated or implied in that passage.

(This passage is from a book published in 1960.)

When we consider great painters of the past, the study of art and the study of illusion cannot always be separated. By illusion I mean those contrivances of *line* color, line, shape, and so forth that lead us to see (5) marks on a flat surface as depicting three-dimensional objects in space. I must emphasize that I am not making a plea, disguised or otherwise, for the exercise of illusionist tricks in painting today, although I am, in fact, rather critical of certain theories of non- (10) representational art. But to argue over these theories would be to miss the point. That the discoveries and effects of representation that were the pride of earlier artists have become trivial today I would not deny for a moment. Yet I believe that we are in real danger of (15) losing contact with past masters if we accept the fashionable doctrine that such matters never had anything to do with art. The very reason why the representation of nature can now be considered something commonplace should be of the greatest (20) interest to art historians. Never before has there been an age when the visual image was so cheap in every sense of the word. We are surrounded and assailed by posters and advertisements, comics and magazine illustrations. We see aspects of reality represented (25) on television, postage stamps, and food packages. Painting is taught in school and practiced as a pastime, and many modest amateurs have mastered tricks that would have looked like sheer magic to the fourteenth-century painter Giotto. Even the crude (30) colored renderings on a cereal box might have made Giotto's contemporaries gasp. Perhaps there are people who conclude from this that the cereal box is superior to a Giotto; I do not. But I think- that the victory and vulgarization of representational skills (35) create a problem for both art historians and critics.

In this connection it is instructive to remember the Greek saying that to marvel is the beginning of knowledge and if we cease to marvel we may be in

- danger of ceasing to know. I believe we must restore
- (40) our sense of wonder at the capacity to conjure up by forms, lines, shades, or colors those mysterious phantoms of visual reality we call "pictures." Even comics and advertisements, rightly viewed, provide food for thought. Just as the study of poetry remains
- (45) incomplete without an awareness of the language of prose, so, I believe, the study of art will be increasingly supplemented by inquiry into the "linguistics" of the visual image. The way the language of art refers to the visible world is both so obvious and so mysterious
- (50) that it is still largely unknown except to artists who use it as we use all language – without needing to know its grammar and semantics.
17. The author of the passage explicitly, disagrees with which of the following statements'
- (A) In modern society even nonartists can master techniques that great artists of the fourteenth century did not employ.
- (B) The ability to represent a three-dimensional object on a flat surface has nothing to do with art.
- (C) In modern society the victory of representational skills has created a problem for art critics.
- (D) The way that artists are able to represent the visible world is an area that needs a great deal more study before it can be fully understood.
- (E) Modern painters do not frequently make use of illusionist tricks in their work.
18. The author suggests which of the following about art historians?
- (A) They do not believe that illusionist tricks have become trivial.
- (B) They generally spend little time studying contemporary artists.
- (C) They have not given enough consideration to how the representation of nature has become commonplace.
- (D) They generally tend to argue about theories rather than address substantive issues.
- (E) They are less likely than art critics to study comics or advertisements.
19. Which of the following best states the author's attitude toward comics, as expressed in the passage?
- (A) They constitute an innovative art form.
- (B) They can be a worthwhile subject for study.
- (C) They are critically important to an understanding of modern art.
- (D) Their -visual structure is more complex than that of medieval art.
- (E) They can be understood best if they are examined in conjunction with advertisements.
20. The author's statement regarding how artists use the language of art (lines 48-52) implies that
- (A) artists are better equipped than are art historians to provide detailed evaluations of other artists' work
- (B) many artists have an unusually quick, intuitive understanding of language
- (C) artists can produce works of art even if they cannot analyze their methods of doing so
- (D) artists of the past, such as Giotto, were better educated about artistic issues than were artists of the author's time
- (E) most artists probably consider the processes involved in their work to be closely akin to those involved in writing poetry
21. The passage asserts which of the following about commercial art?
- (A) There are many examples of commercial art whose artistic merit is equal to that of great works of art of the past.
- (B) Commercial art is heavily influenced by whatever doctrines are fashionable in the serious art world of the time.
- (C) The line between commercial art and great art lies primarily in how an image is used, not in the motivation for its creation.
- (D) The level of technical skill required to produce representational imagery in commercial art and in other kinds of art cannot be compared.
- (E) The pervasiveness of contemporary commercial art has led art historians to undervalue representational skills.

22. Which of the following can be inferred from the passage, about the adherents of "certain theories of nonrepresentational art" (lines 9-10) ?

- (A) They consider the use of illusion to be inappropriate in contemporary art.
- (B) They do not agree that marks on a flat surface can ever satisfactorily convey the illusion of three-dimensional space.
- (C) They do not discuss important works of art created in the past.
- (D) They do not think that the representation of nature was ever the primary goal of past painters.
- (E) They concern themselves more with types art such as advertisements and magazine illustrations than with traditional art.

23. It can be inferred from the passage that someone who wanted to analyze the "grammar and semantics" (line 52) of the language of art would most appropriately comment on which of the following?

- (A) The relationship between the drawings in a comic strip and the accompanying text
- (B) The amount of detail that can be included in a tiny illustration on a postage stamp
- (C) The sociological implications of the images chosen to advertise a particular product
- (D) The degree to which various colors used in different versions of the same poster would attract the attention of passersby
- (E) The particular juxtaposition of shapes in an illustration that makes one shape look as though it were behind another

The 1973 Endangered Species Act made into legal policy the concept that endangered species of wildlife are precious as part of a natural ecosystem. The nearly unanimous passage of this act in the United States

- (5) Congress, reflecting the rising national popularity of environmentalism, masked a bitter debate. Affected industries clung to the former wildlife policy of valuing individual species according to their economic usefulness. They fought to minimize the law's impact (10) by limiting definitions of key terms, but they lost on

nearly every issue. The act defined "wildlife" as almost all kinds of animals—from large mammals to invertebrates—and plants. "Taking" wildlife was defined broadly as any action that threatened an

- (15) endangered species; areas vital to a species' survival could be federally protected as "critical habitats" Though these definitions legislated strong environmentalist goals, political compromises made in the enforcement of the act were to determine just what (20) economic interests would be set aside for the sake of ecological stabilization.

24. According to the passage, which of the following does the Endangered Species Act define as a "critical habitat"?

- (A) A natural ecosystem that is threatened by imminent development
- (B) An industrial or urban area in which wildlife species have almost ceased to live among humans
- (C) A natural area that is crucial to the survival of a species and thus eligible for federal protection
- (D) A wilderness area in which the "taking" of wildlife species is permitted rarely and only under strict federal regulation
- (E) A natural environment that is protected under law because its wildlife has a high economic value

25. According to the passage, which of the following is an explanation for the degree of support that the Endangered Species Act received in Congress?

- (A) Concern for the environment had gained increasing national popularity.
- (B) Ecological research had created new economic opportunities dependent on the survival of certain species.
- (C) Congress had long wanted to change the existing wildlife policy.
- (D) The growth of industry had endangered increasing numbers of wildlife species.
- (E) Legislators did not anticipate that the act could be effectively enforced.

26. It can be inferred from the passage that if business interests had won the debate on provisions of the 1973 Endangered Species Act, which of the following would have resulted?

- (A) Environmentalist concepts would not have become widely popular.
- (B) The definitions of key terms of the act would have been more restricted.
- (C) Enforcement of the act would have been more difficult.
- (D) The act would have had stronger support from Congressional leaders.
- (E) The public would have boycotted the industries that had the greatest impact in defining the act.

27. The author refers to the terms "wildlife" (line 11), "taking" (line 13), and "critical habitats" (line 16) most likely in order to

- (A) illustrate the misuse of scientific language and concepts in political processes
- (B) emphasize the importance of selecting precise language in transforming scientific concepts into law
- (C) represent terminology whose definition was crucial in writing environmentalist goals into law
- (D) demonstrate the triviality of the issues debated by industries before Congress passed the Endangered Species Act
- (E) show that broad definitions of key terms in many types of laws resulted in ambiguity and thus left room for disagreement about how the law should be enforced

Since some of the questions require you to distinguish fine shades of meaning, be sure to consider all the choices before deciding which one is best.

28. SWERVE:

- (A) maintain direction
- (B) resume operation
- (C) slow down
- (D) divert
- (E) orient

29. HUSBAND:

- (A) rearrange
- (B) alarm
- (C) assist
- (D) prize
- (E) squander

30. DEACTIVATE:

- (A) palpate
- (B) alleviate
- (C) inhale
- (D) articulate
- (E) potentiate

31. INTRANSIGENT:

- (A) accustomed to command
- (B) qualified to arbitrate
- (C) open to compromise
- (D) resigned to conflict
- (E) opposed to violence

32. OCCLUDED:

- (A) unvaried
- (B) entire
- (C) functional
- (D) inverted
- (E) unobstructed

33. ASSUAGE:

- (A) intensify
- (B) accuse
- (C) correct
- (D) create
- (E) assert

34. QUIXOTIC:

- (A) displaying consistently practical behavior
- (B) considering several points of view
- (C) expressing dissatisfaction
- (D) suggesting uneasiness
- (E) acting decisively

35. PELLUCID:

- (A) stagnant

- (B) murky
(C) glutinous
(D) noxious
(E) rancid

36. LACONISM:

- (A) temerity
(B) vacuity
(C) dishonesty
(D) immaturity
(E) verbosity

37. REFRACTORY:

- (A) active
(B) productive
(C) energetic
(D) responsive
(E) powerful

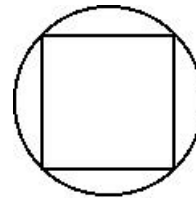
38. DEFINITIVE:

- (A) prosaic
(B) convoluted
(C) unusual
(D) provisional
(E) vast

SECTION 2

Time –30 minutes

30 Questions



The square is inscribed in the circle.

- | | |
|---|--|
| 1. The length of a
diameter of the
circle | The length of a
diagonal of the
square |
|---|--|
-

$$\frac{3}{4}k = \frac{1}{8}$$

- | | | |
|----|-----|---------------|
| 2. | k | $\frac{1}{6}$ |
|----|-----|---------------|
-

- | | | |
|----|-----------|---|
| | $x-y=y-x$ | |
| 3. | $x-y$ | 0 |
-

In a certain store, each record costs X dollars and each tape costs Y dollars. the total cost of 3 records and 2 tapes is \$39.

- | | | |
|----|----------------------|--------------------|
| 4. | The cost of 1 record | The cost of 1 tape |
|----|----------------------|--------------------|
-

The perimeter of rectangle $RSTU$ is 750, and $RS=350$.

- | | | |
|----|------|----|
| 5. | ST | 50 |
|----|------|----|
-

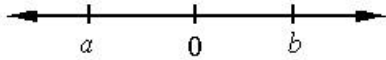
- | | | |
|----|-----------------|-----|
| 6. | $\frac{1}{2.5}$ | 0.4 |
|----|-----------------|-----|
-



- | | | |
|----|-----|-----|
| 7. | x | 140 |
|----|-----|-----|
-

$$S = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \frac{1}{7} - \frac{1}{8} + \frac{1}{9} - \frac{1}{10}$$

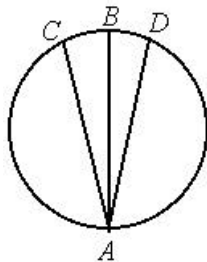
8. S $\frac{1}{2}$



9. a $-b$

$$2 - \frac{1}{x} > 2$$

10. x 1



AB is a diameter of the circle.

11. The length of AB The average (arithmetic mean) of the lengths of AC and AD

$$0 < x < y < 1$$

12. $1 - y$ $y - x$

13. $\frac{1}{\sqrt{5}} \left[\left(\frac{1+\sqrt{5}}{2} \right) - \left(\frac{1-\sqrt{5}}{2} \right) \right]$ 1

$$y = x + \frac{1}{x}$$

$$0 < x < 10$$

14. The value of y 100

At a sale, the cost of each tie was reduced by 20 percent and the cost of each belt was reduced by 30 percent.

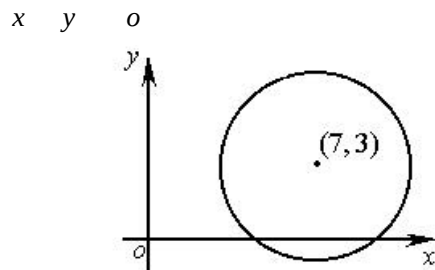
15. The percent reduction on the total cost of 1 tie and 2 belts 25%

16. $\frac{18}{60}(0.1254) =$

- (A) 0.00522
(B) 0.03135
(C) 0.03762
(D) 0.0418
(E) 0.0627

17. What percent of the integers between 100 and 999, inclusive, have all three digits the same?

- (A) 1%
(B) 2%
(C) 3%
(D) 4%
(E) 5%



18. If $(7, 3)$ is the center of the circle above, then the radius of the circle could be equal to which of the following?

- (A) 2
(B) 3
(C) 5
(D) 7
(E) 9

19. If revenues \$196,000 from division A of Company X represent 28 percent of the total revenues of Company X for the year, What were the total revenues of Company X for the year?

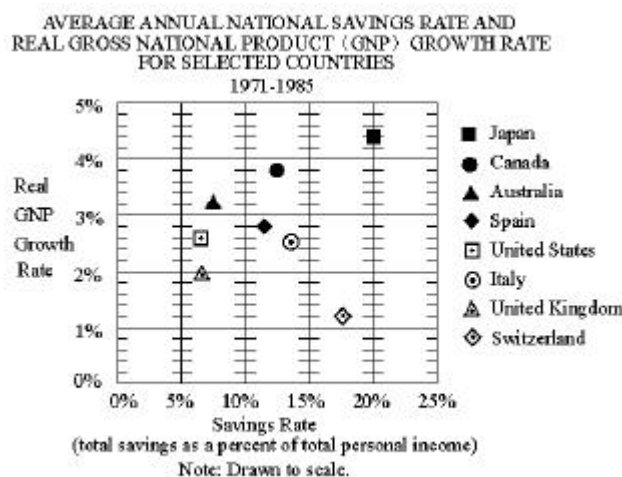
- (A) \$141,100
(B) \$272,000
(C) \$413,300
(D) \$596,100
(E) \$700,000

20. If $xy \neq 0$, which of the following is equivalent to

$$\left(\frac{x}{y}\right)^3 \left(\frac{2y}{x}\right)^4 ?$$

- (A) $2xy$
 (B) $8xy^2$
 (C) $16x^2y^3$
 (D) $\frac{2y}{x}$
 (E) $\frac{16y}{x}$

Questions 21-25 refer to the following graph.



21. What was the savings rate for the country that had the greatest real GNP growth rate?

- (A) 25%
 (B) 20%
 (C) 18%
 (D) 12.5%
 (E) 4.5%

22. For which country was the ratio of its savings rate to its real GNP growth rate greatest?

- (A) Japan
 (B) Canada
 (C) Australia
 (D) Italy
 (E) Switzerland

23. The savings rate for Canada was approximately how many times that of the United States?

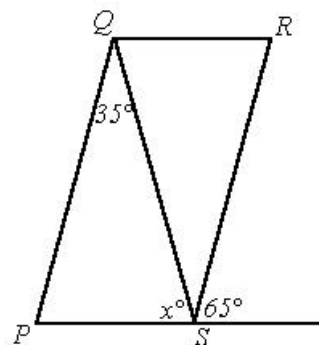
- (A) $1\frac{1}{2}$
 (B) 2
 (C) $2\frac{1}{2}$
 (D) 3
 (E) $3\frac{1}{2}$

24. For how many of the countries shown was the savings rate more than 5 times the real GNP growth rate?

- (A) Five
 (B) Four
 (C) Three
 (D) Two
 (E) One

25. Which of the following statement can be inferred from the graph?

- I. On the average, people in the United States saved about the same amount as people in the United Kingdom.
 II. The median of the savings rates for the eight countries was greater than 11 percent
 III. Only two of the countries had a higher savings rate than Italy.
- (A) I only
 (B) II only
 (C) III only
 (D) I and II
 (E) II and III



26. In the figure above, if $PQRS$ is a parallelogram, then $x =$

- (A) 35
- (B) 65
- (C) 75
- (D) 80
- (E) 100

27. A certain doctor suggests that an individual's daily water intake be $\frac{1}{2}$ ounce per pound of body weight plus 8 ounces for every 25 pounds by which the individual exceeds his or her ideal weight. If this doctor suggests a daily water intake of 136 ounces for a particular 240-pound individual, how many pounds above his or her ideal weight is that individual?

- (A) $12\frac{1}{2}$
- (B) 16
- (C) 30
- (D) 50
- (E) 120

28. A political poll showed that 80 percent of those polled said they would vote for proposition P . Of those who said they would vote for proposition P 70 percent actually voted for P , and of those who did not say they would vote for P , 20 percent actually voted for P . What percent of those polled voted for P ?

- (A) 56%
- (B) 60%
- (C) 64%
- (D) 76%
- (E) 90%

29. If $x \neq 1$ and $x \neq 0$, then $\frac{1 - \frac{1}{x}}{x - 1}$ is equivalent to

- (A) $\frac{1}{x}$
- (B) x

(C) $\frac{x}{1 - x}$

(D) $\frac{x - 1}{x}$

(E) $\frac{(x - 1)^2}{x}$

30. In a group of 80 students, 24 are enrolled in geometry, 40 in biology, and 20 in both. If a student were randomly selected from the 80 students, what is the probability that the student selected would not be enrolled in either course?

- (A) 0.20
- (B) 0.25
- (C) 0.45
- (D) 0.55
- (E) 0.60

SECTION 3

Time – 30 Minutes

25 questions

Questions 1-7

Seven flags will be flown on seven poles, one flag per pole. The poles are arranged in a row and numbered consecutively 1 through 7. Three flags are green, two are white, and two are yellow. The arrangement of flags will conform to the following conditions:

No two green flags can be flown on poles that are next to each other.

None of the green flags can be flown on pole 3.

Neither yellow flag can be flown on pole 5.

1. If the two yellow flags are flown on poles that are next to each other and the two white flags are flown on poles that are next to each other, then the flags flown on poles 2 and 6 must be
 - (A) both green
 - (B) green and white, respectively
 - (C) White and green, respectively
 - (D) yellow and green, respectively
 - (E) yellow and white, respectively
2. If the white flags are flown on poles 2 and 3, which of the following must be true?
 - (A) A green flag is flown on pole 4.
 - (B) A green flag is flown on pole 6.
 - (C) A green flag is flown on pole 7.
 - (D) A yellow flag is flown on pole 1.
 - (E) A yellow flag is flown on pole 7.
3. Which of the following, CANNOT be true?
 - (A) Green flags are flown on poles 1 and 4.
 - (B) White flags are flown on poles 1 and 7.
 - (C) Yellow flags are flown on poles 1 and 7.
 - (D) A green flag is flown on pole 1 and a yellow flag is flown on pole 7.
 - (E) a white flag is flown on pole 1 and a yellow flag is flown on pole 7.
4. If a green flag is flown on pole 5, which of the following must be true?
 - (A) a green flag is flown on pole 1.
 - (B) a green flag is flown on pole 2.
 - (C) A green flag is flown on pole 7.
 - (D) a white flag is flown on pole 3.
 - (E) A white flag is flown on pole 4.
5. If the yellow flags are flown on poles 1 and 6, which of the following must be true?
 - (A) A green flag is flown on pole 4.
 - (B) A green flag is flown on pole 5.
 - (C) The white flags are flown on poles that are next to each other.
 - (D) A white flag and a yellow flag are flown on poles that are next to each other.
 - (E) Each white flag is flown on a pole that is next to a pole on which a green flag is flown.
6. If green flags are flown on poles 2 and 6, which of the following can be true?
 - (A) A green flag is flown on pole 1.
 - (B) A green flag is flown on pole 5.
 - (C) A white flag is flown on pole 3.
 - (D) A white flag is flown on pole 4.
 - (E) A yellow flag is flown on pole 4.
7. If the yellow flags are flown on poles 2 and 4, which of the following must be true?
 - (A) The white flags are flown on poles that are next to each other.
 - (B) a green flag is flown on pole that is next to two poles on which white flags are flown.
 - (C) A white flag is flown on a pole that is next to a pole on which a green flag is flown, and also next to a pole on which a yellow flag is flown.
 - (D) Each green flag is flown on a pole that is next to a pole on which a white flag is flown.
 - (E) Each yellow flag is flown on a pole that is next to a pole on which a green flag is flown.
8. At the Shadybrook dog kennel, all the adult animals were given a new medication designed to reduce a dog's risk of contracting a certain common infection. Several days after the medication was administered, most of the puppies of these dogs had elevated temperatures. Since raised body temperature is a side effect of this medication, the kennel owner hypothesized that the puppies'
 - (A) a green flag is flown on pole 1.
 - (B) a green flag is flown on pole 2.
 - (C) A green flag is flown on pole 7.
 - (D) a white flag is flown on pole 3.
 - (E) A white flag is flown on pole 4.

elevated temperatures resulted from the medication's being passed to them through their mothers' milk.

Which of the following, if true, provides the most support for the kennel owner's hypothesis?

- (A) Some puppies have been given the new medication directly but have not suffered elevated temperatures as a side effect.
- (B) The new medication has been well received by dog breeders as a safe and effective way of preventing the spread of certain common canine infections.
- (C) None of the four puppies in the kennel who had been bottle-fed with formula had elevated temperatures.
- (D) an elevated temperature is a side effect of a number of medications for dogs other than the new medication administered at the kennel.
- (E) Elevated temperatures such as those suffered by most of the puppies in the kennel rarely have serious long-term effects on a puppy's health.

9. Which of the following most logically completes the argument?

Alivia's government has approved funds for an electricity-generation project based on the construction of a pipeline that will carry water from Lake Cylus, in the mountains, to the much smaller Lake Tifele, in a nearby valley. The amount of electricity generated will be insufficient by itself to justify the project's cost, even if the price of imported oil—Alivia's primary source of electricity—increases sharply. Nonetheless, the pipeline project is worth its cost, because —

- (A) the price of oil, once subject to frequent sharp increases, has fallen significantly and is now fairly stable
- (B) the project could restore Lake Tifele, which is currently at risk of drying up and thus of being lost as a source of recreation income for Alivia
- (C) the government of Alivia is currently on excellent terms with the governments of most of the countries from which it purchases oil
- (D) it would cost less to generate electricity by moving water from Lake Cylus to lake Tifele

than to do so by moving water from Lake Cylus to another valley lake

- (E) Alivian officials do not expect that the amount of electricity used in Alivia will increase substantially within the next ten years

10. Amusement rides at permanent fairgrounds are dismantled once a year for safety inspections by independent consultants. Traveling fairs, which relocate each month, can slip past the net of safety inspections and escape independent inspection for several years. Therefore, the rides at traveling fairs are less safe than the rides at permanent fairs. Which of the following, if true about traveling fairs, most seriously weakens the argument?

- (A) Before each relocation, the operators dismantle their rides, observing and repairing potential sources of danger, such as worn ball bearings.
- (B) their managers have less capital to spend on the safety and upkeep of the rides than do managers of permanent fairs.
- (C) Since they can travel to new customers, they rely less on keeping up a good reputation for safety.
- (D) While they are traveling, the fairs do not receive notices of equipment recalls sent out by the manufacturers of their rides.
- (E) The operators of the rides often do not pay careful attention to the instructions for operating their rides.

Questions 11-15

A candidate for mayor will visit six institutions—a factory, a hospital, a mall, a police station, a shelter, and a university — on six consecutive days. On each day, the candidate will visit exactly one of the institutions. The visits will conform to the following restrictions:

The visit to the factory must occur on some day before the visit to the university.

The visit to the hospital must occur on the second day after the visit to the university.

Visits to exactly two of the institutions must occur between the visit to the police station and the visit to the shelter, whether the visit to the police station

occurs before or after the visit to the shelter.

11. Which of the following is an acceptable order in which the institutions can be visited?

(A) Factory, shelter, university, hospital, police station, mall
 (B) Factory, university, shelter, hospital, police station, mall
 (C) Shelter, factory, university, mall, police station, hospital
 (D) Shelter, factory, university, police station, hospital, mall.
 (E) University, shelter, hospital, factory, police station, mall.

12. If the university is visited on the second day, then the mall must be visited on the

(A) first day
 (B) third day
 (C) fourth day
 (D) fifth day
 (E) sixth day

13. If the factory is not visited on the day immediately preceding the visit to the university, then the police station can be visited on the

(A) first day
 (B) second day
 (C) third day
 (D) fourth day
 (E) sixth day

14. If the mall is visited either on the day immediately preceding or on the day immediately following the visit to the shelter, which of the following must be true?

(A) The factory is visited on the day immediately preceding the visit to the police station.
 (B) The factory is visited on the day immediately preceding the visit to the shelter.
 (C) The factory is visited on the day immediately preceding the visit to the university.
 (D) The hospital is visited on the day immediately preceding the visit to the shelter.
 (E) The university is visited on the day

immediately preceding the visit to the police station.

15. If the hospital is visited on the day immediately preceding the visit to the mall, which of the following can be true?

(A) The factory is visited on the second day.
 (B) The factory is visited on the third day.
 (C) The mall is visited on the third day.
 (D) the mall is visited on the fourth day.
 (E) The university is visited on the fourth day.

Questions 16-22

The organizer of a publishing conference is scheduling workshops to be given on three days-Wednesday, Thursday, and Friday. On each day, three day-long workshops-Graphics, Marketing, and Research-will be given in three rooms-room I through room 3-subject to the following constraints:

On each day, there is one workshop given per room.
 Over the course of the three days, each workshop is given in each of the three rooms.

16. Which of the following can be the workshops given in room 1 and room 2 on each of the three days?

	<u>Wednesday</u>	<u>Thursday</u>	<u>Friday</u>
(A)	1:Graphics 2:Research	Marketing Graphics	Graphics Marketing
(B)	1:Graphics 2:Marketing	Research Marketing	Marketing Research
(C)	1:Marketing 2:Graphics	Graphics Research	Research Marketing
(D)	1:Marketing 2:Research	Research Marketing	Graphics Graphics
(E)	1:Research 2:Graphics	Marketing Research	Research Marketing

17. If Graphics is scheduled for room I on Wednesday and Marketing is scheduled for room 2 on Thursday, then which of the following must be scheduled for Friday?

(A) Graphics in room 3
 (B) Marketing in room 1

- (C) Marketing in room 3
(D) Research in room 1
(E) Research in room 2
18. If Graphics is scheduled for room 1 on Wednesday and Marketing is scheduled for room 3 on Friday, then which of the following must be true?
- (A) Graphics is scheduled for room 2 on Thursday.
(B) Marketing is scheduled for room 2 on Thursday.
(C) Marketing is scheduled for room 3 on Wednesday.
(D) Research is scheduled for room 2 on Wednesday.
(E) Research is scheduled for room 2 on Thursday.
19. If Marketing is scheduled for room 3 on Thursday and Research is scheduled for room 2 on Friday, then which of the following must be true?
- (A) Graphics is scheduled for room 2 on Wednesday.
(B) Graphics is scheduled for room 1 on Thursday.
(C) Marketing is scheduled for room 1 on Wednesday.
(D) Marketing is scheduled for room 2 on Wednesday.
(E) Research is scheduled for room 2 on Thursday.
20. If Research is scheduled for room 1 on Wednesday and Graphics is scheduled for room 1 on Thursday, then which of the following can be true?
- (A) Graphics is scheduled for room 2 on Wednesday and Marketing is scheduled for room 3 on Thursday.
(B) Graphics is scheduled for room 2 on Thursday and room 1 on Friday.
(C) Marketing is scheduled for room 3 on Thursday and room 2 on Friday.
(D) Research is scheduled for room 3 on Thursday and Marketing is scheduled for room 1 on Friday.
(E) Research is scheduled for room 3 on Thursday and Marketing is scheduled for room 2 on Friday.
21. If Marketing is to be scheduled for room 1 on Wednesday, then the schedule of workshops would be completely determined if which of the following were true?
- (A) Graphics is scheduled for room 1 on Thursday.
(B) Graphics is scheduled for room 2 on Thursday.
(C) Graphics is scheduled for room 3 on Wednesday.
(D) Research is scheduled for room 1 on Thursday.
(E) Research is scheduled for room 2 on Wednesday.
22. If on Wednesday Graphics, Marketing, and Research are scheduled for room 1, room 2, and room 3, respectively, then any of the following can be true EXCEPT:
- (A) Graphics is scheduled for room 2 on Thursday and Research is scheduled for room 1 on Friday.
(B) Graphics is scheduled for room 3 on Thursday and Marketing is scheduled for room 3 on Friday.
(C) Marketing is scheduled for room 1 on Thursday and Graphics is scheduled for room 2 on Friday.
(D) Marketing is scheduled for room 3 on Thursday and Research is scheduled for room 2 on Friday.
(E) Research is scheduled for room 2 on Thursday and Marketing is scheduled for room 3 on Friday.
23. When cut, the synthetic material fiberglass, like asbestos, releases microscopic fibers into the air. It is known that people who inhale asbestos, fibers suffer impairment of lung functions. A study of 300 factory workers who regularly cut fiberglass showed that their lung capacity is, on average, only 90 percent of that of a comparable group of people who do not cut fiberglass. The statements above, if true, most strongly support which of the following hypotheses?
- (A) People who work with fiberglass are likely also to work with asbestos.

- (B) Fiberglass fibers impair lung function in people who inhale them.
- (C) Fiberglass releases as many fibers into the air when cut as does asbestos.
- (D) Coarse fibers do not impair lung function in people who inhale them.
- (E) If uncut, fiberglass poses no health risk to people who work with it.

24. Politician: Pundits claim that by voting for candidates who promise to cut taxes, people show that they want the government to provide fewer services than it has been providing. By that reasoning, however, people who drink too much alcohol at a party in the evening want a headache the next morning.

Which of the following could replace the statement about people who drink too much without undermining the force of the politician's argument?

- (A) People who spend more money than they can afford want the things they spend that money on.
- (B) People who seek different jobs than they currently have do not want to work at all.
- (C) People who buy new cars want to own cars that are under manufacturer's warranty.
- (D) People who decide to stay in bed a few extra minutes on a workday morning want to have to rush to arrive at work on time.
- (E) People who buy lottery tickets want the economic freedom that winning the lottery would bring.

25. Like most other coastal towns in Norway, the town of Stavanger was quiet and peaceful until the early 1960's, when it became Norway's center for offshore oil exploration. Between then and now, violent crime and vandalism in Stavanger have greatly increased. Stavanger's social problems probably resulted from the oil boom, since violent crime and vandalism have remained low in coastal towns in Norway that have had no oil boom. Which of the following most accurately describes the method of reasoning employed in the argument?

- (A) Arguing that a circumstance is not a precondition for a phenomenon on the grounds that the phenomenon sometimes occurs where the circumstance is not present
- (B) Arguing that a circumstance is a cause of a phenomenon on the grounds that the phenomenon has not occurred where the circumstance is not present
- (C) Arguing that a particular thing cannot have caused a phenomenon because that thing was not present before the phenomenon occurred
- (D) Attempting to establish a claim by arguing that the denial of the claim is inconsistent with the observed facts
- (E) Attempting to establish that certain circumstances that would have had to occur for a particular explanation to be correct could not have occurred

SECTION 4

Time-30minutes

38 Questions

1. That she seemed to prefer ----- to concentrated effort is undeniable; nevertheless, the impressive quality of her finished paintings suggests that her actual relationship to her art was anything but-----.
 (A) preparation...passionate
 (B) artfulness...disengaged
 (C) dabbling...superficial
 (D) caprice...considered
 (E) indecision...lighthearted
2. Because of the excellent preservation of the fossil, anatomical details of early horseshoe crabs were ---- for the first time, enabling experts to ---- the evolution of the horseshoe crab.
 (A) scrutinized...ensure
 (B) verified...advance
 (C) identified...distort
 (D) obscured...illustrate
 (E) clarified...reassess
3. The philosopher claimed that a person who must consciously ----- his or her own indifference before helping another is behaving more nobly than one whose basic disposition allows such an act to be performed without-----.
 (A) feign...enthusiasm
 (B) censure...comment
 (C) embrace...duplicity
 (D) suffer...effort
 (E) overcome...deliberation
4. The senator's attempt to convince the public that he is not interested in running for a second term is ----- given the extremely----- fund-raising activities of his campaign committee.
 (A) futile...clandestine
 (B) sincere...visible
 (C) specious...apathetic
 (D) disingenuous...public
 (E) straightforward... dubious
5. Although a change in management may appear to ----- a shift in a company's fortunes, more often than not its impact is -----
 (A) hinder...measurable
 (B) promote...demonstrable
 (C) accelerate...profound
 (D) betray...fundamental
 (E) augur...inconsiderable
6. The skeleton of ----- bird that was recently discovered indicated that this ancient creature ----- today's birds in that, unlike earlier birds and unlike reptilian ancestors, it had not a tooth in its head.
 (A) a primeval... obscured
 (B) a unique... preempted
 (C) a primitive...anticipated
 (D) a contemporary... foreshadowed
 (E) an advanced...differed from
7. While many people utilize homeopathic remedies to treat health problems, other people do not -----such alternative treatments, ----- conventional medical treatments instead.
 (A) distrust...employing
 (B) embrace...eschewing
 (C) reject...envisioning
 (D) countenance... relying on
 (E) recommend... turning from
8. PROGRAM: CONCERT::
 (A) bibliography : book
 (B) menu : entree
 (C) questionnaire : poll
 (D) platform : campaign
 (E) agenda : meeting
9. EMBRACE: AFFECTION::
 (A) prediction : memory
 (B) innuendo : secrecy
 (C) shrug : indifference
 (D) conversation : familiarity
 (E) vote : unanimity
10. ENTHUSIASM : MANIA::
 (A) idea : inspiration

(B) nightmare : hallucination (C) failure : disgust (D) suspicion : paranoia (E) energy : fitness 11. ANONYMOUS : IDENTIFY:: (A) nonchalant : excite (B) repressed : constrain (C) misled : trust (D) annoying : assist (E) unremarkable : please 12. CARTOGRAPHER : MAP:: (A) astronomer : stars (B) carpenter : wood (C) lumberjack : saw (D) tailor : clothing (E) weaver : loom 13. EXEMPLARY : IMITATION:: (A) venerable : denigration (B) novel : duplication (C) redoubtable : regard (D) challenging : determination (E) creditable : verification 14. INSENSITIVE : BOOR:: (A) spontaneous : extrovert (B) mischievous : imp (C) conformist : ally (D) officious : zealot (E) extravagant : miser 15. LABYRINTHINE : SIMPLICITY:: (A) epic : scope (B) digressive : motive (C) heretical : sincerity (D) austere : design (E) jejune : interest 16. EUPHEMISM : OFFENSIVE:: (A) rhetoric : persuasive (B) aphorism : diffuse (C) metaphor : descriptive (D) repetition : fatiguing	(E) conciliation : appeasing From the 1900's through the 1950's waitresses in the United States developed a form of unionism based on the unions' defending the skills that their occupation <i>line</i> included and enforcing standards for the performance (5) of those skills. This "occupational unionism" differed substantially from the "worksite unionism" prevalent among factory workers. Rather than unionizing the workforces of particular employers, waitress locals sought to control their occupation throughout a city. (10) Occupational unionism operated through union hiring halls, which provided free placement services to employers who agreed to hire their personnel only through the union. Hiring halls offered union waitresses collective employment security, not individual (15) job security—a basic protection offered by worksite unions. That is, when a waitress lost her job, the local did not intervene with her employer but placed her elsewhere; and when jobs were scarce, the work hours available were distributed fairly among all members rather than being assigned according to seniority. 17. The primary purpose of the passage is to (A) analyze a current trend in relation to the past (B) discuss a particular solution to a long-standing problem (C) analyze changes in the way that certain standards have been enforced (D) apply a generalization to an unusual situation (E) describe an approach by contrasting it with another approach 18. Which of the following statements best summarizes a distinction mentioned in the passage between waitress unions and factory workers' unions? (A) Waitress unions were more successful than factory workers' unions in that they were able to unionize whole cities. (B) Waitress unions had an impact on only certain local areas, whereas the impact of factory workers' unions was national. (C) Waitress union members held primarily part-time positions, whereas factory workers'
--	---

- unions placed their members in full-time jobs.
- (D) Waitress unions emphasized the occupation of workers, whereas factory workers' unions emphasized the worksite at which workers were employed.
- (E) Waitress unions defined the skills of their trade, whereas the skills of factory trades were determined by employers' groups.
19. According to the passage, which of the following was characteristic of the form of union that United States waitresses developed in the first half of the twentieth century?
- (A) The union represented a wide variety of restaurant and hotel service occupations.
- (B) The union defined the skills required of waitresses and disciplined its members to meet certain standards.
- (C) The union billed employers for its members' work and distributed the earnings among all members.
- (D) The union negotiated the enforcement of occupational standards with each employer whose workforce joined the union.
- (E) The union ensured that a worker could not be laid off arbitrarily by an employer.
20. The author of the passage mentions "particular employers' (line 8) primarily in order to
- (A) suggest that occupational unions found some employers difficult to satisfy
- (B) indicate that the occupational unions served some employers but not others
- (C) emphasize the unique focus of occupational unionism
- (D) accentuate the hostility of some employers toward occupational unionism
- (E) point out a weakness of worksite unionism
- In prehistoric times brachiopods were one of the most abundant and diverse forms of life on Earth: more than 30,000 species of this clamlike creature have been cataloged from fossil records. Today
- (5) brachiopods are not as numerous, and existing species are not well studied, partly because neither

- the animal's fleshy inner tissue nor its shell has any commercial value. Moreover, in contrast to the greater diversity of the extinct species, the approxi-
- (10) mately 300 known surviving species are relatively, uniform in appearance. Many zoologists have interpreted this as a sign that the animal has been unable to compete successfully with other marine organisms in the evolutionary struggle.
- (15) Several things, however, suggest that the conventional view needs revising. For example, the genus *Lingula* has an unbroken fossil record extending over more than half a billion years to the present. Thus, if longevity is any measure, brachiopods are the most
- (20) successful organisms extant. Further, recent studies suggest that diversity among species is a less important measure of evolutionary success than is the ability to withstand environmental change, such as when a layer of clay replaces sand on the ocean
- (25) bottom. The relatively greater uniformity among the existing brachiopod species may offer greater protection from environmental change and hence may reflect highly successful adaptive behavior.
- The adaptive advantages of uniformity for brachio-
- (30) pods can be seen by considering specialization, a process that occurs as a result of prolonged colonization of a uniform substrate. Those that can survive on many surfaces are called generalists, while those that can survive on a limited range of substrates are called
- (35) specialists. One specialist species, for example, has valves weighted at the base, a characteristic that assures that the organism is properly positioned for feeding in mud and similar substrates; other species secrete glue allowing them to survive on the face of
- (40) underwater cliffs. The fossil record demonstrates that most brachiopod lineages have followed a trend toward increased specialization. However, during periods of environmental instability, when a particular substrate to which a specialist species has
- (45) adapted is no longer available, the species quickly dies out. Generalists, on the other hand are not dependent on a particular substrate, and are thus less vulnerable to environmental change. One study of the fossil record revealed a mass extinction of brachiopods
- (50) following a change in sedimentation from chalk to clay. Of the 35 brachiopod species found in the chalk, only 6 survived in the clay, all of them generalists.

- As long as enough generalist species are maintained, and studies of arctic and subarctic seas (55) suggest that generalists are often dominant members of the marine communities there, it seems unlikely that the phylum is close to extinction.
21. In the passage, the author is primarily concerned with
- (A) rejecting an earlier explanation for the longevity of certain brachiopod species
 - (B) reevaluating the implications of uniformity among existing brachiopod species
 - (C) describing the varieties of environmental change to which brachiopods are vulnerable
 - (D) reconciling opposing explanations for brachiopods' lack of evolutionary success
 - (E) elaborating the mechanisms responsible for the tendency among brachiopod species toward specialization
22. It can be inferred from the passage that many zoologists assume that a large diversity among species of a given class of organisms typically leads to which of the following?
- (A) Difficulty in classification
 - (B) A discontinuous fossil record
 - (C) A greater chance of survival over time
 - (D) Numerical abundance
 - (E) A longer life span
23. The second paragraph makes use of which of the following?
- (A) Specific examples
 - (B) Analogy
 - (C) Metaphor
 - (D) Quotation
 - (E) Exaggeration
24. The author suggests that the scientists holding the conventional view mentioned in lines 15-16 make which of the following errors?
- (A) They mistakenly emphasize survival rather than diversity.
 - (B) They misunderstand the causes of specialization.
 - (C) They misuse zoological terminology.
 - (D) They catalog fossilized remains improperly.
 - (E) They overlook an alternative criterion of evolutionary success.
25. It can be inferred from the passage that the decision to study an organism may sometimes be influenced by
- (A) its practical or commercial benefits to society
 - (B) the nature and prevalence of its fossilized remains
 - (C) the relative convenience of its geographical distribution
 - (D) its similarity to one or more better-known species
 - (E) the degree of its physiological complexity
26. Which of the following, if true, would most strengthen the author's claim (lines 56-57) that "it seems unlikely that the phylum is close to extinction"?
- (A) Generalist species now living in arctic water give few if any indications of a tendency towards significant future specialization.
 - (B) Zoologists have recently discovered that a common marine organism is a natural predator of brachiopods.
 - (C) It was recently discovered that certain brachiopod species are almost always concentrated near areas rich in offshore oil deposits.
 - (D) The ratio of specialist to Generalist species is slowly but steadily increasing.
 - (E) It is easier for a brachiopod to survive a change in sedimentation than a change in water temperature.
27. Information in the passage supports which of the following statements about brachiopods?
- I. Few brachiopods living in prehistoric times were specialists.
 - II. A tendency toward specialization, though typical, is not inevitable.
 - III. Specialist species dominate in all but arctic and subarctic waters.

- (A) I only
- (B) II only
- (C) II and III only
- (D) I and III only
- (E) I, II and III

Since some of the questions require you to distinguish fine shades of meaning, be sure to consider all the choices before deciding which one is best.

28. MISREAD:

- (A) refocus
- (B) approve
- (C) predict
- (D) explain succinctly
- (E) interpret correctly

29. DISSIPATE:

- (A) gather
- (B) seethe
- (C) relax
- (D) exert
- (E) incite

30. ENUNCIATE:

- (A) mumble
- (B) disclaim
- (C) dissuade
- (D) bluster
- (E) commend

31. TAUTEN:

- (A) rarefy
- (B) coarsen
- (C) force
- (D) loosen
- (E) constrain

32. ZEALOTRY:

- (A) pessimism
- (B) generosity
- (C) gullibility
- (D) lack of fervor
- (E) excess of confidence

33. REDOLENT

- (A) cheerful
- (B) resolute
- (C) unscented
- (D) uncovered
- (E) untainted

34. GLUTINOUS:

- (A) nonviscous
- (B) nonporous
- (C) antitoxic
- (D) catalytic
- (E) alkaline

35. PANEGYRIC:

- (A) covenant
- (B) recantation
- (C) enigma
- (D) termination
- (E) anathema

36. AWASH:

- (A) fouled
- (B) quenched
- (C) rigid
- (D) dry
- (E) sturdy

37. UNTOWARD:

- (A) direct
- (B) fortunate
- (C) tangential
- (D) decisive
- (E) effective

38. SUPERCILIOUS.

- (A) castigating
- (B) obsequious
- (C) reclusive
- (D) rambunctious
- (E) abrasive

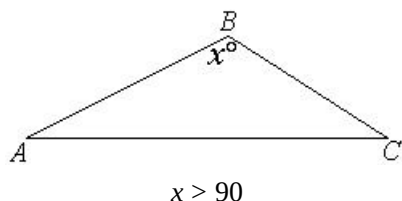
SECTION 5

Time –30 minutes

30 Questions

$$2x - 1 > 0$$

$$1. \quad x \quad \frac{1}{4}$$



$$2. \quad \begin{array}{ll} \text{The perimeter} & 3 \text{ times the length} \\ \text{of } \triangle ABC & \text{of } AC \end{array}$$

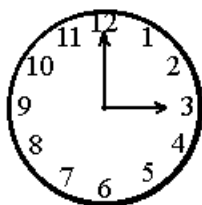
$$3. \quad \begin{array}{ll} \text{The total area of 18 non-} & 36 \text{ square inches} \\ \text{overlapping circular} & \\ \text{regions, each having a} & \\ \text{diameter of 2 inches} & \end{array}$$

$$0 < p < 1$$

$$4. \quad \begin{array}{ll} \text{The greatest value of} & \frac{1}{2} \\ p(1-p) & \end{array}$$

S is the sum of the first n negative integer powers of 2;
i.e., $S = 2^{-1} + 2^{-2} + \dots + 2^{-n}$

$$5. \quad s \quad 1$$

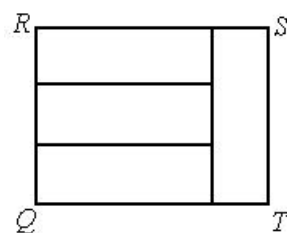


The minute hand of a tower clock is 3 feet long and the hour hand is 2 feet long. The two hands are in the three o'clock position, as shown.

$$6. \quad \begin{array}{ll} \text{The distance between} & 4 \text{ feet} \\ \text{the tips of the two hands} & \\ \text{of the clock} & \end{array}$$

s and t are integers, $s > t$, and $t \neq 0$.

$$7. \quad \frac{s}{t} \quad s$$



Rectangular region $QRST$ is divided into four smaller rectangular regions, each with length l and width w .

$$8. \quad \frac{QR}{RS} \quad \frac{3}{4}$$

$$9. \quad (5)^0(-3)^0 \quad 0$$

$$10. \quad \frac{6}{\sqrt{3}} \quad 2\sqrt{3}$$

$$xy + y^2 = 3$$

$$11. \quad x \quad y$$

$$12. \quad \frac{0.205}{0.305} \quad \frac{2}{3}$$

$$x - 1 = y$$

$$13. \quad 1-x \quad y$$

The 20 people at a party are divided into n mutually exclusive groups in such a way that the number of people in any group does not exceed the number in any other group by more than 1.

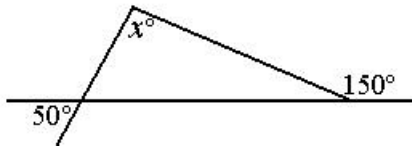
$$14. \quad \begin{array}{ll} \text{The value of } n \text{ if at} & 6 \\ \text{least one of the groups} & \\ \text{consists of 3 people} & \end{array}$$

For the line with equation $y = ax + b$, $ab \neq 0$, the x -intercept is twice the y intercept.

$$15. \quad \text{The slope of the line} \quad \frac{1}{2}$$

$$16. \quad \text{If } x + y = x, \text{ what is the value of } y?$$

- (A) -2
 (B) -1
 (C) 0
 (D) 2
 (E) It cannot be determined from the information given.



17. In the figure above, $x =$

- (A) 30
 (B) 80
 (C) 100
 (D) 130
 (E) 160

18. The average (arithmetic mean) number of trees per acre in a 40-acre plot is 140. If a 10-acre section of the plot contains 90 trees per acre, how many trees are there in the remaining 30 acres?

- (A) 5,700
 (B) 4,700
 (C) 4,200
 (D) 3,600

(E) 2,700

19. Which of the following sums is greater than 1?

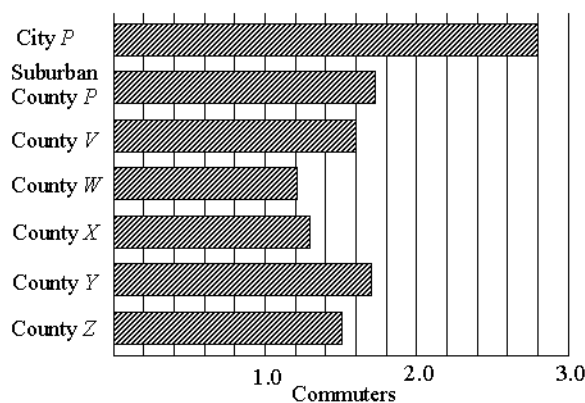
- (A) $\frac{1}{2} + \frac{1}{3}$
 (B) $\frac{7}{8} + \frac{3}{30}$
 (C) $\frac{15}{16} + \frac{2}{40}$
 (D) $\frac{12}{25} + \frac{12}{30}$
 (E) $\frac{35}{102} + \frac{2}{3}$

20. If the vertices of a triangle have rectangular coordinates (0,0), (8,0), and (8,6), respectively, then the perimeter of the triangle is

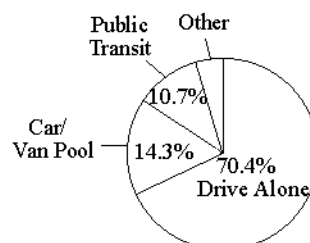
- (A) 10
 (B) 14
 (C) 24
 (D) 36
 (E) 48

Questions 21-25 refer to the following graphs.

AVERAGE VEHICLE OCCUPANCY RATE FOR COMMUTERS TO CITY P AND ITS SIX SUBURBAN COUNTIES



MEANS OF TRAVEL FOR COMMUTERS TO CITY P



Note: Drawn to scale.

21. For how many of the areas listed was the average vehicle occupancy rate for commuters less than 1.8?
- (A) seven
(B) Six
(C) Five
(D) Four
(E) Three
22. The average vehicle occupancy rate for commuters to County Y is most nearly
- (A) 1.3
(B) 1.4
(C) 1.5
(D) 1.6
(E) 1.7
23. Of the 2 million people who commute to City P, approximately how many travel by public transit?
- (A) 21,400
(B) 140,000
(C) 214,000
(D) 286,000
(E) 2,140,000
24. If the average vehicle occupancy rate for commuters to County X were to increase to 2.8, what would be the approximate percent increase in the occupancy rate?
- (A) 46%
(B) 54%
(C) 87%
(D) 115%
(E) 215%
25. If the total number of commuters to County W is twice the number to County Z, and if the average number of vehicles that transport commuters daily to County W is 30,000, what is the approximate average number of vehicles that transport commuters daily to County Z?
- (A) 12,000
(B) 15,000
(C) 18,000
(D) 27,000
(E) 36,000
26. If the average (arithmetic mean) of k and $7k$ is 60, then $k =$
- (A) 6
(B) 7.5
(C) 8
(D) 9.5
(E) 15
27. In a crate of fruit that contained strawberries, blueberries, and raspberries, the ratio of the number of pints of strawberries to the number of pints of blueberries to the number of pints of raspberries was 6 to 4 to 5, respectively. If the crate contained a total of 45 pints of these fruits, how many more pints of strawberries than blueberries were there in the crate?
- (A) 2
(B) 3
(C) 4
(D) 5
(E) 6
28. For a project, a square piece of cloth is folded in half and sewed together to form a rectangle that has a perimeter of 36 centimeters. What was the area in square centimeters of the piece of cloth before it was folded?
- (A) 16
(B) 36
(C) 81
(D) 108
(E) 144
29. How many positive 4-digit integers begin (on the left) with an odd digit and end with an even digit?
- (A) 250

- (B) 500
- (C) 2,000
- (D) 2,500
- (E) 5,000

30. For a certain farm, soybean production increased by 25 percent from year X to year Y , and the selling price of soybeans decreased by 25 percent from year X to year Y . If the entire soybean production was sold each year, approximately what was the percent change in the revenues from the sale of the soybeans from year X to year Y ?

- (A) 56% decrease
- (B) 6% decrease
- (C) No change
- (D) 6% increase
- (E) 56% increase

SECTION 6

Time-30 minutes

25 questions

1. Excavations at a Mayan site have uncovered jewelry workshops located some distance from the center of the site on roads radiating outward from the center. Since the nobility lived only in the area of the center, archaeologists conclude that these workshops made jewelry, not for the nobility, but for a middle class that must have been prosperous enough to afford it.

The archaeologists' argument assumes which of the following about the artisans who worked in the workshops'

- (A) They were themselves prosperous members of a middle class.
- (B) They lived near their workshops.
- (C) Their products were not made from the same materials as was jewelry for the nobility.
- (D) They worked full-time at making jewelry and did not engage in farming
- (E) They did not take the jewelry they had made in the workshops to clients who were members of the nobility.

2. Over the last 40 years there has been a great increase not only in the number of agricultural pesticides in use but also in the care and sophistication with which they are used by farmers. Nevertheless, the proportion of agricultural crops lost to certain pests worldwide has increased over the same period, even when the pests concerned have not developed resistance to existing pesticides.

Which of the following, if true, best explains how improvements in pesticide use have been accompanied by greater losses to certain pests?

- (A) Some dangerous but relatively ineffective pesticides common 40 years ago are no longer in widespread use.
- (B) As pesticides have become increasingly pest-specific, controlling certain pests with pesticides has turned out to cost more in many cases than the value of crop losses caused by

those pests.

- (C) Because today's pesticides typically have more specific application conditions than did pesticides in use 40 years ago, today's farmers observe their fields more closely than did farmers 40 years ago.
- (D) Certain pest-control methods that some farmers use today do not involve the use of chemical pesticides but are just as effective in eliminating insect pests as those that do.
- (E) Forty years ago, much less was known about the effects of pesticides on humans and other mammalian species than is now known.

Questions 3-8

During a single day, exactly seven paintings-G, H, J, K, L, M, and O-will, be auctioned one at a time. Each painting will be auctioned only once. The order in which the paintings are auctioned will comply with the following constraints:

J must be auctioned at some time before M is auctioned.

G and J must each be auctioned at some time before K is auctioned.

H must be the painting auctioned immediately before or immediately after L is auctioned.

3. If K is the seventh painting auctioned, the latest in the order that J can be auctioned is
- (A) second
(B) third
(C) fourth
(D) fifth
(E) sixth
4. If the fourth, fifth, and sixth paintings auctioned are O, M, and H, respectively, which of the following must be true?
- (A) G is the first painting auctioned.
(B) G is the second painting auctioned.
(C) J is the third painting, auctioned.
(D) K is the third painting auctioned.
(E) K is the seventh painting auctioned.
5. If O is the first painting auctioned and G is the fifth painting auctioned, then J must be either the

- (A) second or the third painting auctioned
(B) second or the fourth painting auctioned
(C) third or the fourth painting auctioned
(D) third or the sixth painting auctioned
(E) fourth or the seventh painting auctioned

6. If M is auctioned as early in the order as possible, which of the following must be true?

- (A) G is the first painting auctioned.
(B) H is the fourth painting auctioned.
(C) J is the first painting auctioned.
(D) K is the fourth painting auctioned.
(E) O is the seventh painting auctioned.

7. If O is the third painting, auctioned and J is the fourth painting auctioned, Which of the following can be auctioned later in the order than G is auctioned'

- (A) H
(B) J
(C) L
(D) M
(E) O

8. If G is the second painting auctioned and if J is auctioned as late in the order as possible, then O must be the painting auctioned

- (A) first
(B) third
(C) fourth
(D) fifth
(E) seventh

9. Authorities in California required drivers to use their headlights on a certain road during the daytime as well as at night and found that annual accident rates on the road fell 15 percent from the previous level. They concluded that applying the daytime rule statewide would lead to a similar reduction in accidents.

Which of the following, if true, most strengthens the authorities' argument?

- (A) Because an alternate route became available, the volume of traffic on the test road decreased

during the test period.

- (B) Drivers were informed of the requirement to use their headlights on the test road by means of a series of three conspicuous signs in each direction of travel.
- (C) Under certain conditions, among them fog and heavy rain, most drivers in California already use their headlights during the daytime.
- (D) Full-scale application of the daytime rule would cause headlight bulbs to burn out sooner than they currently do and thus to require more frequent replacement.
- (E) The test road was selected to include a great variety of the sorts of road conditions that drivers in California are likely to encounter.

10. Which of the following, if true, most logically completes the passage?

Every fusion reaction releases neutrinos. To test a hypothesis about the frequency of fusion reactions in the Sun, physicists calculated the number of neutrinos the Sun would produce annually if the hypothesis were correct. From this they estimated how many neutrinos should pass through a particular location on Earth. The fact that far fewer neutrinos were counted than were predicted to pass through the location would seem to prove that the hypothesis is wrong, except that-----.

- (A) the physicists, using a different method for estimating how many neutrinos should reach the location, confirmed their original estimate
- (B) there are several competing hypotheses about the frequency of solar fusion reactions
- (C) there is not enough energy in the Sun to destroy a neutrino once it is released
- (D) the method used to count neutrinos detects no more than approximately ten percent of the neutrinos that pass through
- (E) neutrinos released in the fusion reactions of other stars also reach the Earth

11. An economist concluded that Kregg Company deliberately discriminated against people with a history of union affiliation in hiring workers for its new plant. The economist's evidence is that, of the 1,500 people hired to work at the new plant, only

100 had ever belonged to a labor union, whereas in Kregg Company's older plants, a much higher proportion of workers have a history of union affiliation.

Which of the following is an assumption on which the economist's argument depends?

- (A) None of the people with a history of union affiliation who were hired to work at the new plant were union organizers.
- (B) Applicants for jobs at the new plant were not asked by Kregg's recruiters whether they had ever belonged to a labor union.
- (C) In the plants of some of Kregg's competitors, the workforce consists predominantly of union members.
- (D) The company believes that the cost of running the new plant will be lower if labor unions are not represented in the workforce.
- (E) The pool of potential candidates for jobs at the new plant included some people, in addition to those Kregg hired, with a history of union affiliation.

12. Hastings' contracture is a disorder of the connective tissue in one or both hands, most commonly causing loss of mobility. A survey of thousands of medical-insurance claims found that over 30 percent of people who had one hand operated on for Hastings' contracture underwent surgery a second time for this disorder within three years. Clearly, therefore, a single surgical treatment of Hastings' contracture is often ineffective at providing long-term correction of the disorder.

Which of the following, if true, most seriously weakens the argument?

- (A) The medical-insurance claims did not specify whether the surgery was on the patient's right or left hand.
- (B) The surgical techniques used to treat Hastings' contracture are identical to those used successfully to treat certain work-related injuries to the hand.
- (C) A separate survey found that 90 percent of patients operated on for Hastings' contracture

report increased hand mobility within one month after the surgery.

- (D) All of the patients in the survey were required by their insurance companies to seek a second opinion from a qualified surgeon before undergoing the operation.
- (E) Many people who have Hastings' contracture choose to tolerate its effects rather than undergo the risks of surgery.

Questions 13-16

Each of two colored lights on a control panel—light X and light Z—can be any one of exactly four colors: green, orange, purple, or red. The color of either light can change, but X cannot be the same color as Z at any one time. Only one light can change color at any one time. Each change of color is instantaneous.

The following changes are the eight changes possible:

- red to green, and vice versa
- red to orange, and vice versa
- red to purple, and vice versa
- purple to orange, and vice versa

Each change of color is followed by a period during which both lights are steady.

13. Which of the following color sequences could one of the lights go through as its color changes?
- (A.) From green to red to orange to green
 - (B) From green to red to green to purple
 - (C) From purple to orange to green to red
 - (D) From purple to red to green to orange
 - (E) From red to orange to red to purple
14. If X is red and Z is purple, the next color change to occur can be a change from
- (A) green to red
 - (B) orange to red
 - (C) purple to orange
 - (D) purple to red
 - (E) red to purple
15. If X and Z each change color exactly once during a certain period and if, as a result of these changes X is green and Z is orange, which of the following must be true about that period?

- (A) X changed color before Z did.
- (B) Z changed color before X did.
- (C) X was purple before it changed color.
- (D) Z was purple before it changed color.
- (E) Z was red before it changed color.

- 16 If X is orange and Z is purple, an interchange of these colors between X and Z requires a total of at least how many color changes?

- (A) Two
- (B) Three
- (C) Four
- (D) Five
- (E) Six

Questions 17-22

A research laboratory has a contract to do exactly Five experiments—M, P, R, S, and T. Each experiment must be performed separately, one after the other, according to the following restrictions:

- T cannot be done fourth unless S is done second.
- If M is done first, either S or R must be done fifth.
- If R is done fifth, P cannot be done third.
- P must be done sometime before T is done.

17. Which of the following is an acceptable sequence of experiments, from first to last?
- (A) M, P, R, T, S
 - (B) M, P, S, R, T
 - (C) M, S, P, T, R
 - (D) P, M, T, R, S
 - (E) S, M, T, P, R
18. Which of the following experiments can be done fourth and fifth, respectively, in the sequence of experiments?
- (A) M and P
 - (B) P and M
 - (C) P and T
 - (D) T and S
 - (E) T and P

19. If P is done first and S is done second, then T can be done in

- (A) third position, but in neither fourth nor fifth position
- (B) fourth position, but in neither third nor fifth position
- (C) fifth position, but in neither third nor fourth position
- (D) either third or fourth position, but not in fifth position
- (E) either third, fourth, or fifth position

20. If S is to be done at some time after T, then T can be done in

- (A) first position, but in no other position
- (B) second position, but in no other position
- (C) third position, but in no other position
- (D) either second or third position, but in no other position
- (E) either first, second, or third position, but in no other position

21. If M is done first, which of the following true?

- (A) P is done second.
- (B) P is done third.
- (C) R is done fourth.
- (D) R is done fifth.
- (E) T is done second

22. If as many of the experiments as possible must be performed between R and S, which of the following must be true?

- (A) M is done fourth.
- (B) P is done third.
- (C) P is done fourth.
- (D) R is done fifth.
- (E) T is done fourth.

23. The most widely used therapy for a certain type of ulcer completely heals such ulcers in 44 percent of patients within six months. In a six-month trial of a new therapy for this type of ulcer, 80 percent of ulcers treated achieved significant healing and 61 percent were completely healed. Since the trial treated only ulcers of this type that were worse

than average, the new therapy clearly promotes healing more effectively than the most widely used therapy.

The answer to which of the following would be most useful in evaluating the argument given?

- (A) What differences are there, if any, in the ways that the two therapies are administered?
- (B) Is there any significant difference between the costs associated with the two therapies?
- (C) What percentage of people with ulcers of this type who were treated with the most widely used therapy for six months experienced significant healing?
- (D) How quickly do ulcers of this type, if left untreated, become significantly worse.
- (E) What percentage of patients involved in the six-month trial of the new therapy were disappointed at the rate of healing were experiencing?

24. A society can achieve a fair distribution of resources only under conditions of economic growth. There can be no economic growth unless the society guarantees equality of economic opportunity to all of its citizens. Equality of economic opportunity cannot be guaranteed unless a society's government actively works to bring it about.

If the statements given are true, it can be properly concluded from them that

- (A) no government can achieve a fair distribution of resources under conditions of economic growth
- (B) all societies that guarantee equality of economic opportunity to all of their members are societies that distribute resources fairly
- (C) a society can achieve a fair distribution of resources only if its government actively works to bring about equality of economic opportunity
- (D) there can be no economic growth in a society unless that society guarantees a fair distribution of resources
- (E) some societies that experience economic growth fail to guarantee equality of

opportunity to all of their citizens

25. High Towers, a company that occupies several office buildings, is considering installing new energy-efficient lightbulbs in its buildings. The new bulbs require less than half the electricity consumed by the conventional bulbs currently used to produce the same amount of light. The new bulbs also last considerably longer. It follows that by replacing old bulbs as they burn out with the new kind of bulb, High Towers would significantly reduce its overall lighting costs. Which of the following, if true, most strengthens the argument given?

- (A) If the new bulbs are widely adopted, as seems likely, they will be produced in large enough quantities to be offered at prices comparable to those of conventional bulbs.
- (B) The utility that supplies High Towers with electricity offers discount rates to its largest customers.
- (C) High Towers has recently signed a contract to occupy an additional small office building.
- (D) High Towers has begun a campaign to encourage its employees to turn off lights whenever they leave a room.
- (E) The company that manufactures the new bulbs has been granted a patent on the innovative technology used in the bulbs and thus has exclusive rights to manufacture them.

SECTION 1

Time –30 minutes

18 Questions

1. While many Russian composers of the nineteenth century contributed to an emerging national style, other composers did not---- idiomatic Russian musical elements, ---- instead the traditional musical vocabulary of Western European Romanticism.
(A) utilize ..rejecting
(B) incorporate.. preferring
(C) exclude.. avoiding
(D) repudiate.. expanding
(E) esteem.. disdaining
2. Because the painter Albert Pinkham Ryder was obsessed with his ----perfection, he was rarely ----a painting, creating endless variations of a scene on one canvas, one on top of another.
(A) quest for.. satisfied with
(B) insistence on .. displeased with
(C) contempt for.. disconcerted by
(D) alienation from.. immersed in
(E) need for.. concerned with
3. Objectively set standards can serve as a ----for physicians, providing them ----unjustified malpractice claims.
(A) trial.. evidence of
(B) model.. experience with
(C) criterion.. reasons for
(D) test.. questions about
(E) safeguard.. protection from
4. In spite of ----reviews in the press, the production of her play was ----almost certain oblivion by enthusiastic audiences whose acumen was greater than that of the critics.
(A) lukewarm.. condemned to
(B) scathing.. exposed to
(C) lackluster.. rescued from
(D) sensitive.. reduced to
(E) admiring.. insured against
5. The passions of love and pride are often found in the same individual, but having little in common, they mutually ----, not to say destroy, each other.
(A) reinforce
(B) annihilate
(C) enhance
(D) weaken
(E) embrace
6. The necessity of establishing discrete categories for observations frequently leads to attempts to make absolute ----when there are in reality only----.
(A) analyses.. hypotheses
(B) correlations.. digressions
(C) distinctions.. gradations
(D) complications.. ambiguities
(E) conjectures.. approximations
7. A unique clay disk found at the Minoan site of Phaistos is often ----as the earliest example of printing by scholars who have defended its claim to this status despite equivalent claims put forward for other printing artifacts.
(A) questioned
(B) overlooked
(C) adduced
(D) conceded
(E) dismissed
8. EXEMPT: LIABILITY::
(A) flout: authority
(B) bestow: reward
(C) permit: request
(D) restrain: disorder
(E) pardon: penalty
9. FULL-BODIED: FLAVOR::
(A) penetrating: vision
(B) humorous: character
(C) salacious: language
(D) nostalgic: feeling
(E) resonant: sound

10. LEGACY: PREDECESSOR::

- (A) gift: donor
- (B) gratuity: service
- (C) contribution: charity
- (D) receipt: customer
- (E) loan: collector

11. HERO: ADMIRABLE::

- (A) critic: capricious
- (B) braggart: surly
- (C) eccentric: unconventional
- (D) anarchist: powerful
- (E) enemy: immoral

12. GALVANIZE: STIMULATE::

- (A) agitate: occlude
- (B) incubate: humidify
- (C) sterilize: separate
- (D) irrigate: flush
- (E) purify: amalgamate

13. MANIFEST: PERCEIVE::

- (A) porous: tear
- (B) renovated: improve
- (C) doubtful: assess
- (D) brittle: break
- (E) elite: qualify

14. LOOSE: CONFINEMENT::

- (A) forgive: injury
- (B) promulgate: rule
- (C) disabuse: misconception
- (D) redress: allegation
- (E) disengage: independence

15. BLANDISHMENT: COAX::

- (A) prevarication: deceive
- (B) reverie: dream
- (C) persuasion: coerce
- (D) enticement: impoverish
- (E) explanation: mislead

16. CONVULSION: CONTRACTION::

- (A) aggression: attack
- (B) sulkiness: punishment

(C) persistence: acquiescence

(D) frenzy: emotion

(E) indifference: greeting

Much of the research on hallucinogenic drugs such as LSD has focused on the neurotransmitter serotonin, a chemical that when released from a presynaptic serotonin-secreting neuron causes the transmission of

- (5) a nerve impulse across a synapse to an adjacent postsynaptic, or target, neuron. There are two major reasons for this emphasis. First, it was discovered early on that many of the major hallucinogens have a molecular structure similar to that of serotonin. In (10) addition, animal studies of brain neurochemistry following administration of hallucinogens invariably reported changes in serotonin levels.

- Early investigators correctly reasoned that the structural similarity to the serotonin molecule might (15) imply that LSD's effects are brought about by an action on the neurotransmission of serotonin in the brain. Unfortunately, the level of technical expertise in the field of brain research was such that this hypothesis had to be tested on peripheral tissue (20) (tissue outside the brain). Two different groups of scientists reported that LSD powerfully blockaded serotonin's action. Their conclusions were quickly challenged, however. We now know that the action of a drug at one site in the body does not necessarily (25) correspond to the drug's action at another site, especially when one site is in the brain and the other is not.

- By the 1960's, technical advances permitted the direct testing of the hypothesis that LSD and related (30) hallucinogens act by directly suppressing the activity of serotonin-secreting neurons themselves—the so-called presynaptic hypothesis. Researchers reasoned that if the hallucinogenic drugs act by suppressing the activity of serotonin-secreting neurons, then drugs (35) administered after these neurons had been destroyed should have no effect on behavior, because the system would already be maximally suppressed. Contrary to their expectations, neuron destruction enhanced the effect of LSD and related hallucinogens (40) on behavior. Thus, hallucinogenic drugs apparently do not act directly on serotonin-secreting neurons.

However, these and other available data do support an alternative hypothesis, that LSD and related drugs act directly at receptor sites on serotonin target (45) neurons (the postsynaptic hypothesis). The fact that LSD elicits “serotonin syndrome”—that is, causes the same kinds of behaviors as does the administration of serotonin—in animals whose brains are depleted of serotonin indicates that LSD acts directly (50) on serotonin receptors, rather than indirectly through the release of stores of serotonin. The enhanced effect of LSD reported after serotonin depletion could be due to a proliferation of serotonin receptor sites on serotonin target neurons. This phenomenon often (55) follows neuron destruction or neurotransmitter depletion; the increase in the number of receptor sites appears to be a compensatory response to decreased input. Significantly, this hypothesis is supported by data from a number of different laboratories.

17. According to the passage, which of the following is one of the primary factors that led researchers studying hallucinogenic drugs to focus on serotonin?
- (A) The suppression of the activity of serotonin-secreting neurons by the administration of hallucinogens
 - (B) The observed similarities in the chemical structures of serotonin and hallucinogens
 - (C) The effects the administration of hallucinogens has on serotonin production in the human brain
 - (D) Serotonin-induced changes in the effects of hallucinogens on behavior
 - (E) Hallucinogen-induced changes in the effects of serotonin on behavior
18. It can be inferred that researchers abandoned the presynaptic hypothesis because
- (A) a new and more attractive hypothesis was suggested
 - (B) no research was reported that supported the hypothesis
 - (C) research results provided evidence to counter the hypothesis
 - (D) the hypothesis was supported only by

studies of animals and not by studies of human beings

- (E) the level of technical expertise in the field of brain research did not permit adequate testing of the hypothesis

19. Which of the following best expresses the main idea of the passage?
- (A) Research has suggested that the neurotransmitter serotonin is responsible for the effects of hallucinogenic drugs on the brain and on behavior.
 - (B) Researchers have spent an inadequate amount of time developing theories concerning the way in which the effects of hallucinogenic drugs occur.
 - (C) Research results strongly suggest that hallucinogenic drugs create their effects by acting on the serotonin receptor sites located on target neurons in the brain.
 - (D) Researchers have recently made valuable discoveries concerning the effects of depleting the amount of serotonin in the brain.
 - (E) Researchers have concluded that hallucinogenic drugs suppress the activity of serotonin-secreting neurons.
20. The research described in the passage is primarily concerned with answering which of the following questions?
- (A) How can researchers control the effects that LSD has on behavior?
 - (B) How are animals’ reactions to LSD different from those of human beings?
 - (C) What triggers the effects that LSD has on human behavior?
 - (D) What technical advances would permit researchers to predict more accurately the effects of LSD on behavior?
 - (E) What relationship does the suppression of neuron activity have to the occurrence of “serotonin syndrome”?

21. Which of the following best defines “serotonin syndrome” (line 46) as the term is used in the passage?
- (A) The series of behaviors, usually associated with the administration of serotonin, that also occurs when LSD is administered to animals whose brains are depleted of serotonin
- (B) The series of behaviors, usually associated with the administration of LSD, that also occurs when the amount of serotonin in the brain is reduced
- (C) The maximal suppression of neuron activity that results from the destruction of serotonin-secreting neurons
- (D) The release of stores of serotonin from serotonin-secreting neurons in the brain
- (E) The proliferation of serotonin receptor sites that follows depletion of serotonin supplies in the brain
22. Which of the following best describes the organization of the argument that the author of the passage presents in the last two paragraphs?
- (A) Two approaches to testing a hypothesis are described, and the greater merits of one approach are indicated.
- (B) The assumptions underlying two hypotheses are outlined, and evidence for and against each hypothesis is discussed.
- (C) A phenomenon is described, and hypotheses concerning its occurrence are considered and rejected.
- (D) The reasoning behind a hypothesis is summarized, evidence supporting the hypothesis is presented, and research that counters the supporting evidence is described.
- (E) A hypothesis is discussed, evidence undermining the hypothesis is revealed, and a further hypothesis based on the undermining evidence is explained
23. The author’s attitude toward early researchers’ reasoning concerning the implications of similarities in the structures of serotonin and

LSD molecules can best be described as one of

- (A) complete agreement
- (B) reluctant support
- (C) subtle condescension
- (D) irreverent dismissal
- (E) strong opposition

- When literary periods are defined on the basis of men’s writing, women’s writing must be forcibly assimilated into an irrelevant grid: a Renaissance that is not a renaissance for women, a Romantic period in which women played very little part, a modernism with which women conflict. Simultaneously, the history of women’s writing has been suppressed, leaving large, mysterious gaps in accounts of the development of various genres. Feminist criticism is beginning to correct this situation. Margaret Anne Doody, for example, suggests that during “the period between the death of Richardson and the appearance of the novels of Scott and Austen,” which has “been regarded as a dead period.” Late-eighteenth-century women writers actually developed “the paradigm for women’s fiction of the nineteenth century—something hardly less than the paradigm of the nineteenth-century novel itself.” Feminist critics have also pointed out that the twentieth-century writer Virginia Woolf belonged to a tradition other than modernism and that this tradition surfaces in her work precisely where criticism has hitherto found obscurities, evasions, implausibilities, and imperfections.
24. It can be inferred from the passage that the author views the division of literature into periods based on men’s writing as an approach that
- (A) makes distinctions among literary periods ambiguous
- (B) is appropriate for evaluating only premodern literature
- (C) was misunderstood until the advent of feminist criticism
- (D) provides a valuable basis from which feminist criticism has evolved
- (E) obscures women’s contributions to literature

25. The passage suggests which of the following about Virginia Woolf's work?

- I. Nonfeminist criticism of it has been flawed.
- II. Critics have treated it as part of modernism.
- III. It is based on the work of late-eighteenth-century women writers.

- (A) I only
- (B) II only
- (C) I and II only
- (D) II and III only
- (E) I, II and III

26. The author quotes Doody most probably in order to illustrate

- (A) a contribution that feminist criticism can make to literary criticism
- (B) a modernist approach that conflicts with women's writing
- (C) writing by a woman which had previously been ignored
- (D) the hitherto overlooked significance of Scott's and Austen's novels
- (E) a standard system of defining literary periods

27. The passage provides information that answers which of the following questions?

- (A) In what tradition do feminist critics usually place Virginia Woolf?
- (B) What are the main themes of women's fiction of the nineteenth century?
- (C) What events motivated the feminist reinterpretation of literary history?
- (D) How has the period between Richardson's death and Scott's and Austen's novels traditionally been regarded by critics?
- (E) How was the development of the nineteenth-century novel affected by women's fiction in the same century?

28. GROUNDED:

- (A) attendant
- (B) flawless
- (C) effective
- (D) aloft
- (E) noteworthy

29. DISCHARGE:

- (A) retreat
- (B) hire
- (C) insist
- (D) circulate
- (E) pause

30. INTERMITTENT:

- (A) compatible
- (B) constant
- (C) neutral
- (D) unadulterated
- (E) indispensable

31. APT:

- (A) exceptionally ornate
- (B) patently absurd
- (C) singularly destructive
- (D) extremely inappropriate
- (E) fundamentally insensitive

32. JUSTIFY:

- (A) misjudge
- (B) ponder
- (C) terminate
- (D) argue against
- (E) select from

33. TEDIOUS:

- (A) intricate
- (B) straightforward
- (C) conspicuous
- (D) entertaining
- (E) prominent

34. INTEGRAL:

- (A) profuse
- (B) superfluous
- (C) meritorious
- (D) neutral
- (E) displaced

35. COWED:

- (A) unencumbered
- (B) untired

- (C) unversed
- (D) unworried
- (E) undaunted

36. CONCORD:

- (A) continuance
- (B) severance
- (C) dissension
- (D) complex relationship
- (E) unrealistic hypothesis

37. FRIABLE :

- (A) substantial
- (B) inflexible
- (C) easily contained
- (D) slow to accelerate
- (E) not easily crumbled

38. DERACINATE:

- (A) illuminate
- (B) quench
- (C) amplify
- (D) polish
- (E) plant

SECTION 2

Time –30 minutes

25 Questions

Questions 1-7

A developer is assigning six rectangular building lots in an industrial park to six companies—Handicorp, Instantype, Kleentek, Linostyle, Messagex, and Nanoware. Each company will be assigned a different one of the six lots, which are arranged in two blocks—block 700 and block 800. Each block includes three lots, numbered consecutively lot 1 through lot 3. Within each block, lot 1 is adjacent to lot 2 and lot 2 is adjacent to lot 3; no other lots are adjacent to each other. In assigning lots, the developer will obey the following restrictions:

Instantype's lot cannot be in the same block as

Linostyle's lot.

Linostyle's lot must be adjacent to Messagex' lot.

Messagex' lot cannot be lot 2 of either block.

Nanoware's lot cannot be adjacent to

Handicorp's lot.

1. Which of the following is an acceptable

assignment of companies to block 700, listed from lot 1 through lot 3?

- (A) Instantype, Handicorp, Nanoware
- (B) Instantype, Linostyle, Messagex
- (C) Linostyle, Messagex, Kleentek
- (D) Messagex, Linostyle, Nanoware
- (E) Nanoware, Instantype, Messagex

2. If Messagex is assigned a lot in block 800, which of the following must be assigned a lot in block 700?

- (A) Handicorp
- (B) Instantype
- (C) Kleentek
- (D) Linostyle
- (E) Nanoware

3. If Handicorp and Nanoware are each assigned a lot in block 700, which of the following must be true?

- (A) Instantype is assigned lot 2 in block 700.
 (B) Instantype is assigned lot 3 in block 700.
 (C) Kleentek is assigned lot 1 in block 800.
 (D) Linostyle is assigned lot 3 in block 800.
 (E) Messagex is assigned lot 1 in block 800.
4. If Nanoware is assigned lot 2 in block 700, which of the following is a pair of companies that must be assigned lots that are adjacent to each other?
 (A) Handicorp and Instantype
 (B) Handicorp and Messagex
 (C) Instantype and Kleentek
 (D) Kleentek and Nanoware
 (E) Messagex and Nanoware
5. If Kleentek is assigned lot 2 in block 800, which of the following can be true?
 (A) Handicorp is assigned a lot that is adjacent to the lot assigned to Instantype.
 (B) Messagex is assigned a lot that is adjacent to the lot assigned to Nanoware.
 (C) Instantype is assigned a lot on block 700.
 (D) Linostyle is assigned lot 1 on block 800.
 (E) Messagex is assigned lot 1 on block 700.
6. If Instantype and Handicorp are assigned lots 1 and 2, respectively, in block 700, which of the following must be true?
 (A) Kleentek is assigned lot 3 on block 700.
 (B) Linostyle is assigned lot 3 on block 700.
 (C) Messagex is assigned lot 1 on block 800.
 (D) Nanoware is assigned lot 3 on block 700.
 (E) Nanoware is assigned lot 1 on block 800.
7. If Linostyle and Kleentek are assigned lots that are on the same block as each other, which of the following must be true?
 (A) Handicorp is assigned lot 2 of one of the blocks.
 (B) Instantype is assigned lot 2 of one of the blocks.
 (C) Kleentek is assigned lot 1 of one of the blocks.
 (D) Linostyle is assigned lot 1 of one of the blocks.
 (E) Messagex is assigned lot 3 of one of the blocks.
8. Auditor from Acme Industries: Last week at Acme Bakery, about six percent of the pastries

baked during the night shift were found to be imperfect, but no imperfect pastries were found among those baked during the day shift. Pastries are inspected during the same shift in which they are baked, so clearly the night-shift quality control inspectors were more alert, despite their nighttime work hours, than the dayshift quality control inspectors.

The argument depends on the assumption that
 (A) at least some imperfect pastries were baked during the day shift at Acme Bakery last week

- (B) not all of the pastries that the night-shift quality control inspectors judged to be imperfect were in fact imperfect
 (C) the night-shift quality control inspectors received more training in quality control procedures than did the day-shift quality control inspectors
 (D) in a normal week, fewer than six percent of the pastries baked during the night shift at Acme Bakery are found to be imperfect
 (E) there are only two shifts per day at Acme Bakery, a day shift and a night shift

9. Spiders of many species change color to match the pigmentation of the flowers they sit on. The insects preyed on by those spiders, unlike human beings, possess color discrimination so acute that they can readily see the spiders despite the seeming camouflage. Clearly, then, it must be in evading their own predators that the spiders' color changes are useful to them.

Which of the following, if true, most strengthens the argument?

- (A) Among the animals that feed on color-changing spiders are a few species of bat, which find their prey through sound echoes.
 (B) Certain animals that feed on color-changing spiders do so only sparingly in order to keep from ingesting harmful amounts of spider venom.
 (C) Color-changing spiders possess color

discrimination that is more acute than that of spiders that lack the ability to change color.

- (D) Color-changing spiders spin webs that are readily seen by the predators of those spiders.
 (E) The color discrimination of certain birds that feed on color-changing spiders is no more acute than that of human beings.

10. Which of the following most logically completes the argument below?

Each year every employee of SAI Corporation must enroll in one of the two health insurance plans offered by SAI. One plan requires a sizable monetary contribution from employees; the other plan is paid for entirely by SAI. Many SAI employees enroll in the plan requiring employee contributions. This fact does not show that they feel that this plan's benefits are superior to those provided by the plan requiring no employee contribution since----.

- (A) the plan that requires an employee contribution costs and enrolled employee significantly less per year than do typical health insurance plans offered by corporations other than SAI
 (B) only SAI employees who have worked for SAI for at least fifteen years are eligible to enroll in the plan paid for entirely by SAI
 (C) the two health insurance plans currently offered by SAI are substantially the same plans SAI has offered for the past ten years
 (D) most of the SAI employees enrolled in the plan paid for entirely by SAI are under 50 years old
 (E) both plans offered by SAI provide benefits not only for employees of SAI but also for children and spouses of enrolled employees

Questions 11-17

Frank, Greg, Julia, Laura, Magda, and Nick will be the speakers at an all-day meeting of a local astronomy club. Each speaker will speak once, for one hour, with no other speakers in that hour. Three speakers will be scheduled to speak before lunch

and three will be scheduled to speak after lunch.

The following conditions must be observed in planning the schedule for the meeting:

Greg must speak before lunch.

There must be exactly one speaker scheduled to speak between Magda and Nick, whether or not Magda is earlier in the order of speakers than Nick, and whether or not lunch intervenes between Magda's and Nick's speeches.

Frank must speak either first or third.

11. Which of the following is an acceptable order of the speakers from first to last?

- | | | | | | | |
|-----|----------|----------|----------|----------|----------|----------|
| | <u>1</u> | <u>2</u> | <u>3</u> | <u>4</u> | <u>5</u> | <u>6</u> |
| (A) | Frank | Greg | Magda | Nick | Julia | Laura |
| (B) | Frank | Julia | Greg | Nick | Magda | Laura |
| (C) | Greg | Julia | Frank | Nick | Laura | Magda |
| (D) | Julia | Laura | Frank | Magda | Greg | Nick |
| (E) | Nick | Julia | Greg | Frank | Magda | Laura |

12. If Julia is the first speaker, which of the following must be the second speaker?

- (A) Frank
 (B) Greg
 (C) Laura
 (D) Magda
 (E) Nick

13. If Julia speaks fourth, the third speaker must be either

- (A) Frank or Greg
 (B) Frank or Magda
 (C) Greg or Laura
 (D) Laura or Nick
 (E) Magda or Nick

14. If Laura speaks in the morning and Magda is not the sixth speaker, which of the following must be the next speaker after Magda?

- (A) Frank
 (B) Greg
 (C) Julia
 (D) Laura
 (E) Nick

- | | |
|--|--|
| <p>15. If lunch intervenes between Magda's and Nick's speeches, which of the following lists all the speakers who can be the speaker scheduled between Magda and Nick?</p> <p>(A) Frank, Greg
(B) Greg, Julia
(C) Julia, Laura
(D) Frank, Greg, Julia
(E) Frank, Greg, Julia, Laura</p> <p>16. If Julia is scheduled to speak at some time before Frank, in which of the following positions in the order of speakers can Nick be scheduled to speak?</p> <p>(A) First</p> | <p>(B) Second
(C) Third
(D) Fourth
(E) Fifth</p> <p>17. If Laura is scheduled to be the next speaker after Julia, how many speakers in total must be scheduled to speak before Laura?</p> <p>(A) One
(B) Two
(C) Three
(D) Four
(E) Five</p> |
|--|--|

Questions 18-22

A kennel worker must both walk and groom each of three dogs—Lucky, Muffin, and Rover. The worker is scheduling these activities into exactly six time slots—slot 1 through slot 6. During each time slot, the worker will be scheduled to perform only one activity, either walking or grooming, with only one dog. The following conditions also apply to the schedule:

Neither Lucky nor Muffin can be walked until both of them have been groomed.

Rover must be walked at some time before being groomed.

The worker cannot perform the same activity, either walking or grooming, for more than two consecutive time slots.

18. Which of the following is an acceptable schedule for the first four time slots?

	<u>Slot 1</u>	<u>Slot 2</u>	<u>Slot 3</u>	<u>Slot 4</u>
(A)	Grooming Lucky	Grooming Muffin	Walking Lucky	Grooming Rover
(B)	Grooming Lucky	Walking Muffin	Walking Lucky	Grooming Muffin
(C)	Grooming Muffin	Walking Rover	Grooming Lucky	Walking Rover
(D)	Grooming Muffin	Walking Rover	Grooming Lucky	Walking Muffin
(E)	Walking Rover	Grooming Muffin	Grooming Rover	Grooming Lucky

- | | |
|--|--|
| <p>19. The earliest time slot for which Muffin can be scheduled for walking is slot</p> <p>(A) 2 (B) 3 (C) 4 (D) 5 (E) 6</p> <p>20. If Rover is scheduled for walking in the time slot immediately before the time slot for which Muffin is scheduled for walking, then which of the following must be scheduled for slot 5?</p> <p>(A) Grooming Lucky
(B) Grooming Rover
(C) Walking Lucky
(D) Walking Muffin</p> | <p>(E) Walking Rover</p> <p>21. If Lucky is scheduled for walking in the time slot immediately before the time slot for which Rover is scheduled for walking, then which of the following must be true?</p> <p>(A) Grooming is scheduled for slot 3.
(B) Walking is scheduled for slot 2.
(C) Walking is scheduled for slot 6.
(D) Lucky is scheduled for grooming in slot 1.
(E) Muffin is scheduled for walking in slot 4.</p> <p>22. If Rover is scheduled for walking in slot 1,</p> |
|--|--|

- which of the following must be true?
- (A) Grooming is scheduled for slot 5.
 - (B) Walking is scheduled for slot 2.
 - (C) Walking is scheduled for slot 4.
 - (D) Muffin is scheduled for grooming in slot 3.
 - (E) Rover is scheduled for grooming in slot 6.

23. V-shaped walled structures in central Asia were used by prehistoric hunters who drove hoofed animals into an enclosure at the point of the V. The central Asians who built these structures probably learned this hunting technique from invaders from southwest Asia, because the arrival of invaders from a region in southwest Asia where similar structures had long been used coincides roughly with the building of the earliest of such structures in central Asia.

Which of the following, if true, most strengthens the argument?

- (A) Excavations in the central Asian region do not indicate whether invaders from southwest Asia settled permanently in central Asia.
- (B) The V-shaped structures in central Asia were roughly 70 meters long, whereas the similar structures in southwest Asia were usually over 300 meters long.
- (C) The walls of the structures in central Asia were made from earth, whereas the walls of the structures in southwest Asia were made of rock.
- (D) The earliest examples of V-shaped walled structures in central Asia were of an advanced design.
- (E) Some of the walled structures used for hunting in southwest Asia were built well after the earliest such structures were built in central Asia.

24. Which of the following most logically completes the argument?

Virtually all respondents to a recent voter

survey reported allegiance to one of the two major political parties. But over a third of the voters from each party reported being so disenchanted with the governing philosophies of both parties that they might join a third major party if one were formed. Even if this poll reflects general voter sentiment, however, there is no chance that a new party could attract a third of all voters, since----

- (A) the current level of disenchantment with the governing philosophies of the two major parties is unprecedented
- (B) the disenchanted members of the two major parties are attracted to very different governing philosophies
- (C) most respondents overestimated the proportion of voters disenchanted with both parties, saving that the proportion was more than 50 percent
- (D) nearly half of all respondents reported that they would be more likely to cease voting altogether than to switch their party affiliation
- (E) any new party would be likely to inspire citizens who have not voted before to join and to become regular voters

25. When amphibians first appeared on Earth millions of years ago, the amount of ultraviolet radiation penetrating Earth's atmosphere was much greater than it is today. Therefore, current dramatic decreases in amphibian populations cannot be the result of recent increases in ultraviolet radiation penetrating Earth's atmosphere.

Which of the following is an assumption on which the argument depends?

- (A) The eggs of modern amphibians are not significantly more vulnerable to ultraviolet radiation than the eggs of the first amphibians were.
- (B) Modern amphibians are not as likely as the first amphibians were to live in habitats that shield them from ultraviolet radiation.
- (C) Populations of modern amphibians are not able to adapt to changing levels of radiation as readily as populations of early amphibians

were.

(D) The skin of amphibians is generally more sensitive to ultraviolet radiation than the skin of other animals is.

(E) The skin of amphibians is less sensitive to ultraviolet radiation than to other types of radiation.

SECTION 3

Time—30 minutes

30 Questions

The rent for each room at Hotel X was \$ 120 before it was increased 10 percent.

1. The rent for each room at Hotel X immediately after the increase

\$132

2. $\frac{1}{22.03 - 1.03}$

$\frac{1}{2}$

Train X traveled away from station A , and train Y traveled toward station A . The trains traveled toward each other on parallel tracks and passed each other at 10:30 A.M.

3. The number of minutes before 10:30 A.M. that train X traveled after leaving station A

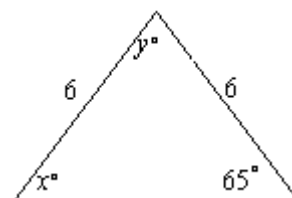
The number of minutes after 10:30 A.M. that train Y traveled before arriving at station A

4. The length of the hypotenuse of a right triangle with legs of lengths 3 and 4

The length of the hypotenuse of a right triangle with legs of lengths 2 and 5

The average (arithmetic mean) of the numbers 1, 2, 3, 4, and n is equal to 2.

5. $\frac{n}{2}$



6. $\frac{x}{y}$

7. $\frac{1}{m} + \frac{1}{n}$

$\frac{m+n}{mn}$

Points P , Q , and R have rectangular coordinates $(0,8)$, $(4,0)$, and $(0,-3)$, respectively.

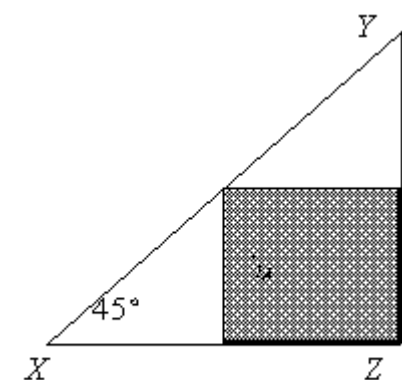
8. The perimeter of $\triangle PQR$ 25

The result of multiplying z by $\frac{4}{5}$ is $\frac{6}{7}$

9. z 1

10. $xyz < 0$
 $yz > 0$
 x y

11. $(0.01)^5$ $\frac{1}{10^{10}}$



The shaded region is rectangular.

12. The perimeter of the shaded region $XZ + YZ$

13. 3^5 0

The sum of x and y is less than the product of x and y .

$$y = \frac{1}{2}$$

14. x 0

15. $\frac{10^7 - 10^6}{9}$ 100^3

16. If the range of the six measurements 140, 125, 180, 110, 165, and x is 80, which of the following could be the value of x ?
- (A) 60
(B) 85
(C) 190
(D) 220
(E) 245

k	a_k	p_k
1	100	0.10
2	200	0.25
3	300	0.20
4	400	0.25
5	500	0.20

17. If in an experiment the probabilities of obtaining the values a_1, a_2, a_3, a_4 , and a_5 are p_1, p_2, p_3, p_4 , and p_5 , respectively, then the expected value is defined as $a_1p_1 + a_2p_2 + a_3p_3 + a_4p_4 + a_5p_5$. For the values and their corresponding probabilities in the table above, what is the expected value?
- (A) 350
(B) 320
(C) 300
(D) 270
(E) 250

18. If m is an integer, for what value of m is

$$3^m < 100 < 3^{m+1}?$$

- (A) 0
(B) 1
(C) 2
(D) 3
(E) 4

19. A certain club is collecting money for a charity. A local company has agreed to contribute \$1 to the charity for every \$3

collected by the club. How much money must the club collect in order for the total amount for the charity, including the company contribution, to equal \$24,000?

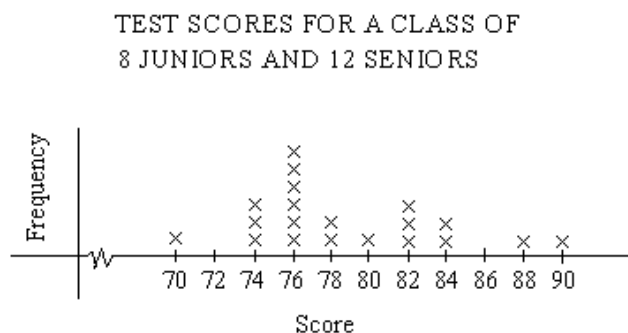
- (A) \$18,000
- (B) \$16,000
- (C) \$15,000
- (D) \$12,000
- (E) \$10,000

20. A corner of a square tile is cut off, leaving the piece shown above. What is the area of this piece?

- (A) 90 sq in
- (B) 85 sq in
- (C) 80 sq in
- (D) 75 sq in
- (E) 70 sq in

Questions 21-25 refer to the following distribution.

TEST SCORES FOR A CLASS OF
8 JUNIORS AND 12 SENIORS



21. If 5 seniors have scores of 82 or above, how many juniors have scores below 82?

- (A) 7
- (B) 6
- (C) 5
- (D) 4
- (E) 3

22. If 76 is the lowest passing score, what percent of the class did not get a passing score?

- (A) 8%

- (B) 10%
- (C) 12%
- (D) 20%
- (E) 25%

23. The median score for the class is

- (A) 76
- (B) 77
- (C) 78
- (D) 79
- (E) 80

24. If 5 points were added to each score, which of the following would NOT be affected?

- (A) The highest score
- (B) The mean for all scores
- (C) The median for the seniors' scores
- (D) The mode for the juniors' scores
- (E) The standard deviation for all scores

25. If the mean score for the juniors were known, which of the following could be calculated from the information given?

- I. The range of the scores for the seniors
- II. The median score for the juniors
- III. The mean score for the seniors

- (A) None
- (B) I only
- (C) III only
- (D) I and II
- (E) II and III

26. A membership list of 620 people shows that 31 of them have first and last names that begin with the same letter. If a person is selected at random from the list, what is the probability that the person's first and last names do not begin with the same letter?

- (A) 0.05
- (B) 0.25
- (C) 0.50
- (D) 0.75
- (E) 0.95

27. If p and r are prime numbers, which of the following must also be prime?

- (A) pr

- (B) $p + r$
 (C) $pr + 1$
 (D) $p^2 + r^2$
 (E) None of the above

28. For what ordered pair (x, y) on the graph of $y = \frac{1}{2}x - 1$ does the x - coordinate equal the y - coordinate?

- (A) $(-2, -2)$
 (B) $(\frac{1}{2}, \frac{1}{2})$
 (C) $(1, -\frac{1}{2})$

- (D) $(2, 1)$
 (E) $(2, 2)$

29. David and Michael charged Mr. Jimenez \$3,000 to remodel his basement. To complete the project, David worked 4 days alone, Michael worked 1 day alone, and they worked 10 days together. If they each received the same amount of money for each day that they worked, how much of the \$3,000 did David receive?

- (A) \$1,800
 (B) \$1,750
 (C) \$1,680
 (D) \$1,575
 (E) \$1,200

30. If x is positive and $6 - x^2 = \frac{15}{16}$, then $\sqrt{x} =$

- (A) $\frac{81}{16}$
 (B) $\frac{9}{10}$
 (C) $\frac{\sqrt{3}}{2}$
 (D) $\frac{2}{3}$
 (E) $\frac{3}{2}$

SECTION 4

Time – 30 minutes

38 Questions

1. Punishment for violating moral rules is much more common than reward for following them; thus, ----- the rules goes almost ----- in society.

- (A) association with .. undefended
 (B) adherence to .. unnoticed
 (C) affiliation of .. uncorrected
 (D) opposition to .. unchecked
 (E) ignorance of .. unresolved

2. Compassion is a great respecter of justice: we pity those who suffer -----.

- (A) shamelessly
 (B) unwittingly
 (C) vicariously
 (D) intensively
 (E) undeservedly

3. No work illustrated his disdain for a systematic approach to research better than his dissertation, which was rejected primarily because his bibliography constituted, at best, ----- survey of the major texts in his field.

- (A) an unimaginative
 (B) an orthodox
 (C) a meticulous
 (D) a comprehensive
 (E) a haphazard

4. In contrast to the----- with which the acquisition of language by young children was once regarded, the process by which such learning occurs has now become the object of -----.

- (A) intensity .. fascination
 (B) incuriosity .. scrutiny
 (C) anxiety .. criticism
 (D) reverence .. admiration
 (E) impatience .. training

5. The senator's remark that she is ambivalent about running for a second term is -----given

the extremely ----- fund-raising activities of her campaign committee.

- (A) disingenuous .. reluctant
- (B) futile .. clandestine
- (C) sincere .. visible
- (D) persuasive .. apathetic
- (E) straightforward .. energetic

6. Until quite recently research on diabetes had, as a kind of holding action, attempted to refine the -----of the disease, primarily because no preventive strategy seemed at all likely to be ----- .

- (A) definition .. necessary
- (B) anticipation .. acceptable
- (C) understanding .. costly
- (D) treatment .. practicable
- (E) symptoms .. feasible

7. Most plant species exhibit ----- in their geographical distribution: often, a given species is found over a large geographical area, but individual populations within that range are widely ----- .

- (A) discontinuity .. separated
- (B) density .. dispersed
- (C) symmetry .. observed
- (D) uniformity .. scattered
- (E) concentration .. adaptable

Directions: In each of the following questions, a related pair of words or phrases is followed by five lettered pairs of words or phrases. Select the lettered pair that best expresses a relationship similar to that expressed in the original pair.

8. FATIGUE: REST::

- (A) gravity: weight
- (B) friction: heat
- (C) dehydration: water
- (D) dizziness: vertigo
- (E) radiation: light

9. RECYCLE: DISPOSAL::

- (A) recommend: insistence
- (B) reciprocate: treatment
- (C) rehabilitate: demolition
- (D) attach: conquest
- (E) offer: sale

10. DICTIONARY: ALPHABETICAL::

- (A) map: contoured
- (B) diary: anecdotal
- (C) outline: detailed
- (D) narrative: prosaic
- (E) annals: chronological

11. ATTENUATE: THICKNESS::

- (A) separate: substance
- (B) ventilate: circulation
- (C) vaccinate: immunity
- (D) relocate: site
- (E) debilitate: strength

12. SATIRE: RIDICULE::

- (A) oration: enmity
- (B) lullaby: dream
- (C) parody: praise
- (D) elegy: sorrow
- (E) sonnet: remembrance

13. STOIC: PERTURB::

- (A) perplexed: enlighten
- (B) nondescript: neglect
- (C) tranquil: pacify
- (D) avaricious: satisfy
- (E) daunting: bewilder

14. EXCULPATORY: ABSOLVE::

- (A) motivational: stir
- (B) conventional: resist
- (C) rhetorical: speak
- (D) pedantic: learn
- (E) ponderous: choose

15. MODERATE: INTENSITY::

- (A) extenuate: seriousness
- (B) separate: distance
- (C) indulge: chaos

(D) commemorate: memorial (E) disparage: animosity 16. JOLT: MOVE:: (A) possess: acquire (B) arrive: remain (C) check: stop (D) spiral: turn (E) rattle: hear The origin of the theory that major geologic events may occur at regular intervals can be traced back not to a study of volcanism or plate tectonics but to an investigation of marine extinctions. In the early 1980's, (5) scientists began to look closely at the question of how these extinctions occur. Two paleontologists, Raup and Sepkoski, compiled a master list of marine species that died out during the past 268 million years and noted that there were brief periods during which (10) many species disappeared at once. These mass extinctions occurred at surprisingly regular intervals. Later studies revealed that extinctions of terrestrial reptiles and mammals also occurred periodically. These findings, combined with the research of Raup (15) and Sepkoski, led scientists to hypothesize the existence of some kind of cyclically recurring force powerful enough to affect living things profoundly. Speculation that so powerful a force might affect geologic events as well led geologists to search for (20) evidence of periodicity in episodes of volcanism, seafloor spreading, and plate movement. 17. According to the passage, Raup and Sepkoski's research was concerned with (A) learning more about the habitats of marine species (B) studying plate tectonics and the occurrence of volcanism over the past 268 million years (C) examining extinctions of marine species over the past 268 million years (D) finding out whether a rhythmically recurring geologic force exists (E) confirming previous evidence suggesting that extinction of terrestrial species occurred regularly	18. The author of the passage would most likely describe the findings of Raup and Sepkoski as (A) plausible, because the findings supported the theories of previous researchers (B) significant, because the findings were an impetus for subsequent research (C) controversial, because the findings contradicted the theories of previous researchers (D) questionable, because the authors were not working in their field of expertise (E) definitive, because the findings confirmed the existence of a rhythmically recurring force 19. The author of the passage is primarily concerned with (A) determining the dates of various geologic events (B) defending the conclusions reached by Raup and Sepkoski (C) establishing a link between the disciplines of paleontology and geology (D) proving that mass extinctions of marine animals occur periodically (E) explaining how a theory concerning geologic events was formulated 20. The passage suggests which of the following about the "force" mentioned in lines 16 and 18 ? (A) It is responsible for most of the major geologic events that have occurred. (B) It is responsible for most of the marine extinctions that have occurred. (C) Its recurrence is unlikely to be able to be predicted by scientists. (D) Its existence was not seriously considered by scientists before Raup and Sepkoski did their research. (E) Its existence was confirmed by the research of Raup and Sepkoski. A recent history of the Chicago meat-packing industry and its workers examines how the industry grew from its appearance in the 1830's through the
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early 1890's. Meat-packers, the author argues, had
 (5) good wages, working conditions, and prospects for advancement within the packinghouses, and did not cooperate with labor agitators since labor relations were so harmonious. Because the history maintains that conditions were above standard for the era, the
 (10) frequency of labor disputes, especially in the mid-1880's, is not accounted for. The work ignores the fact that the 1880's were crucial years in American labor history, and that the packinghouse workers' efforts were part of the national movement for labor
 (15) reform.

In fact, other historical sources for the late nineteenth century record deteriorating housing and high disease and infant mortality rates in the industrial community, due to low wages and unhealthy working
 (20) conditions. Additional data from the University of Chicago suggest that the packinghouses were dangerous places to work. The government investigation commissioned by President Theodore Roosevelt which eventually led to the adoption of the 1906
 (25) Meat Inspection Act found the packinghouses unsanitary, while social workers observed that most of the workers were poorly paid and overworked. The history may be too optimistic because most of its data date from the 1880's at the latest, and the infor-
 (30) mation provided from that decade is insufficiently analyzed. Conditions actually declined in the 1880's, and continued to decline after the 1880's, due to a reorganization of the packing process and a massive influx of unskilled workers. The deterioration in
 (35) worker status, partly a result of the new availability of unskilled and hence cheap labor, is not discussed. Though a detailed account of work in the packinghouses is attempted, the author fails to distinguish between the wages and conditions for skilled workers
 (40) and for those unskilled laborers who comprised the majority of the industry's workers from the 1880's on. While conditions for the former were arguably tolerable due to the strategic importance of skilled workers in the complicated slaughtering, cutting, and
 (45) packing process (though worker complaints about the rate and conditions of work were frequent), pay and conditions for the latter were wretched.

The author's misinterpretation of the origins of the

feelings the meat-packers had for their industrial
 (50) neighborhood may account for the history's faulty generalizations. The pride and contentment the author remarks upon were, arguably, less the products of the industrial world of the packers ---- the giant yards and the intricate plants ---- than of the unity and vibrance
 (55) of the ethnic cultures that formed a viable community on Chicago's South Side. Indeed, the strength of this community succeeded in generating a social movement that effectively confronted the problems of the industry that provided its livelihood.

21. The passage is primarily concerned with discussing

- (A) how historians ought to explain the origins of the conditions in the Chicago meat-packing industry
- (B) why it is difficult to determine the actual nature of the conditions in the Chicago meat-packing industry
- (C) why a particular account of the conditions in the Chicago meat-packing industry is inaccurate
- (D) what ought to be included in any account of the Chicago meat-packers' role in the national labor movement
- (E) what data are most relevant for an accurate account of the relations between Chicago meat-packers and local labor agitators

22. The author of the passage mentions all of the following as describing negative conditions in the meat-packing industry EXCEPT

- (A) data from the University of Chicago
- (B) a recent history of the meat-packing industry
- (C) social workers
- (D) historical sources for the late nineteenth century
- (E) government records

23. The author of the passage mentions the "social movement" (line 57) generated by Chicago's South Side community primarily in order to

- (A) inform the reader of events that occurred in the meat-packing industry after the period

- of time covered by the history
- (B) suggest the history's limitations by pointing out a situation that the history failed to explain adequately
- (C) salvage the history's point of view by suggesting that there were positive developments in the meat-packing industry due to worker unity
- (D) introduce a new issue designed to elaborate on the good relationship between the meat-packers and Chicago's ethnic communities
- (E) suggest that the history should have focused more on the general issue of the relationship between labor movements and healthy industrial communities
24. According to the passage, the working conditions of skilled workers in the meat-packing industry during the 1880's were influenced by
- (A) the workers' determined complaints about the rate and conditions of their work
- (B) the efforts of social workers to improve sanitation in the packinghouses
- (C) the workers' ability to perform the industry's complex tasks
- (D) improvements in the industry's packing process that occurred in the 1880's
- (E) opportunities for job advancement due to the filling of less desirable positions by increasing numbers of unskilled workers
25. The author of the passage uses the second paragraph to
- (A) summarize the main point of the history discussed in the passage
- (B) explain why the history discussed in the passage has been disparaged by critics
- (C) evaluate the findings of recent studies that undermine the premises of the history discussed in the passage
- (D) introduce a hypothesis that will be discussed in detail later in the passage
- (E) present evidence that is intended to refute the argument of the history discussed in the passage
26. The tone of the author of the passage in discussing the meat-packer community on Chicago's South Side can best be described as one of
- (A) appreciation of the community's ability to cope with difficult conditions
- (B) admiration for the community's refusal to cooperate with labor agitators
- (C) indignation at the kinds of social conditions the community faced
- (D) annoyance at the community's inability to abolish discrimination in the meat-packing industry
- (E) concern that the meat-packers' feelings for their community have not been documented
27. The information in the passage suggests that the author of the history discussed in the passage made which of the following errors?
- (A) Failing to recognize the effect of the diversity of the South Side community on the meat-packers' efforts to reform the industry
- (B) Attributing good working conditions in the meat-packing industry to the efforts of labor agitators
- (C) Overemphasizing the importance of the availability of unskilled labor as an influence on conditions in the meat packing industry
- (D) Interpreting the meat-packers' feelings for their community as appreciation of their industry
- (E) Failing to observe the pride and contentment felt by the meat-packers
28. CELEBRITY:
- (A) eccentricity
- (B) informality
- (C) obscurity
- (D) aloofness
- (E) nonchalance
29. CHRONIC:
- (A) imminent
- (B) asynchronous
- (C) sequential

- | | |
|--|--|
| <p>(D) sporadic
(E) spontaneous</p> <p>30. ACCUMULATION:
(A) severance
(B) dissipation
(C) reciprocity
(D) absolution
(E) remuneration</p> <p>31. CALCIFICATION:
(A) forgetfulness
(B) abundance
(C) streamlining
(D) clairvoyance
(E) flexibility</p> <p>32. MIGRATORY:
(A) speculative
(B) transitory
(C) sedentary
(D) kinetic
(E) convergent</p> <p>33. CIVILITY:
(A) impassivity
(B) rudeness
(C) indiscretion
(D) dubiety
(E) indolence</p> <p>34. VARIANCE:
(A) contingency
(B) congruity
(C) encumbrance
(D) usefulness
(E) distinctness</p> <p>35. GENIAL:
(A) dyspeptic
(B) ceremonious
(C) wistful
(D) ravishing
(E) variable</p> | <p>36. DIURNAL:
(A) predictable
(B) ephemeral
(C) primitive
(D) nocturnal
(E) vestigial</p> <p>37. APOLOGIST:
(A) accompanist
(B) protagonist
(C) suppliant
(D) critic
(E) conspirator</p> <p>38. VIRULENT:
(A) auspicious
(B) polite
(C) salubrious
(D) vanquished
(E) intermittent</p> |
|--|--|

SECTION 5

Time – 30 minutes

25 Questions

1. Chris: Hundreds of traffic accidents annually are attributable to the poor condition of our city's streets. The streets must therefore be repaired to save lives.

Leslie: For less than the cost of those repairs, the city could improve its mass transit system and thus dramatically reduce traffic congestion, which contributes significantly to those traffic accidents. The city cannot afford to do both, so it should improve mass transit, because reduced traffic congestion has additional advantages.

Which of the following best describes the point at issue between Chris and Leslie?

- (A) Whether a certain problem in fact exists
 - (B) How a certain problem came into being
 - (C) Who is responsible for addressing a certain problem
 - (D) Whether the city has sufficient financial resources to address a certain problem
 - (E) How the city can best address a certain problem
2. According to ancient records, the first tax that the government of Selea imposed on a basic commodity was a tax of two centima coins on every jar of cooking oil sold in Selea. Tax records show that despite a stable population and strict enforcement of tax laws, revenues from the oil tax declined steeply over the first two years that the tax was in effect.

Which of the following, if true, most helps to explain the decline in Selean oil-tax revenues?

- (A) During the decade following the implementation of the tax, the average household income in Selea rose steadily.
- (B) Two years after implementing the tax on cooking oil, the Selean government began to implement taxes on numerous other

basic commodities.

- (C) Jars of cooking oil were traditionally bought as wedding gifts in Selea at the time the tax went into effect, and gifts of cooking oil increased after the implementation of the tax.
- (D) After the tax was imposed., Selean merchants began selling cooking oil in larger jars than before.
- (E) Few Selean households began to produce their own cooking oil after the tax was imposed.

Questions 3-8

A small ski resort's ski patrol has exactly six members ---- F, G, J, K, M, and O ---- who communicate with each other by means of one-way and two-way radios. Each patrol member has a oneway radio capable of transmitting signals to exactly one other patrol member: F signals G, G signals J, J signals K, K signals M, M signals O, and O signals F. In addition, F and K have two-way radios enabling each to signal the other, and J and O have two-way radios enabling each to signal the other.

No other transmissions of signals between ski patrol members are possible.

No patrol member can transmit a particular message more than once.

Transmission of a message ceases once the intended recipient receives the message.

3. Which of the following patrol members can signal M directly?
- (A) F
 - (B) G
 - (C) J
 - (D) K
 - (E) O
4. If O wishes to send a communication to G using the fewest possible intermediaries, the person O signals directly must be
- (A) F
 - (B) G

- (C) J
(D) K
(E) M
5. Which of the following communications would require a minimum of two intermediaries?
- (A) From F to K
(B) From G to K
(C) From J to F
(D) From K to G
(E) From M to K
6. If a communication is to be sent from K to G via the fewest possible intermediaries, that communication must be sent
- (A) from K to F to G
(B) from K to J to G
(C) from K to O to G
(D) from K to O to J to G
(E) from K to M to O to F to G
7. A message could be sent by either of two different routes, each using the same number of intermediaries, from
- (A) F to O
(B) G to M
(C) G to K
(D) J to O
(E) M to G
8. A message could be sent by any of three different routes from
- (A) F to M
(B) G to F
(C) G to O
(D) J to K
(E) K to G
9. Housing construction materials give off distinctive sounds when exposed to high temperatures. Acoustic sensors accurately detect such sounds and fire alarms incorporating acoustic sensors can provide an early warning of house fires, allowing inhabitants to escape before being overcome by smoke. Since smoke inhalation is the most common cause of fatalities in house fires, mandating acoustic -sensor-based alarms instead of smoke detectors will eliminate house fire as a major cause of death.
- Which of the following, if true, most weakens the argument given?
- (A) The present high cost of acoustic-sensor-based alarm systems will decline if their use becomes widespread.
(B) When fully ignited, many materials used in housing construction give off sounds that are audible even from several hundred yards away.
(C) Many fires begin in cushions or in mattresses, producing large amounts of smoke without giving off any sounds.
(D) Two or more acoustic -sensor-based alarms would be needed to provide adequate protection in some larger houses.
(E) Smoke detectors have been responsible for saving many lives since their use became widespread.
10. In December 1992 Tideville Shopping Mall repaired and improved the lighting in the mall's parking lots, and in 1993 car thefts and attempted car thefts from those lots decreased by 76 percent from the previous year. Since potential car thieves are generally deterred by good lighting, the decrease can be attributed to these improvements.
- Which of the following, if true, most helps to strengthen the argument above?
- (A) Both in 1992 and in 1993, most of the cars stolen from the mall's parking lots were relatively new and expensive luxury models.
(B) Most of the cars that were stolen from the mall in 1992 were stolen between 11 A. M. and 4 P.M.
(C) Tideville Shopping Mall is one of only three shopping malls in the Tideville area.
(D) In the town of Tideville, where the mall is located, the number of car thefts was about the same in 1993 as in 1992.
(E) In 1993 the number of security officers patrolling the mall's parking lots at night was doubled.

11. Legislator: We should not waste any more of the taxpayers' money on the government's job-creation program. The unemployment rate in this country has actually risen since the program was begun, so the program has clearly been a failure.

Which of the following is an assumption on which the legislator's argument depends?

- (A) The budget of the job-creation program has typically increased every year.
- (B) The unemployment rate would not have risen even more than it has if the job-creation program had not been in existence.
- (C) The unemployment rate is higher now than at any time before the inception of the job-creation program.
- (D) If the job-creation program had been run more efficiently, it could have better served its purpose.
- (E) Other government programs are no more effective in reducing unemployment than is the job-creation program.

12. Which of the following most logically completes the argument?

Each year a consumer agency ranks all domestic airlines for on-time performance during the previous year, using as its sole criterion the percentage of each airline's flights that left no more than fifteen minutes late. The agency does not count delays due to mechanical reasons, but the fact that the percentage of delayed flights that were delayed for mechanical reasons was approximately the same for all domestic airlines last year means that ---- .

- (A) including delays for mechanical reasons in calculating the airline rankings for on-time performance would have had little, if any, effect on last year's rankings
- (B) airlines would work harder to reduce delays if delays for mechanical reasons were included in the determination of on-time performance rankings
- (C) the agency's rankings do not give consumers an accurate idea of how a given

airline compares to other airlines with respect to the percentage of flights delayed last year

- (D) those airlines with the best on-time performance record last year also had the greatest number of delays for mechanical reasons
- (E) on-time performance was approximately the same for all domestic airlines last year

Questions 13-18

Four types of vines – *F*, *G*, *H*, and *I* – are indigenous to an area that contains a small forest. The forest has three habitats – 1, 2, and 3 – which are located so that 1 is adjacent to 2 and 2 is adjacent to 3, but 1 is not adjacent to 3. In the forest, the nature of the habitats and of the vines causes the growth of the vines to conform to the following conditions:

Each habitat has either *F* vines or *H* vines or both.

G vines do not grow in habitat 2.

If *F* vines grow in a habitat, then *G* vines also grow in that habitat.

If *H* vines grow in a habitat, then *I* vines do not grow in any adjacent habitat.

If *I* vines grow in a habitat, then *F* vines grow in at least one adjacent habitat.

13. Which of the following can be a distribution of vine types in habitats?

<u>Habitat 1</u>	<u>Habitat 2</u>	<u>Habitat 3</u>
(A) <i>F</i> , <i>G</i>	<i>F</i> , <i>H</i>	<i>F</i> , <i>G</i> , <i>H</i>
(B) <i>F</i> , <i>G</i>	<i>G</i> , <i>H</i>	<i>G</i> , <i>H</i>
(C) <i>F</i> , <i>G</i> , <i>I</i>	<i>H</i> , <i>I</i>	<i>F</i> , <i>G</i>
(D) <i>G</i> , <i>H</i> , <i>I</i>	<i>H</i>	<i>F</i> , <i>G</i> , <i>H</i>
(E) <i>H</i>	<i>H</i>	<i>F</i> , <i>G</i> , <i>H</i>

14. Which of the following CANNOT be true?

- (A) *F* vines grow in habitat 2.
- (B) *F* vines grow in habitat 3.
- (C) *G* vines grow in habitat 1.
- (D) *G* vines grow in habitat 3.
- (E) *I* vines grow in habitat 2.

15. Which of the following must be true?

- (A) F vines grow in habitat 1.
- (B) F vines grow in habitat 3.
- (C) H vines grow in habitat 1.
- (D) H vines grow in habitat 2.
- (E) H vines grow in habitat 3.

16. Which of the following is a complete and accurate list of the habitats in which I vines can grow?

- (A) 2
- (B) 3
- (C) 1,2
- (D) 2,3
- (E) 1,2,3

17. If I vines grow in habitat 2, which of the following must be true?

- (A) F vines grow in 1.
- (B) F vines grow in 2.
- (C) H vines grow in 1.
- (D) H vines grow in 3.
- (E) I vines grow in 1.

18. If F vines do not grow in any of the habitats, which of the following must be true?

- (A) G vines grow in habitat 1.
- (B) G vines grow in habitat 3.
- (C) H vines grow in habitat 1.
- (D) H vines do not grow in any habitat.
- (E) I vines grow in exactly one of the habitats.

Questions 19-22

A team consisting of seven members – Fran, Irma, Jean, Karen, Lois, Maria, and Ruth – is scheduled to hold a special practice session in which four team members are to be assigned as players, one as the umpire, and two as spectators. The assignments will hold throughout the session and are to be made in accordance with the following conditions:

If Fran is a player, Irma must be a player.
 Karen and Lois cannot both be players.
 Neither Jean nor Lois nor Ruth can be the

umpire.

Jean and Karen cannot both be spectators.

19. If Karen is a spectator, which of the following must be true?

- (A) Irma is a player.
- (B) Jean is a player.
- (C) Lois is a player.
- (D) Maria is the umpire.
- (E) Ruth is a spectator.

20. If Karen is among the players, the two spectators can be

- (A) Fran and Irma
- (B) Fran and Jean
- (C) Irma and Jean
- (D) Irma and Lois
- (E) Maria and Ruth

21. Which of the following can be assigned as the group of players?

- (A) Fran, Irma, Karen, and Lois
- (B) Fran, Irma, Karen, and Maria
- (C) Fran, Jean, Lois, and Maria
- (D) Jean, Karen, Lois, and Ruth
- (E) Jean, Karen, Maria, and Ruth

22. If Maria is the umpire, the spectators can be

- (A) Fran and Irma
- (B) Fran and Jean
- (C) Irma and Karen
- (D) Jean and Ruth
- (E) Karen and Lois

23. No one can be licensed as an electrician in Parker County without first completing a certain course in electrical safety procedures. All students majoring in computer technology at Parker County Technical College must complete that course before graduating. Therefore, any of the college's graduates in computer technology can be licensed as an electrician in Parker County.

The answer to which of the following would be most helpful in evaluating the argument?

- (A) Is a college degree a requirement for being

licensed as an electrician in Parker County? (B) Do all students majoring in computer technology who complete the course in electrical safety procedures at Parker County Technical College eventually graduate? (C) Is completion of a course in electrical safety procedures the only way a person licensed as an electrician in Parker County can have learned those procedures? (D) Is a period of practical apprenticeship a requirement for becoming a licensed electrician in Parker County but not for graduating from the college in computer technology? (E) Do any of the students at Parker County Technical College who are not majoring in computer technology take the course in electrical safety procedures? 24. Pollutants in the atmosphere can cause acid rain (rain with high acidity levels). While acid rain in itself cannot significantly affect the acidity of bodies of water into which it falls, it can greatly increase the acidity of nearby lakes by increasing the amount of decaying matter on a forest floor. A recent increase in the acidity of the water in Forest Lake, therefore, surely indicates that the rain falling nearby has become more acid. Which of the following, if true, most seriously weakens the argument? (A) Even in areas without significant amounts of acid rain, most lakes in regions with vegetation similar to the vegetation around Forest Lake have acidity levels higher than those of other lakes. (B) Recent air-quality tests in the region around Forest Lake have revealed a slight increase in the amount of pollutants in the air. (C) Large-scale logging, which was recently begun in the forest surrounding Forest Lake, has increased the amount of decaying matter on the forest floor. (D) There is some disagreement among scientists about exactly how pollutants in the atmosphere cause acid rain.	(E) Decaying matter exists on all forest floors and is an important factor in maintaining the healthy growth of the forests. 25. Most of Earth's surface is ocean. The ocean floor is inaccessible for extensive research without equipment of greater technological sophistication than is currently available. It must therefore be true that scientists know less about the ocean floor environment than about almost any other environment on Earth. Which of the following, if true, provides the most support for the conclusion? (A) Many mountain ranges lie entirely beneath the ocean surface, yet new underwater surveying equipment has produced three-dimensional charts of them that are as accurate as those available for mountain ranges on land. (B) Strong water currents circulate on the ocean floor, but the general pattern of their movement is not so well understood as is the pattern of air currents that circulate over land. (C) In contrast to most land environments, temperature conditions at the ocean floor are generally stable and uniform, since sunlight does not penetrate far below the ocean surface. (D) Very few people have seen detailed maps of extended regions of the ocean floor, even though such maps are available in almost all large libraries. (E) Animals living on the ocean floor must be able to withstand water pressure that is far greater than the atmospheric pressure with which land animals live.
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SECTION 6

Time—30 minutes

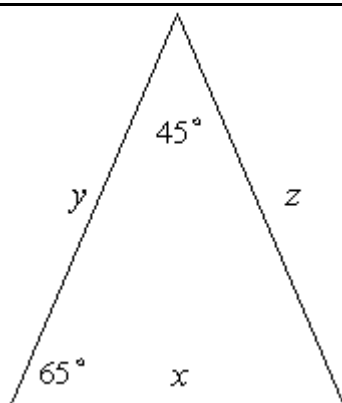
30 Questions

1. $\frac{2}{3}$ $\frac{5}{6}$

$$r = 7$$

$$s = -7$$

2. $2r - 2s + r^2$ $2s - 2r + s^2$



3. y z

$$0.1234 < n < 0.1245$$

4. The thousandths digit of n 4

5. The perimeter of a square with side of length $4x$ Four times the perimeter of a square with side of length x

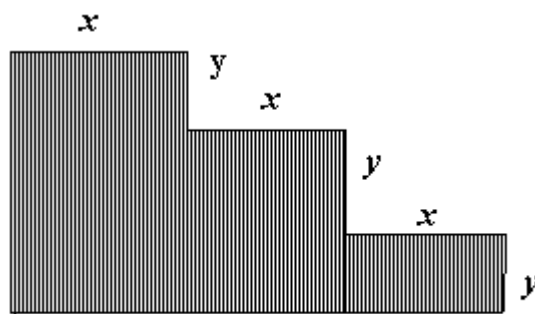
$$m - n = 0$$

$$mn \neq 0$$

6. $\frac{m}{m+n}$ $\frac{1}{3}$

$$10^x - 125y$$

7. x y



In the figure, all intersecting segments meet at right angles.

8. The area of the shaded region $6xy$

A total of 600 tickets to a concert were sold at prices ranging from \$10 to \$50 each.

9. The average (arithmetic mean) price per ticket $\$30$

10. $\frac{6}{\sqrt{3}}$ $2\sqrt{3}$

11. C 5

A light-year is the distance that light travels in one year, or approximately 9.5×10^{12} kilometers. A certain star is 2.4×10^{15} kilometers away from the Sun.

12. The number of light-years that the star is away from the Sun $1,000$

E is the sum of the first 40 positive even integers.

K is the sum of the first 40 positive odd integers.

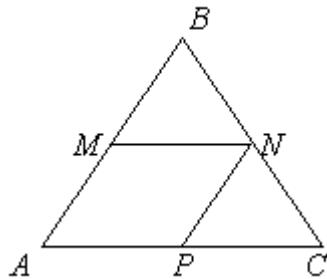
13. $E - K$ 40

14. $5(y - 4)$ $4(x - 5)$

The probability that event R will occur is 0.38.

15. The probability that events R and W will both occur 0.40

16. For a certain telephone company, it is projected that next year 2 out of every 3 new telephone numbers will be assigned to cellular telephones. If the company projects that a total of 1,200,000 new telephone numbers will be assigned next year, how many of these numbers are projected to be assigned to cellular telephones?
- (A) 800,000
(B) 600,000
(C) 400,000
(D) 360,000
(E) 240,000



17. In the figure above, M , N , and P are the midpoints on the respective sides of equilateral triangle ABC . If the perimeter of triangle ABC is 24, what is the perimeter of quadrilateral $AMNP$?
- (A) 8 (B) 16 (C) 32
(D) 48 (E) 64
18. Of the 500 delegates attending a convention, 200 are Republicans and the rest are Democrats. One hundred of the delegates are vegetarians and, of those who are not vegetarians, 270 are Democrats. How many of the vegetarian delegates are Republicans?
- (A) 30 (B) 40
(C) 50 (D) 60 (E) 70
19. If $3x = (x+2)(x-2)$, then x could be

- (A) 2
(B) 1
(C) $\frac{1}{4}$
(D) -1
(E) -4

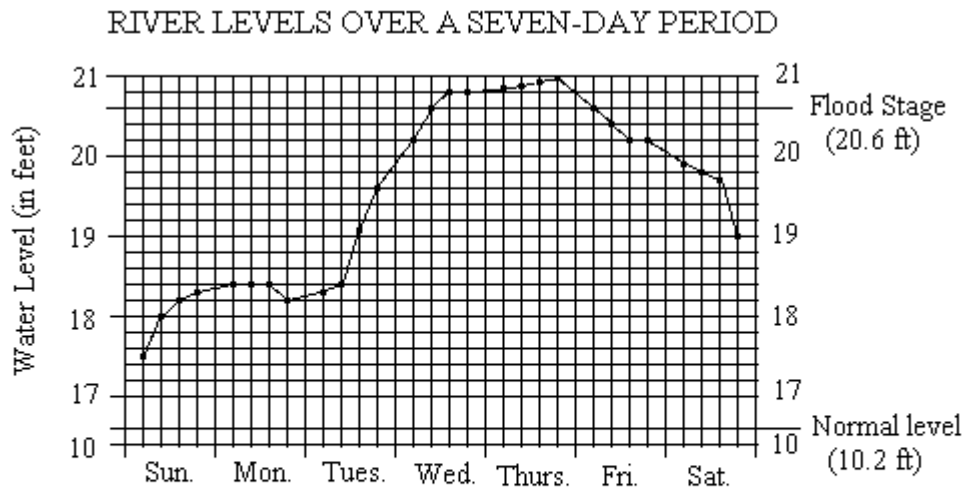
20. Art, Bob, and Carmen share a prize of \$400.

Bob receives $\frac{1}{2}$ as much as Carmen, how

If Art receives twice as much as Bob, and if much does Carmen receive?

- (A) \$20
(B) \$40
(C) \$80
(D) \$140
(E) \$160

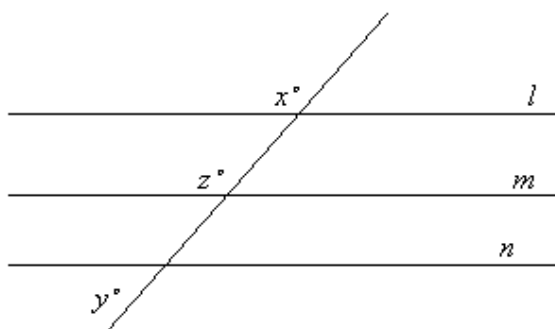
Questions 21-25 refer to the following graph.



Note: Drawn to scale:

Water level readings taken at the same spot of the river each day at 7:00 and 11:00 in the morning, at 3:00 in the afternoon, and at 7:00 in the evening, respectively

21. Approximately how many feet from flood stage was the water level at 11:00 in the morning on Tuesday?
- (A) 1.2
(B) 1.6
(C) 2.2
(D) 2.6
(E) 3.2
22. Which reading indicated the greatest rise in water level from the previous reading?
- (A) Sunday at 11:00 in the morning
(B) Tuesday at 3:00 in the afternoon
(C) Tuesday at 7:00 in the evening
(D) Wednesday at 7:00 in the morning
(E) Thursday at 7:00 in the evening
23. Approximately what was the average rise in water level per hour from the last reading on Tuesday to the first reading on Wednesday?
- (A) 0.05 feet
(B) 0.10 feet
(C) 0.15 feet
(D) 0.20 feet
(E) 0.25 feet
24. For how many successive pairs of readings was there a drop in water level of at least $\frac{1}{2}$ foot?
- (A) None
(B) One
(C) Two
(D) Three
(E) Four
25. The highest water level reading was approximately what percent greater than the lowest water level reading?
- (A) 4%
(B) 13%
(C) 16%
(D) 20%
(E) 24%



26. In the figure above, if lines l , m , and n are parallel and $z = x + 60$, what is the value of $x + y + z$?
- (A) 120
(B) 150
(C) 180
(D) 210
(E) 240
27. The average (arithmetic mean) of 6 numbers is 49. How much greater would the average of the 7 numbers consisting of the original 6 numbers and the number 112 be than the average of the original 6 numbers?
- (A) 9
(B) 31.5
(C) 55
(D) 63
(E) 80.5
28. According to a newspaper article about employment in Country J, 30.2 million workers, or 20.9 percent of the labor force of Country J, were union members. Which of the following equals the number of workers in millions in the labor force of Country J?
- (A) (30,2) (20.9)
(B) (30.2) (0.209)
(C) $\frac{20.9}{0.302}$
(D) $\frac{30.2}{20.9}$
(E) $\frac{30.2}{0.209}$

29. If $x = \frac{y}{7}$, $z = \frac{x}{y}$, and $y \neq 0$, what is the Value of $(z-1)^2$?

- (A) $\frac{1}{49}$
(B) $\frac{36}{49}$
(C) $\frac{8}{7}$
(D) $\frac{64}{49}$
(E) $\frac{36}{7}$
30. Which of the following CANNOT be expressed as the product of exactly 2 consecutive integers?
- (A) (2) (3) (7)
(B) (2) (3) (7) (11)
(C) (2) (5^2) (13)
(D) (2^2) (3) (5) (7)
(E) (3^2) (11) (13)

SECTION 1

Time—30 minutes

25 Questions

1. The Environmental Protection Agency must respond to the hazard to children's health posed by exposure to asbestos fibers released in the air in school classrooms. Since it is impossible to close school buildings, the best plan would be to initiate programs that mandate the immediate removal of asbestos from all the school buildings that are found to contain asbestos, regardless of whether or not the buildings are in use.

Which of the following, if true, is the strongest reason for the Environmental Protection Agency not to follow the plan outlined above?

- (A) The techniques available for removing asbestos often increase the level of airborne asbestos.
- (B) Schools are places where asbestos is especially likely to be released into the air by the action of the occupants.
- (C) Children exposed to airborne asbestos run a greater risk of developing cancer than do adults exposed to airborne asbestos.
- (D) The cost of removing asbestos varies from school to school, depending on accessibility and the quantity of asbestos to be removed.
- (E) It is impossible to determine with any degree of certainty if and when construction materials that contain asbestos will break down and release asbestos fibers into the air.

2. *Aedes albopictus*, a variety of mosquito that has recently established itself in the southeastern United States, is less widespread than the indigenous swamp mosquito. Both the swamp mosquito and *A. albopictus* can carry viruses that are sometimes fatal to humans, but *A. albopictus* is a greater danger to public health.

Each of the following, if true, provides additional information that strengthens the judgment given about the danger to public health EXCEPT:

- (A) Unlike the swamp mosquito, *A. albopictus* originated in Asia, and larvae of it were not observed in the United States before the mid-1980's.
- (B) Unlike the swamp mosquito, *A. albopictus* tends to spend most of its adult life near human habitation.
- (C) Unlike swamp mosquito larvae, *A. albopictus* larvae survive in flower pots, tin cans, and many small household objects that hold a little water.
- (D) In comparison with the swamp mosquito, *A. albopictus* hosts a much wider variety of viruses known to cause serious diseases in humans.
- (E) *A. albopictus* seeks out a much wider range of animal hosts than does the swamp mosquito, and it is more likely to bite humans.

Questions 3-8

The manager of a horse show is placing seven obstacles—one chicken coop, one gate, two stone walls, and three fences—on a jumping course that consists of seven positions, numbered and arranged consecutively from 1 to 7. The placement of the obstacles in the seven positions must conform to the following conditions:

- No two fences can be placed in consecutive positions.
- The stone walls must be placed in consecutive positions.

3. Which of the following is an acceptable placement of obstacles in the seven positions, in order from the first position to the last position on the course?
- (A) Chicken coop, fence, gate, stone wall, fence, stone wall, fence
 - (B) Fence, gate, fence, fence, chicken coop, stone wall, stone wall
 - (C) Fence, stone wall, stone wall, gate, chicken coop, fence, fence
 - (D) Gate, stone wall, stone wall, fence, fence, chicken coop, fence
 - (E) Stone wall, stone wall, fence, chicken coop, fence, gate, fence

4. If one of the fences is in the third position and another is in the sixth position, which of the following must be true?
- (A) The chicken coop is in the seventh position.
 (B) The gate is in the second position.
 (C) The gate is in the seventh position.
 (D) One of the stone walls is in the first position.
 (E) One of the stone walls is in the fourth position.
5. If one of the stone walls is in the seventh position, which of the following must be FALSE?
- (A) The chicken coop is in the second position.
 (B) The chicken coop is in the fourth position.
 (C) One of the fences is in the first position.
 (D) One of the fences is in the second position.
 (E) The gate is in the fourth position.
6. Which of the following CANNOT be the positions occupied by the three fences?
- (A) First, third, and fifth
 (B) First, third, and sixth
 (C) Second, fourth, and sixth
 (D) Second, fourth, and seventh
 (E) Third, fifth, and seventh
7. If a stone wall is placed immediately after the gate, which of the following is a complete and accurate list of the positions in which the gate can be placed?
- (A) Second, third (B) Second, fourth
 (C) Third, fourth (D) Second, third, fourth
 (E) Third, fourth, fifth
8. If the chicken coop is not placed immediately after any fence, which of the following is a complete and accurate list of the positions in which the chicken coop can be placed?
- (A) First, second, third
 (B) First, third, fourth
 (C) First, fourth, sixth
 (D) First, second, third, fourth
 (E) First, third, fourth, sixth
9. A person's cholesterol level will decline significantly if that person increases the

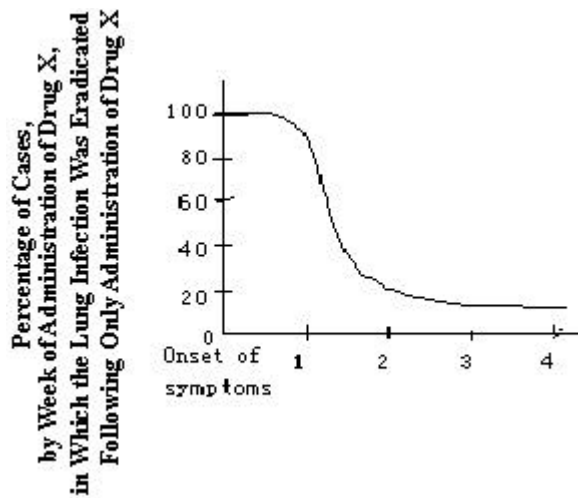
number of meals eaten per day, but only if there is no significant increase in the amount of food eaten. However, most people who increase the number of meals they eat each day will eat a lot more food as well.

If the statements above are true, which of the following is most strongly supported by them?

- (A) For most people, cholesterol level is not significantly affected by the amount of food eaten per day.
 (B) For most people, the amount of food eaten per meal is most strongly affected by the time of day at which the meal is eaten.
 (C) For most people, increasing the number of meals eaten per day will not result in a significantly lower cholesterol level.
 (D) For most people, the total amount of food eaten per day is unaffected by the number of meals eaten per day.
 (E) For most people, increasing the number of meals eaten per day will result in a significant change in the types of food eaten.
10. A certain type of dinnerware made in Ganandia contains lead. Lead can leach into acidic foods, and Ganandians tend to eat highly acidic foods. However, the extreme rarity of lead poisoning in Ganandia indicates that the dinnerware does not contain dangerous amounts of lead.
- Which of the following, if true, most seriously weakens the argument above?
- (A) The dinnerware is produced exclusively for sale outside Ganandia.
 (B) Ganandian foods typically are much more acidic than foods anywhere else in the world.
 (C) The only source of lead poisoning in Ganandia is lead that has leached into food.
 (D) Most people who use the dinnerware are not aware that it contains lead.
 (E) Acidic foods can leach lead from dinnerware even if that dinnerware has a protective coating.

Question 11 is based on the following graph.

EFFECTIVENESS OF DRUG X IN ERADICATING
A BACTERIAL LUNG INFECTION IN ADULT
PATIENTS



Point During the Course of the Infection
at Which Drug X Was First Administered to Patients
(in weeks following the onset of symptoms)

11. Drug X, which kills on contact the bacteria that cause the infection, is administered to patients by means of an aerosol inhaler.

Which of the following, if true, contributes most to explaining the change in drug X's effectiveness during the course of the infection?

- (A) Symptoms of the infection usually become evident during the first 48 hours following infection.
- (B) Most patients with lung infections say they prefer aerosol inhalers to other means of administering antibacterial drugs.
- (C) In most patients taking drug X, the dosage administered is increased slightly each week until symptoms disappear.
- (D) In patients who have the infection, the ability to inhale becomes increasingly impaired beginning in the second week after the onset of symptoms.
- (E) Drug X is not administered to any patient who shows signs of suffering from secondary infections.

12. Sergeant

Our police academy no longer requires its applicants to pass a physical examination before being admitted to the academy. As a result, several candidates with weak hearts and high blood pressure have been admitted. Hence, we can expect our future police force to have more health problems than our current police force.

Knowledge of each of the following would be relevant to determining the reliability of the sergeant's prediction EXCEPT whether

- (A) police officer candidates are screened for high blood pressure before joining the police force
- (B) the police officer candidates who are not healthy now are likely to be unhealthy as police officers
- (C) graduates of the police academy are required to pass a physical examination
- (D) the health of the current police officer candidates is worse than was the health of police officer candidates in the past
- (E) a police officer's health is a reliable indicator of the officer's performance

Questions 13-16

A transcontinental railroad train has exactly eight cars—J, K, L, M, N, O, P, and R—bound for several different destinations. The positions of the cars in any ordering of the train are numbered first through eighth from the front of the train. Because the cars will be detached at different points, certain ordering requirements must be met, as follows:

J must be somewhere behind M in the train.

K must be immediately in front of or immediately behind P.

O must be in front of N, and exactly one car must be between them.

R must be among the frontmost four cars and somewhere behind O.

13. Which of the following represents a possible order for the cars, from the front to the rear of the train?
 (A) L, M, O, R, N, J, K, P
 (B) M, K, P, O, R, N, L, J
 (C) M, L, O, R, N, K, J, P
 (D) O, R, M, N, P, K, J, L
 (E) P, K, R, L, O, M, N, J
14. If K is the first car, then the last car must be either
 (A) J or L (B) J or M (C) L or M
 (D) L or N (E) M or N
15. Which of the following can be neither the first nor the last car?
 (A) J (B) K (C) L (D) M (E) N
16. If R is somewhere behind N, which of the following must be true?
 (A) O is the first car.
 (B) M is the second car.
 (C) Either K or P is the last car.
 (D) L is one of the last four cars.
 (E) J is somewhere in front of K.

Questions 17-22

A mining company is planning a survey of exactly six regions-F, G, H, I, K, and L-for deposits of platinum and uranium. Each region will contain one of four possible combinations of minerals-both platinum and uranium, neither platinum nor uranium, platinum and no uranium, or uranium and no platinum. Prior to conducting a detailed survey, the mining company has the following information:

Exactly as many of the regions contain platinum deposits as contain uranium deposits.

Region F contains exactly the same deposits as does region H.

Regions G and I both contain uranium deposits.

Regions H and K both contain platinum deposits.

Regions G and L either both contain platinum deposits or neither of them does.

17. If there are exactly four regions that contain platinum deposits, these four could be
 (A) F, G, H, and K (B) F, G, H, and L
 (C) F, H, I, and K (D) F, H, K, and L
 (E) G, H, K, and L
18. If some region contains neither platinum deposits nor uranium deposits, it must be
 (A) F (B) G (C) H (D) I (E) L
19. If one of the six regions contains deposits of neither platinum deposits nor uranium deposits, which of the following CANNOT contain platinum deposits?
 (A) F (B) G (C) H (D) I (E) K
20. If exactly one region contains no platinum deposits, it must be
 (A) F (B) G (C) I (D) K (E) L
21. If K is the only region containing platinum deposits but no uranium deposits, which of the following must be two of the regions that contain both platinum deposits and uranium deposits?
 (A) F and G (B) F and H (C) G and L
 (D) H and I (E) I and L
22. If no region contains deposits of both platinum and uranium, which of the following must be true?
 (A) F contains uranium deposits.
 (B) G contains platinum deposits.
 (C) I contains platinum deposits.
 (D) K contains uranium deposits.
 (E) L contains uranium deposits.
23. Because adult iguanas on Plazos Island are much smaller than adult iguanas of the same species on nearby islands, researchers assumed that environmental conditions on Plazos favor the survival of relatively smaller baby iguanas (hatchlings) in each yearly brood. They discovered instead that for each of the past three years, 10 percent of the smaller and 40 percent of the larger hatchlings survived, because larger

hatchlings successfully evade their predators.

Which of the following, if true about Plazos but not about nearby islands, contributes most to an explanation of the long-standing tendency of iguanas on Plazos to be smaller than those of the same age on nearby islands?

- (A) Periodic wind shifts cause extended dry spells on Plazos every year, putting the larger iguanas, whose bodies require relatively more water, at a great disadvantage.
- (B) There are exactly three species of iguanas on Plazos but only two species of seagulls that feed on iguanas, and a relatively small percentage of each year's hatchlings are consumed by seagulls.
- (C) Wild cats, which were introduced as pets by early settlers and which were formerly major predators of Plazos iguanas, were recently killed off by a disease specific to cats.
- (D) The iguanas on Plazos are a relatively ancient part of the island's animal life.
- (E) Both land and marine iguanas live on Plazos, and the land iguanas tend to be larger than marine iguanas of the same age.

24. Every human being who has ever lived had two parents. Therefore, more people were alive three thousand years ago than are alive now.

The reasoning in the argument is flawed because it

- (A) overlooks the number of people in each generation during the last three thousand years who left no descendants
- (B) disregards possible effects of disasters such as famines and plagues on human history
- (C) overestimates the mathematical effect of repeated doublings on population size
- (D) fails to take into account that people now alive have overlapping sets of ancestors
- (E) fails to consider that accurate estimation of the number of people alive three thousand years ago might be impossible

25. Each of the academic journals *Thought* and *Ergo* has a review committee to prevent misattributed quotations from appearing in its published articles. Nevertheless, about ten percent of the quotations in *Thought*'s published articles are misattributed, whereas *Ergo* contains no misattributions. *Ergo*'s committee is more effective, therefore, than *Thought*'s at finding misattributed quotations.

The argument above assumes that

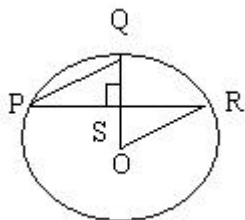
- (A) most of the articles submitted to *Thought* for publication contain misattributed quotations
- (B) there are at least some misattributed quotations in articles submitted to *Ergo* for publication
- (C) the members of *Ergo*'s committee are, on the whole, more knowledgeable than are the members of *Thought*'s committee
- (D) the number of misattributed quotations in a journal is an accurate measure of how carefully that journal is edited
- (E) the authors who submit articles to *Ergo* for publication are more thorough in attributing quotations than are the authors who submit articles to *Thought*

SECTION 2

Time—30 minutes

25 Questions

1. 60 percent of 16 10

O is the center of the circle and $OS = SQ$.

2. PQ OR

n is an integer such that $1 < n < 4$.

- 3.
- $n-1$
- 2

$$\frac{1}{2}X + 9 = 20$$

- 4.
- $x+19$
- 40

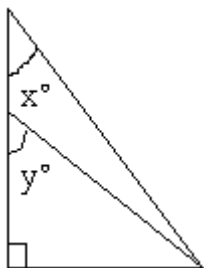
Board A measures between 2.15 feet and 2.25 feet in length; board B measures between 2.20 feet and 2.30 feet in length.

5. The length of board A The length of board B

$$t^2 - 2t = 0$$

$$t \neq 0$$

6. t 2



7. x y

- 8.
- $x(y+z)$
- $xy+z$

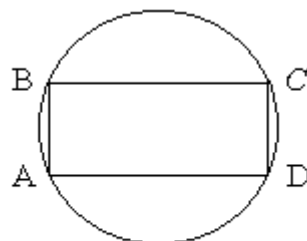
- 9.
- 10^4
- 5^6

10. x 120

- 11.
- $\left(\frac{1.5}{2.5}\right)^2$
- 0.36

For each positive integer n, $a_n = \frac{1}{n} - \frac{1}{n+1}$

- 12.
- $a_1 + a_2 + a_3 + a_4$
- $\frac{4}{5}$



The radius of the circle is r.

13. The area of rectangular region ABCD
- r^2

S is a set of n consecutive integers.

14. The mean of S The median of S

The length of a rectangular box is 4 inches longer than the depth, and the width of the box is 1 inch less than the length. The depth of the box is between 2 inches and 4 inches.

15. The volume of the box in cubic inches 200

16. In a circle graph used to represent a budget totaling \$600, the measure of the central angle associated with a \$120 item in the budget is
-
- (A)
- 72°
- (B)
- 108°
- (C)
- 120°
- (D)
- 144°
- (E) 216

- 17.
- $\left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) + \left(\frac{1}{4} + \frac{1}{2}\right) =$

- (A) 0 (B) $\frac{1}{4}$ (C) $\frac{1}{2}$
(D) 1 (E) $\frac{5}{4}$

18. The vertices of square S have coordinates $(-1,-2)$, $(-1,1)$, $(2,1)$, and $(2,-2)$, respectively. What are the coordinates of the point where the diagonals of S intersect?

- (A) $(\frac{1}{2}, \frac{1}{2})$ (B) $(\frac{1}{2}, -\frac{1}{2})$
 (C) $(\frac{3}{2}, \frac{1}{2})$ (D) $(\frac{3}{2}, -\frac{1}{2})$
 (E) $(\frac{\sqrt{3}}{2}, \frac{1}{2})$

19. The admission price per child at a certain

amusement park is $\frac{7}{12}$ of the admission price per adult. If the admission price for 4 adults and 6 children is \$112.50, what is the admission price per adult?

- (A) \$15.00 (B) \$13.50 (C) \$12.75
 (D) \$11.25 (E) \$8.75

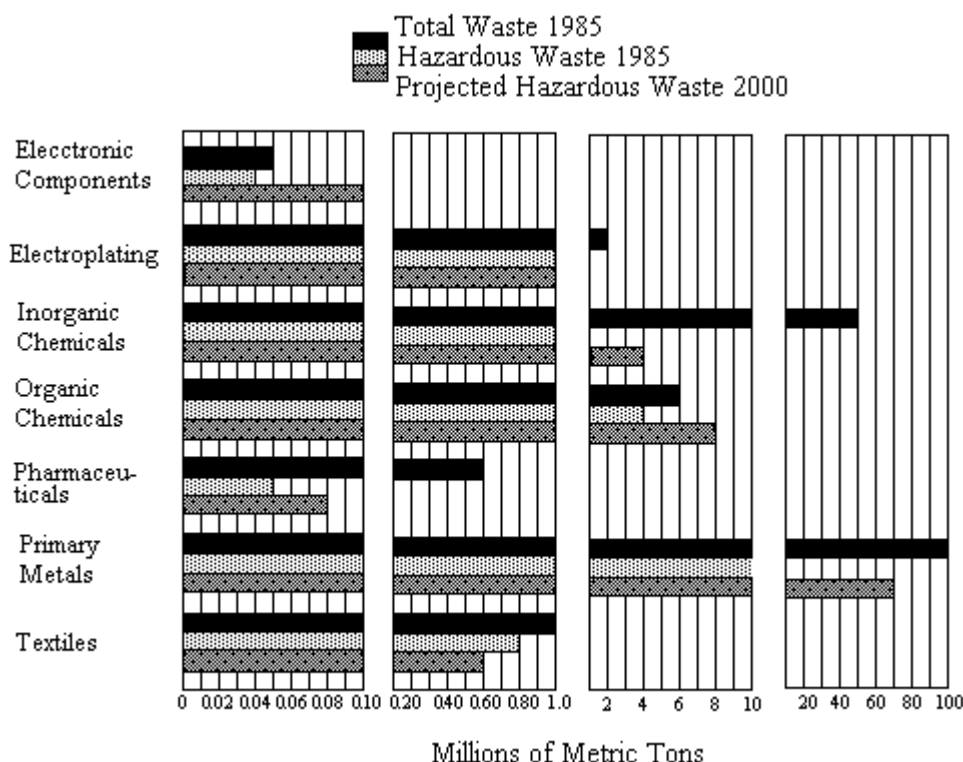
20. If $x=2y$ and $y=2z/3$, what is the value of z in terms of x ?

- (A) $\frac{2x}{3}$ (B) $\frac{3x}{4}$ (C) $\frac{4x}{3}$
 (D) $\frac{3x}{2}$ (E) $3x$

Questions 21-25 refer to the following graph.

INDUSTRIAL WASTE GENERATED BY SPECIFIC INDUSTRIES OF COUNTRY X

Note: Because of the great disparity in the amounts of waste generated by different industries, the graph is broken in three places, and after each break, a new and more appropriate scale is introduced. As usual, the value represented by a bar is read only at its far right end.



21. How many million metric tons of hazardous waste was produced in 1985 by the inorganic and organic chemicals industries combined?
- (A) 66 (B) 16 (C) 10
 (D) 5 (E) 3

22. For those industries that generated a total of more than a million metric tons of waste in 1985, what was the approximate average (arithmetic mean) total waste, in millions of metric tons, generated per industry?
- (A) 42 (B) 34 (C) 28
 (D) 25 (E) 23

23. In 1985 hazardous waste in electroplating exceeded hazardous waste in electronic components by how many million metric tons?
 (A) 1.96 (B) 1.50 (C) 0.96
 (D) 0.80 (E) 0.60
24. In 1985 the pharmaceuticals industry generated total waste equal to how many times the hazardous waste in the same industry?
 (A) 1.2 (B) 2.5 (C) 3
 (D) 6 (E) 12
25. For which of the following industries is the hazardous waste projection for the year 2000 at least double its 1985 level?
 I. Electronic components
 II. Electroplating
 III. Inorganic chemicals
 (A) I only
 (B) I and II only
 (C) I and III only
 (D) II and III only
 (E) I, II, and III
26. Sixty-eight people are sitting in 20 cars and each car contains at most 4 people. What is the maximum possible number of cars that could contain exactly 1 of the 68 people?
 (A) 2 (B) 3 (C) 4
 (D) 8 (E) 12
27. The width of a rectangular playground is 75 percent of the length. If the perimeter of the playground is 280 meters, how long, in meters, is a straight path that cuts diagonally across the playground from one corner to another?
 (A) 60
 (B) 70
 (C) 80
 (D) 90
 (E) 100
28. Which of the following numbers is NOT the sum of three consecutive odd integers?
 (A) 15
- (B) 75
 (C) 123
 (D) 297
 (E) 313
29. If $72.42 = k(24 + \frac{n}{100})$, where k and n are positive integers and $n < 100$, then $k + n =$
 (A) 17
 (B) 16
 (C) 15
 (D) 14
 (E) 13
30. Which of the following pairs of numbers has an average (arithmetic mean) of 2?
 (A) $2 - \sqrt{2}$, $4 - \sqrt{2}$
 (B) $2\sqrt{3}$, $2 - 2\sqrt{3}$
 (C) $\frac{1}{0.5}$, $\frac{2.4}{1.6}$
 (D) $\sqrt{5}$, $\sqrt{3}$
 (E) $\frac{1}{2}$, $\frac{1}{2}$

SECTION 3

Time-30 minutes

Questions

1. What these people were waiting for would not have been apparent to others and was perhaps not very---their own minds.
(A) obscure to
(B) intimate to
(C) illusory to
(D) difficult for
(E) definite in
2. The attempt to breed suitable varieties of jojoba by using hybridization to---favorable traits was finally abandoned in favor of a simpler and much faster---: the domestication of flourishing wild strains.
(A) eliminate.. alternative
(B) reinforce.. method
(C) allow.. creation
(D) reduce.. idea
(E) concentrate.. theory
3. According to one political theorist, a regime that has as its goal absolute---, without any---law or principle, has declared war on justice.
(A) respectability.. codification of
(B) supremacy .. suppression of
(C) autonomy .. accountability to
(D) fairness .. deviation from
(E) responsibility .. prioritization of
4. Despite its ---, the book deals---with a number of crucial issues.
(A) optimism .. cursorily
(B) importance .. needlessly
(C) virtues .. inadequately
(D) novelty .. strangely
(E) completeness .. thoroughly
5. Although frequent air travelers remain unconvinced, researchers have found that, paradoxically, the---disorientation inherent in jet lag also may yield some mental health---.
(A) temporal.. benefits
(B) acquired.. hazards
(C) somatic .. disorders
(D) random .. deficiencies
(E) typical .. standards
6. Ironically, the proper use of figurative language must be based on the denotative meaning of the words, because it is the failure to recognize this--- meaning that leads to mixed metaphors and their attendant incongruity.
(A) esoteric
(B) literal
(C) latent
(D) allusive
(E) symbolic
7. Although it seems---that there would be a greater risk of serious automobile accidents in densely populated areas, such accidents are more likely to occur in sparsely populated regions.
(A) paradoxical
(B) axiomatic
(C) anomalous
(D) irrelevant
(E) portentous
8. CATASTROPHE: MISHAP::
(A) prediction: recollection
(B) contest: recognition
(C) humiliation: embarrassment
(D) reconciliation: solution
(E) hurdle: challenge
9. SONNET: POET::
(A) stage: actor
(B) orchestra: conductor
(C) music: dancer
(D) canvas: painter
(E) symphony: composer
10. LOQUACIOUS: SUCCINCT::
(A) placid: indolent
(B) vivacious: cheerful
(C) vulgar: offensive
(D) pretentious: sympathetic
(E) adroit: ungainly

11. DEPORTATION: COUNTRY::

- (A) evacuation: shelter
- (B) abdication: throne
- (C) extradition: court
- (D) eviction: dwelling
- (E) debarkation: destination

12. MAELSTROM: TURBULENT::

- (A) stricture: imperative
- (B) mirage: illusory
- (C) antique: rare
- (D) myth: authentic
- (E) verdict: fair

13. ABSTEMIOUS: INDULGE::

- (A) affectionate: embrace
- (B) austere: decorate
- (C) articulate: preach
- (D) argumentative: harangue
- (E) affable: jest

14. BLUSTERING: SPEAK::

- (A) grimacing: smile
- (B) blinking: stare
- (C) slouching: sit
- (D) jeering: laugh
- (E) swaggering: walk

15. SOLACE: GRIEF::

- (A) rebuke: mistake
- (B) mortification: passion
- (C) encouragement: confidence
- (D) justification: action
- (E) pacification: anger

16. INDELIBLE: FORGET::

- (A) lucid: comprehend
- (B) astounding: expect
- (C) inconsequential: reduce
- (D) incorrigible: agree
- (E) fearsome: avoid

Investigators of monkeys' social behavior have always been struck by monkeys' aggressive potential and the consequent need for social control of their aggressive behavior. Studies directed at describing aggressive behavior and the situations that elicit it, as well as the social mechanisms that control it, were therefore among the first investigations of monkeys' social behavior.

- (5) Investigators initially believed that monkeys would compete for any resource in the environment: hungry monkeys would fight over food, thirsty monkeys would fight over water, and, in general, any time more than one monkey in a group sought the same incentive simultaneously, a dispute would result and would be resolved through some form of aggression. However, the motivating force of competition for incentives began to be doubted when experiments like Southwick's on the reduction of space or the withholding of food failed to produce more than temporary increases in intragroup aggression. Indeed, food deprivation not only failed to increase aggression but in some cases actually resulted in decreased frequencies of aggression.

- (20) Studies of animals in the wild under conditions of extreme food deprivation likewise revealed that starving monkeys devoted almost all available energy to foraging, with little energy remaining for aggressive interaction. Furthermore, accumulating evidence from later studies of a variety of primate groups, for example, the study conducted by Bernstein, indicates that one of the most potent stimuli for eliciting aggression is the introduction of an intruder into an organized group. Such introductions result in far more serious aggression than that produced in any other types of experiments contrived to produce competition.

- (35) These studies of intruders suggest that adult members of the same species introduced to one another for the first time show considerable hostility because, in the absence of a social order, one must be established to control interanimal relationships. When a single new animal is introduced into an existing social organization, the newcomer meets even more serious aggression. Whereas in the first case aggression establishes a social order, in the second case resident animals mob the intruder, thereby initially excluding the new animal from the existing social unit. The simultaneous introduction of several animals lessens the effect, if only because the group divides its attention among the multiple targets. If, however, the several animals introduced to a group constitute their own social unit, each group may fight the opposing group as a unit; but, again, no individual is subjected to mass attack, and the very cohesion of the groups precludes prolonged individual combat. The submission of the defeated group, rather than unleashing unchecked aggression on the part of the victorious group, reduces both the intensity and frequency of further attack. Monkey groups therefor see to be organized primarily to maintain their established social order rather than to engage in hostilities per se.

17. The author of the passage is primarily concerned with
- (A) advancing a new methodology for changing a monkey's social behavior
 - (B) comparing the methods of several research studies on aggression among monkeys
 - (C) explaining the reasons for researchers' interest in monkeys' social behavior
 - (D) discussing the development of investigators' theories about aggression among monkeys
 - (E) examining the effects of competition on monkeys' social behavior
18. Which of the following best summarizes the findings reported in the passage about the effects of food deprivation on monkeys' behavior?
- (A) Food deprivation has no effect on aggression among monkeys.
 - (B) Food deprivation increases aggression among monkeys because one of the most potent stimuli for eliciting aggression is the competition for incentives.
 - (C) Food deprivation may increase long-term aggression among monkeys in a laboratory setting, but it produces only temporary increases among monkeys in the wild.
 - (D) Food deprivation may temporarily increase aggression among monkeys, but it also leads to a decrease in conflict.
 - (E) Food deprivation decreases the intensity but not the frequency of aggressive incidents among monkey.
19. According to the author, studies such as Southwick's had which of the following effects on investigators theories about monkeys' social behavior?
- (A) They suggested that existing theories about the role of aggression among monkeys did not fully account for the monkeys' ability to maintain an established social order.
 - (B) They confirmed investigators' theories about monkeys' aggressive response to competition for food and water.
 - (C) They confirmed investigators' beliefs about the motivation for continued aggression among monkeys in the same social group.
 - (D) They disproved investigators' theory that the introduction of intruders in an organized monkey group elicits intragroup aggressive behavior.
 - (E) They cast doubt on investigators' theories that could account for observed patterns of aggression among monkeys.
20. The passage suggests that investigators of monkeys social behavior have been especially interested in aggressive behavior among monkeys because
- (A) aggression is the most common social behavior among monkeys
 - (B) successful competition for incentives determines the social order in a monkey group
 - (C) situations that elicit aggressive behavior can be studied in a laboratory
 - (D) most monkeys are potentially aggressive, yet they live in social units that could not function without control of their aggressive impulses
 - (E) most monkeys are social, yet they frequently respond to newcomers entering existing social units by attacking them
21. It can be inferred from the passage that the establishment and preservation of social order among a group of monkeys is essential in order to
- (A) keep the monkeys from straying and joining other groups
 - (B) control aggressive behavior among group members
 - (C) prevent the domination of that group by another
 - (D) protect individuals seeking to become members of that group from mass attack
 - (E) prevent aggressive competition for incentives between that group and another
22. The passage supplies information to answer which of the following questions?
- (A) How does the reduction of space affect

intragroup aggression among monkeys in an experimental setting?

- (B) Do family units within a monkey social group compete with other family units for food?
- (C) What are the mechanisms by which the social order of an established group of monkeys controls aggression within that group?
- (D) How do monkeys engaged in aggression with other monkeys signal submission?
- (E) Do monkeys of different species engage in aggression with each other over food?

23. Which of the following best describes the organization of the second paragraph?

- (A) A hypothesis is explained and counter evidence is described.
- (B) A theory is advanced and specific evidence supporting it is cited.
- (C) Field observations are described and a conclusion about their significance is drawn.
- (D) Two theories are explained and evidence supporting each of them is detailed.
- (E) An explanation of a general principle is stated and specific examples of its operation are given.

Analysis of prehistoric air trapped in tiny bubbles beneath the polar ice sheets and of the composition of ice surrounding those bubbles suggests a correlation between carbon dioxide levels in the Earth's atmosphere and global

- (5) temperature over the last 160,000 years. Estimates of global temperature at the time air in the bubbles was trapped rely on measuring the relative abundances of hydrogen and its heavier isotope, deuterium, in the ice surrounding the bubbles. When global temperatures are relatively low,
- (10) water containing deuterium tends to condense and precipitate before reaching the poles; thus, ice deposited at the poles when the global temperature was cooler contained relatively less deuterium than ice deposited at warmer global temperatures. Estimates of global temperature based
- (15) on this information, combined with analysis of the carbon dioxide content of air trapped in ice deep beneath the polar surface, suggest that during periods of postglacial warming carbon dioxide in the Earth's atmosphere increased by approximately 40 percent.

24. In the passage, the author is primarily concerned with doing which of the following?

- (A) Describing a new method of estimating decreases in global temperature that have occurred over the last 160,000 years
- (B) Describing a method of analysis that provides information regarding the relation between the carbon dioxide content of the Earth's atmosphere and global temperature
- (C) Presenting information that suggests that global temperature has increased over the last 160,000 years.
- (D) Describing the kinds of information that can be gleaned from a careful analysis of the contents of sheets
- (E) Demonstrating the difficulty of arriving at a firm conclusion regarding how increases in the amount of carbon dioxide in the Earth's atmosphere affect global temperature

25. It can be inferred from the passage that during periods of post glacial warming, which of the following occurred?

- (A) The total volume of air trapped in bubbles beneath the polar ice sheets increased.
- (B) The amount of deuterium in ice deposited at the poles increased.
- (C) Carbon dioxide levels in the Earth atmosphere decreased.
- (D) The amount of hydrogen in the Earth's atmosphere decreased relatively the amount of deuterium
- (E) The rate at which ice was deposited at the poles increased

26. The author states that there is evidence to support which of the following assertions?

- (A) Estimates of global temperature that rely on measurements of deuterium in ice deposited at the poles are more reliable than those based on the amount of carbon dioxide contained in air bubbles beneath the polar surface.
- (B) The amount of deuterium in the Earth's atmosphere tends to increase as global temperature decreases.
- (C) Periods of post glacial warming are

- characterized by the presence of increased levels of carbon dioxide in the Earth's atmosphere.
- (D) Increases in global temperature over the last 160,000 years are largely the result of increases in the ratio of deuterium to hydrogen in the Earth's atmosphere.
- (E) Increases in global temperature over the last 160,000 years have been accompanied by decreases in the amount of deuterium in the ice deposited at the poles.
27. It can be inferred from the passage that the conclusion stated in the last sentence would need to be reevaluated if scientists discovered that which of the following were true?
- (A) The amount of deuterium in ice deposited on the polar surface is significantly greater than the amount of deuterium in ice located deep beneath the polar surface.
- (B) Both the air bubbles trapped deep beneath the polar surface and the ice surrounding them contain relatively low levels of deuterium.
- (C) Air bubbles trapped deep beneath the polar surface and containing relatively high levels of carbon dioxide are surrounded by ice that contained relatively low levels of deuterium.
- (D) The current level of carbon dioxide in the Earth's atmosphere exceeds the level of carbon dioxide in the prehistoric air trapped beneath the polar surface.
- (E) Increases in the level of carbon dioxide in the Earth's atmosphere are accompanied by increases in the amount of deuterium in the ice deposited at the poles.
28. CUMBERSOME:
- (A) likely to succeed
- (B) reasonable to trust
- (C) valuable to have
- (D) easy to handle
- (E) important to know
29. INDUCEMENT:
- (A) reproof
- (B) deterrent
- (C) partiality
- (D) distinction
- (E) consideration
30. STARTLE :
- (A) appease
- (B) lull
- (C) reconcile
- (D) dally
- (E) slumber
31. ANOMALY:
- (A) derivation from estimates
- (B) conformity to norms
- (C) return to origins
- (D) adaptation to stresses
- (E) repression of traits
32. RECIPROCATING:
- (A) releasing slowly
- (B) calculating approximately
- (C) accepting provisionally
- (D) moving unidirectionally
- (E) mixing thoroughly
33. MOLLYCODDLE:
- (A) talk boastfully
- (B) flee swiftly
- (C) treat harshly
- (D) demand suddenly
- (E) adjust temporarily
34. SURFEIT:
- (A) affirmation
- (B) compromise
- (C) dexterity
- (D) deficiency
- (E) languor
35. SANGUINE:
- (A) morose
- (B) puzzled
- (C) gifted
- (D) witty
- (E) persistent

36. RETROSPECTIVE:

- (A) irresolute
 (B) hopeful
 (C) unencumbered
 (D) evanescent
 (E) anticipatory

37. ENCOMIUM:

- (A) biased evaluation
 (B) polite response
 (C) vague description
 (D) harsh criticism
 (E) sorrowful expression

38. FACTIONAL:

- (A) excessive
 (B) undistinguished
 (C) disdainful
 (D) disinterested
 (E) disparate

SECTION 4

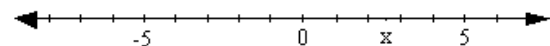
Time—30 minutes

30 Questions

1. 1 $\frac{1}{2^1} + \frac{1}{2^1} + \frac{1}{2^3}$

r and s are integers, and $r < s$.

2. The number of odd integers between r and s The number of even integers between r and s.

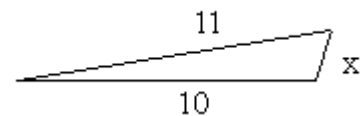


3. $x-y$ $x+y$

4. $10x-x$ 10

On a drawing done to scale, $\frac{1}{4}$ inch represents 5 feet.

5. The number of inches on the drawing that represents 150 feet



6. x 3

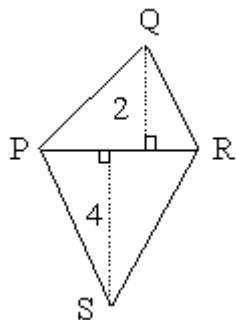
$$y > 0$$

$$x = 3y$$

7. 20 percent of x 50 percent of y

In a certain order of goods, $\frac{1}{3}$ of the items are shirts costing \$18 each and $\frac{2}{3}$ of the items are hats costing \$12 each.

8. The average (arithmetic mean) cost per item in the order \$15



9. The area of triangular region PRS is twice the area of triangular region PQR

$$xy < 0$$

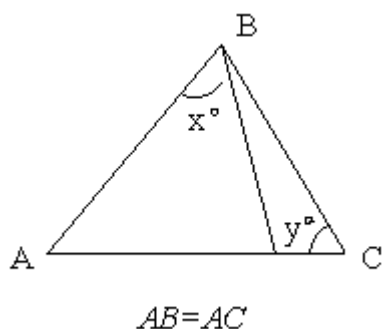
10. $xy > 0$

11. The average (arithmetic mean) of $\frac{10}{11}$ and $\frac{11}{10}$ is 1

$$d > 0$$

12. The area of a circular region with diameter d is equal to the area of a square region with diagonal of length d

13. $\frac{(0.83)^6}{(0.83)^7} = 1$



14. $x > y$

15. $(2x+3y)^2 = 4x^2 + 6xy + 9y^2$

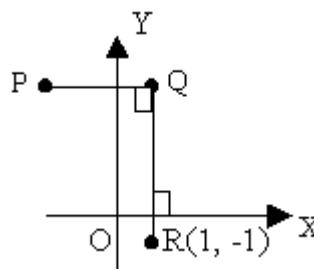
**TEMPERATURES IN DEGREES FAHRENHEIT
RECORDED AT NOON ON THE FIRST FOUR
DAYS OF
CERTAIN MONTHS**

Month	Temperatures
January	32,14,24,28
April	45,50,58,47
June	76,80,74,79
August	84,95,100,89
November	48,43,39,42

16. In a set of measurements, the range is defined as the greatest measurement minus the least measurement.

According to the table above, during the first four days of which month was the range of temperatures at noon the greatest?

- (A) January
(B) April
(C) June
(D) August
(E) November



17. In the figure above, if $QR=4$ and $PQ=3$, then the (x,y) coordinates of point P are

- (A) $(-4, 4)$
(B) $(-3,4)$
(C) $(-3,3)$
(D) $(-2,3)$
(E) $(-2,4)$

18. If $x^2=18$, then $|x| =$

- (A) -9
(B) 9

19. If $y-2x=-6$, then $8x-4y=$

- (A) 24

- (B) $\frac{3}{2}$
 (C) 0
 (D) $-\frac{3}{2}$
 (E) -24

20. A car gets 22 miles per gallon using gasoline

costing \$1.10 per gallon. What is the approximate cost, in dollars, for driving the car x miles using this gasoline?

- (A) $0.50x$
 (B) $0.30x$
 (C) $0.11x$
 (D) $0.10x$
 (E) $0.05x$

Questions 21-25 refer to the following table.

POPULATION DATA FOR TEN SELECTED STATES IN 1980 AND 1987

State	Population(in thousands)		Percent Change in Population, 1980-1987	Population Per Square Mile in 1987
	1980	1987		
A	23, 668	27, 663	16. 9	177
B	17, 558	17, 825	1. 5	372
C	14, 229	16, 789	18. 0	64
D	9, 746	12, 023	23. 4	222
E	11, 864	11, 936	0. 6	266
F	11, 427	11, 582	1. 4	208
G	10, 798	10, 784	-0. 1	263
H	9, 262	9, 200	-0. 7	162
I	7, 365	7, 672	4. 2	1, 027
J	5, 882	6, 413	9. 0	131

21. Which of the following states had the most land area in 1987?

- (A) A (B) B (C) C (D) D (E) E

22. In 1987 the average (arithmetic mean) population of the three most populous of the ten selected states was most nearly equal to.

- (A) 18 million (B) 19 million
 (C) 20 million (D) 21 million (E) 22 million

23. If the land area of State J was the same in 1980 as it was in 1987, then the population square mile of State J in 1980 was most nearly equal to

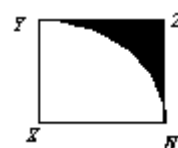
- (A) 140 (B) 130 (C) 120 (D) 110 (E) 100

24. If ranked from highest to lowest according to population, how many of the ten states changed in rank from 1980 to 1987?

- (A) One (B) Two (C) Three
 (D) Four (E) Five

25. Of the following expressions, which represents the population per square mile of the region consisting of states B and E in 1987?

- (A) $\frac{372 + 266}{2}$
 (B) $\frac{17,825 + 11,936}{372 + 266}$
 (C) $\frac{17,825}{372} + \frac{11,936}{266}$
 (D) $\frac{372}{17,825} + \frac{266}{11,936}$
 (E) $\frac{17,825 + 11,936}{372 + 266}$



26. In the figure above, XYZW is a square with sides of length s . If YW is the arc of a circle

with center X , which of the following is the area of the shaded region in terms of s ?

(A) $px^2 - (\frac{s}{2})^2$

(B) $s^2 - ps^2$

(C) $s^2 - \frac{1}{4}ps^2$

(D) $4s - ps$

(E) $4s - \frac{1}{4}ps$

27. In a graduating class of 236 students, 142 took algebra and 121 took chemistry. What is the greatest number of students that could have taken both algebra and chemistry?

- (A) 21 (B) 27 (C) 37 (D) 121 (E) 142

28. If one number is chosen at random from the first 1,000 positive integers, what is the probability that the number chosen is multiple of both 2 and 8?

(A) $\frac{1}{125}$

(B) $\frac{1}{8}$

(C) $\frac{1}{2}$

(D) $\frac{9}{16}$

(E) $\frac{5}{8}$

29. The price of product R is 20 percent higher than the price of product S, which in turn is 30 percent higher than the price of product T. The price of product R is what percent higher than the price of product T?

- (A) 60% (B) 56% (C) 50% (D) 44% (E) 25%

YX7

+6Y

Y7X

30. In the sum above, if X and Y each denote one of the digits from 0 to 9, inclusive, then $X =$

- (A) 9 (B) 5 (C) 3 (D) 1 (E) 0

SECTION 5

Time —30 minutes

25 Questions

Questions 1-6

Each of seven worker trainees —F, G, H, J, K, L, and P—will be assigned to one of four branch offices: Iowa, Maine, Texas, or Utah. The assignments will be subject to the following constraints:

Each office must be assigned at least one trainee.

Utah must be assigned exactly two trainees.

F must be assigned to the same office as K.

L cannot be assigned to the same office as J.

If G is assigned to Utah, P must also be assigned to Utah.

1. Which of the following is an acceptable assignment of trainees to offices?

	<u>Iowa</u>	<u>Maine</u>	<u>Texas</u>	<u>Utah</u>
(A)	G	J	F, L	H, K, P
(B)	G, J	H, L	F, K	P
(C)	H	F, K	J, L	G, P
(D)	L, H	F, K	J	G, P
(E)	L, P	J	F, K	G, H

2. If G is assigned to Utah and if both F and H are assigned to Texas, which of the following lists all those trainees and only those trainees who will be assigned to an office in which there is no other trainee?

- (A) J
(B) K
(C) L
(D) J, L
(E) K, L

3. If J is assigned to Utah and H is assigned to Maine, which of the following must also be assigned to Utah?

- (A) F
(B) G
(C) K
(D) L
(E) P

4. If K is assigned to Utah, each of the following could be assigned to Maine EXCEPT

- (A) F
- (B) G
- (C) H
- (D) L
- (E) P

5. If F is assigned to Iowa, L is assigned to Maine, and G is assigned to Utah, then J must be assigned to either

- (A) Iowa or Maine
- (B) Iowa or Texas
- (C) Maine or Texas
- (D) Maine or Utah
- (E) Texas or Utah

6. If J is to be assigned to Texas, G is to be assigned to Utah, and none of the offices is to be assigned three trainees, how many acceptable combinations of assignments are there to select from?

- (A) One
- (B) Two
- (C) Three
- (D) Four
- (E) Five

Question 7 is based on the following table.

**RESULTS OF A CONTROLLED STUDY OF THE
EFFECTS OF CHEWING GUM ON TOOTH DECAY IN
CHILDREN**

Average total number of new cavities
per child over the course of three years

Children who regularly chewed gum sweetened with X	4.0
Children who regularly chewed gum sweetened with Y	1.5
Children who did not chew gum	2.5

7. Which of the following, if true, most helps to explain the difference among the children who chewed gum sweetened with X, the children who chewed gum sweetened with Y, and the children who did not chew any gum?

- (A) X, but not Y, consists of a substance that helps to protect teeth against harmful substances present in other foods.
- (B) The children who did not chew any gum during the study ate fewer sweet foods than did either the children who chewed gum sweetened with X or the children who chewed gum sweetened with Y.
- (C) The action of chewing gum stimulates the production of saliva, which contains a substance that helps fight tooth decay, but X, unlike Y, is a contributing factor to tooth decay.
- (D) Each group of children who chewed gum during the study brushed their teeth more often than did the children who did not chew gum during the study, but the children who chewed gum sweetened with Y brushed their teeth less often than did the children who chewed gum sweetened with X.
- (E) The action of chewing gum improves the circulation of blood in the jaw and strengthens the roots of adult teeth, but it also causes baby teeth to fall out more quickly than they would otherwise.

8. In the last few decades, grassy wetlands, essential to the nesting and breeding of ducks, geese, swans, and most other species of waterfowl, have been extensively drained and cultivated in southern Canada and the northern United States. Duck populations in North America have plummeted during this time, but populations of swans and geese have been affected less dramatically.

Which of the following, if true, most helps to explain the difference mentioned above?

- (A) Prohibition of hunting of waterfowl is easier to enforce in areas under cultivation than in wild lands.
- (B) Most geese and swans nest and breed farther north than ducks do, in areas that still are not cultivated.
- (C) Land that has been harvested rarely provides food suitable for waterfowl.

- (D) Goose and swan populations decline in periods of drought, when breeding sites are fewer.
- (E) Because they are larger than ducks, geese and swans have a harder time finding protected nesting sites in areas that are cultivated.

9. A researcher found that, in proportion to their body weights, children eat more carbohydrates than adults do. Children also exercise more than adults do. The researcher hypothesized that carbohydrate consumption varies in direct proportion to the calorie demands associated with different levels of exercise.

Which of the following, if true, most seriously undermines the researcher's hypothesis?

- (A) More carbohydrates are eaten per capita in nations where the government spends more per capita on public exercise programs.
- (B) Children who do not participate in organized sports tend to eat fewer carbohydrates than children who participate in organized sports.
- (C) Consumption of increased amounts of carbohydrates is a popular tactic of runners preparing for long-distance races.
- (D) Periods of physical growth require a relatively higher level of carbohydrate consumption than otherwise.
- (E) Though carbohydrates are necessary for the maintenance of good health, people who consume more carbohydrates are not necessarily healthier.

Questions 10-16

Each day of a seven-day flower show is featured all display of one type of flower: daisy, geranium, iris, petunia, rose, tulip, zinnia. Each type of flower will be featured on exactly one of the seven days. The flower show will begin on Sunday and run through the following Saturday. Because of suppliers' schedules, the order of the featured displays is subject to the following restrictions:

The iris display and the tulip display must be featured on consecutive days, beginning with either the iris display or the tulip display.

The daisy display and the zinnia display must be featured on consecutive days, beginning with either the daisy display or the zinnia display.

Sunday, Monday, and Tuesday are the only days available for the tulip display.

Sunday and Saturday are the only days available for the rose display.

10. The following can be the schedule of displays featured in the first five days of the show

	<u>Sunday</u>	<u>Monday</u>	<u>Tuesday</u>	<u>Wednesday</u>	<u>Thursday</u>
(A)	Daisy	Zinnia	Tulip	Iris	Rose
(B)	Geranium	Tulip	Petunia	Iris	Zinnia
(C)	Iris	Tulip	Zinnia	Geranium	Daisy
(D)	Tulip	Iris	Petunia	Daisy	Zinnia
(E)	Rose	Zinnia	Daisy	Tulip	Iris

11. If the rose display is featured on Sunday and the geranium display on Monday, which of the following displays must be featured on Wednesday?

- (A) Daisy (B) Iris (C) Petunia
(D) Tulip (E) Zinnia

12. If the daisy display and the petunia display are featured on Friday and Saturday, respectively, geraniums must be the featured display on either

- (A) Monday or Tuesday
(B) Monday or Wednesday
(C) Tuesday or Wednesday
(D) Tuesday or Thursday
(E) Wednesday or Thursday

13. If the daisy display is featured on Tuesday, which of the following must be true about the order of the displays?
- (A) Geraniums are featured on Thursday or on Friday.
 (B) Irises are featured on Friday or on Saturday.
 (C) Roses are featured on Sunday.
 (D) Tulips are featured on Sunday.
 (E) Zinnias are featured on Monday or on Tuesday.
14. If the daisy display is featured on Sunday, then the petunia display must be featured on either
- (A) Monday or Tuesday
 (B) Tuesday or Wednesday
 (C) Wednesday or Thursday
 (D) Thursday or Friday
 (E) Friday or Saturday
15. If the geranium display is featured on Saturday, there is a total of how many different flower displays any one of which could be featured on Wednesday?
- (A) Two
 (B) Three
 (C) Four
 (D) Five
 (E) Six
16. If the zinnia display is featured on Wednesday and the petunia display is featured on Thursday, which of the following displays must be featured immediately before the geranium display?
- (A) Daisy
 (B) Iris
 (C) Petunia
 (D) Rose
 (E) Tulip

Questions 17-22

Exactly seven radio advertisements—F, G, H, K, L, M and P—are to be aired once each during one radio program. The advertisements are to be aired in two groups, group 1 and group 2, according to the following conditions:

One of the groups must have three consecutive

advertisements; the other must have four.

K, a longer advertisement, must be the middle advertisement in the group with three advertisements.

G must be the first advertisement in its group.

G must be in a different group from H.

H must be in the same group as L and must be sometime after L.

17. Which of the following could be the division of advertisements into groups, with the advertisements listed in order within each group?

<u>Group 1</u>	<u>Group 2</u>
(A) F, K, G	M, L, P, H
(B) G, K, F	L, P, H, M
(C) G, K, L	H, M, P, F
(D) G, L, M, H	F, K, P
(E) P, F, L, K	G, F, M

18. If F is in the same group as L, which of the following must be in the same group as G?

- (A) H
 (B) K
 (C) L
 (D) M
 (E) P

19. Which of the following is a pair of advertisements that CANNOT be in a group together?

- (A) G and L
 (B) G and M
 (C) H and K
 (D) K and P
 (E) L and P

20. If G is in the group with four advertisements, which of the following is a pair of advertisements that must be aired in immediately adjacent positions?

- (A) F and M
 (B) G and P
 (C) K and L
 (D) K and P
 (E) M and P

21. If L is the third advertisement in a group, which of the following lists two advertisements that could be immediately adjacent to each other in a group?

- (A) F and G
- (B) G and P
- (C) H and P
- (D) K and L
- (E) K and M

22. If F is the next advertisement after K in a group, which of the following is a pair of advertisements that must be in a group together?

- (A) F and M
- (B) G and M
- (C) G and P
- (D) K and P
- (E) L and M

23. Experts removed a layer of eighteenth-century red paint from a figure in a painting by a sixteenth-century Italian artist, revealing a layer of green paint underneath. Since the green paint dates from the sixteenth century, the figure must have been green, not red, when the painting was completed in 1563. Which of the following, if true, most seriously weakens the argument?

- (A) The experts had been commissioned to restore the painting to the colors it had when it was completed.
- (B) X-rays reveal an additional layer of paint beneath the green paint on the figure.
- (C) Chemical analyses were used to determine the ages of the red paint and the green paint.
- (D) The red paint was added in the eighteenth century in an attempt to repair damage done in the late seventeenth century.
- (E) Red paint on the robe of another figure in the painting dates from the sixteenth century.

24. Although it is assumed that peacocks' magnificent tails function essentially to attract peahens, no one knows why it should be magnificent tails that give a competitive advantage in securing mates. One explanation is

that peahens are more likely to mate with peacocks with magnificent tails than with peacocks that lack magnificent tails.

Which of the following is an error of reasoning exemplified by the explanation?

- (A) Attributing to animals qualities that are characteristically human
- (B) Extending a conclusion that is true of only one species of a genus to all species of the genus
- (C) Offering as an explanation a hypothesis that in principle can be neither verified nor proved false
- (D) Offering the phenomenon that is to be explained as the explanation of that phenomenon
- (E) Assuming without warrant that peacocks with magnificent tails are likely to have other features strongly attractive to peahens

25. Whenever a French novel is translated into English, the edition sold in Britain should be in British English. If the edition sold in Britain were in American English, its idioms and spellings would appear to British readers to be strikingly American and thus to conflict with the novel's setting.

The recommendation is based on which of the following assumptions?

- (A) The authors of French novels are usually native speakers of French.
- (B) A non-British reader of a novel written in British English will inevitably fail to understand the meanings of some of the words and idioms in the novel.
- (C) No French novel that is to be sold in Britain in English translation is set in the United States.
- (D) A British reader of a British novel will notice that the idioms and spellings used in the novel are British.
- (E) Most French novels are not translated into both British English and American English.

SECTION 6

Time—30 minutes

38 Questions

1. If the theory is self-evidently true, as its proponents assert, then why does---it still exist among well-informed people?
 - (A) support for
 - (B) excitement about
 - (C) regret for
 - (D) resignation about
 - (E) opposition to
2. Although the ---of cases of measles has ---, researchers fear that eradication of the disease, once believed to be imminent, may not come soon.
 - (A) occurrence.. continued
 - (B) incidence.. declined
 - (C) prediction.. resumed
 - (D) number.. increased
 - (E) study.. begun
3. Nothing---his irresponsibility better than his---delay in sending us the items he promised weeks ago.
 - (A) justifies.. conspicuous
 - (B) characterizes.. timely
 - (C) epitomizes.. unnecessary
 - (D) reveals.. conscientious
 - (E) conceals.. inexplicable
4. The author did not see the---inherent in her scathing criticism of a writing style so similar to her own.
 - (A) disinterest
 - (B) incongruity
 - (C) pessimism
 - (D) compliment
 - (E) symbolism
5. Whereas the Elizabethans struggled with the transition from medieval--experience to modern individualism, we confront an electronic technology that seems likely to reverse the trend, rendering individualism obsolete and

interdependence mandatory.

- (A) literary
 - (B) intuitive
 - (C) corporate
 - (D) heroic
 - (E) spiritual
6. Our biological uniqueness requires that the effects of a substance must be verified by---experiments, even after thousands of tests of the effects of that substance on animals.
 - (A) controlled
 - (B) random
 - (C) replicated
 - (D) human
 - (E) evolutionary
 7. Today water is more---in landscape architecture than ever before, because technological advances have made it easy, in some instances even ---to install water features in public places.
 - (A) conspicuous.. prohibitive
 - (B) sporadic.. effortless
 - (C) indispensable.. intricate
 - (D) ubiquitous.. obligatory
 - (E) controversial.. unnecessary

Directions: In each of the following questions, a related pair of words or phrases is followed by five lettered pairs of words or phrases. Select the lettered pair that best expresses a relationship similar to that expressed in the original pair.

8. TERROR: FEAR::
 - (A) craving: desire
 - (B) inclination: liking
 - (C) sympathy: empathy
 - (D) urgency: lack
 - (E) alibi: excuse
9. FEED: HUNGER::
 - (A) reassure: uneasiness
 - (B) penetrate: inclusion
 - (C) abandon: desolation
 - (D) transfer: location
 - (E) fertilize: growth

10. PESTLE: GRIND:: (A) scissors: sharpen (B) spice: flavor (C) spoon: stir (D) hammer: swing (E) fan: rotate	(C) election: legislation (D) prosecution: conviction (E) conclusion: evaluation
11. DISSEMBLE: HONESTY:: (A) smile: amiability (B) snub: politeness (C) disagree: error (D) flee: furtiveness (E) elate: exuberance	14. DEMOGRAPHY: POPULATION:: (A) agronomy: farm (B) astronomy: planets (C) chemistry: heat (D) meteorology: weather (E) genetics: adaptation
12. SYNOPSIS: CONCISENESS:: (A) distillate: purity (B) mutation: viability (C) replication: precedence (D) illusion: quickness (E) icon: charity	15. EQUIVOCATION: TRUTH (A) rhetoric: persuasion (B) obfuscation: clarity (C) metaphor: description (D) repetition: boredom (E) conciliation: appeasement
13. MEDIATION : COMPROMISE:: (A) exclamation: remark (B) approbation: acclaim	16. CRAVEN: ADMIRABLE:: (A) unruly: energetic (B) listless: attractive (C) deft: awkward (D) trifling: amusing (E) volatile: passionate

Bracken fern has been spreading from its woodland strongholds for centuries, but the rate of encroachment into open countryside has lately increased alarmingly throughout northern and western Britain. A tough competitor, 5) bracken reduces the value of grazing land by crowding out other vegetation. The fern is itself poisonous to livestock, and also encourages proliferation of sheep ticks, which not only attack sheep but also transmit diseases. No less important to some people are bracken's effects on threatened 10) habitats and on the use of uplands for recreational purposes, even though many appreciate its beauty.

Biological controls may be the only economic solution. One potentially cheap and self-sustaining method of halting the spread of bracken is to introduce natural enemies of the 15) plant. Initially unrestrained by predators of their own, foreign predators are likely to be able to multiply rapidly and overwhelm intended targets. Because bracken occurs throughout the world, there is plenty of scope for this approach. Two candidates, both moths from the Southern 20) Hemisphere, are now being studied.

Of course, biological control agents can safely be

released only if it can be verified that they feed solely on the target weed. The screening tests have so far been fraught with difficulties. The first large shipment of moths 25) succumbed to a disease. Growing enough bracken indoors is difficult, and the moths do not readily exploit cut stems. These are common problems with rearing insects for biological control.

Other problems can be foreseen. Policymakers need to 30) consider many factors and opinions such as the cost of control compared to existing methods, and the impact of the clearance of bracken on the landscape, wildlife, and vegetation. In fact, scientists already have much of the information needed to assess the impact of biological 35) control of bracken, but it is spread among many individuals, organizations, and government bodies. The potential gains for the environment are likely to outweigh the losses because few plants, insects, mammals, and birds live associated only with bracken, and many would benefit 40) from a return of other vegetation or from a more diverse mosaic of habitats. But legal consequences of attempts at biological control present a potential minefield. For example, many rural tenants still have the right of “estoyers” the right to cut bracken as bedding for livestock and 45) uses. What would happen if they were deprived of these rights? Once a biological control agent is released, it is difficult to control its spread. What consideration is due landowners who do not want to control bracken? According to law, the release of the biological control agents must be 50) authorized by the secretary of state for the environment. But Britain lacks the legal and administrative machinery to assemble evidence for and against release.

17. Which of the following best states the main idea of the passage?

(A) Studies suggest that biological control of bracken will not be technically feasible.

(B) Although biological control appears to be the best solution to bracken infestation, careful assessment of the consequences is required.

(C) Environmentalists are hoping that laboratory technicians will find a way to raise large numbers of moths in captivity.

(D) Bracken is currently the best solution to the proliferation of nonnative moth species.

(E) Even after researchers discover the most economical method of pest control, the government has no authority to implement a control

program.

18. According to the passage, which of the following can be inferred about sheep ticks?
- (A) They increase where bracken spreads.
- (B) They are dangerous only to sheep.
- (C) They are especially adapted to woodland.
- (D) They have no natural enemies.
- (E) They cause disease among bracken.
19. The author cites all of the following as disadvantages of bracken encroachment EXCEPT:
- (A) Bracken is poisonous to farm animals.
- (B) Bracken inhibits the growth of valuable vegetation.
- (C) Bracken indirectly helps spread certain diseases.
- (D) Bracken is aesthetically objectionable.
- (E) Bracken disturbs habitats that some people would like to protect.
20. The final paragraph can best be described as
- (A) a summation of arguments presented in previous paragraphs
- (B) the elimination of competing arguments to strengthen a single remaining conclusion
- (C) an enumeration of advantages to biological control
- (D) an expansion of the discussion from the particular example of bracken control to the general problem of government regulation
- (E) an overview of the variety of factors requiring further assessment

- | | |
|--|--|
| <p>21. It can be inferred from the passage that it is advantageous to choose as the biological control agent a predator that is foreign to the targeted environment for which of the following reasons?</p> <p>(A) Conservation groups prefer not to favor one native species over another.</p> <p>(B) All local predators have already been overwhelmed by the target species.</p> <p>(C) Local predators cannot be effectively screened since they already exist in the wild.</p> <p>(D) There is little risk of an artificially introduced foreign predator multiplying out of control.</p> <p>(E) Native predator species are generally limited by their own predators.</p> <p>22. It can be inferred from the passage that the screening tests performed on the biological control agent are designed primarily to determine</p> <p>(A) its effectiveness in eliminating the target species</p> <p>(B) the response of local residents to its introduction</p> <p>(C) the risk it poses to species other than the target</p> <p>(D) its resistance to the stress of shipment</p> <p>(E) the likelihood of its survival indoors</p> <p>23. As it is discussed in the passage, the place of bracken within the forest habitat can best be described as</p> <p>(A) rapidly expanding</p> <p>(B) the subject of controversy</p> <p>(C) well established</p> <p>(D) circumscribed by numerous predators</p> <p>(E) a significant nutrient source</p> | <p>Allen and Wolkowitz's research challenges the common claim that homework-waged labor performed at home for a company-is primarily a response to women workers' needs and preferences. By focusing on a limited geographical area in order to gather in-depth information, the authors have avoided the methodological pitfalls that have plagued earlier research on homework. Their findings disprove accepted notions about homeworkers: that they are unqualified for other jobs and that they use homework as a short-term strategy for dealing with child care. The authors conclude that the persistence of homework cannot be explained by appeal to such notions, for, in fact, homeworkers do not differ sharply from other employed women. Most homeworkers would prefer to work outside the home but are constrained from doing so by lack of opportunity. In fact, homework is driven by employers' desires to minimize fixed costs: homeworkers receive no benefits and are paid less than regular employees.</p> <p>24. The passage is primarily concerned with</p> <p>(A) advocating a controversial theory</p> <p>(B) presenting and challenging the results of a study</p> <p>(C) describing a problem and proposing a solution</p> <p>(D) discussing research that opposes a widely accepted belief</p> <p>(E) comparing several explanations for the same phenomenon</p> <p>25. According to the passage, which of the following has been generally believed about homework?</p> <p>(A) The benefits of homework accrue primarily to employers rather than to homeworkers.</p> <p>(B) Homework is prevalent predominantly in rural areas.</p> <p>(C) Homework is primarily a response to the preferences of women workers.</p> <p>(D) Few homeworkers rely on homework for the majority of their family income.</p> <p>(E) Most homework is seasonal and part-time rather than full-time and year-round.</p> |
|--|--|

26. Allen and Wolkowitz's research suggests that each of the following is true of most homeworkers EXCEPT:
- (A) They do not necessarily resort to homework as a strategy for dealing with child care.
 - (B) Their family situations are not unlike those of other employed women.
 - (C) They are as well qualified as women who work outside the home.
 - (D) They perform professional-level duties rather than manual tasks or piecework.
 - (E) They do not prefer homework to employment outside the home.

27. The passage suggests which of the following about previous research on homework?
- (A) It was conducted primarily with women who did not have extensive household responsibilities or care for small children at home.
 - (B) It was conducted with homeworkers and companies over a large geographical area.
 - (C) It indicated that women homeworkers had numerous opportunities to work outside the home.
 - (D) It indicated that homeworkers usually work for companies that are close to their homes.
 - (E) It indicated that homework was financially advantageous to large companies.

28. FLIPPANCY:

- (A) temperance
- (B) reliability
- (C) seriousness
- (D) inflexibility
- (E) reticence

32. INIMITABLE:

- (A) enviable
- (B) reparable
- (C) amicable
- (D) unwieldy
- (E) commonplace

36. APPOSITE:

- (A) disposable
- (B) adjacent
- (C) vicarious
- (D) parallel
- (E) extraneous

29. FACETIOUS:

- (A) uncomplicated
- (B) prideful
- (C) earnest
- (D) laconic
- (E) forbearing

33. SERE:

- (A) lush
- (B) obstinate
- (C) immersed
- (D) fortunate
- (E) antiquated

37. BOMBAST:

- (A) kindness
- (B) nonthreatening motion
- (C) great effort
- (D) down-to-earth language
- (E) good-natured approval

30. BUNGLE:

- (A) bring off
- (B) bail out:
- (C) give in
- (D) pull through
- (E) put together

34. VACUOUS:

- (A) courteous
- (B) exhilarated
- (C) modest
- (D) intelligent
- (E) emergent

38. LIMPID:

- (A) unfading
- (B) coarse
- (C) elastic
- (D) murky
- (E) buoyant

31. STODGY:

- (A) nervous
- (B) incisive
- (C) exciting
- (D) talkative
- (E) happy

35. PEDESTRIAN:

- (A) concise
- (B) attractive
- (C) mobile
- (D) delicate
- (E) imaginative

SECTION 1

Time—30 minutes

25 Questions

1. The Environmental Protection Agency must respond to the hazard to children's health posed by exposure to asbestos fibers released in the air in school classrooms. Since it is impossible to close school buildings, the best plan would be to initiate programs that mandate the immediate removal of asbestos from all the school buildings that are found to contain asbestos, regardless of whether or not the buildings are in use.

Which of the following, if true, is the strongest reason for the Environmental Protection Agency not to follow the plan outlined above?

- (A) The techniques available for removing asbestos often increase the level of airborne asbestos.
- (B) Schools are places where asbestos is especially likely to be released into the air by the action of the occupants.
- (C) Children exposed to airborne asbestos run a greater risk of developing cancer than do adults exposed to airborne asbestos.
- (D) The cost of removing asbestos varies from school to school, depending on accessibility and the quantity of asbestos to be removed.
- (E) It is impossible to determine with any degree of certainty if and when construction materials that contain asbestos will break down and release asbestos fibers into the air.

2. *Aedes albopictus*, a variety of mosquito that has recently established itself in the southeastern United States, is less widespread than the indigenous swamp mosquito. Both the swamp mosquito and *A. albopictus* can carry viruses that are sometimes fatal to humans, but *A. albopictus* is a greater danger to public health.

Each of the following, if true, provides additional information that strengthens the judgment given about the danger to public health EXCEPT:

- (A) Unlike the swamp mosquito, *A. albopictus* originated in Asia, and larvae of it were not observed in the United States before the mid-1980's.
- (B) Unlike the swamp mosquito, *A. albopictus* tends to spend most of its adult life near human habitation.
- (C) Unlike swamp mosquito larvae, *A. albopictus* larvae survive in flower pots, tin cans, and many small household objects that hold a little water.
- (D) In comparison with the swamp mosquito, *A. albopictus* hosts a much wider variety of viruses known to cause serious diseases in humans.
- (E) *A. albopictus* seeks out a much wider range of animal hosts than does the swamp mosquito, and it is more likely to bite humans.

Questions 3-8

The manager of a horse show is placing seven obstacles—one chicken coop, one gate, two stone walls, and three fences—on a jumping course that consists of seven positions, numbered and arranged consecutively from 1 to 7. The placement of the obstacles in the seven positions must conform to the following conditions:

No two fences can be placed in consecutive positions.
The stone walls must be placed in consecutive positions.

3. Which of the following is an acceptable placement of obstacles in the seven positions, in order from the first position to the last position on the course?
- (A) Chicken coop, fence, gate, stone wall, fence, stone wall, fence
- (B) Fence, gate, fence, fence, chicken coop, stone wall, stone wall
- (C) Fence, stone wall, stone wall, gate, chicken coop, fence, fence
- (D) Gate, stone wall, stone wall, fence, fence, chicken coop, fence
- (E) Stone wall, stone wall, fence, chicken coop, fence, gate, fence

4. If one of the fences is in the third position and another is in the sixth position, which of the following must be true?
- (A) The chicken coop is in the seventh position.
 (B) The gate is in the second position.
 (C) The gate is in the seventh position.
 (D) One of the stone walls is in the first position.
 (E) One of the stone walls is in the fourth position.
5. If one of the stone walls is in the seventh position, which of the following must be FALSE?
- (A) The chicken coop is in the second position.
 (B) The chicken coop is in the fourth position.
 (C) One of the fences is in the first position.
 (D) One of the fences is in the second position.
 (E) The gate is in the fourth position.
6. Which of the following CANNOT be the positions occupied by the three fences?
- (A) First, third, and fifth
 (B) First, third, and sixth
 (C) Second, fourth, and sixth
 (D) Second, fourth, and seventh
 (E) Third, fifth, and seventh
7. If a stone wall is placed immediately after the gate, which of the following is a complete and accurate list of the positions in which the gate can be placed?
- (A) Second, third (B) Second, fourth
 (C) Third, fourth (D) Second, third, fourth
 (E) Third, fourth, fifth
8. If the chicken coop is not placed immediately after any fence, which of the following is a complete and accurate list of the positions in which the chicken coop can be placed?
- (A) First, second, third
 (B) First, third, fourth
 (C) First, fourth, sixth
 (D) First, second, third, fourth
 (E) First, third, fourth, sixth
9. A person's cholesterol level will decline significantly if that person increases the

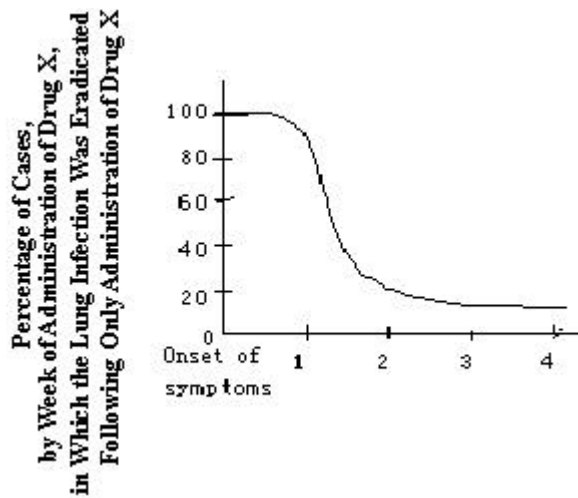
number of meals eaten per day, but only if there is no significant increase in the amount of food eaten. However, most people who increase the number of meals they eat each day will eat a lot more food as well.

If the statements above are true, which of the following is most strongly supported by them?

- (A) For most people, cholesterol level is not significantly affected by the amount of food eaten per day.
 (B) For most people, the amount of food eaten per meal is most strongly affected by the time of day at which the meal is eaten.
 (C) For most people, increasing the number of meals eaten per day will not result in a significantly lower cholesterol level.
 (D) For most people, the total amount of food eaten per day is unaffected by the number of meals eaten per day.
 (E) For most people, increasing the number of meals eaten per day will result in a significant change in the types of food eaten.
10. A certain type of dinnerware made in Ganandia contains lead. Lead can leach into acidic foods, and Ganandians tend to eat highly acidic foods. However, the extreme rarity of lead poisoning in Ganandia indicates that the dinnerware does not contain dangerous amounts of lead.
- Which of the following, if true, most seriously weakens the argument above?
- (A) The dinnerware is produced exclusively for sale outside Ganandia.
 (B) Ganandian foods typically are much more acidic than foods anywhere else in the world.
 (C) The only source of lead poisoning in Ganandia is lead that has leached into food.
 (D) Most people who use the dinnerware are not aware that it contains lead.
 (E) Acidic foods can leach lead from dinnerware even if that dinnerware has a protective coating.

Question 11 is based on the following graph.

EFFECTIVENESS OF DRUG X IN ERADICATING
A BACTERIAL LUNG INFECTION IN ADULT
PATIENTS



Point During the Course of the Infection
at Which Drug X Was First Administered to Patients
(in weeks following the onset of symptoms)

11. Drug X, which kills on contact the bacteria that cause the infection, is administered to patients by means of an aerosol inhaler.

Which of the following, if true, contributes most to explaining the change in drug X's effectiveness during the course of the infection?

- (A) Symptoms of the infection usually become evident during the first 48 hours following infection.
- (B) Most patients with lung infections say they prefer aerosol inhalers to other means of administering antibacterial drugs.
- (C) In most patients taking drug X, the dosage administered is increased slightly each week until symptoms disappear.
- (D) In patients who have the infection, the ability to inhale becomes increasingly impaired beginning in the second week after the onset of symptoms.
- (E) Drug X is not administered to any patient who shows signs of suffering from secondary infections.

12. Sergeant

Our police academy no longer requires its applicants to pass a physical examination before being admitted to the academy. As a result, several candidates with weak hearts and high blood pressure have been admitted. Hence, we can expect our future police force to have more health problems than our current police force.

Knowledge of each of the following would be relevant to determining the reliability of the sergeant's prediction EXCEPT whether

- (A) police officer candidates are screened for high blood pressure before joining the police force
- (B) the police officer candidates who are not healthy now are likely to be unhealthy as police officers
- (C) graduates of the police academy are required to pass a physical examination
- (D) the health of the current police officer candidates is worse than was the health of police officer candidates in the past
- (E) a police officer's health is a reliable indicator of the officer's performance

Questions 13-16

A transcontinental railroad train has exactly eight cars—J, K, L, M, N, O, P, and R—bound for several different destinations. The positions of the cars in any ordering of the train are numbered first through eighth from the front of the train. Because the cars will be detached at different points, certain ordering requirements must be met, as follows:

J must be somewhere behind M in the train.

K must be immediately in front of or immediately behind P.

O must be in front of N, and exactly one car must be between them.

R must be among the frontmost four cars and somewhere behind O.

13. Which of the following represents a possible order for the cars, from the front to the rear of the train?
- (A) L, M, O, R, N, J, K, P
 (B) M, K, P, O, R, N, L, J
 (C) M, L, O, R, N, K, J, P
 (D) O, R, M, N, P, K, J, L
 (E) P, K, R, L, O, M, N, J
14. If K is the first car, then the last car must be either
- (A) J or L (B) J or M (C) L or M
 (D) L or N (E) M or N
15. Which of the following can be neither the first nor the last car?
- (A) J (B) K (C) L (D) M (E) N
16. If R is somewhere behind N, which of the following must be true?
- (A) O is the first car.
 (B) M is the second car.
 (C) Either K or P is the last car.
 (D) L is one of the last four cars.
 (E) J is somewhere in front of K.

Questions 17-22

A mining company is planning a survey of exactly six regions-F, G, H, I, K, and L-for deposits of platinum and uranium. Each region will contain one of four possible combinations of minerals-both platinum and uranium, neither platinum nor uranium, platinum and no uranium, or uranium and no platinum. Prior to conducting a detailed survey, the mining company has the following information:

Exactly as many of the regions contain platinum deposits as contain uranium deposits.

Region F contains exactly the same deposits as does region H.

Regions G and I both contain uranium deposits.

Regions H and K both contain platinum deposits.

Regions G and L either both contain platinum deposits or neither of them does.

17. If there are exactly four regions that contain platinum deposits, these four could be
- (A) F, G, H, and K (B) F, G, H, and L
 (C) F, H, I, and K (D) F, H, K, and L
 (E) G, H, K, and L
18. If some region contains neither platinum deposits nor uranium deposits, it must be
- (A) F (B) G (C) H (D) I (E) L
19. If one of the six regions contains deposits of neither platinum deposits nor uranium deposits, which of the following CANNOT contain platinum deposits?
- (A) F (B) G (C) H (D) I (E) K
20. If exactly one region contains no platinum deposits, it must be
- (A) F (B) G (C) I (D) K (E) L
21. If K is the only region containing platinum deposits but no uranium deposits, which of the following must be two of the regions that contain both platinum deposits and uranium deposits?
- (A) F and G (B) F and H (C) G and L
 (D) H and I (E) I and L
22. If no region contains deposits of both platinum and uranium, which of the following must be true?
- (A) F contains uranium deposits.
 (B) G contains platinum deposits.
 (C) I contains platinum deposits.
 (D) K contains uranium deposits.
 (E) L contains uranium deposits.
23. Because adult iguanas on Plazos Island are much smaller than adult iguanas of the same species on nearby islands, researchers assumed that environmental conditions on Plazos favor the survival of relatively smaller baby iguanas (hatchlings) in each yearly brood. They discovered instead that for each of the past three years, 10 percent of the smaller and 40 percent of the larger hatchlings survived, because larger

hatchlings successfully evade their predators.

Which of the following, if true about Plazos but not about nearby islands, contributes most to an explanation of the long-standing tendency of iguanas on Plazos to be smaller than those of the same age on nearby islands?

- (A) Periodic wind shifts cause extended dry spells on Plazos every year, putting the larger iguanas, whose bodies require relatively more water, at a great disadvantage.
- (B) There are exactly three species of iguanas on Plazos but only two species of seagulls that feed on iguanas, and a relatively small percentage of each year's hatchlings are consumed by seagulls.
- (C) Wild cats, which were introduced as pets by early settlers and which were formerly major predators of Plazos iguanas, were recently killed off by a disease specific to cats.
- (D) The iguanas on Plazos are a relatively ancient part of the island's animal life.
- (E) Both land and marine iguanas live on Plazos, and the land iguanas tend to be larger than marine iguanas of the same age.

24. Every human being who has ever lived had two parents. Therefore, more people were alive three thousand years ago than are alive now.

The reasoning in the argument is flawed because it

- (A) overlooks the number of people in each generation during the last three thousand years who left no descendants
- (B) disregards possible effects of disasters such as famines and plagues on human history
- (C) overestimates the mathematical effect of repeated doublings on population size
- (D) fails to take into account that people now alive have overlapping sets of ancestors
- (E) fails to consider that accurate estimation of the number of people alive three thousand years ago might be impossible

25. Each of the academic journals *Thought* and *Ergo* has a review committee to prevent misattributed quotations from appearing in its published articles. Nevertheless, about ten percent of the quotations in *Thought*'s published articles are misattributed, whereas *Ergo* contains no misattributions. *Ergo*'s committee is more effective, therefore, than *Thought*'s at finding misattributed quotations.

The argument above assumes that

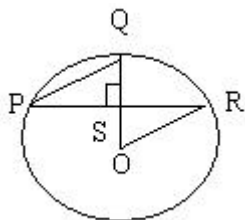
- (A) most of the articles submitted to *Thought* for publication contain misattributed quotations
- (B) there are at least some misattributed quotations in articles submitted to *Ergo* for publication
- (C) the members of *Ergo*'s committee are, on the whole, more knowledgeable than are the members of *Thought*'s committee
- (D) the number of misattributed quotations in a journal is an accurate measure of how carefully that journal is edited
- (E) the authors who submit articles to *Ergo* for publication are more thorough in attributing quotations than are the authors who submit articles to *Thought*

SECTION 2

Time—30 minutes

25 Questions

1. 60 percent of 16 10

O is the center of the circle and $OS = SQ$.

2. PQ OR

n is an integer such that $1 < n < 4$.

- 3.
- $n-1$
- 2

$$\frac{1}{2}X + 9 = 20$$

- 4.
- $x+19$
- 40

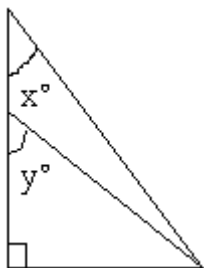
Board A measures between 2.15 feet and 2.25 feet in length; board B measures between 2.20 feet and 2.30 feet in length.

5. The length of board A The length of board B

$$t^2 - 2t = 0$$

$$t \neq 0$$

6. t 2



7. x y

- 8.
- $x(y+z)$
- $xy+z$

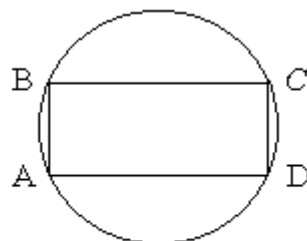
- 9.
- 10^4
- 5^6

10. x 120

- 11.
- $\left(\frac{1.5}{2.5}\right)^2$
- 0.36

For each positive integer n, $a_n = \frac{1}{n} - \frac{1}{n+1}$

- 12.
- $a_1 + a_2 + a_3 + a_4$
- $\frac{4}{5}$



The radius of the circle is r.

13. The area of rectangular region ABCD
- r^2

S is a set of n consecutive integers.

14. The mean of S The median of S

The length of a rectangular box is 4 inches longer than the depth, and the width of the box is 1 inch less than the length. The depth of the box is between 2 inches and 4 inches.

15. The volume of the box in cubic inches 200

16. In a circle graph used to represent a budget totaling \$600, the measure of the central angle associated with a \$120 item in the budget is (A)
- 72°
- (B)
- 108°
- (C)
- 120°
- (D)
- 144°
- (E)
- 216°

- 17.
- $\left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) + \left(\frac{1}{4} + \frac{1}{2}\right) =$

- (A) 0 (B) $\frac{1}{4}$ (C) $\frac{1}{2}$
- (D) 1 (E) $\frac{5}{4}$

18. The vertices of square S have coordinates $(-1,-2)$, $(-1,1)$, $(2,1)$, and $(2,-2)$, respectively. What are the coordinates of the point where the diagonals of S intersect?

(A) $(\frac{1}{2}, \frac{1}{2})$ (B) $(\frac{1}{2}, -\frac{1}{2})$
 (C) $(\frac{3}{2}, \frac{1}{2})$ (D) $(\frac{3}{2}, -\frac{1}{2})$
 (E) $(\frac{\sqrt{3}}{2}, \frac{1}{2})$

19. The admission price per child at a certain

amusement park is $\frac{7}{12}$ of the admission price per adult. If the admission price for 4 adults and 6 children is \$112.50, what is the admission price per adult?

(A) \$15.00 (B) \$13.50 (C) \$12.75
 (D) \$11.25 (E) \$8.75

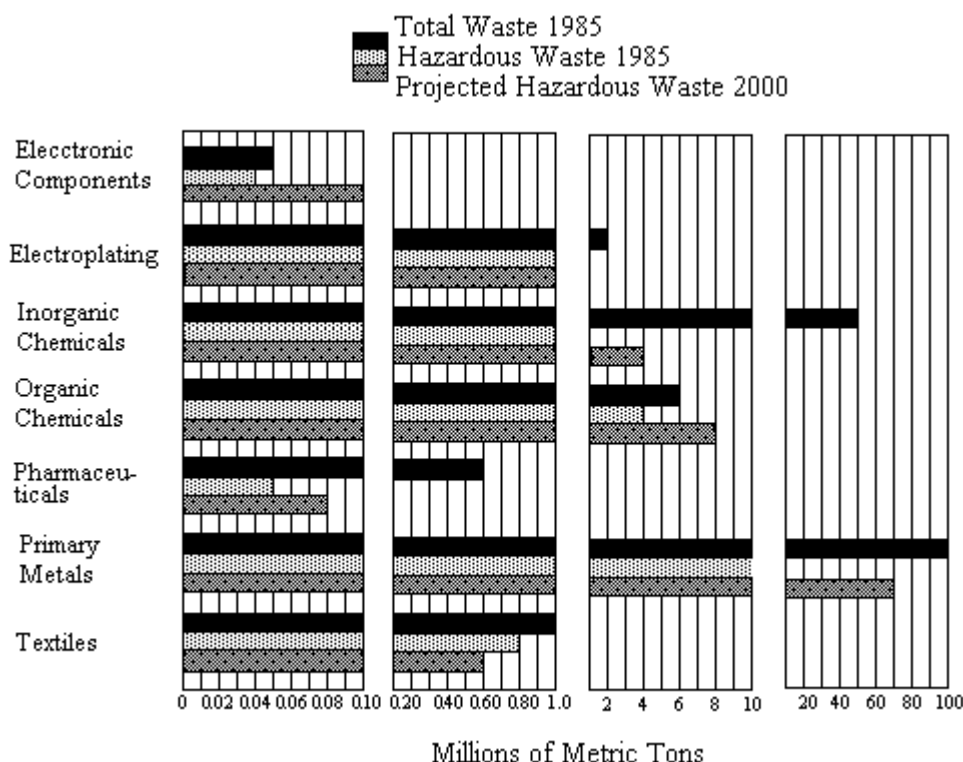
20. If $x=2y$ and $y=2z/3$, what is the value of z in terms of x ?

(A) $\frac{2x}{3}$ (B) $\frac{3x}{4}$ (C) $\frac{4x}{3}$
 (D) $\frac{3x}{2}$ (E) $3x$

Questions 21-25 refer to the following graph.

INDUSTRIAL WASTE GENERATED BY SPECIFIC INDUSTRIES OF COUNTRY X

Note: Because of the great disparity in the amounts of waste generated by different industries, the graph is broken in three places, and after each break, a new and more appropriate scale is introduced. As usual, the value represented by a bar is read only at its far right end.



21. How many million metric tons of hazardous waste was produced in 1985 by the inorganic and organic chemicals industries combined?
- (A) 66 (B) 16 (C) 10
 (D) 5 (E) 3

22. For those industries that generated a total of more than a million metric tons of waste in 1985, what was the approximate average (arithmetic mean) total waste, in millions of metric tons, generated per industry?
- (A) 42 (B) 34 (C) 28
 (D) 25 (E) 23

23. In 1985 hazardous waste in electroplating exceeded hazardous waste in electronic components by how many million metric tons?
 (A) 1.96 (B) 1.50 (C) 0.96
 (D) 0.80 (E) 0.60
24. In 1985 the pharmaceuticals industry generated total waste equal to how many times the hazardous waste in the same industry?
 (A) 1.2 (B) 2.5 (C) 3
 (D) 6 (E) 12
25. For which of the following industries is the hazardous waste projection for the year 2000 at least double its 1985 level?
 I. Electronic components
 II. Electroplating
 III. Inorganic chemicals
 (A) I only
 (B) I and II only
 (C) I and III only
 (D) II and III only
 (E) I, II, and III
26. Sixty-eight people are sitting in 20 cars and each car contains at most 4 people. What is the maximum possible number of cars that could contain exactly 1 of the 68 people?
 (A) 2 (B) 3 (C) 4
 (D) 8 (E) 12
27. The width of a rectangular playground is 75 percent of the length. If the perimeter of the playground is 280 meters, how long, in meters, is a straight path that cuts diagonally across the playground from one corner to another?
 (A) 60
 (B) 70
 (C) 80
 (D) 90
 (E) 100
28. Which of the following numbers is NOT the sum of three consecutive odd integers?
 (A) 15
- (B) 75
 (C) 123
 (D) 297
 (E) 313
29. If $72.42 = k(24 + \frac{n}{100})$, where k and n are positive integers and $n < 100$, then $k + n =$
 (A) 17
 (B) 16
 (C) 15
 (D) 14
 (E) 13
30. Which of the following pairs of numbers has an average (arithmetic mean) of 2?
 (A) $2 - \sqrt{2}$, $4 - \sqrt{2}$
 (B) $2\sqrt{3}$, $2 - 2\sqrt{3}$
 (C) $\frac{1}{0.5}$, $\frac{2.4}{1.6}$
 (D) $\sqrt{5}$, $\sqrt{3}$
 (E) $\frac{1}{2}$, $\frac{1}{2}$

SECTION 3

Time-30 minutes

Questions

1. What these people were waiting for would not have been apparent to others and was perhaps not very---their own minds.
(A) obscure to
(B) intimate to
(C) illusory to
(D) difficult for
(E) definite in
2. The attempt to breed suitable varieties of jojoba by using hybridization to---favorable traits was finally abandoned in favor of a simpler and much faster---: the domestication of flourishing wild strains.
(A) eliminate.. alternative
(B) reinforce.. method
(C) allow.. creation
(D) reduce.. idea
(E) concentrate.. theory
3. According to one political theorist, a regime that has as its goal absolute---, without any---law or principle, has declared war on justice.
(A) respectability.. codification of
(B) supremacy .. suppression of
(C) autonomy .. accountability to
(D) fairness .. deviation from
(E) responsibility .. prioritization of
4. Despite its ---, the book deals---with a number of crucial issues.
(A) optimism .. cursorily
(B) importance .. needlessly
(C) virtues .. inadequately
(D) novelty .. strangely
(E) completeness .. thoroughly
5. Although frequent air travelers remain unconvinced, researchers have found that, paradoxically, the---disorientation inherent in jet lag also may yield some mental health---.
(A) temporal.. benefits
(B) acquired.. hazards
(C) somatic .. disorders
(D) random .. deficiencies
(E) typical .. standards
6. Ironically, the proper use of figurative language must be based on the denotative meaning of the words, because it is the failure to recognize this--- meaning that leads to mixed metaphors and their attendant incongruity.
(A) esoteric
(B) literal
(C) latent
(D) allusive
(E) symbolic
7. Although it seems---that there would be a greater risk of serious automobile accidents in densely populated areas, such accidents are more likely to occur in sparsely populated regions.
(A) paradoxical
(B) axiomatic
(C) anomalous
(D) irrelevant
(E) portentous
8. CATASTROPHE: MISHAP::
(A) prediction: recollection
(B) contest: recognition
(C) humiliation: embarrassment
(D) reconciliation: solution
(E) hurdle: challenge
9. SONNET: POET::
(A) stage: actor
(B) orchestra: conductor
(C) music: dancer
(D) canvas: painter
(E) symphony: composer
10. LOQUACIOUS: SUCCINCT::
(A) placid: indolent
(B) vivacious: cheerful
(C) vulgar: offensive
(D) pretentious: sympathetic
(E) adroit: ungainly

11. DEPORTATION: COUNTRY::

- (A) evacuation: shelter
- (B) abdication: throne
- (C) extradition: court
- (D) eviction: dwelling
- (E) debarkation: destination

12. MAELSTROM: TURBULENT::

- (A) stricture: imperative
- (B) mirage: illusory
- (C) antique: rare
- (D) myth: authentic
- (E) verdict: fair

13. ABSTEMIOUS: INDULGE::

- (A) affectionate: embrace
- (B) austere: decorate
- (C) articulate: preach
- (D) argumentative: harangue
- (E) affable: jest

14. BLUSTERING: SPEAK::

- (A) grimacing: smile
- (B) blinking: stare
- (C) slouching: sit
- (D) jeering: laugh
- (E) swaggering: walk

15. SOLACE: GRIEF::

- (A) rebuke: mistake
- (B) mortification: passion
- (C) encouragement: confidence
- (D) justification: action
- (E) pacification: anger

16. INDELIBLE: FORGET::

- (A) lucid: comprehend
- (B) astounding: expect
- (C) inconsequential: reduce
- (D) incorrigible: agree
- (E) fearsome: avoid

Investigators of monkeys' social behavior have always been struck by monkeys' aggressive potential and the consequent need for social control of their aggressive behavior. Studies directed at describing aggressive behavior and the situations that elicit it, as well as the social mechanisms that control it, were therefore among the first investigations of monkeys' social behavior.

- (5) Investigators initially believed that monkeys would compete for any resource in the environment: hungry monkeys would fight over food, thirsty monkeys would fight over water, and, in general, any time more than one monkey in a group sought the same incentive simultaneously, a dispute would result and would be resolved through some form of aggression. However, the motivating force of competition for incentives began to be doubted when experiments like Southwick's on the reduction of space or the withholding of food failed to produce more than temporary increases in intragroup aggression. Indeed, food deprivation not only failed to increase aggression but in some cases actually resulted in decreased frequencies of aggression.

- (20) Studies of animals in the wild under conditions of extreme food deprivation likewise revealed that starving monkeys devoted almost all available energy to foraging, with little energy remaining for aggressive interaction. Furthermore, accumulating evidence from later studies of a variety of primate groups, for example, the study conducted by Bernstein, indicates that one of the most potent stimuli for eliciting aggression is the introduction of an intruder into an organized group. Such introductions result in far more serious aggression than that produced in any other types of experiments contrived to produce competition.

- (35) These studies of intruders suggest that adult members of the same species introduced to one another for the first time show considerable hostility because, in the absence of a social order, one must be established to control interanimal relationships. When a single new animal is introduced into an existing social organization, the newcomer meets even more serious aggression. Whereas in the first case aggression establishes a social order, in the second case resident animals mob the intruder, thereby initially excluding the new animal from the existing social unit. The simultaneous introduction of several animals lessens the effect, if only because the group divides its attention among the multiple targets. If, however, the several animals introduced to a group constitute their own social unit, each group may fight the opposing group as a unit; but, again, no individual is subjected to mass attack, and the very cohesion of the groups precludes prolonged individual combat. The submission of the defeated group, rather than unleashing unchecked aggression on the part of the victorious group, reduces both the intensity and frequency of further attack. Monkey groups therefor see to be organized primarily to maintain their established social order rather than to engage in hostilities per se.

17. The author of the passage is primarily concerned with
- (A) advancing a new methodology for changing a monkey's social behavior
 - (B) comparing the methods of several research studies on aggression among monkeys
 - (C) explaining the reasons for researchers' interest in monkeys' social behavior
 - (D) discussing the development of investigators' theories about aggression among monkeys
 - (E) examining the effects of competition on monkeys' social behavior
18. Which of the following best summarizes the findings reported in the passage about the effects of food deprivation on monkeys' behavior?
- (A) Food deprivation has no effect on aggression among monkeys.
 - (B) Food deprivation increases aggression among monkeys because one of the most potent stimuli for eliciting aggression is the competition for incentives.
 - (C) Food deprivation may increase long-term aggression among monkeys in a laboratory setting, but it produces only temporary increases among monkeys in the wild.
 - (D) Food deprivation may temporarily increase aggression among monkeys, but it also leads to a decrease in conflict.
 - (E) Food deprivation decreases the intensity but not the frequency of aggressive incidents among monkey.
19. According to the author, studies such as Southwick's had which of the following effects on investigators theories about monkeys' social behavior?
- (A) They suggested that existing theories about the role of aggression among monkeys did not fully account for the monkeys' ability to maintain an established social order.
 - (B) They confirmed investigators' theories about monkeys' aggressive response to competition for food and water.
 - (C) They confirmed investigators' beliefs about the motivation for continued aggression among monkeys in the same social group.
 - (D) They disproved investigators' theory that the introduction of intruders in an organized monkey group elicits intragroup aggressive behavior.
 - (E) They cast doubt on investigators' theories that could account for observed patterns of aggression among monkeys.
20. The passage suggests that investigators of monkeys social behavior have been especially interested in aggressive behavior among monkeys because
- (A) aggression is the most common social behavior among monkeys
 - (B) successful competition for incentives determines the social order in a monkey group
 - (C) situations that elicit aggressive behavior can be studied in a laboratory
 - (D) most monkeys are potentially aggressive, yet they live in social units that could not function without control of their aggressive impulses
 - (E) most monkeys are social, yet they frequently respond to newcomers entering existing social units by attacking them
21. It can be inferred from the passage that the establishment and preservation of social order among a group of monkeys is essential in order to
- (A) keep the monkeys from straying and joining other groups
 - (B) control aggressive behavior among group members
 - (C) prevent the domination of that group by another
 - (D) protect individuals seeking to become members of that group from mass attack
 - (E) prevent aggressive competition for incentives between that group and another
22. The passage supplies information to answer which of the following questions?
- (A) How does the reduction of space affect

intragroup aggression among monkeys in an experimental setting?

- (B) Do family units within a monkey social group compete with other family units for food?
- (C) What are the mechanisms by which the social order of an established group of monkeys controls aggression within that group?
- (D) How do monkeys engaged in aggression with other monkeys signal submission?
- (E) Do monkeys of different species engage in aggression with each other over food?

23. Which of the following best describes the organization of the second paragraph?

- (A) A hypothesis is explained and counter evidence is described.
- (B) A theory is advanced and specific evidence supporting it is cited.
- (C) Field observations are described and a conclusion about their significance is drawn.
- (D) Two theories are explained and evidence supporting each of them is detailed.
- (E) An explanation of a general principle is stated and specific examples of its operation are given.

Analysis of prehistoric air trapped in tiny bubbles beneath the polar ice sheets and of the composition of ice surrounding those bubbles suggests a correlation between carbon dioxide levels in the Earth's atmosphere and global

- (5) temperature over the last 160,000 years. Estimates of global temperature at the time air in the bubbles was trapped rely on measuring the relative abundances of hydrogen and its heavier isotope, deuterium, in the ice surrounding the bubbles. When global temperatures are relatively low,
- (10) water containing deuterium tends to condense and precipitate before reaching the poles; thus, ice deposited at the poles when the global temperature was cooler contained relatively less deuterium than ice deposited at warmer global temperatures. Estimates of global temperature based
- (15) on this information, combined with analysis of the carbon dioxide content of air trapped in ice deep beneath the polar surface, suggest that during periods of postglacial warming carbon dioxide in the Earth's atmosphere increased by approximately 40 percent.

24. In the passage, the author is primarily concerned with doing which of the following?

- (A) Describing a new method of estimating decreases in global temperature that have occurred over the last 160,000 years
- (B) Describing a method of analysis that provides information regarding the relation between the carbon dioxide content of the Earth's atmosphere and global temperature
- (C) Presenting information that suggests that global temperature has increased over the last 160,000 years.
- (D) Describing the kinds of information that can be gleaned from a careful analysis of the contents of sheets
- (E) Demonstrating the difficulty of arriving at a firm conclusion regarding how increases in the amount of carbon dioxide in the Earth's atmosphere affect global temperature

25. It can be inferred from the passage that during periods of post glacial warming, which of the following occurred?

- (A) The total volume of air trapped in bubbles beneath the polar ice sheets increased.
- (B) The amount of deuterium in ice deposited at the poles increased.
- (C) Carbon dioxide levels in the Earth atmosphere decreased.
- (D) The amount of hydrogen in the Earth's atmosphere decreased relatively the amount of deuterium
- (E) The rate at which ice was deposited at the poles increased

26. The author states that there is evidence to support which of the following assertions?

- (A) Estimates of global temperature that rely on measurements of deuterium in ice deposited at the poles are more reliable than those based on the amount of carbon dioxide contained in air bubbles beneath the polar surface.
- (B) The amount of deuterium in the Earth's atmosphere tends to increase as global temperature decreases.
- (C) Periods of post glacial warming are

- characterized by the presence of increased levels of carbon dioxide in the Earth's atmosphere.
- (D) Increases in global temperature over the last 160,000 years are largely the result of increases in the ratio of deuterium to hydrogen in the Earth's atmosphere.
- (E) Increases in global temperature over the last 160,000 years have been accompanied by decreases in the amount of deuterium in the ice deposited at the poles.
27. It can be inferred from the passage that the conclusion stated in the last sentence would need to be reevaluated if scientists discovered that which of the following were true?
- (A) The amount of deuterium in ice deposited on the polar surface is significantly greater than the amount of deuterium in ice located deep beneath the polar surface.
- (B) Both the air bubbles trapped deep beneath the polar surface and the ice surrounding them contain relatively low levels of deuterium.
- (C) Air bubbles trapped deep beneath the polar surface and containing relatively high levels of carbon dioxide are surrounded by ice that contained relatively low levels of deuterium.
- (D) The current level of carbon dioxide in the Earth's atmosphere exceeds the level of carbon dioxide in the prehistoric air trapped beneath the polar surface.
- (E) Increases in the level of carbon dioxide in the Earth's atmosphere are accompanied by increases in the amount of deuterium in the ice deposited at the poles.
28. CUMBERSOME:
- (A) likely to succeed
- (B) reasonable to trust
- (C) valuable to have
- (D) easy to handle
- (E) important to know
29. INDUCEMENT:
- (A) reproof
- (B) deterrent
- (C) partiality
- (D) distinction
- (E) consideration
30. STARTLE :
- (A) appease
- (B) lull
- (C) reconcile
- (D) dally
- (E) slumber
31. ANOMALY:
- (A) derivation from estimates
- (B) conformity to norms
- (C) return to origins
- (D) adaptation to stresses
- (E) repression of traits
32. RECIPROCATING:
- (A) releasing slowly
- (B) calculating approximately
- (C) accepting provisionally
- (D) moving unidirectionally
- (E) mixing thoroughly
33. MOLLYCODDLE:
- (A) talk boastfully
- (B) flee swiftly
- (C) treat harshly
- (D) demand suddenly
- (E) adjust temporarily
34. SURFEIT:
- (A) affirmation
- (B) compromise
- (C) dexterity
- (D) deficiency
- (E) languor
35. SANGUINE:
- (A) morose
- (B) puzzled
- (C) gifted
- (D) witty
- (E) persistent

36. RETROSPECTIVE:

- (A) irresolute
 (B) hopeful
 (C) unencumbered
 (D) evanescent
 (E) anticipatory

37. ENCOMIUM:

- (A) biased evaluation
 (B) polite response
 (C) vague description
 (D) harsh criticism
 (E) sorrowful expression

38. FACTIONAL:

- (A) excessive
 (B) undistinguished
 (C) disdainful
 (D) disinterested
 (E) disparate

SECTION 4

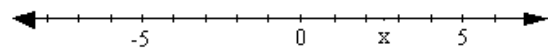
Time—30 minutes

30 Questions

1. 1 $\frac{1}{2^1} + \frac{1}{2^1} + \frac{1}{2^3}$

r and s are integers, and $r < s$.

2. The number of odd integers between r and s The number of even integers between r and s.

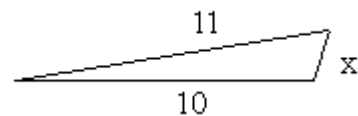


3. $x-y$ $x+y$

4. $10x-x$ 10

On a drawing done to scale, $\frac{1}{4}$ inch represents 5 feet.

5. The number of inches on the drawing that represents 150 feet 7.5



6. x 3

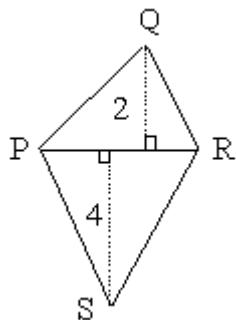
$$y > 0$$

$$x = 3y$$

7. 20 percent of x 50 percent of y

In a certain order of goods, $\frac{1}{3}$ of the items are shirts costing \$18 each and $\frac{2}{3}$ of the items are hats costing \$12 each.

8. The average (arithmetic mean) cost per item in the order \$15



9. The area of triangular region PRS is twice the area of triangular region PQR

$$xy < 0$$

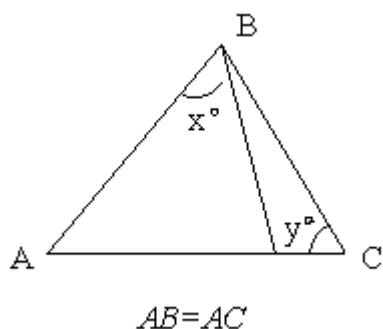
10. $xy > 0$

11. The average (arithmetic mean) of $\frac{10}{11}$ and $\frac{11}{10}$ is 1

$$d > 0$$

12. The area of a circular region with diameter d is the same as the area of a square region with diagonal of length d

13. $\frac{(0.83)^6}{(0.83)^7}$ is 1



14. $x > y$

15. $(2x+3y)^2 = 4x^2 + 6xy + 9y^2$

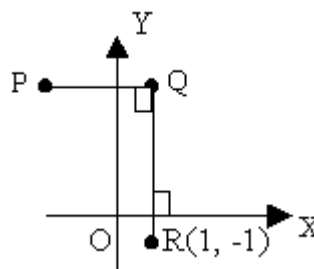
**TEMPERATURES IN DEGREES FAHRENHEIT
RECORDED AT NOON ON THE FIRST FOUR
DAYS OF
CERTAIN MONTHS**

Month	Temperatures
January	32,14,24,28
April	45,50,58,47
June	76,80,74,79
August	84,95,100,89
November	48,43,39,42

16. In a set of measurements, the range is defined as the greatest measurement minus the least measurement.

According to the table above, during the first four days of which month was the range of temperatures at noon the greatest?

- (A) January
(B) April
(C) June
(D) August
(E) November



17. In the figure above, if $QR=4$ and $PQ=3$, then the (x,y) coordinates of point P are

- (A) $(-4, 4)$
(B) $(-3,4)$
(C) $(-3,3)$
(D) $(-2,3)$
(E) $(-2,4)$

18. If $x^2=18$, then $|x| =$

- (A) -9
(B) 9

19. If $y-2x=-6$, then $8x-4y=$

- (A) 24

- (B) $\frac{3}{2}$
 (C) 0
 (D) $-\frac{3}{2}$
 (E) -24

20. A car gets 22 miles per gallon using gasoline

costing \$1.10 per gallon. What is the approximate cost, in dollars, for driving the car x miles using this gasoline?

- (A) $0.50x$
 (B) $0.30x$
 (C) $0.11x$
 (D) $0.10x$
 (E) $0.05x$

Questions 21-25 refer to the following table.

POPULATION DATA FOR TEN SELECTED STATES IN 1980 AND 1987

State	Population(in thousands)		Percent Change in Population, 1980-1987	Population Per Square Mile in 1987
	1980	1987		
A	23, 668	27, 663	16. 9	177
B	17, 558	17, 825	1. 5	372
C	14, 229	16, 789	18. 0	64
D	9, 746	12, 023	23. 4	222
E	11, 864	11, 936	0. 6	266
F	11, 427	11, 582	1. 4	208
G	10, 798	10, 784	-0. 1	263
H	9, 262	9, 200	-0. 7	162
I	7, 365	7, 672	4. 2	1, 027
J	5, 882	6, 413	9. 0	131

21. Which of the following states had the most land area in 1987?

- (A) A (B) B (C) C (D) D (E) E

22. In 1987 the average (arithmetic mean) population of the three most populous of the ten selected states was most nearly equal to.

- (A) 18 million (B) 19 million
 (C) 20 million (D) 21 million (E) 22 million

23. If the land area of State J was the same in 1980 as it was in 1987, then the population square mile of State J in 1980 was most nearly equal to

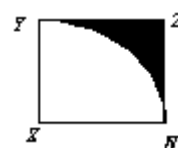
- (A) 140 (B) 130 (C) 120 (D) 110 (E) 100

24. If ranked from highest to lowest according to population, how many of the ten states changed in rank from 1980 to 1987?

- (A) One (B) Two (C) Three
 (D) Four (E) Five

25. Of the following expressions, which represents the population per square mile of the region consisting of states B and E in 1987?

- (A) $\frac{372 + 266}{2}$
 (B) $\frac{17,825 + 11,936}{372 + 266}$
 (C) $\frac{17,825}{372} + \frac{11,936}{266}$
 (D) $\frac{372}{17,825} + \frac{266}{11,936}$
 (E) $\frac{17,825 + 11,936}{372 + 266}$



26. In the figure above, XYZW is a square with sides of length s . If YW is the arc of a circle

with center X , which of the following is the area of the shaded region in terms of s ?

(A) $px^2 - (\frac{s}{2})^2$

(B) $s^2 - ps^2$

(C) $s^2 - \frac{1}{4}ps^2$

(D) $4s - ps$

(E) $4s - \frac{1}{4}ps$

27. In a graduating class of 236 students, 142 took algebra and 121 took chemistry. What is the greatest number of students that could have taken both algebra and chemistry?

- (A) 21 (B) 27 (C) 37 (D) 121 (E) 142

28. If one number is chosen at random from the first 1,000 positive integers, what is the probability that the number chosen is multiple of both 2 and 8?

(A) $\frac{1}{125}$

(B) $\frac{1}{8}$

(C) $\frac{1}{2}$

(D) $\frac{9}{16}$

(E) $\frac{5}{8}$

29. The price of product R is 20 percent higher than the price of product S, which in turn is 30 percent higher than the price of product T. The price of product R is what percent higher than the price of product T?

- (A) 60% (B) 56% (C) 50% (D) 44% (E) 25%

YX7

+6Y

Y7X

30. In the sum above, if X and Y each denote one of the digits from 0 to 9, inclusive, then $X =$

- (A) 9 (B) 5 (C) 3 (D) 1 (E) 0

SECTION 5

Time —30 minutes

25 Questions

Questions 1-6

Each of seven worker trainees —F, G, H, J, K, L, and P—will be assigned to one of four branch offices: Iowa, Maine, Texas, or Utah. The assignments will be subject to the following constraints:

Each office must be assigned at least one trainee.

Utah must be assigned exactly two trainees.

F must be assigned to the same office as K.

L cannot be assigned to the same office as J.

If G is assigned to Utah, P must also be assigned to Utah.

1. Which of the following is an acceptable assignment of trainees to offices?

	<u>Iowa</u>	<u>Maine</u>	<u>Texas</u>	<u>Utah</u>
(A)	G	J	F, L	H, K, P
(B)	G, J	H, L	F, K	P
(C)	H	F, K	J, L	G, P
(D)	L, H	F, K	J	G, P
(E)	L, P	J	F, K	G, H

2. If G is assigned to Utah and if both F and H are assigned to Texas, which of the following lists all those trainees and only those trainees who will be assigned to an office in which there is no other trainee?

- (A) J
(B) K
(C) L
(D) J, L
(E) K, L

3. If J is assigned to Utah and H is assigned to Maine, which of the following must also be assigned to Utah?

- (A) F
(B) G
(C) K
(D) L
(E) P

4. If K is assigned to Utah, each of the following could be assigned to Maine EXCEPT

- (A) F
- (B) G
- (C) H
- (D) L
- (E) P

5. If F is assigned to Iowa, L is assigned to Maine, and G is assigned to Utah, then J must be assigned to either

- (A) Iowa or Maine
- (B) Iowa or Texas
- (C) Maine or Texas
- (D) Maine or Utah
- (E) Texas or Utah

6. If J is to be assigned to Texas, G is to be assigned to Utah, and none of the offices is to be assigned three trainees, how many acceptable combinations of assignments are there to select from?

- (A) One
- (B) Two
- (C) Three
- (D) Four
- (E) Five

Question 7 is based on the following table.

**RESULTS OF A CONTROLLED STUDY OF THE
EFFECTS OF CHEWING GUM ON TOOTH DECAY IN
CHILDREN**

Average total number of new cavities
per child over the course of three years

Children who regularly chewed gum sweetened with X	4.0
Children who regularly chewed gum sweetened with Y	1.5
Children who did not chew gum	2.5

7. Which of the following, if true, most helps to explain the difference among the children who chewed gum sweetened with X, the children who chewed gum sweetened with Y, and the children who did not chew any gum?

- (A) X, but not Y, consists of a substance that helps to protect teeth against harmful substances present in other foods.
- (B) The children who did not chew any gum during the study ate fewer sweet foods than did either the children who chewed gum sweetened with X or the children who chewed gum sweetened with Y.
- (C) The action of chewing gum stimulates the production of saliva, which contains a substance that helps fight tooth decay, but X, unlike Y, is a contributing factor to tooth decay.
- (D) Each group of children who chewed gum during the study brushed their teeth more often than did the children who did not chew gum during the study, but the children who chewed gum sweetened with Y brushed their teeth less often than did the children who chewed gum sweetened with X.
- (E) The action of chewing gum improves the circulation of blood in the jaw and strengthens the roots of adult teeth, but it also causes baby teeth to fall out more quickly than they would otherwise.

8. In the last few decades, grassy wetlands, essential to the nesting and breeding of ducks, geese, swans, and most other species of waterfowl, have been extensively drained and cultivated in southern Canada and the northern United States. Duck populations in North America have plummeted during this time, but populations of swans and geese have been affected less dramatically.

Which of the following, if true, most helps to explain the difference mentioned above?

- (A) Prohibition of hunting of waterfowl is easier to enforce in areas under cultivation than in wild lands.
- (B) Most geese and swans nest and breed farther north than ducks do, in areas that still are not cultivated.
- (C) Land that has been harvested rarely provides food suitable for waterfowl.

- (D) Goose and swan populations decline in periods of drought, when breeding sites are fewer.
- (E) Because they are larger than ducks, geese and swans have a harder time finding protected nesting sites in areas that are cultivated.
9. A researcher found that, in proportion to their body weights, children eat more carbohydrates than adults do. Children also exercise more than adults do. The researcher hypothesized that carbohydrate consumption varies in direct proportion to the calorie demands associated with different levels of exercise.
- Which of the following, if true, most seriously undermines the researcher's hypothesis?
- (A) More carbohydrates are eaten per capita in nations where the government spends more per capita on public exercise programs.
- (B) Children who do not participate in organized sports tend to eat fewer carbohydrates than children who participate in organized sports.
- (C) Consumption of increased amounts of carbohydrates is a popular tactic of runners preparing for long-distance races.
- (D) Periods of physical growth require a relatively higher level of carbohydrate consumption than otherwise.
- (E) Though carbohydrates are necessary for the maintenance of good health, people who consume more carbohydrates are not necessarily healthier.

Questions 10-16

Each day of a seven-day flower show is featured all display of one type of flower: daisy, geranium, iris, petunia, rose, tulip, zinnia. Each type of flower will be featured on exactly one of the seven days. The flower show will begin on Sunday and run through the following Saturday. Because of suppliers' schedules, the order of the featured displays is subject to the following restrictions:

The iris display and the tulip display must be featured on consecutive days, beginning with either the iris display or the tulip display.

The daisy display and the zinnia display must be featured on consecutive days, beginning with either the daisy display or the zinnia display.

Sunday, Monday, and Tuesday are the only days available for the tulip display.

Sunday and Saturday are the only days available for the rose display.

10. The following can be the schedule of displays featured in the first five days of the show

	<u>Sunday</u>	<u>Monday</u>	<u>Tuesday</u>	<u>Wednesday</u>	<u>Thursday</u>
(A)	Daisy	Zinnia	Tulip	Iris	Rose
(B)	Geranium	Tulip	Petunia	Iris	Zinnia
(C)	Iris	Tulip	Zinnia	Geranium	Daisy
(D)	Tulip	Iris	Petunia	Daisy	Zinnia
(E)	Rose	Zinnia	Daisy	Tulip	Iris

11. If the rose display is featured on Sunday and the geranium display on Monday, which of the following displays must be featured on Wednesday?

(A) Daisy (B) Iris (C) Petunia
(D) Tulip (E) Zinnia

12. If the daisy display and the petunia display are featured on Friday and Saturday, respectively, geraniums must be the featured display on either

(A) Monday or Tuesday
(B) Monday or Wednesday
(C) Tuesday or Wednesday
(D) Tuesday or Thursday
(E) Wednesday or Thursday

13. If the daisy display is featured on Tuesday, which of the following must be true about the order of the displays?
- (A) Geraniums are featured on Thursday or on Friday.
 (B) Irises are featured on Friday or on Saturday.
 (C) Roses are featured on Sunday.
 (D) Tulips are featured on Sunday.
 (E) Zinnias are featured on Monday or on Tuesday.
14. If the daisy display is featured on Sunday, then the petunia display must be featured on either
- (A) Monday or Tuesday
 (B) Tuesday or Wednesday
 (C) Wednesday or Thursday
 (D) Thursday or Friday
 (E) Friday or Saturday
15. If the geranium display is featured on Saturday, there is a total of how many different flower displays any one of which could be featured on Wednesday?
- (A) Two
 (B) Three
 (C) Four
 (D) Five
 (E) Six
16. If the zinnia display is featured on Wednesday and the petunia display is featured on Thursday, which of the following displays must be featured immediately before the geranium display?
- (A) Daisy
 (B) Iris
 (C) Petunia
 (D) Rose
 (E) Tulip

Questions 17-22

Exactly seven radio advertisements—F, G, H, K, L, M and P—are to be aired once each during one radio program. The advertisements are to be aired in two groups, group 1 and group 2, according to the following conditions:

One of the groups must have three consecutive

advertisements; the other must have four.

K, a longer advertisement, must be the middle advertisement in the group with three advertisements.

G must be the first advertisement in its group.

G must be in a different group from H.

H must be in the same group as L and must be sometime after L.

17. Which of the following could be the division of advertisements into groups, with the advertisements listed in order within each group?

<u>Group 1</u>	<u>Group 2</u>
(A) F, K, G	M, L, P, H
(B) G, K, F	L, P, H, M
(C) G, K, L	H, M, P, F
(D) G, L, M, H	F, K, P
(E) P, F, L, K	G, F, M

18. If F is in the same group as L, which of the following must be in the same group as G?

- (A) H
 (B) K
 (C) L
 (D) M
 (E) P

19. Which of the following is a pair of advertisements that CANNOT be in a group together?

- (A) G and L
 (B) G and M
 (C) H and K
 (D) K and P
 (E) L and P

20. If G is in the group with four advertisements, which of the following is a pair of advertisements that must be aired in immediately adjacent positions?

- (A) F and M
 (B) G and P
 (C) K and L
 (D) K and P
 (E) M and P

21. If L is the third advertisement in a group, which of the following lists two advertisements that could be immediately adjacent to each other in a group?

- (A) F and G
- (B) G and P
- (C) H and P
- (D) K and L
- (E) K and M

22. If F is the next advertisement after K in a group, which of the following is a pair of advertisements that must be in a group together?

- (A) F and M
- (B) G and M
- (C) G and P
- (D) K and P
- (E) L and M

23. Experts removed a layer of eighteenth-century red paint from a figure in a painting by a sixteenth-century Italian artist, revealing a layer of green paint underneath. Since the green paint dates from the sixteenth century, the figure must have been green, not red, when the painting was completed in 1563. Which of the following, if true, most seriously weakens the argument?

- (A) The experts had been commissioned to restore the painting to the colors it had when it was completed.
- (B) X-rays reveal an additional layer of paint beneath the green paint on the figure.
- (C) Chemical analyses were used to determine the ages of the red paint and the green paint.
- (D) The red paint was added in the eighteenth century in an attempt to repair damage done in the late seventeenth century.
- (E) Red paint on the robe of another figure in the painting dates from the sixteenth century.

24. Although it is assumed that peacocks' magnificent tails function essentially to attract peahens, no one knows why it should be magnificent tails that give a competitive advantage in securing mates. One explanation is

that peahens are more likely to mate with peacocks with magnificent tails than with peacocks that lack magnificent tails.

Which of the following is an error of reasoning exemplified by the explanation?

- (A) Attributing to animals qualities that are characteristically human
- (B) Extending a conclusion that is true of only one species of a genus to all species of the genus
- (C) Offering as an explanation a hypothesis that in principle can be neither verified nor proved false
- (D) Offering the phenomenon that is to be explained as the explanation of that phenomenon
- (E) Assuming without warrant that peacocks with magnificent tails are likely to have other features strongly attractive to peahens

25. Whenever a French novel is translated into English, the edition sold in Britain should be in British English. If the edition sold in Britain were in American English, its idioms and spellings would appear to British readers to be strikingly American and thus to conflict with the novel's setting.

The recommendation is based on which of the following assumptions?

- (A) The authors of French novels are usually native speakers of French.
- (B) A non-British reader of a novel written in British English will inevitably fail to understand the meanings of some of the words and idioms in the novel.
- (C) No French novel that is to be sold in Britain in English translation is set in the United States.
- (D) A British reader of a British novel will notice that the idioms and spellings used in the novel are British.
- (E) Most French novels are not translated into both British English and American English.

SECTION 6

Time—30 minutes

38 Questions

1. If the theory is self-evidently true, as its proponents assert, then why does---it still exist among well-informed people?
 - (A) support for
 - (B) excitement about
 - (C) regret for
 - (D) resignation about
 - (E) opposition to
2. Although the ---of cases of measles has ---, researchers fear that eradication of the disease, once believed to be imminent, may not come soon.
 - (A) occurrence.. continued
 - (B) incidence.. declined
 - (C) prediction.. resumed
 - (D) number.. increased
 - (E) study.. begun
3. Nothing---his irresponsibility better than his---delay in sending us the items he promised weeks ago.
 - (A) justifies.. conspicuous
 - (B) characterizes.. timely
 - (C) epitomizes.. unnecessary
 - (D) reveals.. conscientious
 - (E) conceals.. inexplicable
4. The author did not see the---inherent in her scathing criticism of a writing style so similar to her own.
 - (A) disinterest
 - (B) incongruity
 - (C) pessimism
 - (D) compliment
 - (E) symbolism
5. Whereas the Elizabethans struggled with the transition from medieval--experience to modern individualism, we confront an electronic technology that seems likely to reverse the trend, rendering individualism obsolete and

interdependence mandatory.

- (A) literary
 - (B) intuitive
 - (C) corporate
 - (D) heroic
 - (E) spiritual
6. Our biological uniqueness requires that the effects of a substance must be verified by---experiments, even after thousands of tests of the effects of that substance on animals.
 - (A) controlled
 - (B) random
 - (C) replicated
 - (D) human
 - (E) evolutionary
 7. Today water is more---in landscape architecture than ever before, because technological advances have made it easy, in some instances even ---to install water features in public places.
 - (A) conspicuous.. prohibitive
 - (B) sporadic.. effortless
 - (C) indispensable.. intricate
 - (D) ubiquitous.. obligatory
 - (E) controversial.. unnecessary

Directions: In each of the following questions, a related pair of words or phrases is followed by five lettered pairs of words or phrases. Select the lettered pair that best expresses a relationship similar to that expressed in the original pair.

8. TERROR: FEAR::
 - (A) craving: desire
 - (B) inclination: liking
 - (C) sympathy: empathy
 - (D) urgency: lack
 - (E) alibi: excuse
9. FEED: HUNGER::
 - (A) reassure: uneasiness
 - (B) penetrate: inclusion
 - (C) abandon: desolation
 - (D) transfer: location
 - (E) fertilize: growth

10. PESTLE: GRIND:: (A) scissors: sharpen (B) spice: flavor (C) spoon: stir (D) hammer: swing (E) fan: rotate	(C) election: legislation (D) prosecution: conviction (E) conclusion: evaluation
11. DISSEMBLE: HONESTY:: (A) smile: amiability (B) snub: politeness (C) disagree: error (D) flee: furtiveness (E) elate: exuberance	14. DEMOGRAPHY: POPULATION:: (A) agronomy: farm (B) astronomy: planets (C) chemistry: heat (D) meteorology: weather (E) genetics: adaptation
12. SYNOPSIS: CONCISENESS:: (A) distillate: purity (B) mutation: viability (C) replication: precedence (D) illusion: quickness (E) icon: charity	15. EQUIVOCATION: TRUTH (A) rhetoric: persuasion (B) obfuscation: clarity (C) metaphor: description (D) repetition: boredom (E) conciliation: appeasement
13. MEDIATION : COMPROMISE:: (A) exclamation: remark (B) approbation: acclaim	16. CRAVEN: ADMIRABLE:: (A) unruly: energetic (B) listless: attractive (C) deft: awkward (D) trifling: amusing (E) volatile: passionate

Bracken fern has been spreading from its woodland strongholds for centuries, but the rate of encroachment into open countryside has lately increased alarmingly throughout northern and western Britain. A tough competitor,

- 5) bracken reduces the value of grazing land by crowding out other vegetation. The fern is itself poisonous to livestock, and also encourages proliferation of sheep ticks, which not only attack sheep but also transmit diseases. No less important to some people are bracken's effects on threatened habitats and on the use of uplands for recreational purposes, even though many appreciate its beauty.

Biological controls may be the only economic solution. One potentially cheap and self-sustaining method of halting the spread of bracken is to introduce natural enemies of the plant. Initially unrestrained by predators of their own, foreign predators are likely to be able to multiply rapidly and overwhelm intended targets. Because bracken occurs throughout the world, there is plenty of scope for this approach. Two candidates, both moths from the Southern Hemisphere, are now being studied.

Of course, biological control agents can safely be

released only if it can be verified that they feed solely on the target weed. The screening tests have so far been fraught with difficulties. The first large shipment of moths 25) succumbed to a disease. Growing enough bracken indoors is difficult, and the moths do not readily exploit cut stems. These are common problems with rearing insects for biological control.

Other problems can be foreseen. Policymakers need to 30) consider many factors and opinions such as the cost of control compared to existing methods, and the impact of the clearance of bracken on the landscape, wildlife, and vegetation. In fact, scientists already have much of the information needed to assess the impact of biological 35) control of bracken, but it is spread among many individuals, organizations, and government bodies. The potential gains for the environment are likely to outweigh the losses because few plants, insects, mammals, and birds live associated only with bracken, and many would benefit 40) from a return of other vegetation or from a more diverse mosaic of habitats. But legal consequences of attempts at biological control present a potential minefield. For example, many rural tenants still have the right of “estoyers” the right to cut bracken as bedding for livestock and 45) uses. What would happen if they were deprived of these rights? Once a biological control agent is released, it is difficult to control its spread. What consideration is due landowners who do not want to control bracken? According to law, the release of the biological control agents must be 50) authorized by the secretary of state for the environment. But Britain lacks the legal and administrative machinery to assemble evidence for and against release.

17. Which of the following best states the main idea of the passage?

(A) Studies suggest that biological control of bracken will not be technically feasible.

(B) Although biological control appears to be the best solution to bracken infestation, careful assessment of the consequences is required.

(C) Environmentalists are hoping that laboratory technicians will find a way to raise large numbers of moths in captivity.

(D) Bracken is currently the best solution to the proliferation of nonnative moth species.

(E) Even after researchers discover the most economical method of pest control, the government has no authority to implement a control

program.

18. According to the passage, which of the following can be inferred about sheep ticks?
- (A) They increase where bracken spreads.
- (B) They are dangerous only to sheep.
- (C) They are especially adapted to woodland.
- (D) They have no natural enemies.
- (E) They cause disease among bracken.
19. The author cites all of the following as disadvantages of bracken encroachment EXCEPT:
- (A) Bracken is poisonous to farm animals.
- (B) Bracken inhibits the growth of valuable vegetation.
- (C) Bracken indirectly helps spread certain diseases.
- (D) Bracken is aesthetically objectionable.
- (E) Bracken disturbs habitats that some people would like to protect.
20. The final paragraph can best be described as
- (A) a summation of arguments presented in previous paragraphs
- (B) the elimination of competing arguments to strengthen a single remaining conclusion
- (C) an enumeration of advantages to biological control
- (D) an expansion of the discussion from the particular example of bracken control to the general problem of government regulation
- (E) an overview of the variety of factors requiring further assessment

- | | |
|--|--|
| <p>21. It can be inferred from the passage that it is advantageous to choose as the biological control agent a predator that is foreign to the targeted environment for which of the following reasons?</p> <p>(A) Conservation groups prefer not to favor one native species over another.</p> <p>(B) All local predators have already been overwhelmed by the target species.</p> <p>(C) Local predators cannot be effectively screened since they already exist in the wild.</p> <p>(D) There is little risk of an artificially introduced foreign predator multiplying out of control.</p> <p>(E) Native predator species are generally limited by their own predators.</p> <p>22. It can be inferred from the passage that the screening tests performed on the biological control agent are designed primarily to determine</p> <p>(A) its effectiveness in eliminating the target species</p> <p>(B) the response of local residents to its introduction</p> <p>(C) the risk it poses to species other than the target</p> <p>(D) its resistance to the stress of shipment</p> <p>(E) the likelihood of its survival indoors</p> <p>23. As it is discussed in the passage, the place of bracken within the forest habitat can best be described as</p> <p>(A) rapidly expanding</p> <p>(B) the subject of controversy</p> <p>(C) well established</p> <p>(D) circumscribed by numerous predators</p> <p>(E) a significant nutrient source</p> | <p>Allen and Wolkowitz's research challenges the common claim that homework-waged labor performed at home for a company-is primarily a response to women workers' needs and preferences. By focusing on a limited geographical area in order to gather in-depth information, the authors have avoided the methodological pitfalls that have plagued earlier research on homework. Their findings disprove accepted notions about homeworkers: that they are unqualified for other jobs and that they use homework as a short-term strategy for dealing with child care. The authors conclude that the persistence of homework cannot be explained by appeal to such notions, for, in fact, homeworkers do not differ sharply from other employed women. Most homeworkers would prefer to work outside the home but are constrained from doing so by lack of opportunity. In fact, homework is driven by employers' desires to minimize fixed costs: homeworkers receive no benefits and are paid less than regular employees.</p> <p>24. The passage is primarily concerned with</p> <p>(A) advocating a controversial theory</p> <p>(B) presenting and challenging the results of a study</p> <p>(C) describing a problem and proposing a solution</p> <p>(D) discussing research that opposes a widely accepted belief</p> <p>(E) comparing several explanations for the same phenomenon</p> <p>25. According to the passage, which of the following has been generally believed about homework?</p> <p>(A) The benefits of homework accrue primarily to employers rather than to homeworkers.</p> <p>(B) Homework is prevalent predominantly in rural areas.</p> <p>(C) Homework is primarily a response to the preferences of women workers.</p> <p>(D) Few homeworkers rely on homework for the majority of their family income.</p> <p>(E) Most homework is seasonal and part-time rather than full-time and year-round.</p> |
|--|--|

26. Allen and Wolkowitz's research suggests that each of the following is true of most homeworkers EXCEPT:
- (A) They do not necessarily resort to homework as a strategy for dealing with child care.
 - (B) Their family situations are not unlike those of other employed women.
 - (C) They are as well qualified as women who work outside the home.
 - (D) They perform professional-level duties rather than manual tasks or piecework.
 - (E) They do not prefer homework to employment outside the home.

27. The passage suggests which of the following about previous research on homework?
- (A) It was conducted primarily with women who did not have extensive household responsibilities or care for small children at home.
 - (B) It was conducted with homeworkers and companies over a large geographical area.
 - (C) It indicated that women homeworkers had numerous opportunities to work outside the home.
 - (D) It indicated that homeworkers usually work for companies that are close to their homes.
 - (E) It indicated that homework was financially advantageous to large companies.

28. FLIPPANCY:

- (A) temperance
- (B) reliability
- (C) seriousness
- (D) inflexibility
- (E) reticence

32. INIMITABLE:

- (A) enviable
- (B) reparable
- (C) amicable
- (D) unwieldy
- (E) commonplace

36. APPOSITE:

- (A) disposable
- (B) adjacent
- (C) vicarious
- (D) parallel
- (E) extraneous

29. FACETIOUS:

- (A) uncomplicated
- (B) prideful
- (C) earnest
- (D) laconic
- (E) forbearing

33. SERE:

- (A) lush
- (B) obstinate
- (C) immersed
- (D) fortunate
- (E) antiquated

37. BOMBAST:

- (A) kindness
- (B) nonthreatening motion
- (C) great effort
- (D) down-to-earth language
- (E) good-natured approval

30. BUNGLE:

- (A) bring off
- (B) bail out:
- (C) give in
- (D) pull through
- (E) put together

34. VACUOUS:

- (A) courteous
- (B) exhilarated
- (C) modest
- (D) intelligent
- (E) emergent

38. LIMPID:

- (A) unfading
- (B) coarse
- (C) elastic
- (D) murky
- (E) buoyant

31. STODGY:

- (A) nervous
- (B) incisive
- (C) exciting
- (D) talkative
- (E) happy

35. PEDESTRIAN:

- (A) concise
- (B) attractive
- (C) mobile
- (D) delicate
- (E) imaginative

SECTION 1

Time –30 minutes

38 Questions

1. While scientists dismiss as fanciful the idea of sudden changes in a genetic code (spontaneous mutation), it is possible that nature, like some master musician, ---- on occasion, departing from the expected or predictable.
 - (A) repeats
 - (B) improvises
 - (C) ornaments
 - (D) corrects
 - (E) harmonizes
2. Despite the ---- of time, space, and history, human societies the world over have confronted the same existential problems and have come to remarkably ---- solutions, differing only in superficial details.
 - (A) continuity.. identical
 - (B) uniformity.. diverse
 - (C) actualities.. varied
 - (D) contingencies.. similar
 - (E) exigencies.. unique
3. Although he was known to be extremely ---- in his public behavior, scholars have discovered that his diaries were written with uncommon ----.
 - (A) reserved.. frankness
 - (B) polite.. tenderness
 - (C) modest.. lucidity
 - (D) reticent.. vagueness
 - (E) withdrawn.. subtlety
4. With the ---- of scientific knowledge, work on the new edition of a textbook begins soon after completion of the original.
 - (A) limitation
 - (B) culmination
 - (C) veneration
 - (D) certainty
 - (E) burgeoning
5. She is most frugal in matters of business, but in her private life she reveals a streak of ----.
 - (A) antipathy
 - (B) misanthropy
 - (C) virtuosity
 - (D) equanimity
 - (E) prodigality
6. If the state government's latest budget problems were ----, it would not be useful to employ them as ----examples in the effort to avoid the inevitable effects of shortsighted fiscal planning in the future.
 - (A) typical.. representative
 - (B) exceptional.. aberrant
 - (C) anomalous.. illuminating
 - (D) predictable.. helpful
 - (E) solvable.. insignificant
7. Just as some writers have ---- the capacity of language to express meaning, Giacometti ---- the failure of art to convey reality.
 - (A) scoffed at .. abjured
 - (B) demonstrated.. exemplified
 - (C) denied.. refuted
 - (D) proclaimed.. affirmed
 - (E) despaired of .. bewailed
8. WALLET: MONEY::
 - (A) bank: vault
 - (B) suitcase: clothing
 - (C) checkbook: balance
 - (D) wealth: prestige
 - (E) envelope: stamp
9. INSTRUMENTALIST: SYMPHONY::
 - (A) author: drama
 - (B) photographer: cinema
 - (C) composer: concerto
 - (D) artist: painting
 - (E) dancer: ballet
10. PLATEAU: CHANGE:
 - (A) respite: activity
 - (B) asylum: security
 - (C) terminus: journey
 - (D) interval: time
 - (E) lull: rest
11. ISTHMUS: LAND::
 - (A) peninsula: island
 - (B) canal: river

- (C) stratosphere: air
- (D) strait: water
- (E) tunnel: mountain

12. EMBARGO: COMMERCE::

- (A) abstention: election
- (B) strike: lockout
- (C) boycott: development
- (D) quarantine: contact
- (E) blockade: port

13. DILATORY: PROCRASTINATE::

- (A) recalcitrant: comply
- (B) malcontent: complain
- (C) ambivalent: decide
- (D) inept: modify
- (E) credulous: learn

14. NOMINAL: SIGNIFICANCE::

- (A) titular: honor
- (B) ephemeral: brevity
- (C) divisible: continuity
- (D) anomalous: distinction
- (E) disjunctive: unity

15. PLAGIARISM: IDEAS::

- (A) libel: words
- (B) forgery: documents
- (C) arson: buildings
- (D) kidnapping: ransom
- (E) rustling: cattle

16. POLITIC: OFFEND::

- (A) distressing: terrify
- (B) aloof: associate
- (C) misunderstood: surmise
- (D) vacuous: deplete
- (E) trivial: bore

For many years, Benjamin Quarles' seminal account of the participation of African Americans in the American Revolution has remained the standard work in the field. According to Quarles, the outcome of this conflict was mixed for African American slaves who enlisted in Britain's fight against its rebellious American colonies in return for the promise of freedom:

the British treacherously resold many into slavery in the West Indies, while others obtained freedom in Canada and Africa. Building on Quarles' analysis of the latter group, Sylvia Frey studied the former slaves who emigrated to British colonies in Canada. According to Frey, these refugees -the most successful of the African American Revolutionary War participants-viewed themselves as the ideological heirs of the American Revolution. Frey sees this inheritance reflected in their demands for the same rights that the American revolutionaries had demanded from the British: land ownership, limits to arbitrary authority and burdensome taxes, and freedom of religion.

17. According to the passage, which of the following is true about the African American Revolutionary War participants who settled in Canada after the American Revolution?

- (A) Although they were politically unaligned with either side, they identified more with British ideology than with American ideology.
- (B) While they were not immediately betrayed by the British, they ultimately suffered the same fate as did African American Revolutionary War participants who were resold into slavery in the West Indies.
- (C) They settled in Canada rather than in Africa because of the greater religious freedom available in Canada.
- (D) They were more politically active than were African American Revolutionary War participants who settled in Africa.
- (E) They were more successful than were African American Revolutionary War participants who settled in Africa.

18. Which of the following is most analogous to the relationship between the African American Revolutionary War participants who settled in Canada after the American Revolution and the American revolutionaries, as that relationship is described in the passage?

- (A) A brilliant pupil of a great musician rebels against the teacher, but adopts the teacher's musical style after the teacher's unexpected death.

(B) Two warring rulers finally make peace after a lifetime of strife when they realize that they have been duped by a common enemy. (C) A child who has sided with a domineering parent against a defiant sibling later makes demands of the parent similar to those once made by the sibling. (D) A writer spends much of her life popularizing the work of her mentor, only to discover late in life that much of the older writer's work is plagiarized from the writings of a foreign contemporary. (E) Two research scientists spend much of their careers working together toward a common goal, but later quarrel over which of them should receive credit for the training of a promising student. 19. The author of the passage suggests that which of the following is true of Benjamin Quarles' work? (A) It introduced a new and untried research methodology. (B) It contained theories so controversial that they gave rise to an entire generation of scholarship. (C) It was a pioneering work that has not yet been displaced by subsequent scholarship. (D) It launched the career of a scholar who later wrote even more important works. (E) At the time it appeared, its author already enjoyed a well-established reputation in the field. 20. Which of the following can be inferred from the passage concerning Britain's rule in its Canadian colonies after the American Revolution? (A) Humiliated by their defeat by the Americans, the British sharply curtailed civil rights in their Canadian colonies. (B) The British largely ignored their Canadian colonies. (C) The British encouraged the colonization of Canada by those African Americans who had served on the American side as well as by those who had served on the British side. (D) Some of Britain's policies in its Canadian colonies were similar to its policies in its American colonies before the American Revolution.	(E) To reduce the debt incurred during the war, the British imposed even higher taxes on the Canadian colonists than they had on the American colonists. Over the years, biologists have suggested two main pathways by which sexual selection may have shaped the evolution of male birdsong. In the first, male competition and intrasexual selection produce relatively short, simple songs used mainly in territorial behavior. In the second, female choice and intersexual selection produce longer, more complicated songs used mainly in mate attraction; like such visual ornamentation as the peacock's tail, elaborate vocal characteristics increase the male's chances of being chosen as a mate, and he thus enjoys more reproductive success than his less ostentatious rivals. The two pathways are not mutually exclusive, and we can expect to find examples that reflect their interaction. Teasing them apart has been an important challenge to evolutionary biologists. Early research confirmed the role of intrasexual selection. In a variety of experiments in the field, males responded aggressively to recorded songs by exhibiting territorial behavior near the speakers. The breakthrough for research into intersexual selection came in the development of a new technique for investigating female response in the laboratory. When female cowbirds raised in isolation in sound-proof chambers were exposed to recordings of male song, they responded by exhibiting mating behavior. By quantifying the responses, researchers were able to determine what particular features of the song were most important. In further experiments on song sparrows, researchers found that when exposed to a single song type repeated several times or to a repertoire of different song types, females responded more to the latter. The beauty of the experimental design is that it effectively rules out confounding variables; acoustic isolation assures that the female can respond only to the song structure itself. If intersexual selection operates as theorized, males with more complicated songs should not only attract females more readily but should also enjoy greater reproductive success. At first, however, researchers doing fieldwork with song sparrows found no correlation between larger repertoires and early mating, which has been shown to be one indicator of reproductive success; further, common measures
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of male quality used to predict reproductive success, such as weight, size, age, and territory, also failed to correlate with song complexity.

The confirmation researchers had been seeking was finally achieved in studies involving two varieties of warblers. Unlike the song sparrow, which repeats one of its several song types in bouts before switching to another, the warbler continuously composes much longer and more variable songs without repetition. For the first time, researchers found a significant correlation between repertoire size and early mating, and they discovered further that repertoire size had a more significant effect than any other measure of male quality on the number of young produced. The evidence suggests that warblers use their extremely elaborate songs primarily to attract females, clearly confirming the effect of intersexual selection on the evolution of birdsong.

21. The passage is primarily concerned with

- (A) showing that intrasexual selection has a greater effect on birdsong than does intersexual selection
- (B) contrasting the role of song complexity in several species of birds
- (C) describing research confirming the suspected relationship between intersexual selection and the complexity of birdsong
- (D) demonstrating the superiority of laboratory work over field studies in evolutionary biology
- (E) illustrating the effectiveness of a particular approach to experimental design in evolutionary biology

22. The author mentions the peacock's tail in line 8 most probably in order to

- (A) cite an exception to the theory of the relationship between intrasexual selection and male competition
- (B) illustrate the importance of both of the pathways that shaped the evolution of birdsong
- (C) draw a distinction between competing theories of intersexual selection
- (D) give an example of a feature that may have evolved through intersexual selection by female choice
- (E) refute a commonly held assumption about the role of song in mate attraction

23. According to the passage, which of the following is specifically related to intrasexual selection?

- (A) Female choice
- (B) Territorial behavior
- (C) Complex song types
- (D) Large song repertoires
- (E) Visual ornamentation

24. Which of the following, if true, would most clearly demonstrate the interaction mentioned in lines 11-13?

- (A) Female larks respond similarly both to short, simple songs and to longer, more complicated songs.
- (B) Male canaries use visual ornamentation as well as elaborate song repertoires for mate attraction.
- (C) Both male and female blackbirds develop elaborate visual and vocal characteristics.
- (D) Male jays use songs to compete among themselves and to attract females.
- (E) Male robins with elaborate visual ornamentation have as much reproductive success as rivals with elaborate vocal characteristics.

25. The passage indicates that researchers raised female cowbirds in acoustic isolation in order to

- (A) eliminate confounding variables
- (B) approximate field conditions
- (C) measure reproductive success
- (D) quantify repertoire complexity
- (E) prevent early mating

26. According to the passage, the song sparrow is unlike the warbler in that the song sparrow

- (A) uses songs mainly in territorial behavior
- (B) continuously composes long and complex songs
- (C) has a much larger song repertoire
- (D) repeats one song type before switching to another
- (E) responds aggressively to recorded songs

27. The passage suggests that the song sparrow experiments mentioned in lines 37-43 failed to confirm the role of intersexual selection because

- (A) females were allowed to respond only to the song structure
- (B) song sparrows are unlike other species of birds
- (C) the experiments provided no evidence that

elaborate songs increased male reproductive success	(D) lack of generosity (E) lack of distinction
(D) the experiments included the songs of only a small number of different song sparrows	34. TAME:
(E) the experiments duplicated some of the limitations of previous field studies	(A) resolute (B) ruinous (C) racy (D) erratic (E) experienced
28. STRINGENT:	35. INDURATE:
(A) lax	(A) soften (B) puncture (C) denude (D) immure (E) exchange
(B) elusive	36. PROLIXITY:
(C) impartial	(A) succinctness (B) profundity (C) persuasiveness (D) complacency (E) cleverness
(D) evident	37. CALLOW:
(E) vast	(A) displaying keen intelligence (B) behaving with adult sophistication (C) reacting cheerfully (D) showing foresight (E) deciding quickly
29. INTERIM:	38. FRIABLE:
(A) obscure	(A) not easily crumbled (B) not easily torn (C) not easily melted (D) not easily eroded (E) not easily punctured
(B) permanent	
(C) prudent	
(D) resolute	
(E) secure	
30. SCATHING:	
(A) easily understood	
(B) politely cooperative	
(C) intentionally involuted	
(D) calmly complimentary	
(E) strongly partisan	
31. CAPITULATE:	
(A) enjoin	
(B) resist	
(C) observe closely	
(D) consider carefully	
(E) appraise critically	
32. RECONSTITUTE:	
(A) detail	
(B) invent	
(C) spoil	
(D) conform	
(E) dehydrate	
33. REPUTE:	
(A) lack of caution	
(B) lack of knowledge	
(C) lack of emotion	

SECTION 2

Time –30 inutes

25 Questions

1. The ancient Greek playwright Euripides followed the established conventions of verse composition less rigorously at the end of his career than at the beginning. Since the lines from a recently discovered Euripidean play adhere to those conventions as rigorously as do lines from Euripides' early plays, the recently discovered play must have been composed early in Euripides' career.

Which of the following is an assumption made in the argument?

- (A) All of Euripides' plays were written in verse.
 - (B) Euripides did not write any plays late in his career in which he imitated the style of his early plays.
 - (C) Euripides grew increasingly unaware of the established conventions of verse composition as his career progressed.
 - (D) Late in his career, Euripides was the only playwright of his day who consciously broke with the established conventions of verse composition.
 - (E) Ancient playwrights tended to be less willing to violate certain conventions early in their careers than they were later in their careers
2. In the United States, average fuel efficiency of newly manufactured domestic cars, although remaining worse than that of newly manufactured imported cars, substantially improved between 1983 and 1988. Average fuel efficiency of new domestic cars has not improved since, but the difference in average fuel efficiencies of new domestic cars and new imported cars has steadily decreased.
- If the statements above are true, which of the following must also be true on the basis of them?
- (A) Average fuel efficiency of domestic cars manufactured after 1988 was better than that of imported cars manufactured before 1988.
 - (B) Average fuel efficiency of newly manufactured domestic cars has steadily worsened since 1988.

- (C) Average fuel efficiency of newly manufactured imported cars has steadily worsened since 1988.
- (D) Average fuel efficiency of newly manufactured imported cars has steadily improved since 1983.
- (E) Average fuel efficiency of imported cars manufactured in 1983 was better than that of imported cars manufactured in 1988.

Questions 3-8

In order to remodel a kitchen, exactly six major tasks—installing appliances, plumbing, refinishing, sanding, tiling, and wallpapering? are to be done over six consecutive days? numbered 1 through 6. The order of the tasks is governed by the following conditions :

Exactly one task must be done each day.

Refinishing must be done on the day after sanding is done.

Installing appliances and tiling must each be done on some day after the day on which plumbing is done.

Wallpapering must be done on some day after the day on which refinishing is done.

3. Which of the following is an acceptable order in which the tasks can be done on days 1 through 6?
- (A) Installing appliances, sanding, refinishing, wallpapering, plumbing, tiling
 - (B) Plumbing, installing appliances, wallpapering, sanding, refinishing, tiling
 - (C) Plumbing, sanding, refinishing, installing appliances, wallpapering, tiling
 - (D) Sanding, plumbing, installing appliances, refinishing, tiling, wallpapering
 - (E) Sanding, refinishing, tiling, wallpapering, plumbing, installing appliances
4. The latest day on which plumbing can be done is day
- (A) 1
 - (B) 2
 - (C) 3
 - (D) 4
 - (E) 5
5. If sanding is done on day 1, then wallpapering CANNOT be done on day
- (A) 2

- (B) 3
- (C) 4
- (D) 5
- (E) 6

6.If tiling is done on day 2 and installing appliances is done on day 5, which of the following are the tasks that must be done on days 1 and 6, respectively?

- (A) Plumbing, refinishing
- (B) Plumbing, wallpapering
- (C) Sanding, plumbing
- (D) Sanding, wallpapering
- (E) Wallpapering, sanding

7.If plumbing is done on day 1 and wallpapering is done on day 5, which of the following can be the tasks that are done on days 2 and 6, respectively?

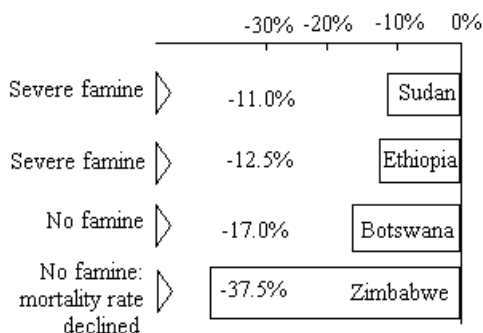
- (A) Refinishing, tiling
- (B) Sanding, installing appliances
- (C) Sanding, refinishing
- (D) Tiling, refinishing
- (E) Tiling, sanding

8.If tiling is done on day 3, which of the following must be done on day 2?

- (A) Installing appliances
- (B) Plumbing
- (C) Refinishing
- (D) Sanding
- (D) Wallpapering

Questions 9-10 are based on the following graph.

EFFECTS OF THE DROUGHT OF 1981-1984
Percentage change in food production per person
from 1979-1981(before the drought)to 1983-1984



9. It can be properly concluded from the graph that
(A) the drought beginning after 1981 affected crops

to a similar degree in the four countries

- (B) a country can withstand a severe famine without a substantial increase in mortality
- (C) a substantial decline in production of food per person in a country does not necessarily result in famine and increased deaths
- (D) the drought was more severe in the four countries by 1984 than it had been in 1982
- (E) there is no way to differentiate between countries that will and countries that will not suffer severe famine when food production drops sharply

10.Which of the following, if true, contributes most to an explanation of differences in the presence of famine that are shown on the graph?

- (A) The drought that began in 1981 was more severe in the countries of Botswana and Zimbabwe than in Sudan and Ethiopia.
- (B) Before the drought, Sudan, Ethiopia, and Botswana were exporters of their major food crops, including grain, but Zimbabwe was not an exporter of its major food crops.
- (C) During 1979-1984, the population of Botswana and Zimbabwe combined was less than the population of either Sudan or Ethiopia.
- (D) At the beginning of the drought, surplus food stocks in Sudan and Botswana were larger, relative to population, than in Ethiopia and Zimbabwe.
- (E) Popular demand for relief elicited a prompt response from the governments of Botswana and Zimbabwe, because they were democracies, but not from the nondemocratic governments of Sudan and Ethiopia.

11. Between 1970 and 1980, energy consumption by United States industry peaked and then declined, so that by 1980 total industrial use of energy was below the 1970 level even though total industrial output had grown substantially in the same period. Industry must have instituted highly effective energy conservation measures in those years to have achieved such impressive results.

Which of the following, if true, most seriously weakens the conclusion of the argument?

- (A) Many industries switched to the greatest extent possible from high-priced oil to lower-priced alternatives throughout the 1970's.
- (B) Total residential energy consumption was higher in the United States in 1980 than it had been in 1970
- (C) Many industrial users of energy had paid little attention to energy conservation prior to 1970.
- (D) Industrial output grew less rapidly from 1970 to 1980 than it had from 1960 to 1970.
- (E) The industries whose production dropped sharply during the 1970's included a disproportionately large number of energy-intensive industries.

12. Many people acquire software programs for their home computers by illegally copying those programs rather than purchasing them. People who own home computers must be making, on average, fewer illegal copies of software programs than before, however, since the average number of software programs that people purchase to use on their home computers has increased substantially over the past five years.

Which of the following, if true, most strengthens the argument?

- (A) The number of home computers in use has increased substantially over the past five years.
- (B) Five years ago, about half of the software programs used on home computers had been illegally copied rather than purchased.
- (C) Most people who have home computers use their computers more frequently the longer they have them.
- (D) Few people who prefer to copy computer software programs illegally cannot copy the software programs they want because they have no acquaintances who have those software programs.
- (E) On average, people with home computers have the same number of software programs today as people with home computers did five years ago.

Questions 13-17

Exactly seven detectives? G, H, J, K, M, O, and P—will investigate two cases—case 1 and case 2. Each of the seven detectives will investigate exactly one of the two cases. Four of the detectives will investigate case 1, and three of the detectives will investigate case 2. Detectives will be assigned to cases in accordance with the following conditions :

- G cannot investigate the same case that J investigates.
- Whichever case K investigates must also be the case that M investigates.
- H must investigate case 1.

13. Which of the following is an acceptable assignment of the detectives to the two cases?

- | <u>Case 1</u> | <u>Case 2</u> |
|----------------|---------------|
| (A) G, H, J, P | K, M, O |
| (B) G, K, M, O | H, J, P |
| (C) H, J, O | G, K, M, P |
| (D) H, J, K, M | G, O, P |
| (E) H, J, K, P | G, M, O |

14. If J investigates the same case that P investigates, which of the following detectives must investigate the same case that K investigates?

- (A) G
- (B) H
- (C) J
- (D) O
- (E) P

15. If O investigates case 2, which of the following must also investigate case 2?

- (A) G
- (B) J
- (C) K
- (D) M
- (E) P

16. If J investigates case 2, which of the following is a pair of detectives who must investigate the same case as each other?

- (A) G and H
- (B) G and P

- (C) H and O
- (D) J and K
- (E) J and M

17. Any of the following can be true EXCEPT :

- (A) G investigates the same case that H investigates.
- (B) H investigates the same case that M investigates.
- (C) K investigates the same case that O investigates.
- (D) J investigates case 1.
- (E) P investigates case 2.

Questions 18-22

Each year, a gardener will plant five kinds of vegetables—F, G, J, K, and M, not necessarily in that order—in a garden consisting of five parallel, adjacent rows, numbered consecutively 1 through 5. One kind of vegetable will be planted per row each year according to the following rules :

- K cannot be planted in the same row in any two successive years.
- If J is planted in a given row in one year, M must be planted in that row the next year.
- Because of nutrient requirements, F and M cannot in any year be planted in rows that are adjacent to each other.
- In any year, J must be planted in a row that is adjacent to the row in which G is planted.

18. Which of the following is an acceptable plan for planting in the first year the garden is planted, with the kinds of vegetables in order from row 1 through row 5?

- (A) F, G, J, M, K
- (B) G, M, J, K, F
- (C) J, K, M, G, F
- (D) K, J, G, M, F
- (E) M, G, K, J, F

19. If in a given year the order of the vegetables planted, from row 1 through row 5, is K, F, G, J, M, then in the next year, F must be planted in row

- (A) 1
- (B) 2
- (C) 3

- (D) 4
- (E) 5

20. Which of the following must be true in a year in which G is planted in row 5?

- (A) F is planted in row 1.
- (B) J is planted in row 3.
- (C) K is planted in row 2.
- (D) M is planted in row 1.
- (E) M is planted in row 3.

21. If M is to be planted in row 5 in the second year the garden is planted, then which of the following must be planted in row 4 in the first year?

- (A) F
- (B) G
- (C) J
- (D) K
- (E) M

22. If in a given year J is planted in row 1, then in the next year K can be planted in row

- (A) 1 or 5 but cannot be planted in any other row
- (B) 2 or 4 but cannot be planted in any other row
- (C) 2 or 5 but cannot be planted in any other row
- (D) 3 or 5 but cannot be planted in any other row
- (E) 4 or 5 but cannot be planted in any other row

23. From a newspaper editorial :

Many people who are addicted to heroin will eventually attempt to overcome their addiction, principally for two reasons : the expense of maintaining a heroin addiction and the fear of arrest. If heroin were legalized and made available cheaply, as some people advocate, neither of these reasons would apply

The considerations above can best serve as part of an argument that

- (A) legalizing the sale of heroin would cause the price of this drug to go down
- (B) making it easier for heroin addicts to obtain treatment for their addiction would encourage many heroin addicts to attempt to overcome their addiction
- (C) legalizing the sale of heroin would increase the

number of crimes committed by heroin addicts to support their addiction

- (D) making heroin available legally and cheaply would make it less likely that heroin addicts will attempt to overcome their addiction
- (E) decreasing the severity of penalties for individuals who use heroin would not increase the number of new heroin addicts

24. Stem borers are insect pests that often ruin North American corn crops. On some other continents, crop damage by stem borers is controlled by a certain species of wasp. Since these wasps eat nothing but stem borers, importing them into North America will keep crop damage from stem borers under control without endangering other North American insect species.

Which of the following is an assumption on which the argument depends?

- (A) Corn is the principal food of stem borers that live on continents other than North America.
- (B) The wasps are capable of surviving in North America long enough to eat significant numbers of stem borers.
- (C) No wasp in North America is closely related to the species of wasp that eats stem borers.
- (D) On continent other than North America, the wasps control stem borers more effectively than does any other pest control measure.
- (E) Corn crops on continents other than North America are not damaged by any insect pests other than stem borers.

25. In the country of Laurelia, legal restrictions on the sale of lock-picking equipment were relaxed ten years ago, and since then Laurelia's burglary rate has risen dramatically. Hence, since legally purchased lock-picking equipment was used in most burglaries, reintroducing strict limits on the sale of this equipment would help to reduce Laurelia's burglary rate.

Which of the following, if true, gives the strongest support to the argument?

- (A) Laurelia's overall crime rate has risen dramatically over the last ten years.

- (B) There is wide popular support in Laurelia for the reintroduction of strict limits on the sale of lock-picking equipment.
- (C) The reintroduction of strict limits on the sale of lock-picking equipment in Laurelia would not prevent legitimate use of this equipment by police and other public safety officials.
- (D) Most lock-picking equipment used in Laurelia is fragile and usually breaks irreparably within a few years of purchase.
- (E) The introduction five years ago of harsher punishments for people convicted of burglary had little effect on Laurelia's burglary rate.

SECTION 3

Time –30 minutes

30 Questions

$$1. \quad \frac{1}{0.82} \qquad \frac{1}{0.81}$$

2. The number of miles the rocket travels in 2 hours 10,000

$$\begin{aligned} -x + 4y &= 20 \\ x - y &= 1 \end{aligned}$$

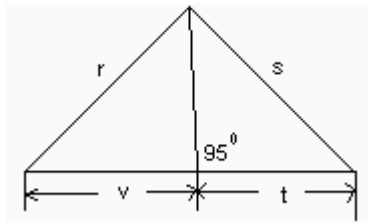
$$3. \quad x \qquad y$$

$$4. \quad 2^{(2^3)} \qquad (2^2)^3$$

$$0 < n < 1$$

$$5. \quad \frac{4}{n+3}$$

$$6. \quad 2 + \sqrt{2} \qquad 2\sqrt{2}$$

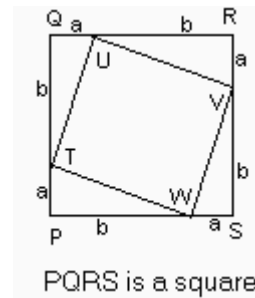


$$7. \quad r + v \qquad s + t$$

A grocer buys apples at the regular price of 38 cents per pound.

8. The amount saved by the grocer on a purchase of 100 pounds of apples if the price per pound is x cents less than the regular price	The additional amount paid by the grocer on a purchase of 100 pounds of apples if the price per pound is x cents more than the regular price
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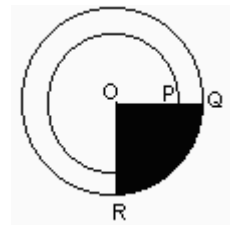
$$9. \quad 2x + y \qquad x + y$$



PQRS is a square.

10. The area of square $a^2 + b^2$
region TUVW

11. The median of the positive integers l, m, n, r , and s is 10, where $l < m < n < r < s$.
 $\frac{s}{2}$ 10



O is the center of both circles.

12. The area of the circular region with radius OP The area of the shaded sector ROQ

$$\begin{aligned} -4 &\leq x \leq 4 \\ -8 &\leq y \leq -4 \end{aligned}$$

13. The greatest possible value of $25x - 12.5y$ 200



p, q, r , and s are the coordinates of the points indicated on the number line.

$$14. \quad p + q \qquad r + s$$

$$15. \quad -|x| \qquad x \neq 0 \qquad -(x)$$

16. If $3x + 9y = 7x + y$, then $8y =$

- (A) $4x$
- (B) $6x$
- (C) $8x$
- (D) $10x$
- (E) $12x$

17. If the number of microbes in a test tube increases by 25 percent per day, how many microbes are there in the test tube at the end of a given day if the number of microbes at the end of the next day is 240,000 ?

- (A) 180,000
- (B) 192,000
- (C) 210,000
- (D) 288,000
- (E) 300,000

18. The average (arithmetic mean) of five numbers is 88.

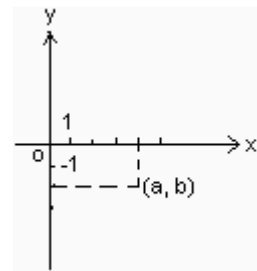
Four of the numbers are 92, 89, 91.....84. What is the fifth number?

- (A) 82
- (B) 84
- (C) 86
- (D) 89
- (E) 92

19. The scores reported for a certain multiple-choice test

were derived by subtracting $\frac{1}{3}$ of the number of wrong answers from the number of right answers. On a 40-question test, if none of the questions was omitted and the score reported was 20, how many wrong answers were there?

- (A) 5
- (B) 10
- (C) 15
- (D) 25
- (E) 30



20. In the figure above, $a - 2b =$

- (A) -10
- (B) -8
- (C) 0
- (D) 8
- (E) 10

ACTUAL ENERGY SAVINGS AS A PERCENT OF PROJECTED ENERGY SAVINGS FOR EACH OF TEN PROGRAMS

Program	Energy Savings	Program	Energy Savings
G	36%	N	62%
H	45%	P	98%
J	107%	Q	72%
K	105%	R	109%
M	-12%	T	59%

Note: Projected savings were based on the expected decrease in use of kilowatt hours of electricity as a result of plans to cut demand for electricity

21. Which program resulted in an increase in energy use instead of a decrease as projected?

- (A) Program G
- (B) Program J
- (C) Program K
- (D) Program M
- (E) Program R

22. For which Program were actual energy savings a closest to $\frac{3}{4}$ of the projected savings?

- (A) Program G
- (B) Program H
- (C) Program P
- (D) Program Q
- (E) Program T

23. How many of the programs resulted in greater energy savings than were projected?

- (A) One
- (B) Three
- (C) Four
- (D) Five
- (E) Eight

24. For which program was the ratio of actual energy savings to projected energy savings closest to 1?

- (A) Program G
- (B) Program K
- (C) Program M
- (D) Program P
- (E) It cannot be determined from the information given.

25. Actual energy savings for Program G were approximately what fraction of actual energy savings for Program T?

- (A) $\frac{1}{5}$
- (B) $\frac{1}{4}$
- (C) $\frac{3}{5}$
- (D) $\frac{5}{3}$
- (E) It cannot be determined from the information given

26. If x is the sum of seven consecutive odd integers beginning with 3 and y is the sum of seven consecutive odd integers beginning with 5, then $y - x$ equals

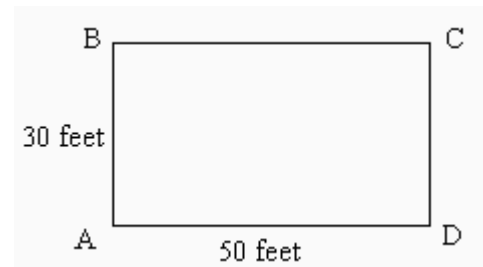
- (A) 2
- (B) 7
- (C) 8
- (D) 12
- (E) 14

27. If $\frac{5^{-1} \times 5^5}{5^4} = 5^y$, what is the value of y ?

- (A) 0
- (B) 1
- (C) 2
- (D) 3
- (E) 4

28. In a rectangular coordinate system, the set of all points (x, y) such that $-2 < x < 2$ and $-2 < y < 2$ comprises

- (A) two perpendicular line segments
- (B) two parallel line segments
- (C) a circular region
- (D) a triangular region
- (E) a square region



29. The figure above shows a rectangular play area in which one child stands at B while another child runs back and forth along the entire side AD. If the running child is in a position randomly located along side AD at a given time, what is the probability that the two children are at most 50 feet apart at that time?

- (A) $\frac{1}{5}$
- (B) $\frac{2}{5}$
- (C) $\frac{3}{5}$
- (D) $\frac{4}{5}$
- (E) 1

30. On a highway there is an electric pole every 96 feet. If the poles are numbered consecutively, what is the number of the pole 2 miles past pole number 56? (1 mile = 5,280 feet)

- (A) 109
- (B) 110
- (C) 152
- (D) 165
- (E) 166

SECTION 4

Time –30 minutes

30 Questions

$$1. \frac{9}{13} \qquad \frac{90}{130}$$

$$x < 10 < y$$

$$2. \quad x - 10 \qquad y - 10$$

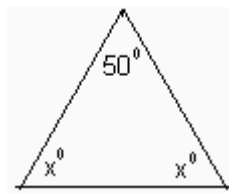
$$3. \quad \begin{array}{ll} 10 \text{ percent of 25 percent} & 35 \text{ percent of} \\ \text{of \$69.97} & \$69.97 \end{array}$$

$$a = -1$$

$$b = 2$$

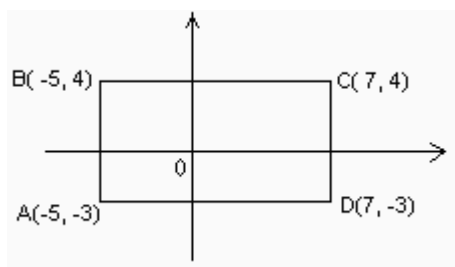
$$c = 3$$

$$4. \quad \frac{a^2 + b^2}{c^2} \qquad \frac{1}{3}$$



$$5. \quad x \qquad 50$$

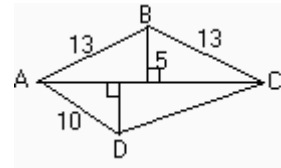
$$6. \quad \sqrt{26} + \sqrt{10} \qquad 8$$



$$7. \quad \begin{array}{ll} \text{The area of rectangular} & 84 \\ \text{region } ABCD & \end{array}$$

$M(r, s, t)$ denotes the average (arithmetic mean) of r , s , and t , and $M(x, y)$ denotes the average of x and y .

$$8. \quad \begin{array}{ll} M(70, 80, 90) & M(x, 90) \text{ where} \\ & x = M(70, 80) \end{array}$$



$$9. \quad \begin{array}{ll} \text{The area of triangular} & \text{The area of triangular} \\ \text{region } ABC & \text{region } ABC \end{array}$$

$$x^2 + (2\sqrt{5})^2 = 6^2$$

$$10. \quad |x| \qquad 4$$

$$\begin{array}{l} x = y^2 \\ 1 = cy \end{array}$$

$$11. \quad xc \qquad y$$

Cylindrical tank X has radius 4 meters and height 3 meters. Cylindrical tank Y has radius 3 meters and height 4 meters.

$$12. \quad \begin{array}{ll} \text{The volume of tank } X & \text{The volume of tank } Y \end{array}$$

$$\frac{x}{y} = \frac{4}{6}$$

$$13. \quad y - x \qquad 1$$

O is the center of the circle.

$$PQ < OP$$

$$14. \quad x \qquad 60$$

COMPARISON OF TEST SCORES
FOR TWO CLASS

	Number of scores	Mean score	Median score
Class A	15	80	84
Class B	30	74	72

$$15. \quad \begin{array}{ll} \text{The mean of the} & \text{The median of} \\ 45 \text{ scores} & \text{the 45 scores} \end{array}$$

16. If the temperature of a compound increases at a constant rate of 15 degrees per minute, how many minutes does it take for the temperature of the compound to increase from 60 degrees to 300 degrees?

- (A) 12
(B) 16

- (C) 20
(D) 24
(E) 30
17. If $\frac{12}{7 - \frac{r}{s}} = 2$, which of the following must be true?
- (A) $r = 0$
(B) $r = 6$
(C) $r = s$
(D) $r = 2s$
(E) $r = 3s$
18. If the edges of a 3-inch by 4-inch rectangular photograph were each lengthened by 50 percent, what would be the area, in square inches, of the enlarged rectangular photograph?
- (A) 18
(B) 20
(C) 24
(D) 27
(E) 30
19. If $\frac{x}{y} = 2$ and $\frac{1}{x} = 3$, then $y =$
- (A) $\frac{1}{6}$
(B) $\frac{2}{3}$
(C) 1
(D) $\frac{3}{2}$
(E) 6
20. Three pumps, X, Y, and Z, removed water from a tank. Pump X removed 550 gallons, pump Y removed 1,250 gallons, and pump Z removed $\frac{1}{3}$ of the total number of gallons removed by the three pumps combined. How many gallons of water did pump Z remove from the tank?
- (A) 450
(B) 600
(C) 900
(D) 1,800
(E) 2,700

UNITED STATES FILM BOX OFFICE RECEIPTS AND AVERAGE ADMISSION CHARGE

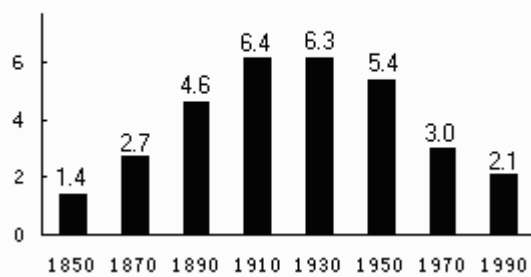
1940-1990

Year	Box Office Receipts* (in millions of dollars)	Average Admission Charge
1940	\$735.0	\$0.24
1950	\$1,376.0	\$0.53
1960	\$951.0	\$0.69
1970	\$1,162.0	\$1.55
1980	\$2,748.5	\$2.69
1990	\$5,021.8	\$4.75

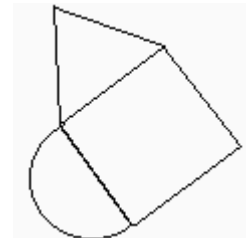
*Box office receipts are the total amounts collected from admission charges

21. For the year after 1940 in which box office receipts were less than they were ten years before, what was the average admission charge?
- (A) \$0.53
(B) \$0.69
(C) \$1.55
(D) \$2.69
(E) \$4.75
22. Which of the following is closest to the ratio of the average admission charge in 1950 to that in 1990?
- (A) $\frac{1}{2}$ 오후 9:59
(B) $\frac{1}{4}$
(C) $\frac{1}{2}$
(D) $\frac{1}{9}$
(E) $\frac{1}{15}$
23. Approximately how many admissions were paid in 1940 ?
- (A) 300,000
(B) 1,800,000
(C) 3,000,000
(D) 177,000,000
(E) 3,000,000,000

NUMBER OF FARMS IN THE UNITED STATES, 1850-1990
(in millions)



24. What was the percent decrease in the number of farms from 1970 to 1990 ?
- (A) 9%
(B) 21%
(C) 30%
(D) 70%
(E) 90%
25. The average acreage per farm was approximately 140 in 1910 and 220 in 1950. The ratio of the total farmland acreage in 1910 to the total in 1950 was most nearly
- (A) $\frac{3}{4}$
(B) $\frac{2}{3}$
(C) $\frac{3}{5}$
(D) $\frac{1}{2}$
(E) $\frac{2}{5}$
26. A research scientist wants to study a certain attribute of dogs. It is estimated that approximately 5 percent of all dogs have this attribute. If the scientist wants to study a sample of N dogs having the attribute, approximately how many dogs should be screened in order to obtain the desired sample size?
- (A) $N/5$
(B) $5N$
(C) $20N$
(D) $105N$
(E) $120N$
27. A square is inscribed in a circle. If the circle has radius 4, what is the perimeter of the square ?
- (A) 8
(B) $8\sqrt{2}$
(C) 16
(D) $16\sqrt{2}$
(E) $32\sqrt{2}$
28. How many 3-digit integers, greater than 100, are there in which the sum of the digits equals 3?
- (A) Three
(B) Four
(C) Six
(D) Nine
(E) Twelve



29. An equilateral triangle with perimeter p , a square and a semicircle were joined to form the figure shown above. What is the perimeter of this figure?
- (A) $3p$
(B) $\frac{p}{2} + pp$
(C) $7p + \frac{p}{2}p$
(D) $\frac{7}{3}p + \frac{p}{3}p$
(E) $\frac{4}{3}p + \frac{p}{6}p$
30. If x is $11\frac{1}{9}$ percent more than y , then y is what percent less than x ?
- (A) 9%
(B) 10%
(C) $11\frac{1}{9}\%$
(D) $12\frac{1}{2}\%$
(E) 15%

SECTION 5

Time —30 minutes

38 Questions

1. In spite of the fact that it is convenient to divide the life span of animals into separate stages such as pre-natal, adolescent, and senescent, these periods are not really——.
 - (A) advanced
 - (B) variable
 - (C) repeatable
 - (D) connected
 - (E) distinct
2. Although the number of reported volcanic eruptions has risen exponentially since 1850, this indicates not —— volcanic activity but rather more widespread and —— record keeping.
 - (A) abating..detailed
 - (B) increasing..systematic
 - (C) substantial..erratic
 - (D) stable..superficial
 - (E) consistent..meticulous
3. The challenge of interpreting fictional works written under politically repressive regimes lies in distinguishing what is —— to an author's beliefs, as opposed to what is —— by political coercion.
 - (A) innate.. understood
 - (B) organic.. imposed
 - (C) contradictory.. conveyed
 - (D) oblique.. captured
 - (E) peripheral.. demanded
4. I am often impressed by my own ---- other people's idiocies: what is harder to —— is that they, in their folly, are equally engaged in putting up with mine.
 - (A) analysis of.. justify
 - (B) forbearance toward.. underestimate
 - (C) exasperation with .. credit
 - (D) involvement in .. allow
 - (E) tolerance of .. appreciate
5. Despite vigorous protestations, the grin on the teenager's face ——her denial that she had known about the practical joke before it was played on her parents.
 - (A) belied
 - (B) illustrated
 - (C) reinforced
 - (D) exacerbated
 - (E) trivialized
6. Far from undermining the impression of permanent decline, the —— statue seemed emblematic of its —— surroundings.
 - (A) indecorous.. opulent
 - (B) grandiose.. ramshackle
 - (C) pretentious.. simple
 - (D) ungainly.. elegant
 - (E) tawdry.. blighted
7. Despite the fact that it is almost universally ——, the practice of indentured servitude still —— in many parts of the world.
 - (A) condemned... abates
 - (B) tolerated.. survives
 - (C) proscribed.. persists
 - (D) mandated.. lingers
 - (E) disdained.. intervenes
8. CANDY: SUGAR::
 - (A) chick: egg
 - (B) tire: rubber
 - (C) pen: ink
 - (D) mushroom: spore
 - (E) rag: scrap
9. SCRIPT: DRAMA::
 - (A) theater: play
 - (B) movement: symphony
 - (C) photograph: scene
 - (D) map: town
 - (E) score: music
10. AMBIGUOUS: UNDERSTAND::
 - (A) veracious: defend
 - (B) blatant: ignore

(C) prosaic: classify (D) arcane: conceal (E) plausible: believe 11. MERCURIAL: MOOD:: (A) callous: emotion (B) doleful: energy (C) jaundiced: attitude (D) whimsical: behavior (E) unversed: experience 12. PRISTINE: DECAY:: (A) adequate: imprecision (B) stable: fluctuation (C) volatile: force (D) symmetric: flaw (E) valid: exception 13. DIGRESS: EXCURSIVE:: (A) improvise: studied (B) reiterate: redundant (C) excise: prolix (D) refute: plausible (E) accede: contentious 14. PONTIFICATE: SPEAK:: (A) indoctrinate: preach (B) impersonate: imitate (C) obey: listen (D) soar: fly (E) strut: walk 15. OFFICIOUS: MEDDLE:: (A) disaffected: rebel (B) bustling: excel (C) profligate: conserve (D) subservient: esteem (E) acrimonious: soothe 16. ATTENUATE: THICKNESS:: (A) separate: substance (B) ventilate: circulation (C) vaccinate: immunity (D) transfer: location (E) cool: temperature	An experiment conducted aboard Space Lab in 1983 was the first attempt to grow protein crystals in the low-gravity environment of space. That experiment is still cited as evidence that growing crystals in microgravity can increase crystal size: the authors reported that they grew lysozyme protein crystals 1,000 times larger than crystals grown in the same device on Earth. Unfortunately, the authors did not point out that their crystals were no larger than the average crystal grown using other, more standard techniques in an Earth laboratory. No research has yet produced results that could justify the enormous costs of producing crystals on a large scale in space. To get an unbiased view of the usefulness of microgravity crystal growth, crystals grown in space must be compared with the best crystals that have been grown with standard techniques on Earth. Given the great expense of conducting such experiments with proper controls, and the limited promise of experiments performed thus far, it is questionable whether further experiments in this area should even be conducted. 17. According to the passage, which of the following is true about the Space Lab experiment conducted in 1983? (A) It was the first experiment to take place in the microgravity environment of space. (B) It was the first experiment in which researchers in space were able to grow lysozyme protein crystals greater in size than those grown on Earth. (C) Its results have been superseded by subsequent research in the field of microgravity protein crystal growth. (D) Its results are still considered by some to be evidence for the advantages of microgravity protein crystal growth. (E) Its results are considered by many to be invalid because nonstandard techniques were employed. 18. It can be inferred from the passage that the author would find the Space Lab experiment more impressive if which of the following were true? (A) The results of the Space Lab experiment could be replicated in producing other kinds of crystals in
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addition to lysozyme protein.

- (B) The device used in the experiment produced larger crystals on Earth than it did in space.
- (C) The size of the crystals produced in the experiment exceeded the size of crystals grown in Earth laboratories using standard techniques.
- (D) The cost of producing the crystals in space exceeded that of producing them using standard laboratory techniques.
- (E) The standard techniques used in Earth laboratories were modified in the Space Lab experiment due to the effects of microgravity.

19. Which of the following can be inferred from the passage about the device used to grow crystals in the Space Lab experiment?

- (A) The device is more expensive to manufacture than are the devices used in standard techniques in an Earth laboratory.
- (B) The device has not been used to grow crystals in space since the Space Lab experiment of 1983.
- (C) Crystals grown in the device on Earth tend to be much smaller than crystals grown in it in space.
- (D) Crystals grown in the device in space have been exceeded in size by crystals grown in subsequent experiments in space using other devices.
- (E) The experiments in which the device was used were conducted with proper controls.

20. The passage suggests that the author would most probably agree with which of the following assessments of the results of the Space Lab experiment?

- (A) Although the results of the experiment are impressive, the experiment was too limited in scope to allow for definitive conclusions.
- (B) The results of the experiment are impressive on the surface, but the report is misleading.
- (C) The results of the experiment convincingly confirm what researchers have long suspected.
- (D) Because of design flaws, the experiment did not yield any results relevant to the issue under investigation.
- (E) The results of the experiment are too contradictory to allow for easy interpretation.

In 1923 the innovative Russian filmmaker Dziga Vertov described filmmaking as a process that leads viewers toward a "fresh perception of the world." Vertov's description of filmmaking should apply to films on the subject of art. Yet films on art have not had a powerful and pervasive effect on the way we see.

Publications on art flourish, but these books and articles do not necessarily succeed in teaching us to see more deeply or more clearly. Much writing in art history advances the discourse in the field but is unlikely to inform the eye of one unfamiliar with its polemics. Films, however, with their capacity to present material visually and to reach a broader audience, have the potential to enhance visual literacy (the ability to identify the details that characterize a particular style) more effectively than publications can. Unfortunately, few of the hundred or so films on art that are made each year in the United States are broadcast nationally on prime-time television.

The fact that films on art are rarely seen on prime-time television may be due not only to limitations on distribution but also to the shortcomings of many such films. Some of these shortcomings can be attributed to the failure of art historians and filmmakers to collaborate closely enough when making films on art. These professionals are able, within their respective disciplines, to increase our awareness of visual forms. For close collaboration to occur, professionals in each discipline need to recognize that films on art can be both educational and entertaining, but this will require compromise on both sides.

A filmmaker who is creating a film about the work of an artist should not follow the standards set by rock videos and advertising. Filmmakers need to resist the impulse to move the camera quickly from detail to detail for fear of boring the viewer, to frame the image for the sake of drama alone, to add music for fear of silence. Filmmakers are aware that an art object demands concentration and, at the same time, are concerned that it may not be compelling enough? and so they hope to provide relief by interposing "real" scenes that bear only a tangential relationship to the subject. But a work of art needs to be explored on its own terms. On the other hand, art historians need to trust that one can indicate and analyze, not solely with words, but also by directing the viewer's gaze. The specialized written language of art history needs to be relinquished or at least tempered for the screen. Only an effective collaboration

between filmmakers and art historians can create films that will enhance viewers' perceptions of art.

21. The passage suggests that a filmmaker desiring to enhance viewers' perceptions of art should do which of the following?

- (A) Rely on the precise language of art history when developing scripts for films on art.
- (B) Rely on dramatic narrative and music to set a film's tone and style.
- (C) Recognize that a work of art by itself can be compelling enough to hold a viewer's attention
- (D) Depend more strongly on narration instead of camera movements to guide the viewer's gaze.
- (E) Emphasize the social and the historical contexts within which works of art have been created.

22. The author of the passage refers to Vertov in the first paragraph most probably in order to

- (A) provide an example of how films can be used to influence perceptions
- (B) present evidence to support the argument that films have been used successfully to influence viewers' perceptions
- (C) introduce the notion that film can influence how viewers see
- (D) contrast a traditional view of the uses of film with a more modern view
- (E) describe how film can change a viewer's perception of a work of art

23. Which of the following best describes the organization of the passage?

- (A) An observation about an unsatisfactory situation is offered, the reasons for the situation are discussed, and then ways to change it are suggested.
- (B) Two opinions regarding a controversial phenomenon are contrasted, supporting evidence for each is presented, and then the two opinions are reconciled.
- (C) Criticism of a point of view is discussed, the criticism is answered, and then the criticism is applied to another point of view.
- (D) A point of view is described, evidence supporting the view is provided, and then a summary is

presented.

- (E) A strategy is presented, reasons for its past failure are discussed, and then a recommendation that will be abandoned is offered.

24. The passage is primarily concerned with

- (A) discussing why film's potential as a medium for presenting art to the general public has not been fully realized and how film might be made more effective in this regard
- (B) discussing the shortcomings of films on art and the technological innovations required to increase the impact of film on visual literacy
- (C) discussing the advantages and the disadvantages of using films rather than publications to present works of art to the general public
- (D) presenting information to support the view that films on art must focus more on education and less on entertainment in order to increase visual literacy
- (E) presenting information to support the view that films on art, because they reach a broader audience than many other kinds of media, have had greater success in promoting visual literacy

25. The author would most likely agree with which of the following statements about film and visual literacy?

- (A) Reading a publication about a work of art and then seeing a film about the same work is the most effective way to develop visual literacy.
- (B) An increase in a viewer's awareness of visual forms will also lead to an increased attention span.
- (C) Film has a great but not yet fully exploited capacity to increase viewers' awareness of visual forms.
- (D) A film that focuses on the details of a work of art will hinder the development of visual literacy.
- (E) Films on art would more effectively enhance the visual literacy of teenagers if filmmakers followed the standards set by rock videos.

26. According to the passage, art historians desiring to work with filmmakers to enhance the public's appreciation of art need to acknowledge which of the fol-

lowing?

- (A) The art historian's role in the creation of a film on art is likely to be a relatively minor one.
- (B) Film provides an ideal opportunity to acquaint viewers with a wide range of issues that relate incidentally to a work of art.
- (C) An in-depth analysis of a work of art is not an appropriate topic for a film on art.
- (D) Although silence may be an appropriate background when viewing a work of art in a museum, it is inappropriate in a film.
- (E) Film can use nonverbal means to achieve some of the same results that a spoken or written discourse can achieve.

27. Which of the following would describe the author's most likely reaction to a claim that films on art would more successfully promote visual literacy if they followed the standards set for rock videos?

- (A) Ambivalence
- (B) Indifference
- (C) Sympathy
- (D) Interest
- (E) Disdain

28. ACCESSORY:

- (A) insubordinate
- (B) invisible
- (C) of high quality
- (D) of massive proportions
- (E) of primary importance

29. CHAMPION:

- (A) emulate
- (B) disparage
- (C) compel
- (D) anticipate
- (E) disappoint

30. DECADENCE:

- (A) cheerfulness
- (B) promptness
- (C) cleanliness
- (D) wholesomeness
- (E) carefulness

31. OPACITY:

- (A) transparency
- (B) smoothness
- (C) colorlessness
- (D) elongation and thinness
- (E) hardness and durability

32. MISGIVING:

- (A) consistency
- (B) certainty
- (C) generosity
- (D) loyalty
- (E) affection

33. HARANGUE:

- (A) overtly envy
- (B) intermittently forget
- (C) gratefully acknowledge
- (D) speak temperately
- (E) sacrifice unnecessarily

34. GERMANE:

- (A) unproductive
- (B) irregular
- (C) indistinguishable
- (D) irrelevant
- (E) unsubstantiated

35. IMPUGN:

- (A) rectify
- (B) classify
- (C) vindicate
- (D) mollify
- (E) chastise

36. INEXORABLE:

- (A) discernible
- (B) quantifiable
- (C) relenting
- (D) inspiring
- (E) revealing

37. RESTIVE:

- (A) necessary

- (B) interesting
 (C) calm
 (D) healthy
 (E) deft

38. BAIT:

- (A) perplex
 (B) disarm
 (C) delude
 (D) release
 (E) fortify

SECTION 6

Time –30 minutes

25 Questions

Questions 1-7

A professor of economics is writing a textbook that will consist of eight chapters, numbered consecutively from 1 through 8, each of which will explain exactly one of eight subjects : N, O, R, S, T, X, Y, and Z. No subject will be explained in more than one chapter. Considerations of background knowledge and relatedness between subjects require that the order in which the eight subjects are explained be subject to the following conditions :

N and O must be explained in two successive chapters, not necessarily in that order.

R, S, and T must be explained in three successive chapters, with S explained in an earlier chapter than T is explained.

X, Y, and Z must be explained in three successive chapters, with Z explained in an earlier chapter than X is explained and in an earlier chapter than Y is explained.

R must be explained in chapter 6.

1. Which of the following is an acceptably ordered list of the subjects to be explained, in order from chapter 1 through chapter 8?

- (A) N, O, Y, Z, X, R, S, T
 (B) O, N, S, T, X, R, Y, Z
 (C) O, N, Z, X, Y, S, T, R
 (D) Z, X, Y, S, T, R, N, O
 (E) Z, X, Y, T, S, R, N, O

2. If O is explained in chapter 5, which of the following could be true?

- (A) S is explained in chapter 4.
 (B) T is explained in chapter 7.
 (C) X is explained in chapter 2.
 (D) Y is explained in chapter 1.
 (E) Z is explained in chapter 3.

3.If S is explained in an earlier chapter than N is explained, which of the following must be true?

- (A) N is explained in chapter 7.
- (B) O is explained in chapter 8.
- (C) X is explained in chapter 3.
- (D) Y is explained in chapter 2.
- (E) Z is explained in chapter 1.

4.If N is explained in the chapter immediately preceding the chapter in which Z is explained, which of the following could be explained in chapter 5?

- (A) N
- (B) O
- (C) S
- (D) T
- (E) X

5.If T is not explained in chapter 5, which of the following must be true?

- (A) N is explained in chapter 3.
- (B) O is explained in chapter 8.
- (C) S is explained in chapter 7.
- (D) X is explained in chapter 2.
- (E) Y is explained in chapter 4.

6.O can be explained in any of the following chapters EXCEPT

- (A) 2
- (B) 3
- (C) 4
- (D) 5
- (E) 7

7.If Y is explained in the chapter immediately preceding the chapter in which O is explained, which of the following must be explained in chapter 5?

- (A) N
- (B) T
- (C) X
- (D) Y
- (E) Z

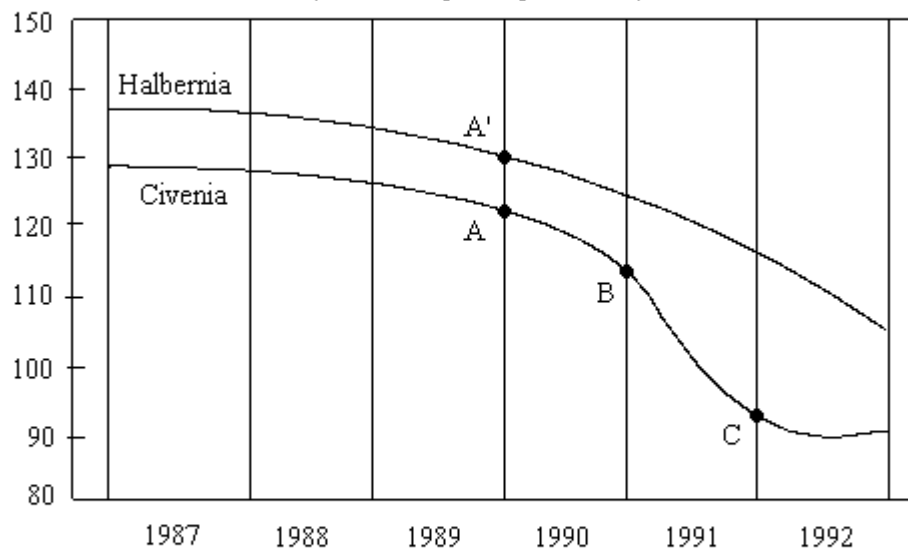
8.Salesperson : The picture tube is the central com-

ponent of any television, and Picturesque televisions use the same high-quality picture tubes as those used in TruVue televisions. Since you pay a much lower price for a Picturesque, you pay a lower price for a television with the same picture quality when you buy a Picturesque instead of a TruVue.

Which of the following is an assumption that, if justified, would allow the conclusion of the salesperson's argument to be properly drawn?

- (A) TruVue televisions are much more widely advertised than are Picturesque televisions.
- (B) The picture quality of a television is determined solely by the quality of its picture tube.
- (C) A salesperson earns much less on the sale of each Picturesque television than on the sale of each TruVue television.
- (D) Many more Picturesque televisions are sold each year than TruVue television.
- (E) Picturesque televisions are assembled in the same factory that assembles TruVue televisions.

CIGARETTE SALES IN TWO NONADJACENT
KIMMERLAND PROVINCES 1987-1992
(millions of packs per month)



- A : Civenia institutes 40 % increase in cigarette tax
 A' : Halbernia institutes 40 % increase in cigarette tax
 B : Civenia begins intensive antismoking television advertising campaign
 C : Civenia concludes intensive antismoking television advertising campaign

9. The claim that Civenia's antismoking television advertising campaign contributed significantly to the steep decline in cigarette purchases in that province during 1991 is best supported if which of the following has been true about the province of Halbernia from the beginning of 1991?
- (A) It kept its cigarette tax at the 1990 level and instituted an antismoking television advertising campaign similar to Civenia's.
- (B) It kept its cigarette tax at the 1990 level and did not institute an antismoking television advertising campaign.
- (C) It rescinded the 1990 cigarette tax increase and did not institute an antismoking television advertising campaign.
- (D) It eliminated all cigarette taxes and did not institute an antismoking television advertising campaign.
- (E) It increased its cigarette tax by an additional 40 percent over the 1990 level and instituted an antismoking television advertising campaign similar to Civenia's.
10. Which of the following, if true, most helps to explain why cigarette purchases in Civenia declined more slowly after the completion of the television antismoking advertising campaign than they did before the campaign began?
- (A) Laws restricting smoking in public places in Civenia went into effect at the beginning of 1992.
- (B) Most of those smokers in Civenia who were likely to quit or cut down on smoking had done so by the beginning of 1992.
- (C) At the beginning of 1992, health insurance companies in Civenia offered lower rates to nonsmoking residents than to those who smoked.
- (D) Cigarette companies increased their prices to Civenia distributors at the beginning of 1992 to offset the decrease in the number of packs sold.
- (E) Some cigarette companies withdrew their advertisements from Civenia newspapers and magazines at the beginning of 1992.

A real estate agent will show exactly five apartments, one after another, in a single afternoon. The apartments will be selected from among four large apartments—G, H, J, and K—and three small apartments—P, Q and R. The agent will show three large apartments and two small apartments. The agent's selection of apartments and the order in which they will be shown are constrained as follows :

The two small apartments selected must be shown second and fourth, respectively.

H and Q cannot both be shown.

If G is shown, it must be shown first.

If P is shown, H must also be shown.

If J and K are both shown, J must be shown at some time before K is shown.

If R and P are both shown, R must be shown at some time before P is shown.

11. Which of the following can be the apartments selected in an order, from first through fifth, in which they can be shown?

- (A) G, H, R, K, P
- (B) G, Q, J, P, K
- (C) G, Q, J, R, K
- (D) H, P, J, R, K
- (E) J, Q, H, P, K

12. Which of the following is a pair of apartments that CANNOT both be shown?

- (A) G and H
- (B) G and P
- (C) H and J
- (D) P and Q
- (E) Q and R

13. If Q and K are both shown, which of the following must be true?

- (A) J is shown third.
- (B) J is shown fifth.
- (C) K is shown third.
- (D) Q is shown second.
- (E) Q is shown fourth.

14. If R is shown fourth, which of the following must be true?

- (A) H is shown first.
- (B) H is shown third.
- (C) K is shown third.
- (D) P is shown second.
- (E) Q is shown second.

15. If J is shown first, then a pair of apartments that can be shown fourth and fifth, respectively, are

- (A) K and P
- (B) K and R
- (C) P and K
- (D) Q and K
- (E) R and K

Out of nine nominees, seven will be selected to form a committee. Exactly four of the nine nominees—two women and two men? belong to party P; another three—one woman and two men—belong to party Q, and the remaining two? one woman and one man—belong to party R. The rules governing the selection of the committee are as follows :

At least three women must be selected.

No party can have more than three of its members selected.

16. Which of the following is an acceptable committee?

<u>Party P</u>	<u>Party Q</u>	<u>Party R</u>
(A) 1 woman	—	1 woman
2 men	2 men	1 man
(B) 2 women	—	1 woman
1 man	2 men	—
(C) 2 women	1 woman	—
2 men	1 man	1 man
(D) 2 women	1 woman	—
1 man	2 men	1 man
(E) —	1 woman	1 woman
2 men	2 men	1 man

17. If both of the men belonging to party P are selected, which of the following must be true?

- (A) More men than women are selected.
- (B) More women than men are selected.

- (C) More members of party P than members of party Q are selected.
 (D) More members of party Q than members of party R are selected.
 (E) More women than members of party Q are selected.
- 18.If more women than men are selected, which of the following must be true?
 (A) The man belonging to party R is selected
 (B) At least one of the men belonging to party Q is selected.
 (C) Members of party Q outnumber members of party P on the committee.
 (D) Members of party P outnumber members of party Q on the committee.
 (E) There are equal numbers of members of party P and members of party Q on the committee.
- 19.Which of the following can be true?
 (A) All of the members of party P are selected.
 (B) The only members of party P who are selected are men.
 (C) The only members of party P who are selected are women.
 (D) The only nominees selected who are not members of party P are men.
 (E) The only nominees selected who are not members of party P are women.
- 20.Which of the following must be true?
 (A) Exactly four women are selected.
 (B) Exactly four men are selected.
 (C) At least one nominee belonging to party R is selected.
 (D) At least three of the nominees belonging to party P are selected.
 (E) All three of the nominees belonging to party Q are selected.
- 21.If one of the people not selected is the woman belonging to party Q, which of the following must be true?
 (A) A man belonging to party P is not selected.
 (B) A man belonging to party Q is not selected.
 (C) The man belonging to party R is not selected.
 (D) A woman belonging to party P is not selected.
 (E) The woman belonging to party R is not selected.
- 22.If members of party Q outnumber members of party P on the committee, which of the following can be true?
 (A) The man belonging to party R is not selected.
 (B) The woman belonging to party R is not selected.
 (C) Both of the men belonging to party P are selected.
 (D) All of the women who were nominated are selected.
 (E) All of the men who were nominated are selected.
- 23.In a recent film set in seventeenth-century Europe, the hero is seen doing the crawl, a swimming stroke not known in Europe before the 1920's.However, since moviegoers obviously are not experts in the history of swimming strokes, for most of the film's audiences this blunder clearly cannot have interfered with whatever sense of historical authenticity the film otherwise achieved.
- Which of the following, if true, most seriously weakens the argument given?
 (A) The film was widely praised for being historically plausible, even though it portrayed many events that were not historically attested.
 (B) The scene that shows the film's hero doing the crawl is a rescue scene pivotal to the film's action, and parts of it are even shown a second time, in a flashback.
 (C) Makers of historical films, even of those set as recently as the nineteenth century, routinely strike compromises between historical authenticity and the need to keep their material accessible to a modern audience, as in the actors' speech patterns.
 (D) The crawl that European swimmers used in the 1920's was much less efficient and more awkward-looking than the crawl that is currently taught.

- (E) A slightly earlier film featuring an eighteenth century sea battle in Europe was ridiculed in numerous popular reviews for the historical lapse of showing a sailor doing the crawl in swimming to safety.

24. The government of Pontran claims that Tor City, one of the six major cities in that country, is alone among Pontran's cities in having sustained strong job growth this year. Clearly, however, any job growth this year. Clearly, however, any job growth there must be purely imaginary; in fact, in Tor City and only there, more people are unemployed this year than were last year.

The argument countering the government's claim depends on the assumption that :

- (A) unemployed workers in Pontran did not, in significant numbers, move to Tor City when the government claimed Tor City had strong job growth
- (B) the unemployment rate in Tor City is higher year than in any previous year
- (C) actions taken by the government of Pontran significantly affect the unemployment rate in Tor City
- (D) the unemployment rate in Tor City, though increased, is still the lowest of any city in Pontran
- (E) there is no significant seasonal variation in the unemployment rate for Pontran as a whole

25. As people age, the number of calories they need each day decreases while their daily requirement of vitamin B₆ increases. Clearly, unless older people take B₆ supplements or eat a diet that contains more B₆ than did the diet they ate when they were young adults, there is little likelihood that they will get B₆ in needed amounts.

Which of the following would be most helpful to know to evaluate the argument?

- (A) Whether the relative decrease in the daily requirement of calories is greater than the relative increase in the daily requirement of vitamin B₆.
- (B) Whether the form of vitamin B₆ found in dietary supplements is more readily absorbed and utilized by the body than is the form of this vitamin that

is found in food.

- (C) Whether the consequences of not getting vitamin B₆ in required daily amounts are more serious for older people than for young adults.
- (D) Whether the diets of most people, when they are young adults, include vitamin B₆ in amounts far in excess of their daily needs.
- (E) Whether the diets of older people are more likely than those of young adults to include one or more meals a day that are devoid of foods containing vitamin B₆.

SECTION 1

Time –30 minutes

38 Questions

1. We first became aware that her support for the new program was less than---- when she declined to make a speech in its favor.
 (A) qualified
 (B) haphazard
 (C) fleeting
 (D) unwarranted
 (E) wholehearted
2. When a person suddenly loses consciousness, a bystander is not expected to ---- the problem but to attempt to ---- its effects by starting vital functions if they are absent
 (A) cure.. precipitate
 (B) minimize.. predict
 (C) determine.. detect
 (D) diagnose.. counter
 (E) magnify.. evaluate
3. The remark was only slightly ----, inviting a chuckle, perhaps, but certainly not a ----.
 (A) audible.. reward
 (B) hostile.. shrug
 (C) amusing.. rebuke
 (D) coherent.. reaction
 (E) humorous.. guffaw
4. Doors were closing on our past, and soon the values we had lived by would become so obsolete that we would seem to people of the new age as ----as travelers from an ancient land.
 (A) elegant
 (B) ambitious
 (C) interesting
 (D) comfortable
 (E) quaint
5. Ability to ---- is the test of the perceptive historian, a history, after all, consists not only of what the historian has included, but also, in some sense, of what has been left out.
 (A) defer
 (B) select
 (C) confer
 (D) devise
 (E) reflect
6. Some artists immodestly idealize or exaggerate the significance of their work; yet others, ---- to exalt the role of the artist, reject a transcendent view of art
 (A) appearing
 (B) disdaining
 (C) seeking
 (D) failing
 (E) tending
7. Estimating the risks of radiation escaping from a nuclear power plant is ---- question, but one whose answer then becomes part of a value-laden, emotionally charged policy debate about whether to construct such a plant.
 (A) an incomprehensible
 (B) an undefined
 (C) an irresponsible
 (D) a divisive
 (E) a technical
8. TREE: FORESTRY::
 (A) tractor: agriculture
 (B) experiment: laboratory
 (C) fuel: combustion
 (D) flower: horticulture
 (E) generator: electricity
9. COMMAND: REQUEST
 (A) presume: inquire
 (B) recommend: propose
 (C) summon: invite
 (D) refuse: rebel
 (E) authorize: permit
10. PESTLE: GRIND::
 (A) whetstone: sharpen
 (B) balloon: float
 (C) mill: turn
 (D) hinge: fasten
 (E) switch: conduct

11. ILLITERACY: EDUCATION

- (A) bureaucracy: clarification
- (B) oppression: agreement
- (C) vagrancy: travel
- (D) inequity: redistribution
- (E) inclemency: evasion

12. REVERENCE: RESPECT::

- (A) resiliency: vitality
- (B) appreciation: dependency
- (C) avidity: enthusiasm
- (D) imagination: creativity
- (E) audacity: sentiment

13. APOSTROPHES:WORD

- (A) letters: alphabet
- (B) verbs: syntax
- (C) ellipses: sentence
- (D) commas: punctuation
- (E) paragraphs: essay

14. EXAGGERATION: CARICATURE::

- (A) craft: art
- (B) detail: sketch
- (C) illusion: story
- (D) brevity: epigram
- (E) sophistication: farce

15. MALLEABLE: SHAPE

- (A) apathetic: emotion
- (B) irresolute: opinion
- (C) demonstrable: evidence
- (D) irredeemable: value
- (E) gustatory: taste

16. BOLSTER: SUPPORT::

- (A) axis: revolve
- (B) spackle: paint
- (C) leakage: caulk
- (D) heat: insulate
- (E) tackle: hoist

Geologists Harris and Gass hypothesized that the Red Sea rift developed along the line of a suture (a splice in the Earth's crust) formed during the late Proterozoic era, and that significant observable differences in the composition of

(5)the upper layers of rocks deposited on either side of the suture give clues to the different natures of the underlying igneous rocks.

Other geologists argued that neither the upper rock layer nor the underlying igneous rocks on the one side of the rift (10)differ fundamentally from the corresponding layers on the other side. These geologists believe, therefore, that there is inadequate evidence to conclude that a suture underlies the rift.

In response, Harris and Gass asserted that the upper rock (15)layers on the two sides of the rift had not been shown to be of similar age, structure, or geochemical content. Furthermore, they cited new evidence that the underlying igneous rocks on either side of the rift contain significantly different kinds of rare metals.

17.Part of the Harris and Gass hypothesis about the Red Sea rift would be weakened if it could be demonstrated that the composition of upper rock layers

- (A) cannot cause a suture to develop
- (B) has no effect on where a suture will occur
- (C) cannot provide information about the nature of underlying rocks
- (D) is similar on the two sides of a rift unless a suture divides the two sides
- (E) is usually different from the composition of underlying rocks

18.It can be inferred from the passage that the "Other geologists" (line 8) would be most likely to agree with which of the following statements?

- (A) Similar geological features along both sides of a possible suture imply the existence of that suture
- (B) Sutures can be discovered only where they are not obscured by superimposed geological features.
- (C) The composition of igneous rocks permits prediction of the likelihood of a rift developing through them.
- (D) It is possible to date igneous rocks by carefully studying the different kinds of rare metals contained in them and by observing their similarity to the layer of rock that lies above them.
- (E) The existence of rock layers on one side of a rift that are similar in composition to rock layers on the other side suggests that no suture exists between the two sides.

19. It can be inferred from the passage that Harris and Gass have done which of the following?

- (A) Drawn detailed diagrams of the Red Sea rift.
- (B) Based their conclusions on the way in which sutures develop in the Earth's crust.
- (C) Rejected other geologists' objections to their hypothesis about the Red Sea rift.
- (D) Suggested that the presence of rare metals in rocks indicates an underlying suture.
- (E) Asserted that rifts usually occur along the lines of sutures

20. According to the passage, Harris and Gass have mentioned all of the following properties of rocks along the Red Sea rift EXCEPT

- (A) age of the upper layers of rock
- (B) structure of the upper layers of rocks
- (C) geochemical content of the upper layers of rocks
- (D) metallic content of the underlying igneous rocks
- (E) age of the underlying igneous rocks

Proponents of different jazz styles have always argued that their predecessors' musical style did not include essential characteristics that define jazz as jazz. Thus, 1940's swing was belittled by beboppers of the 1950's,

(5) who were themselves attacked by free jazzers of the 1960's. The neoboppers of the 1980's and 1990's attacked almost everybody else. The titanic figure of Black saxophonist John Coltrane has complicated the arguments made by proponents of styles from bebop through neobop (10) because in his own musical journey he drew from all those styles. His influence on all types of jazz was immeasurable. At the height of his popularity, Coltrane largely abandoned playing bebop, the style that had brought him fame, to explore the outer reaches of jazz.

(15) Coltrane himself probably believed that the only essential characteristic of jazz was improvisation, the one constant in his journey from bebop to open-ended improvisations on modal, Indian, and African melodies. On the other hand, this dogged student and prodigious technician—who (20) insisted on spending hours each day practicing scales from theory books—was never able to jettison completely the influence of bebop, with its fast and elaborate chains of notes and ornaments on melody.

Two stylistic characteristics shaped the way Coltrane (25) played the tenor saxophone; he favored playing fast runs

of notes built on a melody and depended on heavy, regularly accented beats. The first led Coltrane to "sheets of sound," where he raced faster and faster, pile-driving notes into each other to suggest stacked harmonies. The second (30) meant that his sense of rhythm was almost as close to rock as to bebop.

Three recordings illustrate Coltrane's energizing explorations. Recording *Kind of Blue* with Miles Davis, Coltrane found himself outside bop, exploring modal melodies. Here (35) he played surging, lengthy solos built largely around repeated motifs—an organizing principle unlike that of free jazz saxophone player Ornette Coleman, who modulated or altered melodies in his solos. On *Giant Steps*, Coltrane debuted as leader, introducing his own compositions. Here the sheets of sound, downbeat accents, repetitions, and great speed are part of each solo, and the variety of the shapes of his phrases is unique. Coltrane's searching explorations produced solid achievement. *My Favorite Things* was another kind of watershed. Here (45) Coltrane played the soprano saxophone, an instrument seldom used by jazz musicians. Musically, the results were astounding. With the soprano's piping sound, ideas that had sounded dark and brooding acquired a feeling of giddy fantasy.

(50) When Coltrane began recording for the Impulse! label, he was still searching. His music became raucous, physical. His influence on rockers was enormous, including Jimi Hendrix, the rock guitarist, who following Coltrane, raised the extended guitar solo using repeated motifs to a kind of rock art form.

21. The primary purpose of the passage is to

- (A) discuss the place of Coltrane in the world of jazz and describe his musical explorations
- (B) examine the nature of bebop and contrast it with improvisational jazz
- (C) analyze the musical sources of Coltrane's style and their influence on his work
- (D) acknowledge the influence of Coltrane's music on rock music and rock musicians
- (E) discuss the arguments that divide the proponents of different jazz styles

22. The author implies that which of the following would have been an effect of Coltrane's having chosen to play the tenor rather than the soprano saxophone on

My Favorite Things?

- (A) The tone of the recording would have been more somber.
- (B) The influence of bebop on the recording would have been more obvious
- (C) The music on the recording would have sounded less raucous and physical
- (D) His influence on rock music might have been less pervasive.
- (E) The style of the recording would have been indistinguishable from that on *Kind of Blue*

23. Which of the following best describes the organization of the fourth paragraph?

- (A) A thesis referred to earlier in the passage is mentioned and illustrated with three specific examples
- (B) A thesis is stated and three examples are given each suggesting that a correction needs to be made to a thesis referred to earlier in the passage
- (C) A thesis referred to earlier in the passage is mentioned, and three examples are presented and ranked in order of their support of the thesis.
- (D) A thesis is stated, three seemingly opposing examples are presented, and their underlying correspondence is explained
- (E) A thesis is stated, three dissimilar examples are considered, and the thesis is restated.

24. According to the passage, John Coltrane did all of the following during his career EXCEPT

- (A) improvise on melodies from a number of different cultures
- (B) perform as leader as well as soloist
- (C) spend time improving his technical skills
- (D) experiment with the sounds of various instruments
- (E) eliminate the influence of bebop on his own music

25. The author mentions the work of Ornette Coleman in the fourth paragraph in order to do which of the following?

- (A) Expand the discussion by mentioning the work of a saxophone player who played in Coltrane's style.
- (B) Compare Coltrane's solos with the work of another jazz artist.

- (C) Support the idea that rational organizing principles need to be applied to artistic work .
- (D) Show the increasing intricacy of Coltrane's work after he abandoned bebop
- (E) Indicate disagreement with the way Coltrane modulated the motifs in his lengthy solos.

26. According to the passage, a major difference between Coltrane and other jazz musicians was the

- (A) degree to which Coltrane's music encompassed all of jazz
- (B) repetition of motifs that Coltrane used in his solos
- (C) number of his own compositions that Coltrane recorded
- (D) indifference Coltrane maintained to musical technique
- (E) importance Coltrane placed on rhythm in jazz

27. In terms of its tone and form, the passage can best be characterized as

- (A) dogmatic explanation
- (B) indignant denial
- (C) enthusiastic praise
- (D) speculative study
- (E) lukewarm review

28. RECORD

- (A) postpone
- (B) disperse
- (C) delete
- (D) delay
- (E) devise

29. EMBED

- (A) induce
- (B) extend
- (C) extract
- (D) receive
- (E) diverge

30. WHOLESOME

- (A) deleterious
- (B) submissive
- (C) provoking
- (D) monotonous
- (E) rigorous

31. EXTINCTION:

- (A) immunity
- (B) mutation
- (C) inhibition
- (D) formulation
- (E) perpetuation

32. CURSE:

- (A) exoneration
- (B) untruth
- (C) redress
- (D) benediction
- (E) separation

33. DECORUM

- (A) constant austerity
- (B) false humility
- (C) impropriety
- (D) incompetence
- (E) petulance

34. AGGRANDIZE:

- (A) misrepresent
- (B) disparage
- (C) render helpless
- (D) take advantage of
- (E) shun the company of

35. VIGILANT

- (A) reluctant
- (B) haphazard
- (C) gullible
- (D) ignorant
- (E) oblivious

36. FASTIDIOUS

- (A) coarse
- (B) destructive
- (C) willing
- (D) collective
- (E) secret

37. TRACTABLE

- (A) indefatigable
- (B) incorrigible
- (C) insatiable

(D) impractical

(E) impetuous

38. RESCISSION

- (A) expansion
- (B) enactment
- (C) instigation
- (D) stimulation
- (E) abdication

SECTION 3

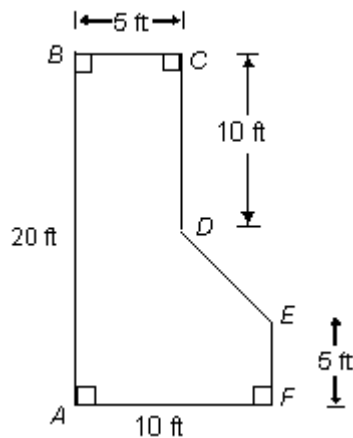
Time—30 minutes

30 Questions

5 is $\frac{1}{5}$ of x 1. x 1

2. A number whose square is 4 less than twice the square of 10 15

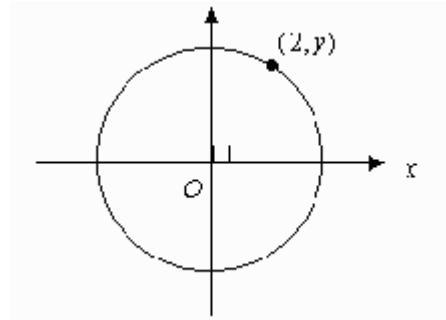
$$\sqrt{x-1} = 4$$

3. x 174. The area of region $ABCDEF$ 150 sqft

$$2x = x + 1$$

5. x y

$$x \neq 0$$

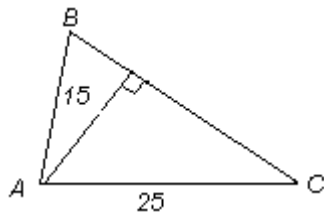
6. $\frac{2}{x}$ $2x$ The circle above with center O has a radius of 57. y $\sqrt{20}$

A government program paid out \$20,000 to each of 60,000 families.

8. The total amount that the program paid out to the families $\$1.2 \times 10^9$ x and y each represent single digits in the decimal $4.3y8$. When the decimal is rounded to the nearest hundredth, the result is 4.36.9. x 5In $\triangle ABC$, $AB = 6$, $BC = 8$, and $CA = 10.5$ 10. The measure of angle ABC 90° 11. $\frac{4^6 - 4^4}{4^5 - 4^3}$ 1

12. The standard deviation of the sample measurements 0, 1, 2, 4, 8 The standard deviation of the sample measurements 1, 2, 3, 5, and 9

 P , Q and R are distinct points on a straight line, with Q between P and R , S is a point not on the line, and $QR = QS$ 13. PR PS



14. The area of triangular region ABC 225

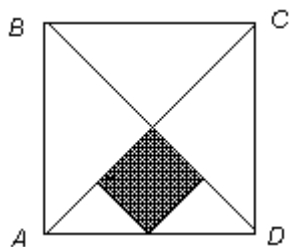
15. t^3 $\frac{r}{1,000}$

16. Diane completed each of her exercise sets in 75 seconds and rested for 55 seconds between the exercise sets. What is the ratio of the amount of time it took her to complete an exercise set to the amount of time she rested between the sets?

- (A) 5:6
(B) 5:11
(C) 6:11
(D) 11:5
(E) 11:6

17. A certain car gets 20 miles per gallon of gasoline for city driving. If the car gets 15 percent more miles per gallon for highway driving, how many miles per gallon does the car get for highway driving?

- (A) 17
(B) 22
(C) 23
(D) 30
(E) 35



18. In the figure above, the area of the shaded square region is what percent of the area of the square region $ABCD$?

- (A) $16\frac{2}{3}\%$
(B) $12\frac{1}{2}\%$

- (C) 12%
(D) $8\frac{1}{2}\%$
(E) 8%

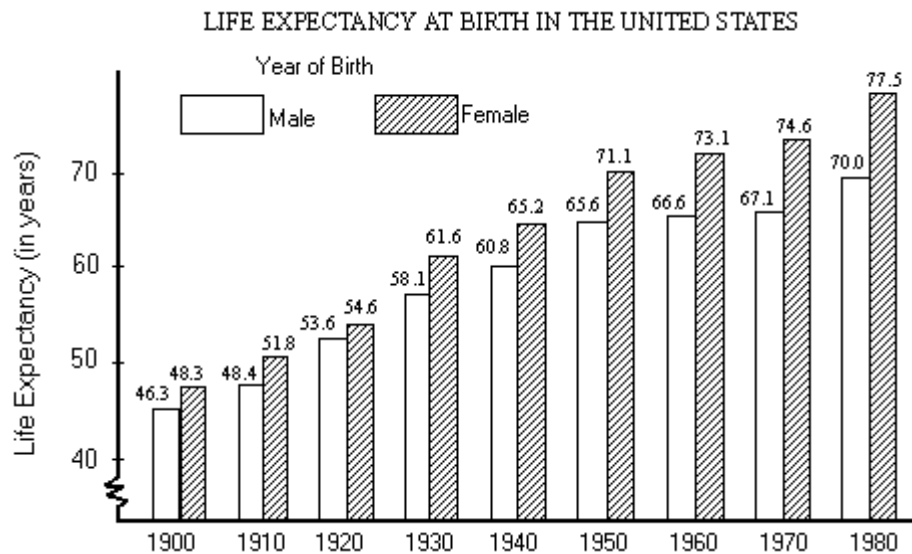
19. The cost of sending a package special delivery is x cents per ounce up to 10 ounces and y cents for each ounce in excess of 10. Which of the following represents the total cost, in cents, of sending special delivery a package weighing w ounces, if $w > 10$?

- (A) $x + (w - 10)y + 10$
(B) $x + (w - 10) + y$
(C) $10x + 10(w - y)$
(D) $10x + y(w - 10)$
(E) $10x + wy$

20. All of the following are equivalent to $\frac{a-b}{c}$ EXCEPT

- (A) $\frac{-(a-b)}{c}$
(B) $\frac{-(a-b)}{-c}$
(C) $-\frac{a-b}{-c}$
(D) $-\frac{b-a}{c}$
(E) $\frac{b-a}{-c}$

Questions 21-25 refer to the following graph. In answering these questions, assume that the male and female populations are equal for each of the years shown.



21. For which of the following years was the difference between the life expectancy at birth for males and the life expectancy at birth for females the least?
- (A) 1900
(B) 1910
(C) 1920
(D) 1930
(E) 1940
22. The average(arithmetic mean) life expectancy at birth for a person born in 1980 was approximately how many more years than for a person born in 1900?
- (A) 20
(B) 22
(C) 24
(D) 26
(E) 30
23. The life expectancy at birth of a male born in 1900 is closest to what fraction of the life expectancy at birth of a male born in 1980?
- (A) $\frac{4}{3}$
(B) $\frac{2}{3}$
(C) $\frac{1}{2}$
(D) $\frac{1}{3}$
(E) $\frac{1}{4}$
24. Which of the following statements can be inferred from the graph?
- I For each year shown, the life expectancy at birth was greater for females than it was for males
II The increase in the life expectancy at birth for males was greater from 1940 to 1980 than it was from 1900 to 1940.
III For each year shown, the difference between the life expectancy at birth for males and that for females was greater than the corresponding difference in the preceding year shown.
- (A) I only
(B) I and II only
(C) I and III only
(D) II and III only
(E) I, II and III
25. From 1900 to 1980, life expectancy at birth for males increased by approximately what percent?
- (A) 24%

- (B) 34%
 (C) 40%
 (D) 51%
 (E) 66%

26 If $\sqrt{7} < x < \sqrt{37}$ and x is an integer, then x can have how many different values?

- (A) Three
 (B) Four
 (C) Five
 (D) Eight
 (E) Ten

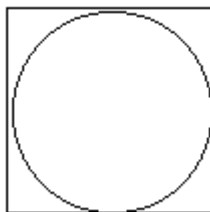
27. For which of the following expressions is the value for $x = 0$ equal to the value for $x = 1$?

- (A) $\frac{x}{x+1}$
 (B) $\frac{x-1}{x+1}$
 (C) $2x - 1$
 (D) $x(x-1) + x$
 (E) $x(x-1) + 1$

28. If the average (arithmetic mean) of x , y , z , 5, and 7 is 8, which of the value for $x = 1$?

- I The median of the five numbers cannot be 5
 II At least one of x , y and z is greater than 9
 III The range of the five numbers is 2 or more

- (A) I only
 (B) II only
 (C) III only
 (D) I and III
 (E) II and III



29. As shown in the figure above a circular flower bed lies in a square garden plot that is 60 meters on each side. What fraction of the garden plot area is not part of the flower bed?

- (A) $\frac{1}{4}$
 (B) $\frac{3}{4}$
 (C) $\frac{4-p}{2}$
 (D) $\frac{4-p}{4}$
 (E) $\frac{p-2}{4}$

30. Which of the following CANNOT be a factor of 2^i and 3^j , where i and j are positive integers?

- (A) 6
 (B) 8
 (C) 27
 (D) 42
 (E) 54

SECTION 4

Time –30 minutes

25 Questions

1. The university's constitution defines the powers of two governing bodies. The general council, which currently retains authority over constitutional matters, consists of all the university's living alumni. The twenty-member senate decides routine matters by majority vote. A new proposal suggests making a unanimous vote in the senate sufficient to change the constitution.

If the statements above are true, which of the following must follow on the basis of them?

- (A) The proposal will take effect only if it receives the senate's unanimous support.
- (B) Currently each member of the senate has power of veto over any proposal that comes before that body.
- (C) Adopting the proposal would allow passage of constitutional changes without the general council's approval.
- (D) Any of the university's alumni can fill any one of the twenty seats on the senate.
- (E) Adopting the new proposal would increase the voice of the alumni in the conduct of university affairs.

2. Comparison of the growth rings of ancient trees enables scientists to determine from a piece of timber the year in which the tree used for the timber was felled. Hence, by analyzing the growth rings in timber surviving from ancient buildings, archaeologists can determine precisely when those buildings were constructed.

Which of the following is an assumption on which the argument depends?

- (A) The timber used for construction purposes in ancient times was made from very old trees.
- (B) The timber that was used in ancient building construction had not, prior to being used in the construction of the buildings, lain unused for an indeterminable amount of time.
- (C) The growth rings of any tree felled in a given

year are identical to the growth rings of any other tree felled in that year.

- (D) The oldest of the ancient buildings that survive to the present day were constructed of the most durable woods.
- (E) Ancient builders did not use more than one type of wood for the construction of a given building.

Questions 3-8

A researcher has designed an experiment to evaluate a theory about the health benefits of exercise at different times during the day. Nine subjects—F, G, H, J, K, M, O, P, and S—must be divided into three groups of three subjects each. Group 1 will exercise in the morning; group 2 will exercise in the afternoon; group 3 will exercise in the evening. The researcher must assign subjects to groups according to the following conditions:

- F must be assigned to group 2.
- H must be assigned to the same group as O.
- J cannot be assigned to group 2.
- M cannot be assigned to group 3.
- P cannot be assigned to the same group as S.

3. Which of the following is an acceptable assignment of subjects to the three groups?

<u>Group1</u>	<u>Group2</u>	<u>Group3</u>
(A) F, M, P	H, J, O	G, K, S
(B) G, J, M	F, H, O	K, P, S
(C) G, K, P	F, M, S	H, J, O
(D) H, J, M	F, G, P	K, O, S
(E) H, O, P	F, G, J	K, M, S

4. If H, J, and O are assigned to group 1, which of the following must be assigned to group 2?

- (A) G
- (B) K
- (C) M
- (D) P
- (E) S

5. If O is assigned to group 2, which of the following could be the subjects assigned to group 1?

- (A) G, J, M

- (B) G, K, M
- (C) G, K, P
- (D) J, M, S
- (E) M, P, S

6.If G, J,and M are assigned to group 1, which of the following must be assigned to group 2 ?

- (A) H
- (B) K
- (C) O
- (D) P
- (E) S

7.If G, K,and S are assigned to group 1, which of the following must be the subjects assigned to group 3 ?

- (A) F, H, O
- (B) H, J, O
- (C) H, J, P
- (D) J, M, P
- (E) M, O, P

8.If G, J,and P are assigned to group 1, which of the following must be true ?

- (A) H is assigned to group 3.
- (B) K is assigned to group 3.
- (C) M is assigned to group 3
- (D) O is assigned to group 2.
- (E) S is assigned to group 2.

9.Fossilized animal bones marked with scratches other than tooth marks were discovered in the sand near some stone tools. When bones were scratched using similar tools, the resulting scratches resembled the scratches found on the fossils. Therefore, stone tools had probably been used on the animal bones that became fossilized.

Which of the following, if true, most seriously undermines the argument ?

- (A) The fossilized bones with scratches are not as old as nearby fossilized bones found to have tooth marks.
- (B) Trampling on sand in which a bone is buried produces scratches on that bone that are similar to those on the fossils.
- (C) Stone tools have been found in areas where nearby

fossilized bones were not scratched.

- (D) The stone tools were too hard to be scratched by animal bones.
- (E) Modern microscopes can clearly reveal the differences between tooth marks and other kinds of scratches.

10.Color-tinting of movies originally photographed in black-and-white is inappropriate. Hundreds of the original artistic choices would have been made differently had these movies been filmed in color. Lighting on the sets of these movies was arranged to make highlights and shadows look right in black-and-white.

Which of the following claims, if substituted for the claim about the lighting of movie sets, would lend the same type of support to the argument above ?

- (A) An important part of what gives these movies their identity is the result of the black-and-white format.
- (B) Color film would have better captured the filmmakers' intentions.
- (C) Color film is superior to black-and-white film for making movies.
- (D) Makeup for the actors was applied so as to look best in black-and-white films.
- (E) The choice of black-and-white film over color film is entirely a matter of taste.

11.In Borkland, university scholarship stipends worth \$ 4.2 million were unclaimed last year because several scholarship programs attracted no applicants who met the programs' qualifying criteria. This is an egregious waste of funds in a country where thousands of promising students each year cannot afford tuition. Qualifying criteria for those scholarships, therefore, should be revised.

Which of the following, if true about Borkland, most strongly supports the conclusion drawn above ?

- (A) Many scholarships are unclaimed not because their qualifying criteria are too restrictive but because the scholarship programs rarely announce the scholarships' availability.
- (B) Because of inflation, the stipends offered by

certain scholarship programs now appear less attractive to potential applicants than they did when they were first offered.

- (C) A significant number of scholarship programs awarded all of their available scholarship stipends last year, yet most of those programs have very restrictive qualifying criteria.
- (D) Certain scholarship programs accept applications only from students enrolled in specific major fields of study, but Borkland's universities no longer offer courses in the fields specified by many of those programs.
- (E) Several scholarship programs have found it impossible to revise their qualifying criteria without engaging in lengthy and costly legal proceedings.

12. In Diersville the new Environmental Action party won two seats on the seven-member town council in 1988. It lost both of those seats in the 1992 election, even though the party's pro-environment platform had essentially remained unchanged. This decline in the party's fortunes clearly demonstrates that in Diersville environmental concerns faded in significance between 1988 and 1992.

Which of the following, if true, most seriously weakens the argument?

- (A) Between 1988 and 1992, the number of eligible voters in Diersville rose, but not the percentage who actually voted.
- (B) Between 1988 and 1992, Diersville's leading political party revised its platform, adopting a strongly environmentalist stance.
- (C) The parties that ran candidates in the 1992 election in Diersville were the same as those that had done so in the 1988 election.
- (D) In 1992 the Environmental Action party won fewer votes in Diersville than it had won in 1988.
- (E) Between 1988 and 1992, some measures intended to benefit the environment had been adopted by the town council, but with inconclusive results.

Questions 13-17

A gardener is landscaping part of a front yard with six types of bushes—forsythia, holly, lilac, oleander, rose, and spirea. The forsythia and the lilac bushes are the only bushes classified as tall bushes. The gardener will plant two clusters of exactly four bushes each, one on the east edge of the property and one on the west edge. The selection of the bushes is subject to the following constraints :

Each type of bush must be used at least once

Each cluster of bushes must contain at least one tall bush.

A cluster that includes one or more forsythia bushes must include at least one oleander bush.

A lilac bush cannot be planted in the same cluster as a spirea bush.

13. If three lilac bushes are selected for one cluster, which of the following can be the bushes selected for the other cluster?

- (A) Forsythia, holly, oleander, rose
- (B) Forsythia, holly, rose, spirea.
- (C) Forsythia, lilac, oleander, spirea.
- (D) Forsythia, oleander, rose, spirea.
- (E) Holly, oleander, rose, spirea.

14. Which of the following can be the selection of bushes for one of the clusters?

- (A) Forsythia, holly, oleander, rose
- (B) Forsythia, holly, oleander, spirea.
- (C) Forsythia, lilac, rose, rose.
- (D) Forsythia, oleander, oleander, oleander.
- (E) Lilac, lilac, rose, spirea.

15. If a total of two forsythia bushes is selected, which of the following must be true?

- (A) Only one holly bush is selected.
- (B) Only one rose bush is selected.
- (C) Only one spirea bush is selected.
- (D) Two oleander bushes are selected.
- (E) The two forsythia bushes are in the same cluster.

16. If one rose bush is selected for each cluster, which of the following combinations must complete one of the

clusters ?

- (A) Forsythia, forsythia, oleander .
- (B) Forsythia, lilac, oleander .
- (C) Forsythia, oleander, spirea.
- (D) Holly, holly,holly.
- (E) Holly, lilac, oleander .

17.If one forsythia bush is selected for the east cluster, and if two holly bushes are selected for the west cluster, then any of the following can be selected for the east cluster EXCEPT

- (A) holly
- (B) lilac
- (C) oleander
- (D) rose
- (E) spirea

Questions 18-22

A ballroom dance competition will be held between two teams, the Green team and the Red team. The Green team consists of couple H, couple I, and couple K. The Red team consists of couple X, couple Y, and couple Z. During the competition, each team will perform three types of dances—the fox-trot, the tango, and the waltz. Each couple on each team will perform exactly one of the three types of dances. At any one time, only one couple will be performing, but the six couples can perform in any order until all dances have been performed. The following rules also apply to the competition :

The same type of dance cannot be performed twice in succession.

Couple H cannot dance the waltz.

Couple I cannot dance the tango.

Couple Y must dance the waltz.

18.If couple K dances a fox-trot, which of the following couples must dance a tango ?

- (A) H
- (B) I
- (C) X
- (D) Y
- (E) Z

19.Which of the following could be the schedule for the competition's first three dances ?

<u>Fox-trot</u>	<u>Tango</u>	<u>Waltz</u>
(A) H	I	Z
(B) K	I	Y
(C) X	Z	H
(D) Y	K	I
(E) Z	H	Y

20.If the competition schedule strictly alternates couples on the Green team with couples on the Red team, and if couple Y dances fourth, then a waltz must be performed

- (A) first
- (B) second
- (C) third
- (D) fifth
- (E) sixth

21.If the third through the sixth dances in the competition are the fox-trot, the tango, the waltz, and the tango, respectively, then couple X can perform in any of the following positions EXCEPT.

- (A) first
- (B) second
- (C) third
- (D) fourth
- (E) sixth

22.Assume that the order of the couples, from first to sixth, has been established as X, Y, Z, H, I, and K. If couple X performs a fox-trot, which of the following couples must also perform a fox-trot ?

- (A) H
- (B) I
- (C) K
- (D) Y
- (E) Z

23.The United States is not usually thought of as a nation of parakeet lovers. Yet in a census of parakeet owners in selected comparable countries, the United States ranked second, with eleven parakeet owners per hundred people. The conclusion can be drawn from this that people in the United States are more likely to own parakeets than are people in most other

countries.

Knowledge of which of the following would be most useful in judging the accuracy of the conclusion?

- (A) The number of parakeets in the United States.
- (B) The number of parakeet owners in the United States.
- (C) The number of parakeet owners per hundred people in the country that ranked first in the census.
- (D) The number of parakeet owners in the United States compared to the numbers of owners of other pet birds in the United States.
- (E) The numbers of parakeet owners per hundred people in the countries not included in the census.

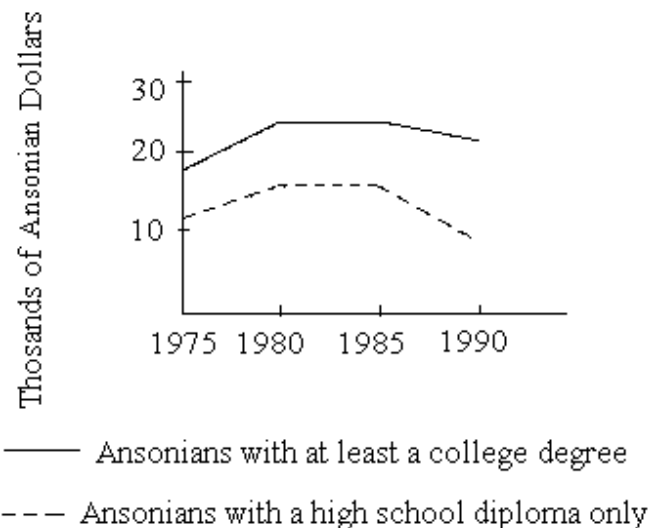
24. Until 1984 only aspirin and acetaminophen shared the lucrative nonprescription pain-reliever market. In 1984, however, ibuprofen was expected to account for fifteen percent of all nonprescription pain-reliever sales. On that basis business experts predicted for 1984 a corresponding fifteen percent decrease in the combined sales of aspirin and acetaminophen.

The prediction mentioned in the last sentence above was based on which of the following assumptions?

- (A) Most consumers would prefer ibuprofen to both aspirin and acetaminophen.
- (B) Aspirin, acetaminophen, and ibuprofen all relieve headache pain and muscular aches, but aspirin and ibuprofen can also cause stomach irritation.
- (C) Before 1984 ibuprofen was available only as a prescription medicine.
- (D) The companies that manufacture and sell aspirin and acetaminophen would not also manufacture and sell ibuprofen.
- (E) The introduction of ibuprofen would not increase total sales of nonprescription pain reliever.

Questions 25 is based on the following graph.

AVERAGE PER CAPITA YEARLY EARNINGS FOR TWO POPULATION GROUPS IN THE COUNTRY OF ANSONIA



Which of the following, if true, utters the best explanation for the growing disparity in the earning levels of the two groups in the graph presented above?

- (A) Since 1985 there has been a decline in the total number of Ansonians who have only a high school diploma.
- (B) Since 1985 the percentage of Ansonians without a college degree who are unemployed has remained relatively constant.
- (C) Government financial support for educational research in Ansonia has declined since 1985.
- (D) The enactment of generous social security legislation by the Ansonian parliament has allowed many Ansonians with only a high school diploma to retire at full salary at age 63.
- (E) The shift in Ansonia from an industrial to a service economy has resulted in a net loss of high-paying jobs traditionally available to Ansonians with only a high school diploma.

SECTION 5

Time –30 minutes

38 Questions

1. Because modern scientists find the ancient Greek view of the cosmos outdated and irrelevant, they now perceive it as only of ----- interest.
(A) historical
(B) intrinsic
(C) astronomical
(D) experimental
(E) superfluous
2. Religious philosopher that he was, Henry More derived his conception of an infinite universe from the Infinite God in whom he believed, a benevolent God of -----whose nature was to create-----
(A) plenitude ..abundance
(B) vengeance .. justice
(C) indifference .. suffering
(D) indulgence .. temperance
(E) rectitude .. havoc
3. While some argue that imposing tolls on highway users circumvents the need to raise public taxes for road maintenance, the phenomenal expense of maintaining a vast network of roads-----reliance on these general taxes.
(A) avoids
(B) diminishes
(C) necessitates
(D) discourages
(E) ameliorates
4. Although they were not direct-----,the new arts of the Classical period were clearly created in the spirit of older Roman models and thus-----many features of the older style.
(A) impressions .. introduced
(B) translations .. accentuated
(C) copies .. maintained
(D) masterpieces .. depicted
(E) borrowings .. improvised
5. In spite of the increasing-----of their opinions, the group knew they had to arrive at a consensus so that the award could be presented
(A) impartiality
(B) consistency
(C) judiciousness
(D) incisiveness
(E) polarity
6. By forcing our surrender to the authority of the clock systematic timekeeping has imposed a form of----- on society.
(A) anarchy
(B) permanence
(C) provincialism
(D) tyranny
(E) autonomy
7. Our high-----vocabulary for street crime contrasts sharply with our-----vocabulary for corporate crime, a fact that corresponds to the general public's unawareness of the extent of corporate crime.
(A) nuanced ..subtle
(B) uninformative ..misleading
(C) euphemistic ..abstract
(D) differentiated ..limited
(E) technical ..jargon-laden
8. DIVERGE : APART::
(A) traverse : across
(B) suspend : around
(C) reverse : beyond
(D) repose : beside
(E) involve : among
9. ATROCIOUS : BAD::
(A) excessive : adequate
(B) momentous : important
(C) unavailing : helpful
(D) contagious : diseased
(E) nominal : satisfactory
- 10.PATRONIZE : CONDESCENSION::
(A) exasperate : anger
(B) deride : mockery
(C) compensate : apology
(D) hurry : decision
(E) encroach : fearlessness

11.FANG : TOOTH::

- (A) gum : mouth
- (B) elbow : arm
- (C) bank : river
- (D) finger : digit
- (E) summit : mountain

12.ANALGESIC : PAIN::

- (A) sedative : sleep
- (B) stimulant : mood
- (C) antiseptic : odor
- (D) anesthetic : sensation
- (E) ointment : skin

13.IMPECCABLE : FLAW::

- (A) foreseeable : outcome
- (B) mundane : substance
- (C) dishonorable : blemish
- (D) ingenuous : guile
- (E) portentous : omen

14.POLEMIC : DISPUTATIOUS::

- (A) anachronism : chronological
- (B) vernacular : unpretentious
- (C) invective : abusive
- (D) platitude : insightful
- (E) eulogy : unrealistic

15.EMBARRASS : SHAME::

- (A) coax : reluctance
- (B) sleep : fatigue
- (C) doubt : uncertainty
- (D) belittle : condescension
- (E) console : comfort

16.ETCH : CORROSIVE::

- (A) shrink : diminutive
- (B) destroy : worthless
- (C) glue : adhesive
- (D) sculpt : malleable
- (E) polish : glossy

A special mucous coating that serves as a chemical camouflage allows clown fish to live among the deadly tentacles of the unsuspecting sea anemone. Utterly dependent on this unlikely host for protection from

(5)predators, clown fish have evolved in isolated communities, a pattern that has led to unusual behavioral adaptations.

The rigidly defined hierarchy of each clown-fish community is dominated by a monogamous breeding pair (10)consisting of the largest fish, a female, and the next largest a male, attended by a fixed number of sexually immature fish ranging in size from large to tiny. A remarkable adaptation is that the development of these juveniles is somehow arrested until the hierarchy changes; then they (15)crow in lockstep, maintaining their relative sizes. While the community thus economizes on limited space and food resources, life is risky for newly spawned clown fish. On hatching, the hundreds of larvae drift off into the plankton. If, within three weeks, the defenseless larval clown fish (20)locates a suitable anemone(either by pure chance or perhaps guided by chemicals secreted by the anemone), it may survive. However, if an anemone is fully occupied, the resident clown fish will repel any newcomer.

Though advantageous for established community mem- (25)bers, the suspended and staggered maturation of juveniles might seem to pose a danger to the continuity of the community: there is only one successor for two breeding fish. Should one of a pair die, the remaining fish cannot swim off in search of a mate, nor is one likely to arrive. It (30)would seem inevitable that reproduction must sometimes have to halt, pending the chance arrival and maturation of a larval fish of the appropriate sex.

This, however, turns out not to be the case. In experiments, vacancies have been contrived by removing an (35)established fish from a community. Elimination of the breeding male triggers the prompt maturation of the largest juvenile. Each remaining juvenile also grows somewhat, and a minuscule newcomer drops in from the plankton. Removal of the female also triggers growth in all (40)remaining fish and acceptance of a newcomer, but the female is replaced by the adult male. Within days, the male's behavior alters and physiological transformation is complete within a few months. Thus, whichever of the breeding pair is lost, a relatively large juvenile can fill (45)the void, and reproduction can resume with a minimal loss of time. Furthermore, the new mate has already proved its ability to survive.

This transformation of a male into a female, or protandrous hermaphroditism, is rare among reef fish. The (50)more common protogynous hermaphroditism, where

females change into males, does not occur among clown fish. An intriguing question for further research is whether a juvenile clown fish can turn directly into a female or whether it must function first as a male.

17. The passage is primarily concerned with

- (A) analyzing the mutually advantageous relationship between two species
- (B) comparing two forms of hermaphroditism among clown fish
- (C) describing and explaining aspects of clown-fish behavior
- (D) outlining proposed research on clown-fish reproduction
- (E) attempting to reconcile inconsistent observations of clown-fish development

18. It can be inferred from the passage that the clown fish is able to survive in close association with the sea anemone because the

- (A) sea anemone cannot detect the presence of the clown fish
- (B) tentacles of the sea anemone cannot grasp the slippery clown fish
- (C) sea anemone prefers other prey
- (D) clown fish does not actually come within the range of the sea anemone's tentacles
- (E) clown fish has developed tolerance to the sea anemone's poison

19. According to the passage, adult clown fish would be at a disadvantage if they were not associated with sea anemones because the clown fish would.

- (A) be incapable of sexual transformation
- (B) be vulnerable to predators
- (C) have no reliable source of food
- (D) have to lay their eggs in the open
- (E) face competition from other clown fish

20. It can be inferred from the passage that sex change would have been less necessary for the clown fish if

- (A) the male clown fish were larger than the female
- (B) each sea anemone were occupied by several varieties of clown fish
- (C) many mature clown fish of both sexes occupied each sea anemone

- (D) juvenile clown fish had a high mortality rate
- (E) both male clown fish and female clown fish were highly territorial

21. The author mentions all of the following as characteristic of the "rigidly defined hierarchy" (line 8) of the clown-fish community EXCEPT:

- (A) At any time only one female clown fish can be reproductively active
- (B) The mature clown fish are monogamous
- (C) The growth of clown fish is synchronized
- (D) The maximum number of clown fish is fixed
- (E) There are equal numbers of male juveniles and female juveniles

22. Which of the following statements about newly hatched clown fish can be inferred from the passage?

- (A) They develop rapidly
- (B) They remain close to the sea anemone occupied by their parents
- (C) They are more sensitive to chemical signals than are adult clown fish.
- (D) They are not protected by their parents
- (E) They are less vulnerable to predation than are adult fish.

23. Which of the following, if true, would be LEAST consistent with the author's explanation of the advantage of hermaphroditism for clown fish?

- (A) The number of individuals in a clown-fish community fluctuates significantly
- (B) Adult clown fish frequently cannibalize their young
- (C) The sea anemone tolerates clown fish only during a specific stage of the anemone's life cycle.
- (D) Juvenile clown fish rarely reach maturity
- (E) Clown-fish communities are capable of efficiently recruiting solitary adult clown fish

Comparing designs in music with visual designs raises interesting questions. We are familiar with the easy transfers of terms denoting qualities from one field to another. The basic problem can be put this way: can music sound the way a design looks? The elements of music are not the same as those of painting. They may be analogous, but to be analogous is not to be identical. Is it possible,

- then, for the same broad characteristics to emerge from different perceptual conditions?
- (10) Two facts about the relation between broad characteristics of a work and their perceptual conditions must be kept distinct. First, the global characteristics of a visual or auditory complex are determined by the discernible parts and their relationships. Thus, any notable change in the (15) parts or their relationships produces a change in some of the global characteristics. Second, a change in the parts or their relationships may leave other global characteristics unchanged.
24. In the first paragraph, the author is primarily concerned with establishing the fact that
- (A) comparisons are not equations
 (B) auditory phenomena are not visual phenomena
 (C) frequently used comparisons are usually inaccurate
 (D) careless perceptions result from careless thought
 (E) questions concerning perception are psychological
25. In the passage, the author is primarily concerned with
- (A) distinguishing mutually exclusive categories
 (B) clarifying an apparent contradiction
 (C) supporting new ideas
 (D) analyzing a problem
 (E) comparing opinions
26. The second paragraph is primarily concerned with establishing the idea that
- (A) different global characteristics of a work result from the same discernible parts
 (B) the parts of a work of art influence the total perception of the work
 (C) visual and auditory characteristics can be combined
 (D) changes in the parts of a work remain isolated from the work as a whole
 (E) the visual complexes in a work of art influence the work's auditory complexes
27. Which of the following statements is most likely be a continuation of the passage?
- (A) The search for broad similarities thus begins by understanding and distinguishing these two facts.
 (B) The search for musical-visual analogies thus depends on the complexity of the works being compared.
 (C) The search for music and art of the highest quality thus depends on very different assumptions
 (D) Thus music and painting exist in mutually exclusive worlds.
 (E) Thus music and painting are too complicated to be evaluated in terms of analogies.
28. COMPRESSION:
- (A) increase in volume
 (B) change of altitude
 (C) loss of stability
 (D) absence of matter
 (E) lack of motion
29. REFINE:
- (A) loosen
 (B) obscure
 (C) destabilize
 (D) decrease size
 (E) reduce purity
30. BALK:
- (A) extend
 (B) derive
 (C) observe
 (D) plan ahead carefully
 (E) move ahead willingly
31. ANTIPATHY:
- (A) affection
 (B) courtesy
 (C) exasperation
 (D) obstinacy
 (E) cynicism
32. PATHOLOGICAL:
- (A) acute
 (B) normal
 (C) adequate
 (D) variable
 (E) temporary
33. REIN:
- (A) prod
 (B) assess

- (C) engulf
(D) commend
(E) affirm

34. MELLIFLUOUS:

- (A) obtuse
(B) ineffable
(C) raspy
(D) deranged
(E) uproarious

35. IMPUGN:

- (A) abandon
(B) anticipate
(C) enable
(D) clarify
(E) endorse

36. PERTINACITY:

- (A) liability
(B) simplicity
(C) vacillation
(D) eccentricity
(E) misrepresentation

37. GAINSAY:

- (A) speak kindly of
(B) tell the truth about
(C) forecast
(D) affirm
(E) reiterate

38. ABSOLUTE:

- (A) inferior
(B) tolerant
(C) qualified
(D) preliminary
(E) immeasurable

SECTION 6

Time –30 minutes

30 Questions

The scale used on a certain map is $\frac{1}{3}$ inch = 12 miles.

1. The number of miles represented by 2 inches on the map

72

2. The number of positive divisors of 24

The number of positive divisors of 50

$$xy = 4$$

3. $(2x)(3y)$

24

x is an integer greater than 500

4. The value of the integer formed if the order of the digits in x is reversed

200

A man drove his automobile 10 kilometers in 10 minutes and then drove an additional 15 kilometers in the next 10 minutes.

5. His average speed during 50 kilometers per hour the 20-minute drive

$$y > 0 \text{ and } \frac{y}{y-1} > 0$$

6. y

1

7. $\frac{1}{2} + \frac{2}{3} + \frac{3}{4}$

$\frac{2}{5} + \frac{5}{3} + \frac{3}{8}$

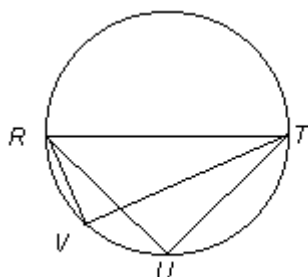
In the rectangular coordinate plane, (x,y) represents a point with coordinates x and y .

8. The distance between the points (1,3) and (1,4)

The distance between the points (2,3) and (3,4)

$$5n + 2 = 7n - 3$$

9. $\left(\frac{n}{5}\right)^2$ $\frac{1}{5}$



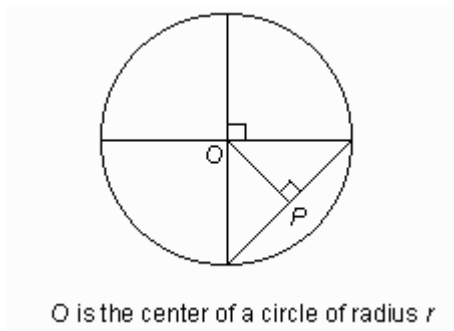
RT is a diameter of the circle above

10. The measure of $\angle RUT$ The measure of $\angle RVT$

11. 2^m $(-2)^m$

$a_k = 1 - (-1)^k$, for positive integers k
 n is a positive integer

12. a_n a_{n-1}



O is the center of a circle of radius r

13. OP $\frac{r}{\sqrt{2}}$

14. The ratio of the area of
 a circular region with
 diameter 3 to the area
 of a circular region with
 diameter 5

$\frac{3}{5}$

SURVEY OF PETS IN HOUSEHOLDS
 IN AN APARTMENT COMPLEX

Kind of Pet	Number of Households
Dog	25
Cat	41
Other	19
No Pet	44

15. The total number of households with at least one pet 85

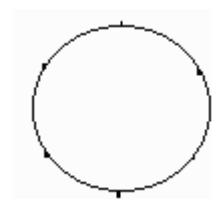
16. If 14 percent of an amount of money is \$420, then 10 percent of the same amount is

- (A) \$224
 (B) \$294
 (C) \$300
 (D) \$378
 (E) \$400

$$n = \frac{k + \frac{r}{s}}{\frac{t}{v}}$$

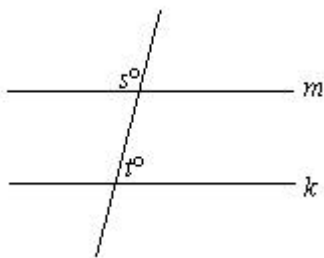
17. In the equation above, k , r , s , t , and v represent positive numbers. Multiplying which one of these numbers by 2 will reduce the value of n to $\frac{1}{2}$ of its present value?

- (A) k (B) r (C) s
 (D) t (E) v



18. There are six marked points on the circle above. How many different lines can be drawn that contain two of the marked points?

- (A) 5
 (B) 6
 (C) 12
 (D) 15
 (E) 30



19. In the figure above, if $m \parallel k$ and $s = t + 30$ then $t =$

- (A) 30
- (B) 60
- (C) 75
- (D) 80
- (E) 105

20. On the real number line, which of the following is halfway between -3.4 and 5.2?

- (A) 0.9
- (B) 1.2
- (C) 1.8
- (D) 2.2
- (E) 4.3

APPARENT FAHRENHEIT TEMPERATURES DUE TO WIND-CHILL FACTOR

(corresponding to actual temperatures and wind speeds)

WIND SPEEDS (miles per hour)	ACTUAL TEMPERATURES (degrees Fahrenheit)				
	-10	0	10	20	30
5	-15	-5	7	16	27
10	-34	-22	-9	3	16
15	-45	-31	-18	-5	9
20	-53	-39	-24	-10	4
25	-59	-44	-29	-15	1
30	-64	-49	-33	-18	-2
35	-67	-52	-35	-20	-4
40	-69	-53	-37	-21	-5

21. If the actual temperature is 20 degrees Fahrenheit and the wind speed increases from 5 miles per hour to 40 miles per hour, what is the corresponding decrease in the apparent temperature?

- (A) 5 degrees Fahrenheit
- (B) 24 degrees Fahrenheit
- (C) 32 degrees Fahrenheit
- (D) 37 degrees Fahrenheit

(E) 44 degrees Fahrenheit

22. If the actual temperature is 13 degrees Fahrenheit and the wind speed is 14 miles per hour, then the apparent temperature could be

- (A) 5 degrees Fahrenheit
- (B) -12 degrees Fahrenheit
- (C) -20 degrees Fahrenheit
- (D) -25 degrees Fahrenheit
- (E) -32 degrees Fahrenheit

23. If the actual temperature is 30 degrees Fahrenheit and the apparent temperature due to the wind-chill factor is 0 degrees Fahrenheit then the wind speed in miles per hour, must be between

- (A) 5 and 10
- (B) 10 and 15
- (C) 15 and 20
- (D) 20 and 25
- (E) 25 and 30

24. At a wind speed of 30 miles per hour, if the actual temperature increases by x degrees Fahrenheit then the apparent temperature increases by approximately how many degrees Fahrenheit?

- (A) 5
- (B) 7
- (C) 10
- (D) 15
- (E) 20

25. Which of the following can be inferred from the table?

- I The apparent temperature for an actual temperature of 20 degrees Fahrenheit and a wind speed of 15 miles per hour is the same as that for an actual temperature of 60 degrees Fahrenheit and a wind speed of 40 miles per hour
- II At a constant wind speed as the actual temperature increases the difference between the actual and the apparent temperatures also increases
- III At a constant actual temperature of -10 degrees Fahrenheit the apparent temperature decreases at a constant rate as the wind speed increases

(A) I only

- (B) II only
(C) I and III only
(D) II and III only
(E) I, II and III
26. If the sum of two positive integers is 43 and the difference of their squares is 43, then the smaller integer is
- (A) 17
(B) 19
(C) 20
(D) 21
(E) 22
27. To obtain an FHA mortgage for \$50,000 or more, the home buyer must have a down payment equal to 4 percent of the first \$25,000 of the mortgage amount and 5 percent of the portion in excess of \$25,000. At settlement the buyer pay a mortgage insurance premium equal to 3 percent of the mortgage amount. What is the maximum FHA mortgage, if any a buyer can obtain if the buyer has only \$6,000 available for the down payment and insurance premium?
- (A) \$62,500
(B) \$71,875
(C) \$78,125
(D) \$125,00
(E) The home buyer cannot obtain an FHA mortgage
28. If two sides of a triangle have lengths 3.2 and 5.4 then the length of the third side must be between
- (A) 0.0 and 2.2
(B) 2.2 and 5.4
(C) 2.2 and 8.6
(D) 3.2 and 5.4
(E) 5.4 and 8.6
29. The odds that a certain event will occur is the ratio of the probability that the event will occur to the probability that it will not occur. If the odds that Pat will win a prize are 4 to 3, what is the probability that Pat will not win the prize ?
- (A) $\frac{1}{4}$
(B) $\frac{1}{3}$
(C) $\frac{3}{7}$
(D) $\frac{4}{7}$
(E) $\frac{3}{4}$
30. A certain money market account that had a balance of \$48,000 during allof last month earned \$360 in interest for the month. At what simple annualinterest rate did the account earn interest last month?
- (A) 7%
(B) 7.5%
(C) 8%
(D) 8.5%
(E) 9%

SECTION 7

Time 30 minutes

25 Questions

Questions 1-7

Seven pedigreed dogs—Frieda, King, Laddie, Max., Pal, Spot, and Toppy—are entered in a dog show. The dogs must be scheduled into seven consecutive time slots for judging, and only one dog can be scheduled into any time slot. The schedule must be based on the following constraints:

Frieda cannot be in a time slot immediately before or immediately after Max's time slot

Laddie cannot be in a time slot immediately before or immediately after Toppy's time slot.

Pal's time slot must be sometime before Spot's time slot.

King must be in the seventh time slot

- If Pal and Max are the third and fourth dogs judged, respectively, Frieda must be scheduled for a time slot selected from which of the following pairs of slots?
 - First and second
 - First and sixth
 - Second and fifth
 - Second and sixth
 - Fifth and sixth
- Which of the following is an acceptable sequence of dogs in the first four time slots?

	<u>First</u>	<u>Second</u>	<u>Third</u>	<u>Fourth</u>
(A)	Frieda	Pal	Toppy	Laddie
(B)	Laddie	Spot	Toppy	Pal
(C)	Max	Frieda	Pal	Spot
(D)	Pal	Frieda	Laddie	King
(E)	Toppy	Max	Pal	Laddie
- Any of the following can be the sixth dog judged EXCEPT
 - Frieda
 - Laddie
 - Max
 - Pal
 - Spot
- If Pal is the first, Laddie the third, and Frieda the fifth dog judged, which of the following must be true?
 - The remaining dogs can be acceptably scheduled in either of exactly two ways
 - The remaining dogs can be acceptably scheduled in only one way.
 - Max is judged immediately before King
 - Spot is judged immediately before Laddie
 - Toppy is judged immediately before Frieda.
- If the first three dogs judged are Max, Pal, and Frieda, respectively, which of the following must be true?
 - King is the sixth dog judged.
 - Laddie is the fourth dog judged.
 - Spot is the fifth dog judged.
 - Toppy is the fifth dog judged.
 - Toppy is the sixth dog judged.
- If Laddie is the fourth dog judged, Toppy can be scheduled into any one of how many different time slots?
 - Two
 - Three
 - Four
 - Five
 - Six
- If the schedule includes Laddie sometime before Pal and Pal sometime before Max, which of the following must be true?
 - Laddie is in one of the first three time slots.
 - Max is in one of the last three time slots
 - Pal is in one of the first three time slots
 - Spot is in the time slot immediately after Max
 - Toppy is in the time slot immediately after Max
- New regulations in Mullentown require manufacturers there to develop five-year pollution-reduction plans. The regulations require that each manufacturer develop a detailed plan for reducing its released pollutants by at least 50 percent. Clearly, the regulations will not result in significant pollution reduction, however, since the regulations do not force manufacturers

to implement their plans.

Which of the following, if true, most weakens the argument?

- (A) Mullentown's manufacturing plants are not the only source of pollution there.
- (B) Detailed plans would reveal that measures to reduce released pollutants would also reduce manufacturers' costs for materials, waste disposal, and legal services
- (C) Pollutants that manufacturing processes create but that are not released directly into the environment must nonetheless be collected and prepared for disposal
- (D) Any reductions in pollutants released from Mullentown's manufacturing plants would not be noticeable for at least five years.
- (E) Each manufacturer will be required to submit its plan to a committee appointed by Mullentown's officials.

9. Not Scored

10. To produce seeds, plants must first produce flowers. Two kinds of tarragon plants, Russian tarragon and French tarragon, look very similar except that Russian tarragon produces flowers and French tarragon does not. The leaves of Russian tarragon, however, lack the distinctive flavor that makes French tarragon a desirable culinary herb

If the information presented is true, which of the following can most reliably be concluded on the basis of it?

- (A) As a decorative plant, French tarragon is more desirable than Russian tarragon.
- (B) The flowers of Russian tarragon plants are probably not flavorful.
- (C) Plants that grow from seeds sold in a packet labeled "tarragon" are not French tarragon.
- (D) There are no other kinds of tarragon besides Russian tarragon and French tarragon
- (E) Garden plants that have flavorful leaves generally do not produce flowers.

Questions 11-16

A historian is attempting to reconstruct the locations of five volcanic islands—Kappa, Lambda, Mu, Pi, and Omega—that once existed off the coast of Penelopia. On the basis of historical documents, the historian has established that the five islands were arranged in a straight line running north to south. The historian also discovered the following:

- Lambda was located immediately north of Pi.
- Omega and kappa were immediately adjacent to each other.
- Mu was located somewhere north of Lambda.

11. The five islands could have been arranged in which of the following north-to-south orderings?
- (A) Kappa, Mu, Omega, lambda, Pi
 - (B) Lambda, Pi, Omega, Kappa, Mu
 - (C) Mu, Kappa, Omega, Lambda, Pi
 - (D) Mu, Pi, lambda, Kappa, Omega
 - (E) Omega, Mu, Lambda, Pi, Kappa
12. If Mu was located immediately north of Omega, which of the following must be true?
- (A) Kappa was the southernmost of the islands.
 - (B) Lambda was the northernmost of the islands
 - (C) Pi was the southernmost of the islands.
 - (D) Omega was the northernmost of the islands
 - (E) Omega was the southernmost of the islands
13. If Omega was located somewhere north of Mu, which of the following must be true?
- (A) Kappa was located immediately north of Mu.
 - (B) Mu was located immediately north of Lambda.
 - (C) Omega was located immediately north of Mu.
 - (D) Kappa was located immediately north of Lambda.
 - (E) Mu was located immediately north of Pi
14. Which of the following is a complete and accurate list of the islands any one of which could be the island that was northernmost?
- (A) Kappa, Lambda
 - (B) lambda, Mu
 - (C) Mu, Omega
 - (D) Kappa, Lambda, Omega
 - (E) Kappa, Mu, Omega

15.If Mu is discovered to have been the northernmost of the islands, there remain how many possible arrangements any one of which could have been the historically accurate arrangement of the island?

- (A) 2
- (B) 3
- (C) 4
- (D) 5
- (E) 6

16.If Mu was adjacent of Kappa, which of the following must be true?

- (A) Kappa was located somewhere north of Mu
- (B) Lambda was located somewhere north of Omega.
- (C) Mu was located somewhere north of Kappa
- (D) Mu was located somewhere north of Omega
- (E) Omega was located somewhere north of Lambda.

Questions 17-22

Six clubs—F, G, H, J, K, and L—have applied for funding from the student council. The council cannot fund all six club immediately. Unfunded clubs go on a waiting list and will be funded, in the order in which they appear on that list, as money becomes available. Funding decisions are subject to the following constraints:

No more than three clubs can be funded immediately.

L can be funded neither before F is funded nor before G is funded.

If J is assigned to the waiting list, K must also be assigned to the waiting list.

H must be on the waiting list.

17.Which of the following could be a complete and accurate list of the unfunded clubs in the order in which they initially appear on the waiting list?

- (A) F, K, L, H
- (B) H, F, L, J
- (C) H, K
- (D) L, H, F
- (E) L, J, K

18.If K receives immediate funding, which of the following clubs must also receive immediate funding?

- (A) F
- (B) G

- (C) H
- (D) J
- (E) L

19.If G is the third club on the waiting list, which of the following must be true?

- (A) Exactly one club receives immediate funding.
- (B) At most two clubs receive immediate funding.
- (C) Three clubs receive immediate funding
- (D) H is ahead of G on the waiting list
- (E) L is after H on the waiting list

20.Which of the following could be a complete list of the clubs that the student council funds immediately?

- (A) F, G, J, and L
- (B) F, G, and K
- (C) F, J, and L
- (D) G and H
- (E) J

21.If K receives immediate funding and L is the second club on the waiting list, which of the following must be true?

- (A) Exactly two clubs receive immediate funding.
- (B) F and G both receive immediate funding.
- (C) F is first on the waiting list.
- (D) G is first on the waiting list.
- (E) H is third on the waiting list.

22.If both F and J are on the waiting list, which of the following must be true?

- (A) No more than one club receives immediate funding.
- (B) Two clubs receive immediate funding
- (C) G is either first or second on the waiting list
- (D) H is last on the waiting list
- (E) K is ahead of L on the waiting list

23. In the United States, the financing of industrial research by private industrial firms remained steady as a percentage of sales during the period between 1968 and 1978 (after correcting for inflation). But slowdowns in the growth of industrial productivity also occurred during that period, a fact that refutes the notion that the growth of industrial productivity is directly proportional to the amount invested in industrial research.

Which of the following, if true for the United States, most weakens the argument above?

- (A) Federal funds, which constituted a significant portion of the support for industrial research from 1968 to 1978, fell annually and substantially during that period.
- (B) The inflation that occurred between 1968 and 1978 was more severe than leading economists had expected.
- (C) Industrial executives generally favor investing an appreciably larger portion of corporate funds in short-term product development than in basic research.
- (D) The scientists and engineers who worked in industry from 1968 to 1978 were, as a group, more experienced in their jobs than were those who worked in industry during the previous ten-year period.
- (E) Corporate financing of industrial research increased in several of the years immediately following 1978 (after correcting for inflation)

24. Although many brands of gasoline are sold on Haibei Island, gasoline companies there get all of the refined gasoline they sell from Haibei seaport's only storage tank, which is always refilled with the same quality of gasoline. Therefore, the brands of gasoline of sale on Haibei may be different in name and price, but they are identical in quality.

The conclusion drawn above depends on which of the following assumptions?

- (A) Consumers are usually unaware of variations in the quality of the gasoline they buy unless those variations are announced by the gasoline companies.
- (B) When tankers make gasoline deliveries at Haibei's seaport, the storage tank on Haibei always receives the same quantity of gasoline as that in the preceding delivery.
- (C) There is a wide variation in the prices at which the different brands of gasoline on Haibei are sold.
- (D) If any gasoline company on Haibei alters the quality of its gasoline before sale, the other

gasoline companies also use methods before sale that result in the same change in the quality of their gasoline .

- (E) The gasoline storage tank on Haibei is large enough to meet the needs of all of Haibei's different gasoline companies.

25. A group of paintings made approximately 15,000 years ago in a cave in the Loire River valley in what is now France depicts a number of different animals. One of the animals depicted seems to resemble the chiru, a rare antelope of the Himalayas.

Which of the following, if true, best supports the hypothesis that in painting the animal that resembles a chiru the cave artist painted a chiru with which she or he was familiar?

- (A) There are numerous representations of imaginary animals in cave paintings of similar age.
- (B) Fossilized remains of a chiru, approximately 16,000 years old, have been found at the northern end of the valley.
- (C) The cave that contains the depiction of an animal that resembles a chiru contains stylized representations of plant life
- (D) Older caves from the same region contain no representations of animals that resemble a chiru.
- (E) The antlers of the animal in the painting are longer than those of the mature Himalayan chiru.

SECTION 1

Time –30 minutes

38 Questions

1. Because the monkeys under study are ---- the presence of human beings, they typically ---- human observers and go about their business
 - (A) ambivalent about .. welcome
 - (B) habituated to .. disregard
 - (C) pleased with .. snub
 - (D) inhibited by .. seek
 - (E) unaware of .. avoid
2. Give he previously expressed interest and the ambitious tone of her recent speeches, the senator's attempt to convince the public that she is not interested in running for a second term is ----.
 - (A) laudable
 - (B) likely
 - (C) authentic
 - (D) futile
 - (E) sincere
3. Many of her followers remain ---- to her, and even those who have rejected her leadership are unconvinced of the ---- of replacing her during the current turmoil.
 - (A) opposed.. urgency
 - (B) friendly.. harm
 - (C) loyal.. wisdom
 - (D) cool.. usefulness
 - (E) sympathetic.. disadvantage
4. Unlike many recent interpretations of Beethoven's piano sonatas, the recitalist's performance was a delightfully free and introspective one; nevertheless, it was also, seemingly paradoxically, quite ----.
 - (A) appealing
 - (B) exuberant
 - (C) idiosyncratic
 - (D) unskilled
 - (E) controlled
5. Species with relatively ---- metabolic rates, including hibernators, generally live longer than those whose metabolic rates are more rapid.
 - (A) prolific
 - (B) sedentary
 - (C) sluggish
 - (D) measured
 - (E) restive
6. Belying his earlier reputation for ---- as a negotiator, Morgan had recently assumed a more ---- stance for which many of his erstwhile critics praised him.
 - (A) intransigence.. conciliatory
 - (B) impropriety.. intolerant
 - (C) inflexibility.. unreasonable
 - (D) success.. authoritative
 - (E) incompetence.. combative
7. Although Irish literature continued to flourish after the sixteenth century, a ---- tradition is ---- in the visual arts: we think about Irish culture in terms of the word, not in terms of pictorial images.
 - (A) rich.. superfluous
 - (B) lively.. found
 - (C) comparable.. absent
 - (D) forgotten.. apparent
 - (E) lost.. extant
8. SILVER: TARNISH::
 - (A) gold: burnish
 - (B) steel: forge
 - (C) iron: rust
 - (D) lead: cast
 - (E) tin: shear
9. DISLIKE: LOATHING::
 - (A) appreciation: gratification
 - (B) hunger: appetite
 - (C) void: dearth
 - (D) pleasure: bliss
 - (E) pain: ache
10. CRAVEN: HEROIC::
 - (A) unruly: energetic
 - (B) listless: attractive
 - (C) volatile: constant
 - (D) deft: trifling
 - (E) awkward: amusing

11. FILLY: HORSE::

- (A) antennae: butterfly
- (B) pullet: chicken
- (C) gaggle: goose
- (D) duck: drake
- (E) wasp: bee

12. PITHINESS: APHORISM::

- (A) craft: art
- (B) detail: sketch
- (C) illusion: story
- (D) exaggeration: caricature
- (E) sophistication: farce

13. EPHEMERAL: ENDURING::

- (A) infirm: healing
- (B) insensitive: cooperating
- (C) inanimate: living
- (D) interminable: continuing
- (E) ineffectual: proceeding

14. POSTURER: UNAFFECTED::

- (A) brat: insolent
- (B) hypocrite: perceptive
- (C) grouch: respected
- (D) bigot: tolerant
- (E) rogue: empathetic

15. FACETIOUS: SPEECH::

- (A) precocious: learning
- (B) unbecoming: color
- (C) exemplary: conduct
- (D) craven: timidity
- (E) antic: behavior

16. VAGARY: PREDICT::

- (A) quotation: misdirect
- (B) investigation: confirm
- (C) stamina: deplete
- (D) turbulence: upset
- (E) impossibility: execute

This is not to deny that the Black gospel music of the early twentieth century differed in important ways from the slave spirituals. Whereas spirituals were created and disseminated in folk fashion, gospel music was composed,

(5) published, copyrighted, and sold by professionals. Nevertheless, improvisation remained central to gospel music. One has only to listen to the recorded repertoire of gospel songs to realize that Black gospel singers rarely sang a song precisely the same way twice and never according to (10) its exact musical notation. They performed what jazz musicians call "head arrangements" proceeding from their own feelings and from the way "the spirit" moved them at the time. This improvisatory element was reflected in the manner in which gospel music was published. Black gospel (15) composers scored the music intended for White singing groups fully, indicating the various vocal parts and the accompaniment, but the music produced for Black singers included only a vocal line and piano accompaniment.

17. Which of the following best describes "head arrangement" as the term is used in line 11?

- (A) A published version of a gospel song produced for use by Black singers
- (B) A gospel song based on a slave spiritual
- (C) A musical score shared by a gospel singer and a jazz musician
- (D) An informally written composition intended for use by a gospel singer
- (E) An improvised performance inspired by the singer's emotions

18. The author mentions "folk fashion" (line 4) most likely in order to

- (A) counter an assertion about the role of improvisation in music created by Black people
- (B) compare early gospel music with gospel music written later in the twentieth century
- (C) make a distinction between gospel music and slave spirituals
- (D) introduce a discussion about the dissemination of slave spirituals
- (E) describe a similarity between gospel music and slave spirituals

19. The passage suggests which of the following about Black gospel music and slave spirituals?

- (A) Both became widely known in the early twentieth century.
- (B) Both had an important improvisatory element.
- (C) Both were frequently performed by jazz

musicians.

- (D) Both were published with only a vocal line and piano accompaniment.
- (E) Both were disseminated chiefly by Black singing groups.

20. Of the following sentences, which is most likely to have immediately preceded the passage?

- (A) Few composers of gospel music drew on traditions such as the spiritual in creating their songs.
- (B) Spirituals and Black gospel music were derived from the same musical tradition.
- (C) The creation and singing of spirituals, practiced by Black Americans before the Civil War, continued after the war.
- (D) Spirituals and gospel music can be clearly distinguished from one another.
- (E) Improvisation was one of the primary characteristics of the gospel music created by Black musicians.

About a century ago, the Swedish physical scientist Arrhenius proposed a law of classical chemistry that relates chemical reaction rate to temperature. According to the Arrhenius equation, chemical reaction are increasingly unlikely to occur as temperatures approach absolute zero, and at absolute zero (zero degrees Kelvin, or minus 273 degrees Celsius) reactions stop. However, recent experimental evidence reveals that although the Arrhenius equation is generally accurate in describing the kind of chemical reaction that occurs at relatively high temperatures, at temperatures closer to zero a quantum-mechanical effect known as tunneling comes into play; this effect accounts for chemical reactions that are forbidden by the principles of classical chemistry. Specifically, entire molecules can "tunnel" through the barriers of repulsive forces from other molecules and chemically react even though these molecules do not have sufficient energy, according to classical chemistry, to overcome the repulsive barrier.

The rate of any chemical reaction, regardless of the temperature at which it takes place, usually depends on a very important characteristic known as its activation energy. Any molecule can be imagined to reside at the bottom of a so-called potential well of energy. A chemical reaction corresponds to the transition of a molecule from the bottom of one potential well to the bottom of another. In classical

chemistry, such a transition can be accomplished only by going over the potential barrier between the wells, the height of which remains constant and is called the activation energy of the reaction. In tunneling, the reacting molecules tunnel from the bottom of one to the bottom of another well without having to rise over the barrier between the two wells. Recently researchers have developed the concept of tunneling temperature: the temperature below which tunneling transitions greatly outnumber Arrhenius transitions, and classical mechanics gives way to its quantum counterpart.

This tunneling phenomenon at very low temperatures suggested my hypothesis about a cold prehistory of life: the formation of rather complex organic molecules in the deep cold of outer space, where temperatures usually reach only a few degrees Kelvin. Cosmic rays (high-energy protons and other particles) might trigger the synthesis of simple molecules, such as interstellar formaldehyde, in dark clouds of interstellar dust. Afterward complex organic molecules would be formed, slowly but surely, by means of tunneling. After I offered my hypothesis, Hoyle and Wickramasinghe argued that molecules of interstellar formaldehyde have indeed evolved into stable polysaccharides such as cellulose and starch. Their conclusions, although strongly disputed, have generated excitement among investigators such as myself who are proposing that the galactic clouds are the places where the prebiological evolution of compounds necessary to life occurred.

21. The author of the passage is primarily concerned with

- (A) describing how the principles of classical chemistry were developed
- (B) initiating a debate about the kinds of chemical reactions required for the development of life
- (C) explaining how current research in chemistry may be related to broader biological concerns
- (D) reconciling opposing theories about chemical reactions
- (E) clarifying inherent ambiguities in the laws of classical chemistry

22. According to the passage, classical chemical reactions and tunneling reactions are alike in which of the following ways?

- (A) In both types of reactions, reacting molecules have to rise over the barrier between the two wells.

- (B) In both types of reactions, a transition is made from the bottom of one potential well to the bottom of another.
- (C) In neither type of reaction does the height of the barrier between the wells remain constant.
- (D) In neither type of reaction does the rate of a chemical reaction depend on its activation energy.
- (E) In both types of reactions, reacting molecules are able to go through the barrier between the two wells.
23. According to the Arrhenius equation as discussed in the passage, which of the following statements about chemical reactions is true?
- (A) Chemical reactions are less likely to occur at temperatures close to absolute zero.
- (B) In some cases the rate of a chemical reaction is related to temperature and in other cases it is not.
- (C) Chemical reactions frequently occur at a few degrees above absolute zero, but they are very unpredictable.
- (D) The rate of a chemical reaction depends on many other factors besides temperature.
- (E) Chemical reaction rate and temperature are not related.
24. The author's attitude toward the theory of a cold prehistory of life can best be described as
- (A) neutral
- (B) skeptical
- (C) mildly positive
- (D) very supportive
- (E) pointedly critical
25. The author's hypothesis concerning the cold prehistory of life would be most weakened if which of the following were true?
- (A) Cosmic rays are unlikely to trigger the formation of simple molecules.
- (B) Tunneling occurs only in a narrow band of temperatures around zero degrees Kelvin.
- (C) The synthesis of interstellar formaldehyde can be activated by means other than cosmic rays.
- (D) Simple molecules can be synthesized by means of tunneling.
- (E) Classical chemical reactions do not occur at temperatures close to absolute zero.
26. Which of the following best describes the hypothesis of Hoyle and Wickramasinghe as it is presented in the passage?
- (A) Cosmic rays can directly synthesize complex organic molecules.
- (B) The galactic clouds are the places where prebiological evolution of compounds necessary to life occurred.
- (C) Interstellar formaldehyde can be synthesized by tunneling.
- (D) Molecules of interstellar formaldehyde can evolve into complex organic molecules.
- (E) Complex organic molecules can be synthesized from stable polysaccharides such as cellulose and starch.
27. Which of the following best describes the organization of the first two paragraphs of the passage?
- (A) The author cites a basic principle of classical chemistry and then describes the research from which that principle was developed.
- (B) The author cites an apparent contradiction to the principles of classical chemistry and then explains the process of a chemical reaction to show there is in fact no contradiction.
- (C) The author describes the role of heat in chemical reactions and then offers a detailed explanation of its function.
- (D) The author presents a law of classical chemistry in order to introduce a kind of chemical reaction that differs from it and then explains the essential difference between the two.
- (E) The author presents the fundamental rules of classical chemistry in order to introduce an explanation of a specific chemical reaction.
28. PREFACE:
- (A) improvisation
- (B) burlesque
- (C) epilogue
- (D) tangent
- (E) backdrop

29. DEBILITATE:

- (A) implicate
- (B) invigorate
- (C) obfuscate
- (D) realign
- (E) encumber

30. TASTY:

- (A) uninteresting
- (B) unfamiliar
- (C) unexpected
- (D) understated
- (E) undervalued

31. ABNEGATE:

- (A) refresh
- (B) reaffirm
- (C) relieve
- (D) react
- (E) reform

32. SERRIED:

- (A) partially formed
- (B) widely separated
- (C) narrowly missed
- (D) extremely grateful
- (E) reasonably clean

33. BOMBASTIC:

- (A) unflappable
- (B) uninspired
- (C) unpretentious
- (D) inscrutable
- (E) incisive

34. BANAL:

- (A) comfortable
- (B) novel
- (C) equal
- (D) fatal
- (E) competent

35. LANGUISH:

- (A) agitate
- (B) wander
- (C) relieve
- (D) discomfit

(E) thrive

36. ENNUI:

- (A) intimidation
- (B) sleaze
- (C) faint recollection
- (D) keen interest
- (E) deep reservation

37. DAUNTLESS:

- (A) sophomoric
- (B) trifling
- (C) pusillanimous
- (D) specious
- (E) parsimonious

38. TEMERITY:

- (A) credibility
- (B) authority
- (C) celebrity
- (D) acrimony
- (E) circumspection

SECTION 2

Time –30 minutes

25 Questions

1. Drug companies lose money when manufacturing drugs that cure those suffering from rare diseases because selling a drug to only a few people usually does not recoup manufacturing expenses. Therefore, a company manufacturing any of the drugs that cure those suffering from loxemia, an extremely rare disease, will undoubtedly lose money.

Which of the following, if true, most seriously weakens the conclusion above?

- (A) Several drugs that cure those suffering from loxemia also cure those suffering from very common illnesses.
- (B) Most of those who contract loxemia also contract another illness concurrently.
- (C) Most of the drug companies that manufacture drugs that cure rare diseases do not manufacture drugs that cure loxemia.
- (D) A sizable number of people are afflicted with one or another rare disease even though each rare disease afflicts only a small number of people.
- (E) The larger the amount of a drug that is manufactured, the lower the manufacturing expense for each unit of the drug that is produced.

2. The tomb of a warrior killed in 1501 bears a sculpted portrait depicting him dressed for battle. Some historians attribute the portrait to an artist from that century, but of the many references to the tomb in surviving documents, none that predates the 1800's mentions the portrait. The portrait is therefore more likely the work of a much later artist.

Which of the following, if true, would also support the conclusion of the argument if substituted for the evidence given concerning the portrait?

- (A) The portrait of the warrior was commissioned by the family of the warrior's widow.
- (B) References in surviving documents mention that an artist was paid in 1525 for an unspecified number of works for the church in which the tomb is located

(C) The warrior is depicted in the portrait as wearing boots made of a material not used for boots until the 1700's.

(D) Some other art treasures from the church in which the tomb is located have been reliably dated to the 1400's.

(E) The portrait of the warrior on the tomb strongly resembles a portrait of him known to have been completed during his lifetime.

Questions 3-7

A florist is designing flower arrangements containing two or more varieties of flowers selected from among six varieties of flowers : freesias, irises, lilies, peonies, tulips, and zinnias. All acceptable arrangements conform to the following conditions :

If an arrangement contains any freesias, it cannot contain any zinnias.

If an arrangement contains any tulips, it cannot contain any zinnias.

If an arrangement contains any peonies, it must also contain at least one zinnia, and there must be exactly as many zinnias as peonies.

If an arrangement contains any irises, it must also contain tulips, and there must be twice as many tulips as irises.

If an arrangement contains freesias, the number of freesias must be greater than the total number of other flowers used.

3. Which of the following flower arrangements could be made acceptable simply by adding a tulip?

- (A) Three freesias, one lily, two tulips
- (B) Four freesias, two peonies, one tulip
- (C) Five freesias, one iris, one tulip
- (D) Two irises, two tulips, two zinnias
- (E) Two lilies, two peonies, two tulips

4. Which of the following, if added to an unacceptable flower arrangement consisting of four tulips and two freesias, would make the arrangement acceptable?

- (A) Four freesias
- (B) Four irises
- (C) Two lilies
- (D) Two peonies
- (E) Two zinnias

5. Each of the following is a pair of varieties of flowers that can be used together in an acceptable flower arrangement EXCEPT
- (A) freesias and irises
 - (B) freesias and tulips
 - (C) irises and lilies
 - (D) irises and peonies
 - (E) lilies and zinnias
6. Which of the following unacceptable flower arrangements could be made acceptable simply by removing some or all of the flowers of one variety?
- (A) Four freesias, one iris, one lily, one peony
 - (B) Four freesias, one iris, two tulips, one zinnia
 - (C) Four freesias, two irises, two tulips, one zinnia
 - (D) Three freesias, one lily, one peony, two zinnias
 - (E) Three freesias, two peonies, one tulip, two zinnias
7. If an unacceptable flower arrangement consisting of four freesias, one lily, one peony, and two tulips is to be made acceptable by adding or removing only one flower, which of the following must be done?
- (A) Add one freesia
 - (B) Add one iris
 - (C) Add one zinnia
 - (D) Remove the peony
 - (E) Remove one tulip
8. Scientist : More than 1, 000 large asteroids regularly cross the Earth's path. Even though the probability of one colliding with the Earth is extremely slight, we should do whatever we can to reduce that probability since any such collision would be catastrophic. The best way to avoid such a disaster is to deflect the asteroids. The only known way of deflecting asteroids is by hitting them with nuclear weapons that would be stored in space stations.

The scientist's claims are structured so as to lead to which of the following conclusions?

- (A) Nuclear technology is the only technology that can plausibly be used to prevent natural catastrophes.
- (B) Nuclear weapons should be deployed in space.
- (C) No catastrophe has yet been caused by the

collision of an asteroid with the Earth.

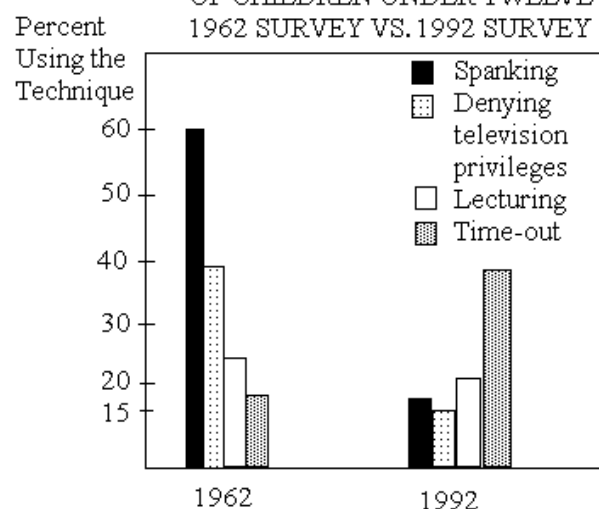
- (D) The 1, 000 large asteroids that cross the Earth's path pose only an extremely slight risk of colliding with the Earth.
- (E) There is currently no acceptable use to which nuclear weapons can be put, aside from protecting the Earth from asteroids.

9. It has long been thought that high levels of the hormone testosterone contribute to the onset of heart disease in men. However, this view cannot be correct, since men who have heart disease typically show significantly lower levels of testosterone than do men who have not had heart disease.

The argument above assumes which of the following?

- (A) Many men who have never had heart disease have unusually low levels of testosterone.
- (B) Having heart disease does not significantly lower the level of testosterone in men.
- (C) Levels of hormones other than testosterone significantly affect the likelihood that a man will develop heart disease.
- (D) Heart disease and lowered testosterone levels in men are the effects of a single common cause.
- (E) High levels of testosterone have never been thought to contribute to a serious disease other than heart disease.

DISCIPLINARY TECHNIQUES
USED BY NORTH AMERICAN PARENTS
OF CHILDREN UNDER TWELVE
1962 SURVEY VS. 1992 SURVEY



The time-out technique involves removing the child from

an undesirable situation in order to let the child think things over. Over the last two decades, family doctors have been advocating this technique as preferable to spanking, which is now known to be potentially injurious and no more effective.

10. Which of the following can properly be concluded from the data presented in the graph ?

- (A) The 1962 survey was based on a larger sample than the 1992 survey was.
- (B) In the period between the surveys, denying television privileges was never the disciplinary technique most popular with parents.
- (C) The four disciplinary techniques featured in the graph were the only disciplinary techniques named by parents in either survey.
- (D) The 1962 survey allowed parents to name more than one disciplinary technique, but the 1992 survey may not have allowed this.
- (E) In the period between the surveys, there were no significant changes in the popularity of lecturing children as a disciplinary method.

11. People who engage in scuba diving are healthier, on average, than people who do not engage in this activity. Therefore, scuba diving tends to promote improved health.

The argument is most vulnerable to criticism on the grounds that it

- (A) presupposes that everyone who takes up scuba diving does so solely for health reasons
- (B) leads to a further and falsifiable conclusion that no one can achieve good health without engaging in scuba diving
- (C) fails to point out that a small number of people are seriously injured in scuba diving accidents each year
- (D) treats a precondition for improving one's health as though it were something that by itself could ensure good health
- (E) overlooks the possibility that people generally do not take up scuba diving unless they are in good health

Questions 12-15

An art museum owns six paintings by an eighteenth-century painter. The paintings, listed in order by estimated value from lowest to highest, are F, G, H, S, T, and U. Paintings F, G, and H are landscapes; S, T, and U are portraits. At any one time, exactly three of the six paintings are on exhibit, subject to the following restrictions:

- The paintings on exhibit cannot all be landscapes.
- If the exhibit includes only one portrait, that portrait must be U.
- H cannot be on exhibit at any time that T is on exhibit.

12. Which of the following could be the three paintings on exhibit at some point ?

- (A) F, G, and H
- (B) F, G, and T
- (C) G, H, and S
- (D) G, S, and U
- (E) H, T, and U

13. Which of the following, if they are the first two paintings selected for inclusion in a future exhibit, leave the widest choice of paintings for the third painting in that future exhibit ?

- (A) F and G
- (B) G and H
- (C) H and U
- (D) S and T
- (E) S and U

14. An exhibit must include S if which of the following is true ?

- (A) T is included in the exhibit.
- (B) T is not included in the exhibit.
- (C) H is the only landscape included in the exhibit.
- (D) U is included in the exhibit.
- (E) The exhibit includes either F or G, but not both.

15. If U is undergoing restoration and is not available to be exhibited, which of the following is a painting that CANNOT then be exhibited ?

- (A) F
- (B) G

- (C)H
(D)S
(E)T

Questions 16-22

In each of the five consecutive days of a cooks' contention, exactly one of five well-known cooks? G, H, J, K, and L? will cook a demonstration meal. Each of the five cooks will cook exactly one of the five meals. The schedule for the cooks is constrained by the following conditions :

H cannot cook on any of the first three days.

L must cook on one of the days before the day on which H cooks.

J must cook on one of the days before the day on which G cooks.

G must cook on one of the days before the day on which K cooks.

16. Which of the following can be the order, from first to fifth, in which the five cooks cook the meals ?
(A) G, K, L, J, H
(B) J, G, K, H, L
(C) J, G, K, L, H
(D) J, K, G, L, H
(E) L, J, H, K, G

17. If K cooks the fourth meal, which of the following must be true ?
(A) G cooks on the third day.
(B) H cooks on the fifth day.
(C) J cooks on the first day.
(D) J cooks on the second day.
(E) L cooks on the third day.

18. Which of the following can be true ?
(A) G cooks the first meal.
(B) J cooks the fourth meal.
(C) L cooks the fifth meal.
(D) H cooks on some day before G cooks.
(E) L cooks on some day after K cooks.

19. If G cooks a meal on some day before L does, then it must be true that
(A) G cooks the second meal
(B) J cooks the third meal
(C) H cooks the fourth meal
(D) K cooks the fifth meal
(E) L cooks the fourth meal

20. If J does not cook on the first day, then it must be true that
(A) G does not cook the third meal
(B) H does not cook the fourth meal
(C) J does not cook the second meal
(D) L does not cook the third meal
(E) K does not cook the fifth meal

21. If H does not cook the fifth meal, which of the following must be true ?
(A) G cooks the second meal.
(B) J cooks the first meal.
(C) J cooks the second meal.
(D) K cooks the fifth meal.
(E) L cooks the first meal.

22. If G cooks the third meal, which of the following is true ?
(A) L is the only one of the five cooks who can cook the first meal.
(B) J is the only one of the five cooks who can cook the second meal.
(C) Any one of exactly three of the five cooks can cook the second meal.
(D) K is the only one of the five cooks who can cook the fourth meal.
(E) Either one of exactly two of the five cooks can cook the fifth meal.

23. Which of the following most logically completes the argument below ?

In recent years, the proportion of car buyers who buy new cars rather than used cars has declined. Some consumers have attributed this change to an increase in new-car prices. As evidence of the price increase, they cite figures that show that, even adjusting for inflation, the price that the buyer of a new car pays,

on average, is far higher now than a few years ago. This evidence is unpersuasive, however, because

- (A) the value of a car that is bought new declines much more rapidly than does the value of a car that is bought used
- (B) after someone has bought a car, it might be several years before that person next buys a car
- (C) a decline in the proportion of car buyers who buy new cars must necessarily mean that the proportion who buy used cars has increased
- (D) the relative increase in used-car sales might be explained by the decisions of only a small proportion of all car buyers
- (E) the change in the average price paid for a new car could result solely from more people's rejecting inexpensive new cars in favor of used cars

24. In Bassaria a group of that country's most senior judges has criticized the uniform mandatory sentences recently introduced for certain specific crimes. The judges argue that such sentences, by depriving them of all discretion in setting sentences, make it impossible for them to consider either aggravating or extenuating circumstances and so make it impossible to achieve true justice? the fitting of the severity of the punishment to the gravity of the particular crime. Which of the following, if true, provides the strongest evidence for the claim that in Bassaria the newly introduced mandatory sentences are not necessarily a change for the worse with respect to achieving true justice as defined in the argument?

- (A) Before mandatory sentencing, judges in eastern Bassaria imposed strikingly different sentences from those in western Bassaria for equally grave instances of the same kind of offense.
- (B) In Bassaria the frequency of crimes that have been made subject to mandatory sentences is lower now than it was just prior to the introduction of mandatory sentencing.
- (C) The law introducing mandatory sentences was passed in the legislature of Bassaria by a large majority and is unlikely to be repealed in the foreseeable future.
- (D) There used to be a wide difference between the

minimum and the maximum sentences allowed by law in cases of crimes now subject to mandatory sentences.

- (E) In Bassaria judges are appointed for life and are thus not easily influenced by political pressure groups.

25. Each of two particular inspection systems that are based on different principles would detect all product flaws but would also erroneously reject three percent of flawless products. Assuming there is no overlap between the products erroneously rejected by the two systems and also no interference between the systems if both operate, using both systems and rejecting only those products found flawed by both would be a way of avoiding all erroneous rejections.

Which of the following most precisely characterizes the reasoning in the argument?

- (A) The reasoning is conclusive, that is, the conclusion cannot be false if the statements offered in its support are true.
- (B) The reasoning is strong but not conclusive, if the statements offered in support of the conclusion are true, they provide good grounds for that conclusion, though it is possible that additional information might weaken the argument.
- (C) The reasoning is weak; the statements offered in support of the conclusion, though relevant to it, by themselves provide at best inadequate grounds for the conclusion.
- (D) The reasoning is flawed in that the conclusion is no more than a paraphrase of one of the pieces of evidence offered in its support.
- (E) The reasoning is flawed in that the argument treats evidence that a factor is necessary to bring about an event as if it were evidence that the factor is sufficient to bring about that event.

SECTION 3

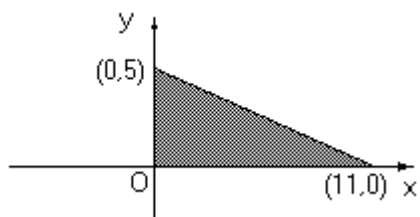
Time 30- minutes

30 Questions

1. $\sqrt{13} + \sqrt{12}$ 5

$$\begin{aligned} x^2 y &< 0 \\ x &> 0 \end{aligned}$$

2. xy 0



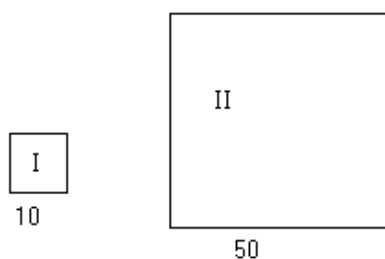
3. The area of the shaded region 55

$$y = x + 1$$

4. $(x - y)^2$ 1

$$xy < 0$$

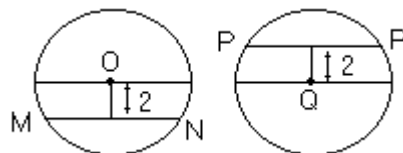
5. $x + y$ 0



6. Area of square region I 0.004
Area of square region s II

The formula $P = 176 - 0.8A$ gives the maximum recommended pulse rate P for a person exercising who is A years of age.

7. The age in years of a person whose maximum recommended pulse rate when exercising is 136 50



8. The length of line segment MN in the circle with center O The length of line segment PR in the circle with center Q

$$x \text{ and } y \text{ are positive and } x^2 + y^2 = 1$$

9. $(x + y)^2$ 1

A certain brand of soap powder is sold in boxes that contain either 8.0 pounds priced at \$2.24 per box or 10.5 pounds priced at \$3.00 per box.
(1 pound = 16 ounces)

10. The price per ounce of the soap powder in the 8.0-pound box The price per ounce of the soap powder in the 10.5-pound box

x and y are integers.

$$0.35 < \frac{x}{y} < 0.45$$

11. $y - x$ 60

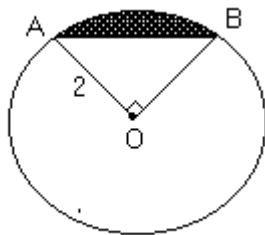
The average (arithmetic mean) of 50 measurements is 34.6, and the least and the greatest of the measurements are 18.3 and 50.9 respectively.

12. The median of the 50 measurements 34.6

$$a \neq 0$$

13. $\frac{1}{a} \left(\frac{1}{a} \left(\frac{1}{a} \left(\frac{1}{a} \cdot a \right) a \right) a \right)$ 1

14. $\sqrt{2}$ $\frac{7}{5}$



O is the center of the circle.

15. Twice the area of the shaded region The area of triangle AOB

16. A ream of paper contains 480 sheets of paper. A certain box holds 16 reams. The number of sheets of paper in 5 of these boxes is

(A) 2,400
(B) 3,840
(C) 7,680
(D) 38,400
(E) 76,800

17. The numbers in a table are arranged in 10 rows and 4 columns such that one number is placed at the intersection of each row and column. How many numbers are contained in the table?

(A) 14
(B) 40
(C) 400
(D) 10^4
(E) 4^{10}

18. A recipe for 42 cookies requires $1\frac{3}{4}$ cups of flour.

How many cups of flour are required to make 210 of these cookies?

(A) 5
(B) $8\frac{3}{4}$
(C) $9\frac{1}{4}$
(D) $10\frac{1}{2}$
(E) $17\frac{1}{2}$

19. If $y = 7x - 5$, then $x =$

(A) $7y + 5$
(B) $\frac{y}{7} - 5$
(C) $\frac{y}{7} + 5$
(D) $\frac{y - 5}{7}$
(E) $\frac{y + 5}{7}$

20. If $2a + 2b = 7$ and $3c + 3d = 15$, then the average (arithmetic mean) of a , b , c , and d is

(A) $1\frac{1}{2}$
(B) $2\frac{1}{8}$
(C) 3
(D) $4\frac{1}{4}$
(E) $8\frac{1}{2}$

Questions 21-25 refer to the following information

A sample of employees were tested on data-entry skills for one hour, and the number of errors (x) they made and the percent of employees (p) making x errors were recorded as follows.

Number of Errors x	Percent of Employees p
0	2%
1	5%
2	10%
3	24%
4	17%
5	20%
6 or more	22%

21. What is the range for the number of errors made by the employees in the sample?

(A) 4
(B) 5
(C) 6
(D) 7

- (E) It cannot be determined from the information given.
22. What percent of the sample made fewer than 4 errors?
- (A) 17%
(B) 24%
(C) 39%
(D) 41%
(E) 58%
23. If 40 employees in the sample made exactly 5 error, what was the total number of employees in the sample?
- (A) 800
(B) 600
(C) 500
(D) 400
(E) 200
24. If those employees who made 6 or more errors were removed from the sample and an employee were selected at random from those remaining, what is the probability that the employee selected made no errors?
- (A) $\frac{1}{11}$
(B) $\frac{1}{22}$
(C) $\frac{1}{39}$
(D) $\frac{1}{50}$
(E) $\frac{1}{78}$
25. What was the median number of errors in the sample?
- (A) 3
(B) 3.5
(C) 4
(D) 4.5
(E) It cannot be determined from the information given.
26. In the rectangular coordinate system above, if P , not shown, is a point on AB and if the x -coordinate of P is 1, what is the y -coordinate of P ?
- (A) $\frac{4}{3}$
(B) $\frac{3}{2}$
(C) $\frac{7}{3}$
(D) $\frac{5}{2}$
(E) $\frac{8}{3}$
27. Circular region C has a radius of 1 centimeter and a circumference of x centimeters: If the region has an area of y square centimeters, what is the ratio of x to y ?
- (A) 2
(B) $\frac{1}{2}$
(C) $2p$
(D) p
(E) $\frac{1}{p}$
28. If $2.5p \times 10^5$ cubic feet of earth are to be removed in constructing a tunnel in the shape of a right circular cylinder 20 feet in diameter, what will be the length, in feet, of the tunnel?
- (A) 25,000
(B) 12,500
(C) 5,000
(D) 2,500
(E) 1,250
29. One adult and 10 children are on an elevator. If the adult's weight is 4 times the average (arithmetic mean) weight of the children, the adult's weight is what fraction of the total weight of the 11 people on the elevator?
- (A) $\frac{1}{11}$
(B) $\frac{1}{4}$
(C) $\frac{2}{7}$

- (D) $\frac{4}{11}$
 (E) $\frac{2}{5}$
30. $\frac{(0.2)^5}{((0.2)^3)^2} =$
 (A) 0.05
 (B) 0.5
 (C) 4.0
 (D) 2.0
 (E) 5.0

SECTION 4

Time-30 minutes

38 Questions

- Although sales have continued to increase since last April, unfortunately the rate of increase has ----.
 (A) resurged
 (B) capitulated
 (C) retaliated
 (D) persevered
 (E) decelerated
- Although the mental process that creates a fresh and original poem or drama is doubtless ---- that which originates and elaborates scientific discoveries, there is clearly a discernible difference between the creators
 (A) peripheral to
 (B) contiguous with
 (C) opposed to
 (D) analogous to
 (E) inconsistent with
- It is disappointing to note that the latest edition of the bibliography belies its long-standing reputation for ---- by ---- some significant references to recent publications.
 (A) imprecision.. appropriating
 (B) relevance.. adding
 (C) timeliness.. updating
 (D) meticulousness.. revising
 (E) exhaustiveness.. omitting
- Although Simpson was ingenious at ---- to appear innovative and spontaneous, beneath the ruse he remained uninspired and rigid in his approach to problem-solving.
 (A) intending
 (B) contriving
 (C) forbearing
 (D) declining
 (E) deserving
- She was criticized by her fellow lawyers not because she was not ----, but because she so ---- prepared her cases that she failed to bring the expected

- number to trial.
 (A) well versed.. knowledgeably
 (B) well trained.. enthusiastically
 (C) congenial.. rapidly
 (D) hardworking.. minutely
 (E) astute.. efficiently
6. Schlesinger has recently assumed a conciliatory attitude that is not ---- by his colleagues, who continue to ---- compromise.
 (A) eschewed.. dread
 (B) shared.. defend
 (C) questioned.. reject
 (D) understood.. advocate
 (E) commended.. disparage
7. The National Archives contain information so ---- that researchers have been known never to publish because they cannot bear to bring their studies to an end.
 (A) divisive
 (B) seductive
 (C) selective
 (D) repetitive
 (E) resourceful
8. HILL: MOUNTAIN::
 (A) grass: rocks
 (B) autumn: winter
 (C) creek: river
 (D) star: sun
 (E) cliff: slope
9. AERATE: OXYGEN::
 (A) eclipse: light
 (B) desiccate: moisture
 (C) precipitate: additive
 (D) hydrate: water
 (E) striate: texture
10. ORCHESTRA: MUSICIAN:
 (A) cube: side
 (B) kilometer: meter
 (C) sonnet: poem
 (D) biped: foot
 (E) pack: wolf

11. EQUIVOCATION: MISLEADING::

- (A) mitigation: severe
 (B) advice: peremptory
 (C) bromide: hackneyed
 (D) precept: obedient
 (E) explanation: unintelligible

12. CENSORSHIP: COMMUNICATION::

- (A) propaganda: ideology
 (B) preservative: decay
 (C) revision: accuracy
 (D) rest: atrophy
 (E) exercise: fitness

13. BUS: PASSENGERS:

- (A) flock: birds
 (B) tanker: liquid
 (C) envelope: letter
 (D) bin: coal
 (E) automobile: gasoline

14. BALLAD: STANZA::

- (A) novel: chapter
 (B) poem: meter
 (C) play: dialogue
 (D) movie: script
 (E) photograph: caption

15. DISABUSE: FALLACY::

- (A) cure: disease
 (B) persevere: dereliction
 (C) belittle: imperfection
 (D) discredit: reputation
 (E) discern: discrimination

16. BLANDISHMENT: CAJOLE::

- (A) prediction: convince
 (B) obstacle: impede
 (C) embellishment: praise
 (D) deficiency: compensate
 (E) compliment: exaggerate

Although the hormone adrenaline is known to regulate memory storage, it does not pass from the blood into brain cells. We are faced with an apparent paradox: how can a

hormone that does not act directly on the brain have such a
(5) large effect on brain function?

Recently, we tested the possibility that one of the hormone's actions outside the brain might be responsible. Since one consequence of adrenaline release in an animal is an increase in blood glucose levels, we examined the
(10) effects of glucose on memory in rats. We found that glucose injected immediately after training enhances memory tested the next day. Additional evidence was provided by negative findings: drugs called adrenergic antagonists, which block peripheral adrenaline receptors, disrupted
(15) adrenaline's ability to regulate memory but did not affect memory enhancements produced by glucose that was not stimulated by adrenaline. These results are as they should be if adrenaline affects memory modulation by increasing blood glucose levels.

17. The primary purpose of the passage is to

- (A) reconcile two opposing theories
- (B) compare two different explanations for a phenomenon
- (C) describe experimental research that appears to support an unpopular theory
- (D) present evidence that may help to resolve an apparent contradiction
- (E) describe a hypothesis that has caused a controversy

18. It can be inferred from the passage that the author would most likely describe the "additional evidence" (line 12) provided by experiments with adrenergic antagonists as

- (A) revolutionary
- (B) disappointing
- (C) incomplete
- (D) unexpected
- (E) corroborative

19. The passage provides information about which of the following topics?

- (A) The mechanism by which glucose affects memory storage
- (B) The evidence that prompted scientist to test the effects of adrenaline on memory regulation
- (C) The reason that the effects of glucose on memory were tested

(D) The ways that memory storage modifies the structure of the brain

(E) The kinds of training used to test memory enhancement in rats

20. The author refers to the results of the experiment using adrenergic antagonists as "negative findings" (line 13) most likely because the adrenergic antagonists

- (A) failed to disrupt adrenaline's effect on memory
- (B) did not affect glucose's ability to enhance memory.
- (C) did not block adrenaline's ability to increase blood glucose levels
- (D) only partially affected adrenaline's ability to enhance memory
- (E) disrupted both adrenaline's and glucose's effect on memory

The age at which young children begin to make moral discriminations about harmful actions committed against themselves or others has been the focus of recent research into the moral development of children. Until recently,
(5) child psychologists supported pioneer developmentalist Jean Piaget in his hypothesis that because of their immaturity, children under age seven do not take into account the intentions of a person committing accidental or deliberate harm, but rather simply assign punishment for transgressions on
(10) the basis of the magnitude of the negative consequences caused. According to Piaget, children under age seven occupy the first stage of moral development, which is characterized by moral absolutism (rules made by authorities must be obeyed) and imminent justice (if rules are broken,
(15) punishment will be meted out). Until young children mature, their moral judgments are based entirely on the effect rather than the cause of a transgression. However, in recent research, Keasey found that six-year-old children not only distinguish between accidental and intentional harm, but
(20) also judge intentional harm as naughtier, regardless of the amount of damage produced. Both of these findings seem to indicate that children, at an earlier age than Piaget claimed, advance into the second stage of moral development, moral autonomy, in which they accept social rules
(25) but view them as more arbitrary than do children in the first stage.

Keasey's research raises two key questions for developmental psychologists about children under age seven: do they recognize justifications for harmful actions, and do

(30)they make distinctions between harmful acts that are preventable and those acts that have unforeseen harmful consequences? Studies indicate that justifications excusing harmful actions might include public duty, self-defense, and provocation. For example, Nesdale and Rule concluded that (35)children were capable of considering whether or not an aggressor's action was justified by public duty: five year olds reacted very differently to "Bonnie wrecks Ann's pretend house" depending on whether Bonnie did it "so somebody won't fall over it" or because Bonnie wanted "to (40)make Ann feel bad." Thus, a child of five begins to understand that certain harmful actions, though intentional, can be justified; the constraints of moral absolutism no longer solely guide their judgments.

Psychologists have determined that during kindergarten (45)children learn to make subtle distinctions involving harm. Darley observed that among acts involving unintentional harm, six-year-old children just entering kindergarten could not differentiate between foreseeable, and thus preventable, harm and unforeseeable harm for which the perpetrator (50)cannot be blamed. Seven months later, however, Darley found that these same children could make both distinctions, thus demonstrating that they had become morally autonomous.

21. Which of the following best describes the passage as a whole?

- (A) An outline for future research
- (B) An expanded definition of commonly misunderstood terms
- (C) An analysis of a dispute between two theorists
- (D) A discussion of research findings in an ongoing inquiry
- (E) A confirmation of an established authority's theory

22. According to the passage, Darley found that after seven months of kindergarten six year olds acquired which of the following abilities?

- (A) Differentiating between foreseeable and unforeseeable harm
- (B) Identifying with the perpetrator of a harmful action
- (C) Justifying harmful actions that result from provocation
- (D) Evaluating the magnitude of negative consequences resulting from the breaking of rules
- (E) Recognizing the difference between moral absolu-

tism and moral autonomy

23. According to the passage, Piaget and Keasey would not have agreed on which of the following points?

- (A) The kinds of excuses children give for harmful acts they commit
- (B) The age at which children begin to discriminate between intentional and unintentional harm
- (C) The intentions children have in perpetrating harm
- (D) The circumstances under which children punish harmful acts
- (E) The justifications children recognize for mitigating punishment for harmful acts

24. It can be inferred that the term "public duty" (line 33) in the context of the passage, means which of the following?

- (A) The necessity to apprehend perpetrators.
- (B) The responsibility to punish transgressors
- (C) An obligation to prevent harm to another
- (D) The assignment of punishment for harmful action
- (E) A justification for punishing transgressions

25. According to the passage, Keasey's findings support which of the following conclusions about six-year-old children?

- (A) They have the ability to make autonomous moral judgments.
- (B) They regard moral absolutism as a threat to their moral autonomy.
- (C) They do not understand the concept of public duty.
- (D) They accept moral judgment made by their peers more easily than do older children.
- (E) They make arbitrary moral judgments.

26. It can be inferred from the passage that Piaget would be likely to agree with which of the following statements about the punishment that children under seven assign to wrongdoing?

- (A) The severity of the assigned punishment is determined by the perceived magnitude of negative consequences more than by any other factor.
- (B) The punishment is to be administered immediately following the transgression.
- (C) The children assign punishment less arbitrarily than they do when they reach the age of moral

- autonomy.
- (D) The punishment for acts of unintentional harm is less severe than it is for acts involving accidental harm.
- (E) The more developmentally immature a child, the more severe the punishment that the child will assign.
27. According to the passage, the research of Nesdale and Rule suggests which of the following about five-year-old children?
- (A) Their reactions to intentional and accidental harm determine the severity of the punishments they assign.
- (B) They, as perpetrators of harmful acts, disregard the feelings of the children they harm.
- (C) They take into account the motivations of actions when judging the behavior of other children.
- (D) They view public duty as a justification for accidental, but not intentional, harm.
- (E) They justify any action that protects them from harm.
28. DEBUT:
- (A) collaboration
- (B) monologue
- (C) farewell performance
- (D) repertoire standard
- (E) starring role
29. WITHER:
- (A) disagree
- (B) shine
- (C) plant
- (D) adhere
- (E) revive
30. BUCK:
- (A) cover over
- (B) assent to
- (C) brag about
- (D) improve
- (E) repair
31. MEAN:
- (A) trusting
- (B) ardent
- (C) clever
- (D) incautious
- (E) noble
32. ADJUNCT:
- (A) expert appraisal
- (B) generous donation
- (C) essential element
- (D) mild reproof
- (E) impartial judgment
33. CANONICAL:
- (A) imprecise
- (B) ubiquitous
- (C) superfluous
- (D) nontraditional
- (E) divisive
34. TICKLISH:
- (A) heavy-handed
- (B) significant
- (C) tolerant
- (D) impartial
- (E) imperturbable
35. PREVALENT:
- (A) invasive
- (B) inconsistent
- (C) indistinct
- (D) unpalatable
- (E) unusual
36. PENURY:
- (A) approbation
- (B) affluence
- (C) objectivity
- (D) compensation
- (E) grandiosity
37. MINATORY:
- (A) convenient
- (B) nonthreatening
- (C) straightforward
- (D) fastidious
- (E) rational

38. CALUMNIOUS:

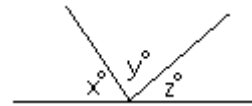
- (A) adept
 (B) aloof
 (C) quaint
 (D) decorous
 (E) flattering

SECTION 5

Time –30 minutes

30 Questions

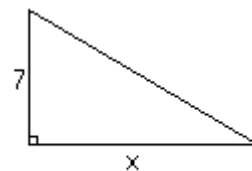
1. 0.125 $\frac{1}{8}$



2. The average (arithmetic mean) of x , y , and z is 75



3. WY XZ



The area of the triangle is 35

4. x 5

5. $x - x^2$ $0 < x < 1$ $x^2 - x^3$

$$\frac{x}{y^2} > 0$$

6. x

7. Area of a square region with side s is twice the area of a square region with side $\frac{s}{2}$

8. The tenths digit of $10^2 d$ is $d = 561.165$. The tenths digit of $\frac{d}{10^2}$ is

$$x = 0.5$$

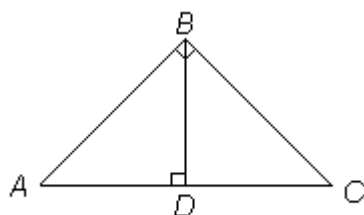
9. $\frac{(1+x)^2}{(1-x)^2}$ 9

Olive oil, vinegar, and water were combined to make 10 cups of salad dressing. The oil and vinegar were mixed in the ratio of 2 parts oil to 3 parts vinegar.

10. The number of cups of water used in the salad dressing 5

After a store had sold k television sets for p dollars each, it reduced the price p of each set by 5 percent and then sold twice as many sets at this reduced price.

11. The total revenue from the sale of all of these television sets $2.95pk$



$\triangle ABC$ is equilateral

12. The degree measure of $\angle DBC$ Half the degree measure of $\angle BDC$

13. $(0.95)^{12} - (0.95)^{11}$ 0.95

$$x^2 + y^2 = \frac{37}{4}$$

$$xy = \frac{3}{2}$$

14. $(x+y)^2$ $\frac{49}{4}$

$$k^2 = 9$$

15. $2k^3$ $3k^2$

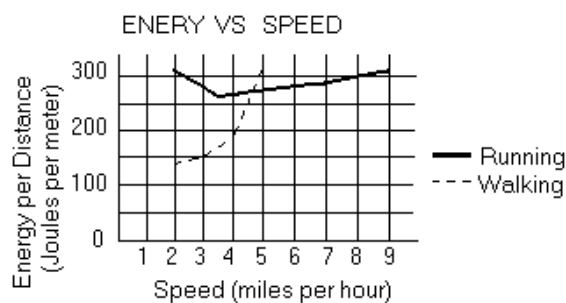
16. For which of the following lists of numbers is the median equal to the average (arithmetic mean)?

- (A) 3, 4, 7
(B) 1, 10, 19
(C) 7, 7, 10, 12
(D) 0, 1, 1, 5
(E) 0, 2, 3, 4

17. $(1-0.05)^3 + 1 =$
(A) 0.0025
(B) 0.9025
(C) 1.4025
(D) 1.9025
(E) 2.1025

18. Which of the following is NOT a solution of the inequality $3x - 4 > -5x + 12$?
(A) 1.8
(B) 2.5
(C) 2.7
(D) 3.0
(E) 4.2

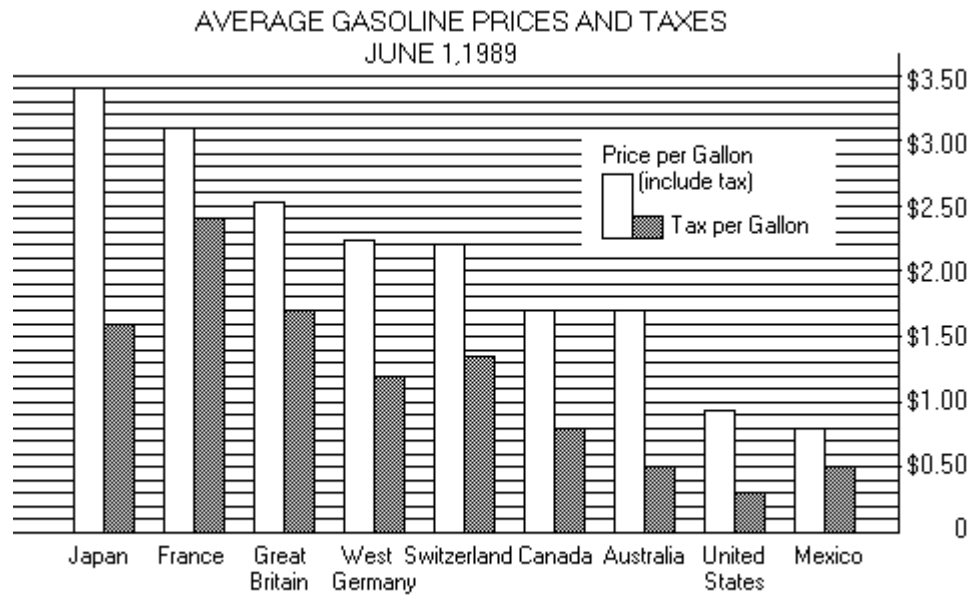
19. What is the area of a circular region with diameter 6?
(A) $6p$
(B) $9p$
(C) $12p$
(D) $24p$
(E) $36p$



Note: Drawn to scale

20. According to the graph above, at approximately what speed, in miles per hour (mph), is the energy per distance used in running approximately twice the energy per distance used in walking?
(A) Between 6.5 and 7 mph
(B) Between 5.5 and 6 mph
(C) Between 4.5 and 5 mph
(D) Between 3.5 and 4 mph
(E) Between 2.5 and 3 mph

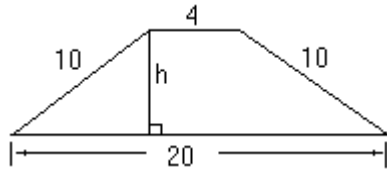
Questions 21-25 refer to the following graph. In these questions all references to gasoline prices and taxes refer to average prices, including tax, and average taxes, in United States dollars, on June 1, 1989.



21. What was the approximate price per gallon of gasoline for the country with the least tax per gallon of gasoline?
- (A) \$0.25
(B) \$0.50
(C) \$0.80
(D) \$0.87
(E) \$0.95
22. For a vacationer who planned to drive 3,500 miles in Mexico in June 1989, using a car with an average fuel efficiency of 25 miles per gallon of gasoline, which of the following would have been the best estimate of the cost of gasoline for the trip?
- (A) \$50
(B) \$70
(C) \$110
(D) \$140
(E) \$170
23. The ratio of the lowest price per gallon of gasoline to the highest price for the countries shown was closest to
- (A) 1:3
(B) 1:4
(C) 1:5
(D) 1:6
(E) 1:7
24. For how many of the countries shown did the tax account for more than 50 percent of the price per gallon of gasoline?
- (A) One
(B) Two
(C) Three
(D) Four
(E) Five
25. If the tax per gallon of gasoline in Canada were doubled and the increase in tax added to the price per gallon of gasoline, what percent of the resulting price per gallon would the tax then be?
- (A) 36%
(B) 50%
(C) 64%
(D) 75%
(E) 90%
26. A gardener wishes to plant 5 bushes in a straight row. Each bush has flowers of a different solid color (white, yellow, pink red, and purple). How many ways can the bushes be arranged so that the middle

bush is the one with red flowers?

- (A) 24
- (B) 30
- (C) 60
- (D) 96
- (E) 120



27. What is the value of h in the trapezoid above?

- (A) $6\sqrt{2}$
- (B) $4\sqrt{3}$
- (C) 8
- (D) 6
- (E) 4

28. The bottom of an underground rectangular tank has an area of 12 square feet and the depth of the tank is 10 feet. A liquid is pumped into the empty tank at the constant rate of 30 cubic feet per minute. After how many minutes will the depth of the liquid in the tank be 8 feet?

- (A) 1.5
- (B) 2.0
- (C) 2.5
- (D) 3.2
- (E) 4.0

29. A candy assortment consists of seven flavors of chocolate-covered creams packed in two-layered boxes with 27 creams in each layer. The flavors are always packed in rows so that the flavor varies with each piece in the following order, vanilla, orange, cherry, vanilla, raspberry, lime, pecan, cherry, lemon. How many chocolate-covered vanilla creams are needed to pack 200 boxes of the assortment?

- (A) 600
- (B) 1,200
- (C) 1,800
- (D) 2,400
- (E) 3,600

30. If $\frac{m}{-19}$ is an even integer, which of the following

- (A) m is negative
- (B) m is positive
- (C) m is a prime number
- (D) m is an odd integer
- (E) m is an even integer

SECTION 6

Time –30 minutes

25 Questions

Questions 1-8

A bakery makes nine kinds of cookies. Of these nine, three kinds are fruit cookies—G, H, and J; three kinds are nut cookies—K, L, and O; and three kinds are plain cookies—X, Y, and Z. Each day of the week, Monday through Sunday, the bakery will feature a special price on exactly three different kinds of cookies. The three featured cookies will be selected according to the following rules:

Each day at least one fruit cookie must be featured, and each day at least one nut cookie must be featured.

On any day on which cookie J is featured, cookie L cannot be featured.

On any day on which cookie K is featured, cookie Y must also be featured.

No kind of cookie can be featured more than three times in a week.

1. Which of the following lists three cookies that can be featured together?

- (A) G, L, Z
- (B) H, K, X
- (C) J, L, Y
- (D) J, O, Z
- (E) K, O, Y

2. On a day on which both cookie L and cookie Z are featured, which of the following can be the third kind of cookie featured?

- (A) H
- (B) J
- (C) O
- (D) X
- (E) Y

3. A partial schedule of featured cookies is shown below.

<u>Monday</u>	<u>Tuesday</u>	<u>Wednesday</u>	<u>Thursday</u>	<u>Friday</u>
L	H	O	G	Z

According to this schedule, which of the following is a day on which cookie X CANNOT be one of the featured cookies?

- (A) Monday
- (B) Tuesday
- (C) Wednesday
- (D) Thursday
- (E) Friday

4. If cookie J is featured on Friday, Saturday, and Sunday; if cookie K is featured on Monday, Tuesday, and Wednesday, and if cookie G is featured only on Thursday, then cookie L can be featured on

- (A) Monday only
- (B) Thursday only
- (C) Monday, Tuesday, and Wednesday only
- (D) Friday, Saturday, and Sunday only
- (E) any two of the first four days of the week

5. If each kind of nut cookie is featured three times in one week, what is the maximum number of days on which plain cookies can be featured during that week?

- (A) Three
- (B) Four
- (C) Five
- (D) Six
- (E) Seven

6. If cookie H and cookie Y are each featured on Monday, Tuesday, and Wednesday, and if cookie G and cookie X are each featured on Thursday, Friday and Saturday, then the cookies featured on Sunday must include both

- (A) J and K
- (B) J and L
- (C) J and O
- (D) K and L
- (E) K and Z

7. If exactly seven kinds of cookies are featured during one week, which of the following must be true about that week?

- (A) X is the only kind of plain cookie that is featured
- (B) Y is the only kind of plain cookie that is featured
- (C) Z is the only kind of plain cookie that is featured.
- (D) On at least one day, both cookie G and cookie Z are featured.
- (E) On at least one day, both cookie J and cookie X are featured.

8.If cookie X is featured exactly twice and cookie Z is featured exactly three times in one week, which of the following must be true?

- (A) Cookie G is featured exactly three times during the week.
- (B) Cookie J is featured at most twice during the week.
- (C) Cookie K is featured at most twice during the week.
- (D) Cookie L is featured at most twice during the week.
- (E) Cookie Y is featured exactly twice during the week.

9. In recent years, there has been a dramatic decline in the population of the shrike, a predatory bird that inhabits flat land, such as farms and pastures. Some ornithologists hypothesize that this decline is due to the introduction of new, more effective pesticides to control the insect species on which shrikes prey.

The answer to which of the following questions is NOT relevant to evaluating the ornithologists' hypothesis?

- (A) Was there a decline in the shrike population before the new pesticides were first used?
- (B) Have shrike populations declined significantly in those habitats where the new pesticides have not been used?
- (C) Have the new pesticides more significantly reduced the population of insect species on which shrikes prey than did the pesticides previously used?
- (D) Are insects that have consumed the new pesticides more toxic to the shrikes that eat those insects than were insects that consumed the less effective pesticides?
- (E) Are the new pesticides considered by most people to be less harmful to the environment than the old pesticides were considered to be?

10. Census data for Prenland show that unmarried Prenlandic men in their thirties outnumber unmarried Prenlandic women in that age group by about ten to One. Most of these men do wish to marry. Clearly,

however, unless many of them marry women who are not Prenlandic, all but a minority will remain unmarried.

The argument makes which of the following assumptions?

- (A) Emigration from Prenland is more common among women than among men.
- (B) A greater proportion of Prenlandic women in their thirties than of Prenlandic men of the same age would prefer to remain unmarried.
- (C) It is unlikely that many of these unmarried Prenlandic men will marry women more than a few years older than themselves.
- (D) Prenland has a high rate of divorce.
- (E) Most of the unmarried Prenlandic men are unwilling to marry women who are not Prenlandic.

11. Certain extremely harmful bacteria found only in sewage are difficult to detect directly. Testing for *E. coli*, an easily detected and less harmful type of bacteria, in ocean water would be a reliable way of determining whether or not these more harmful bacteria are present, since ocean water contains *E. Coli* only if the water is contaminated with sewage that contains the harmful bacteria.

Which of the following, if true, most seriously weakens the argument?

- (A) There are many different strains of the *E. coli* bacteria, and only some of these strains are harmful.
- (B) Some types of bacteria found in sewage are neither disease-causing nor difficult to detect directly.
- (C) Some of the types of bacteria found in sewage along with *E. coli* are not harmful to people unless the bacteria are ingested in large quantities.
- (D) *E. coli* dies out much more quickly than some of the more harmful bacteria found in sewage and then can no longer be easily detected.
- (E) Some of the types of bacteria found in sewage along with *E. coli* reproduce at a slower rate than *E. coli*.

Questions 12-17

A bank has exactly four cashier windows, arranged in a row and numbered consecutively 1 through 4 from one end of the row to the other. The bank has exactly six cashiers: two supervisors (Joan and Karim); and four trainees (Lorraine, Mark, Nora, and Patrick). Throughout a particular peak-hour period, the stationing of cashiers at windows is restricted as follows:

There must be exactly one cashier at each window.

The cashier at window 2 must be a supervisor.

Lorraine must be at a window but cannot be at window

3.

If Mark is at one of the windows, Joan must be at a window immediately adjacent to it.

The cashiers at the windows must include either Nora or Patrick, but they cannot include both Nora and Patrick.

12. Which of the following lists the cashiers who can be stationed at windows 1 through 4 during this period?

- | <u>1</u> | <u>2</u> | <u>3</u> | <u>4</u> |
|-------------|----------|----------|----------|
| (A) Joan | Karim | Mark | Lorraine |
| (B) Joan | Karim | Nora | Lorraine |
| (C) Karim | Nora | Joan | Lorraine |
| (D) Mark | Joan | Lorraine | Patrick |
| (E) Patrick | Joan | Nora | Lorraine |

13. Which of the following must be true about the stationing of the cashiers during this period?

- (A) Joan is at window 1 or at window 2.
- (B) Karim is at window 2 or at window 4.
- (C) Lorraine is at window 1 or window 4.
- (D) Nora is at window 1 or at window 3.
- (E) Patrick is at window 3 or at window 4.

14. If during this period Patrick and Mark, not neces-

sarily in that order, are stationed at immediately adjacent windows, which of the following must be stationed at window 4 during this period?

- (A) Joan
- (B) Karim
- (C) Lorraine
- (D) Mark
- (E) Patrick

15. If during this period Lorraine and Mark, not necessarily in that order, are stationed at immediately adjacent windows, which of the following can be stationed at window 1 during this period?

- (A) Joan
- (B) Karim
- (C) Lorraine
- (D) Mark
- (E) Nora

16. If during this period Lorraine and Nora, not necessarily in that order, are stationed at immediately adjacent windows, which of the following must be true during this period?

- (A) Joan is stationed at window 1.
- (B) Joan is stationed at window 2.
- (C) Karim is stationed at window 2.
- (D) Nora is stationed at windows 3.
- (E) Patrick is stationed at window 1.

17. If during this period Mark is stationed at a window, which of the following CANNOT be stationed at a window during this period?

- (A) Joan
- (B) Karim
- (C) Lorraine
- (D) Nora
- (E) Patrick

Questions 18-22

Seven photographs—three landscapes: F, H, and J; and four still lifes: Q, R, T and W—will appear on the first seven pages—numbered consecutively from page 1 through page 7—of an exhibit catalog. Each page will contain exactly one of the photographs. The ordering of the photographs in the catalog is governed by the following conditions.

J and W, not necessarily in that order, must appear on consecutively numbered pages.

The three landscapes cannot appear on consecutively numbered pages.

Neither page 2 nor page 4 is a page on which a landscape can appear.

A landscape must appear on page 7

18. Which of the following is an acceptable ordering of the photographs in the catalog?

	<u>Page 1</u>	<u>Page 2</u>	<u>page 3</u>	<u>Page 4</u>	<u>page 5</u>	<u>page 6</u>	<u>page 7</u>
(A)	F	T	H	Q	W	R	J
(B)	H	Q	J	W	R	F	T
(C)	J	W	H	R	T	Q	F
(D)	Q	T	R	W	J	F	H
(E)	T	F	Q	W	J	R	H

19. Any of the following can appear on page 3 EXCEPT

- (A) J
- (B) Q
- (C) R
- (D) T
- (E) W

20. If F appears on page 6, H must appear on page

- (A) 1
- (B) 2
- (C) 3
- (D) 5
- (E) 7

21. If the still lifes all appear on consecutive pages, which of the following must be true?

- (A) A still life appears on page 1.
- (B) A still life appears on page 5.
- (C) J appears on page 6.
- (D) T appears on page 4.
- (E) W appears on page 5

22. If F and H, not necessarily in that order, appear on consecutive pages, which of the following can be true?

- (A) J appears on page 1
- (B) J appears on page 5
- (C) R appears on page 6
- (D) W appears on page 1
- (E) W appears on page 5

23. The organizers of tomorrow's outdoor concert announced that it will go on tomorrow on schedule unless bad weather is forecast or too few advance tickets are sold. If the concert is canceled, refunds will be made to ticket holders. Since some ticket holders have already been issued refunds even though more than enough advance tickets were sold, it must be the case that bad weather is forecast.

Which of the following is an error of reasoning contained in the argument?

- (A) It proceeds as if a condition, which by itself is enough to guarantee a certain result, is the only condition under which that result would occur.
- (B) It bases a conclusion that is known to require two conditions on evidence that bears on only one of those conditions.
- (C) It explains one event as being caused by another event, even though both events must actually

have been caused by some third, unidentified event.

- (D) It treats evidence for the absence of one condition under which a circumstance would occur as conclusive evidence that that circumstance will not occur.
- (E) Evidence given to support the conclusion actually undermines it.

24. Although the prevailing supposition has been that it is too hot for microorganisms to survive deep below the Earth's surface, some scientists argue that there are living communities of microorganisms there that have been cut off from surface life for millions of years. These scientists base their argument on the discovery of living microorganisms in samples of material that were taken from holes drilled as deep as 1.74 miles.

The scientists' argument depends on which of the following assumptions?

- (A) The microorganisms brought up were of a species that is related to those previously known to science.
- (B) No holes have been drilled into the Earth's surface to a distance deeper than 1.74 miles
- (C) The microorganisms did not come from surface soil that came into contact with the drilling equipment.
- (D) The stratum from which the samples came has been below the surface of the Earth ever since the Earth came into existence.
- (E) The temperature at the bottom of the holes drilled was not significantly hotter than that of the hottest spots on the Earth's surface.

25. For 20 years all applicants for jobs as technicians at EquipCorp were required to demonstrate that they could operate and repair the machinery that was central to EquipCorp's manufacturing business. Now, however, that particular machinery is obsolete, and very different machinery fills the central role. Therefore, the old requirement is no longer a useful method for evaluating whether applicants for jobs as technicians at EquipCorp have the skills necessary for the job.

Which of the following, if true, most strengthens the

argument?

- (A) The machinery that is now obsolete was used by a large number of manufacturing companies before it became obsolete.
- (B) Among the people already holding jobs as technicians at Equip Corp, those who are most skillful at operating the new machinery had been some of the least skillful at operating the old machinery
- (C) Most people applying for jobs as technicians today have much broader skills than did people applying for jobs as technicians 20 years ago.
- (D) The skills required to operate and repair the obsolete machinery are useful in operating and maintaining many other types of machinery at EquipCorp that are not obsolete.
- (E) Much of the machinery that EquipCorp now uses in manufacturing is very likely to become obsolete within the next 20 years.